Chapter 1

Functional Integrals

Contents

1.1 Introduction . . . 5
1.1.1 Path integral in quantum mechanics . . . 5
1.1.2 Functional integral in statistical physics . . . 19
1.2 Second quantization and Green functions . . . 26
1.2.1 Fock space . . . 26
1.2.2 Operators in second-quantized form . . . 31
1.2.3 Green functions . . . 35
1.3 Coherent states . . . 39
1.3.1 Boson coherent states . . . 40
1.3.2 Grassmann variables . . . 43
1.3.3 Fermion coherent states . . . 44
1.4 Functional integral in many-particle systems . . . 46
1.4.1 Coherent-state functional integral . . . 47
1.4.2 Non-interacting particles . . . 49
1.5 Perturbation theory and Feynman diagrams . . . 54
1.5.1 Wick's theorem . . . 54
1.5.2 Partition function and thermodynamic potential . . . 56
1.5.3 Green functions . . . 61
1.5.4 (Anti)symmetrized vertices . . . 65
1.6 Generating functionals . . . 71
1.6.1 Connected Green functions . . . 72
1.6.2 Effective action and 1PI vertices . . . 76
1.6.3 Luttinger-Ward functional and 2PI vertices . . . 86
1.7 Saddle-point approximation and loop expansion . . . 94
1.7.1 Steepest-descent and stationary-phase approximations . . . 94
1.7.2 Saddle-point approximation and loop expansion in field theory . . . 96
1.8 Perturbation theory at zero temperature . . . 105
1.8.1 Fermions . . . 106
1.8.2 Bosons . . . 109

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1.8.3 The zero-temperature limit . . . 111
1.9 Quantization of the electromagnetic field . . . 112
1.9.1 Lagrangian and Hamiltonian formalism . . . 112
1.9.2 Canonical quantization in the Coulomb gauge . . . 116
1.9.3 Functional integral in the Coulomb gauge . . . 118
Appendix 1.A Review of classical mechanics . . . 121
1.A.1 Lagrangian formalism . . . 122
1.A.2 Hamiltonian formalism . . . 124
1.A.3 Classical field theory . . . 127
Appendix 1.B Review of quantum mechanics . . . 132
1.B.1 The postulates of quantum mechanics . . . 132
1.B.2 Quantum fields . . . 136
Appendix 1.C Review of statistical physics . . . 137
1.C.1 Macrostates . . . 137
1.C.2 Statistical entropy – Boltzmann-Gibbs distribution . . . 139
1.C.3 Statistical ensembles . . . 142
1.C.4 The classical limit . . . 149
1.C.5 Non-interacting quantum gases . . . 151
Appendix 1.D Functionals and functional derivatives . . . 153
1.D.1 Grassmann variables . . . 155
Appendix 1.E Gaussian integrals and Wick's theorem . . . 155
1.E.1 Real variables . . . 155
1.E.2 Complex variables . . . 157
1.E.3 Grassmannian integrals . . . 158
1.E.4 Gaussian functional integrals . . . 161
Appendix 1.F Matsubara frequency sums . . . 162
Appendix 1.G Perturbation theory in the operator formalism . . . 167
1.G.1 Zero-temperature formalism . . . 167
1.G.2 Finite-temperature formalism . . . 171
Bibliography . . . 173

This chapter aims at introducing the general framework for the study of macroscopic quantum systems. The formalism is based on the functional integral, an extension of Feynman's path integral to systems with an infinite number of degrees of freedom. It enables to derive standard results, such as perturbation expansions, in an economical way and set up non-perturbative approaches. It also provides a unified view of many concepts and theoretical methods that can be found in various fields of physics: condensed-matter physics, quantum optics, high-energy physics, etc.

We first derive the functional integral representation of the partition function of a system of interacting quantum particles (bosons or fermions) (Secs. 1.2-1.4). This representation is based on second-quantized fields and coherent states. We discuss the perturbative calculation of the partition function and the correlation functions (Green functions), and its representation in terms of Feynman diagrams (Secs. 1.5 and 1.6). These diagrams provide an intuitive picture of perturbation theory in terms of elementary interaction processes taking place in the system. They are also a very efficient tool to partially resum the perturbation series to infinite order. We show how the perturbation expansion may be organized about a broken-symmetry state when a symmetry is spontaneously broken (Sec. 1.7). We also

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discuss the quantization of the electromagnetic field in the functional integral formalism (Sec. 1.9).[1]

Although Feynman diagrams play an essential role in many-body physics, both as a pictorial representation of perturbation expansions and an efficient computational tool, they also suffer serious drawbacks. On the one hand they often lead to uncontrolled approximations. On the other hand diagrammatic resummations are usually restricted to systems with weak interactions. In many cases, however, one is interested in strongly correlated systems where the interactions are strong and perturbative expansions likely to fail. We shall see how the generating functionals of one-particle and two-particle-irreducible vertices introduced in Sec. 1.6 can be used to set up non-perturbative approaches. The effective action (Sec. 1.6.2) is at the basis of the formulation of the non-perturbative functional renormalization group (RG) approach (Chap. 11) while the Luttinger-Ward functional (Sec. 1.6.3) is the starting point of the dynamical mean-field theory (DMFT) and various cluster expansions (Chap. 9).

This chapter is somewhat formal but constitutes the backbone of the book. The prerequisite knowledge of classical, quantum and statistical mechanics is summarized in Appendices 1.A-1.B.2. Other Appendices provide technical details on functional calculus, Gaussian integrals and Matsubara sums as well as a discussion of perturbation theory in the operator formalism.

1.1 Introduction

Elementary quantum mechanics is usually formulated in the framework of canonical quantization where observables of the classical theory become quantum mechanical operators acting on state vectors in a Hilbert space (Appendix 1.B). In this introductory section, we discuss an alternative formulation of quantum mechanics based on the Feynman path integral. As in classical mechanics, the concept of trajectories (or paths) plays a central role and the Feynman path integral yields a very intuitive picture of quantum mechanics. For a single quantum particle or a system with few degrees of freedom, there is no real advantage to use the path integral formalism, except for pedagogical purposes, the usual approach based on the Schrödinger equation being usually more efficient. However, for systems with a large number of degrees of freedom and in particular in statistical physics, the path integral formalism – or, more precisely, the functional integral formalism – is often more appropriate.

We start the discussion with the study of a quantum particle moving in a static potential (Sec. 1.1.1). Then we consider the functional integral formalism for the quantum harmonic string (Sec. 1.1.2). This example already exhibits many key aspects of the functional integral formalism of quantum many-particle systems that will be introduced in the following sections and used throughout the book.

1.1.1 Path integral in quantum mechanics

To introduce the concept of path integral in quantum mechanics, we consider a particle of mass m in a one-dimensional space. Its classical dynamics is governed by the Lagrangian

$$L(q,\dot{q}) = \frac{1}{2}m\dot{q}^2 - V(q), \tag{1.1}$$

[1] The coherent-state functional integral for quantum spin systems is discussed in Sec. 8.4.

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assuming that the particle moves in a time-independent potential $V(q)$. q denotes the position of the particle and $\dot{q} = \partial_t q$. The trajectory of the particle is obtained from the Euler-Lagrange equation (Appendix 1.A)

$$\frac{d}{dt}\frac{\partial L(q,\dot{q})}{\partial \dot{q}} = \frac{\partial L(q,\dot{q})}{\partial q} \quad \text{i.e.} \quad m\ddot{q} = -\frac{\partial V(q)}{\partial q}. \tag{1.2}$$

Alternatively, one can consider the Hamiltonian $H(p,q)$ defined by

$$\begin{aligned} p &= \frac{\partial L(q,\dot{q})}{\partial \dot{q}} = m\dot{q}, \\ H(p,q) &= p\dot{q} - L(q,\dot{q}) = \frac{p^2}{2m} + V(q), \end{aligned} \tag{1.3}$$

where p is the momentum canonically conjugate to the variable q. The classical dynamics of the particle then follows from Hamilton's equations of motion,

$$\begin{aligned} \dot{p} &= -\frac{\partial H(p,q)}{\partial q} = -\frac{\partial V(q)}{\partial q}, \\ \dot{q} &= \frac{\partial H(p,q)}{\partial p} = \frac{p}{m}. \end{aligned} \tag{1.4}$$

1.1.1.1 Canonical quantization

Canonical quantization is achieved by promoting the canonical variables p and q of the classical theory to operators $\hat{p}$ and $\hat{q}$ satisfying the commutation relations[2]

$$[\hat{q},\hat{p}] = i\hbar. \tag{1.5}$$

A classical variable $A(p,q)$ becomes an operator $\hat{A} \equiv A(\hat{p},\hat{q})$. In particular, the quantum Hamiltonian reads

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{q}). \tag{1.6}$$

A physical state of the system is described by a state vector $|\psi(t)\rangle$ of a Hilbert space $\mathcal{H}$. The time evolution of $|\psi(t)\rangle$ is given by the Schrödinger equation

$$i\hbar\partial_t|\psi(t)\rangle = \hat{H}|\psi(t)\rangle, \tag{1.7}$$

the solution of which can be written in terms of the time-evolution operator $\hat{U}$,

$$|\psi(t)\rangle = \hat{U}(t)|\psi(t=0)\rangle, \qquad i\hbar\partial_t\hat{U}(t) = \hat{H}\hat{U}(t). \tag{1.8}$$

The Hamiltonian being time independent, one simply has

$$\hat{U}(t) = e^{-\frac{i}{\hbar}\hat{H}t}. \tag{1.9}$$

The probability amplitude to find the particle at position q_f at time t_f is given by

$$\psi(q_f,t_f) = \langle q_f|\psi(t_f)\rangle = \langle q_f|\hat{U}(t_f-t_i)|\psi(t_i)\rangle = \int dq_i\, U(q_f,q_i;t_f-t_i)\psi(q_i,t_i), \tag{1.10}$$

[2] See Appendix 1.B.1.4.

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where the ket $|q\rangle$ is defined as the eigenstate of the position operator $\hat{q}$: $\hat{q}|q\rangle = q|q\rangle$. $U(q_f, q_i; t_f - t_i) = \langle q_f|\hat{U}(t_f - t_i)|q_i\rangle$ is referred to as the propagator, since it expresses the probability amplitude for the particle to propagate from point q_i to point q_f in a time $t_f - t_i$. If we know a basis $\{|n\rangle, \epsilon_n\}$ of eigenstates of the Hamiltonian $\hat{H}$, we can write the propagator as

$$\begin{aligned} U(q_f, q_i; t_f - t_i) &= \langle q_f|e^{-\frac{i}{\hbar}\hat{H}(t_f-t_i)}|q_i\rangle = \sum_n \langle q_f|n\rangle e^{-\frac{i}{\hbar}\epsilon_n(t_f-t_i)}\langle n|q_i\rangle \\ &= \sum_n e^{-\frac{i}{\hbar}\epsilon_n(t_f-t_i)}\varphi_n(q_f)\varphi_n^*(q_i) \end{aligned} \tag{1.11}$$

(using the closure relation $\sum_n |n\rangle\langle n| = 1$), where $\varphi_n(q) = \langle q|n\rangle$ denotes the wavefunction in the coordinate representation. $U(q_f, q_i; t_f - t_i)$ provides information about wavefunctions and energy levels of the Hamiltonian.

1.1.1.2 Path integral

The propagator $U(q_f, q_i; t)$ can also be represented as a "path integral". We first consider an infinitesimal time ϵ. To leading order in ϵ, one has[3]

$$U(q_f, q_i; \epsilon) = \langle q_f|e^{-i\hat{H}\epsilon}|q_i\rangle \simeq \langle q_f|e^{-i\epsilon\frac{\hat{p}^2}{2m}}e^{-i\epsilon V(\hat{q})}|q_i\rangle \tag{1.12}$$

(we now set $\hbar = 1$). The propagator is then calculated by inserting the closure relation $\sum_p |p\rangle\langle p| = 1$,[4]

$$\begin{aligned} U(q_f, q_i; \epsilon) &= \sum_p \langle q_f|e^{-i\epsilon\frac{\hat{p}^2}{2m}}|p\rangle\langle p|e^{-i\epsilon V(\hat{q})}|q_i\rangle \\ &= \int \frac{dp}{2\pi} \exp\left\{-i\epsilon\left[\frac{p^2}{2m} + V(q_i)\right] + ip(q_f - q_i)\right\} \\ &= \left(\frac{m}{2\pi i\epsilon}\right)^{1/2} \exp\left\{i\epsilon\left[\frac{m}{2}\frac{(q_f - q_i)^2}{\epsilon^2} - V(q_i)\right]\right\}. \end{aligned} \tag{1.13}$$

In order for the integral over p to converge, we assume ϵ to contain a small negative imaginary part. The argument in the exponential is nothing but i times the infinitesimal action $S(q_f, q_i; \epsilon)$ corresponding to the straight line trajectory (at constant velocity) between q_i and q_f in the infinitesimal time ϵ,

$$U(q_f, q_i; \epsilon) = \left(\frac{m}{2\pi i\epsilon}\right)^{1/2} e^{iS(q_f,q_i;\epsilon)+\mathcal{O}(\epsilon^2)} \tag{1.14}$$

(writing explicitly that the error in the exponential is of order ϵ^2).

[3] This follows from $e^{\epsilon\hat{A}+\epsilon\hat{B}} = 1 + \epsilon(\hat{A} + \hat{B}) + \mathcal{O}(\epsilon^2) = e^{\epsilon\hat{A}}e^{\epsilon\hat{B}}e^{\mathcal{O}(\epsilon^2)}$ where the $\mathcal{O}(\epsilon^2)$ term is given by the commutator $-\frac{\epsilon^2}{2}[\hat{A}, \hat{B}]$ (Baker-Hausdorff formula).

[4] We use the normalization conventions $\langle q|p\rangle = L^{-1/2}e^{ipq}$, $\int dq|q\rangle\langle q| = 1$, and $\sum_p |p\rangle\langle p| = 1$. $q \in [0, L]$ is a continuous position variable and $p = n\frac{2\pi}{L}$ ($n \in \mathbb{Z}$) a discrete momentum variable corresponding to periodic boundary conditions ($e^{ipL} = 1$). In the thermodynamic limit $L \to \infty$, one has $\frac{1}{L}\sum_p \to \int_{-\infty}^{\infty}\frac{dp}{2\pi}$ (Appendix 1.B).

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To calculate the propagator $U(q_f, q_i; t_f - t_i)$ for an arbitrary time $t_f - t_i$, we split $t_f - t_i$ into N equal steps of size $\epsilon = (t_f - t_i)/N$; we shall eventually take the limit $N \to \infty$ ($\epsilon \to 0$). Inserting the closure relation $\int dq |q\rangle\langle q| = 1$ at each intermediate time, we obtain

$$\begin{aligned} U(q_f, q_i; t_f - t_i) &= \langle q_f | e^{-i\hat{H}\epsilon} \cdots e^{-i\hat{H}\epsilon} | q_i \rangle \\ &= \int \prod_{k=1}^{N-1} dq_k \langle q_f | e^{-i\hat{H}\epsilon} | q_{N-1} \rangle \langle q_{N-1} | e^{-i\hat{H}\epsilon} | q_{N-2} \rangle \cdots \langle q_1 | e^{-i\hat{H}\epsilon} | q_i \rangle \\ &= \int \prod_{k=1}^{N-1} dq_k \prod_{k=1}^{N} U(q_k, q_{k-1}; \epsilon), \end{aligned} \tag{1.15}$$

where $q_0 = q_i$ and $q_N = q_f$. The expression (1.14) introduces an error of order ϵ^2 at each time step, i.e. a total error of order ϵ. In the limit $N \to \infty$, one therefore finds[5]

$$U(q_f, q_i; t) = \lim_{N\to\infty} \left(\frac{mN}{2\pi i t} \right)^{N/2} \int \prod_{k=1}^{N-1} dq_k \, e^{iS[q]}, \tag{1.16}$$

where

$$S[q] = \sum_{k=1}^{N} S(q_k, q_{k-1}; \epsilon) = \epsilon \sum_{k=1}^{N} \left[\frac{m}{2} \frac{(q_k - q_{k-1})^2}{\epsilon^2} - V(q_{k-1}) \right] \tag{1.17}$$

is the action associated with the discrete trajectory $(q_0, q_1, \cdots, q_N)$ with $q_0 = q_i$ and $q_N = q_f$ (Fig. 1.1). In the limit $N \to \infty$, we may write symbolically

$$\begin{aligned} \epsilon \sum_{k=1}^{N} \frac{m}{2} \frac{(q_k - q_{k-1})^2}{\epsilon^2} &\to \int_{t_i}^{t_f} dt \frac{m}{2} \dot{q}^2, \\ \epsilon \sum_{k=1}^{N} V(q_{k-1}) &\to \int_{t_i}^{t_f} dt \, V(q), \end{aligned} \tag{1.18}$$

and denote by $q(t)$ the "trajectory" $(q_0, \cdots, q_N)$ with $q(t_i) = q_i$ and $q(t_f) = q_f$. This notation does not imply continuity or differentiability of the trajectory $(q_0, q_1, \cdots, q_N)$ in the limit $N \to \infty$. If we define the integration measure as

$$\mathcal{D}[q] = \lim_{N\to\infty} \left(\frac{mN}{2\pi i \hbar t} \right)^{N/2} \prod_{k=1}^{N-1} dq_k, \tag{1.19}$$

then the propagator reads

$$U(q_f, q_i; t_f - t_i) = \int_{q(t_i)=q_i}^{q(t_f)=q_f} \mathcal{D}[q] \, e^{\frac{i}{\hbar} S[q]}, \tag{1.20}$$

where

$$S[q] = \int_{t_i}^{t_f} dt \, L(q, \dot{q}) \tag{1.21}$$

[5] The extension of (1.16,1.17) to three dimensions is straightforward. The variable q becomes a three-dimensional vector $\mathbf{q}$ and the factor $(mN/2\pi i t)^{N/2}$ must be replaced by $(mN/2\pi i t)^{3N/2}$.

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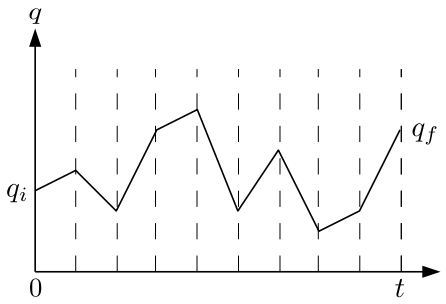


Figure 1.1: A typical discrete trajectory contributing to the action $S[q]$.

is the classical action corresponding to the trajectory $q(t)$. Note that the integration measure $\mathcal{D}[q]$ contains a factor that diverges in the limit $N \to \infty$. In (1.20), we have momentarily restored Planck's constant $\hbar$ since it plays an essential role in the discussion of the classical limit (see below). Thus the matrix element of the evolution operator $\hat{U}(t_f - t_i)$ can be represented as a sum over all possible paths going from $q(t_i) = q_i$ to $q(t_f) = q_f$ and weighted by the classical action $S[q]$ times $i/\hbar$. The Feynman path integral (1.20) provides a nice illustration of the superposition principle of quantum mechanics: When a process can take place in more than one way, its probability amplitude is given by the sum of the amplitudes for each way. Although no explicit reference to quantum mechanical operators is made, the Feynman path integral is an exact representation of the evolution operator $\hat{U}(t) = e^{-i\hat{H}t}$ and may be used as the starting point for the formulation of quantum mechanics [2].

The kinetic term selects sufficiently regular trajectories such that $[q(t+\epsilon) - q(t)]^2/\epsilon$ remains finite in the limit $\epsilon \to 0$. When this condition is not fulfilled, the kinetic energy is large and the exponential factor $e^{\frac{i}{\hbar}S[q]}$ strongly oscillates when considering nearby trajectories and therefore averages to zero. Thus typical paths are continuous but not necessarily differentiable, and reminiscent of Brownian motion.[6] Since there is no reason for $(q_{k+1} - q_k)/\epsilon$ to remain finite in the limit $\epsilon \to 0$, the interpretation of $\dot{q}$ may appear delicate. We show below that for the free particle and the harmonic oscillator, the path integral can be computed from the sum over discrete trajectories (taking the limit $N \to \infty$ at the end of the calculation) or using "blindly" the continuous-time formulation.[7]

1.1.1.3 Classical limit

The important trajectories are those for which the action varies weakly when the path $q(t)$ is slightly deformed, i.e. those which are near the classical trajectory $q_c(t)$ whose action is stationary,

$$\left.\frac{\delta S[q]}{\delta q(t)}\right|_{q=q_c} = 0. \tag{1.22}$$

Non-stationary trajectories imply large oscillations of the action and therefore average to zero (the Feynman paths interfere destructively). More precisely, the propagator $U(q_f, q_i; t_f -$

[6] For a detailed discussion of the relation between path integrals and Brownian motion, see Ref. [3].

[7] We shall see that except for Gaussian actions, we rarely compute path or functional integrals, but rather the ratio I/I_0 between the (path or functional) integral of interest and a reference (path or functional) integral whose calculation is trivial. For instance, in perturbation theory, the ratio I/I_0 is given by Feynman diagrams whose calculation follows well established rules and has nothing to do with the actual computation of the functional integral.

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t_i) is dominated by the trajectories $q(t)$ whose action $S[q]$ differs from the classical action $S_c = S[q_c]$ by a term of order $\hbar$: $|S - S_c| \lesssim \hbar$. When $|S_c| \gg \hbar$, these trajectories are very close to the classical trajectory. In the opposite limit, the condition $|S - S_c| \lesssim \hbar$ is fulfilled by trajectories very different from the classical one. Formally, the classical limit corresponds to the limit $\hbar \to 0$.

To obtain the propagator in the limit $\hbar \to 0$, we write $q(t) = q_c(t) + r(t)$ (assuming there is only one classical trajectory) and expand the action to second order in $r(t)$,

$$U(q_f, q_i; t_f - t_i) \simeq e^{\frac{i}{\hbar}S[q_c]} \int_{r(t_i)=0}^{r(t_f)=0} \mathcal{D}[r] \exp\left\{ \frac{i}{2\hbar} \int_{t_i}^{t_f} dt\, dt' \left. \frac{\delta^2 S[q]}{\delta q(t)\delta q(t')} \right|_{q=q_c} r(t) r(t') \right\}. \quad (1.23)$$

The integral is Gaussian and can be done exactly (Appendix 1.E),

$$U(q_f, q_i; t_f - t_i) \simeq e^{\frac{i}{\hbar}S[q_c]} \det \left(\frac{1}{2\pi i\hbar} \left. \frac{\delta^2 S[q]}{\delta q(t)\delta q(t')} \right|_{q=q_c} \right)^{-1/2}. \quad (1.24)$$

The procedure we have followed to obtain (1.24) is known as the stationary-phase approximation. It gives the leading term of an expansion in powers of $\hbar$ that can be systematically carried out to higher orders (see Sec. 1.7).

1.1.1.4 Time-ordered product of operators

It is also possible to consider matrix elements including the operator $\hat{q}(t) = e^{i\hat{H}t}\hat{q}e^{-i\hat{H}t}$ (in the Heisenberg representation),[8]

$$\langle q_f, t_f | \hat{q}(t) | q_i, t_i \rangle = \langle q_f | e^{-i\hat{H}(t_f - t)} \hat{q} e^{-i\hat{H}(t - t_i)} | q_i \rangle, \quad (1.25)$$

where $|q, t\rangle = e^{i\hat{H}t}|q\rangle$ corresponds to the state $|q\rangle$ in the Heisenberg picture and satisfies $\hat{q}(t)|q(t)\rangle = q|q(t)\rangle$. By splitting the time intervals $t_f - t$ and $t - t_i$ into infinitesimal time steps, one easily obtains the following path integral representation

$$\langle q_f, t_f | \hat{q}(t) | q_i, t_i \rangle = \int_{q(t_i)=q_i}^{q(t_f)=q_f} \mathcal{D}[q]\, q(t) e^{iS[q]}. \quad (1.26)$$

Products of operators at different times are also represented by a path integral. For example[9]

$$\langle q_f, t_f | T\hat{q}(t)\hat{q}(t') | q_i, t_i \rangle = \int_{q(t_i)=q_i}^{q(t_f)=q_f} \mathcal{D}[q]\, q(t) q(t') e^{iS[q]}, \quad (1.27)$$

where T is the time-ordering operator defined by

$$T\hat{q}(t)\hat{q}(t') = \Theta(t - t')\hat{q}(t)\hat{q}(t') + \Theta(t' - t)\hat{q}(t')\hat{q}(t). \quad (1.28)$$

Path integrals naturally represent expectation values of time-ordered products of operators.

[8] More generally, one could consider the matrix element $\langle q_f, t_f | \hat{A}(t) | q_i, t_i \rangle$ with $\hat{A}$ any operator.
[9] Equation (1.27) is easily proven by considering the two cases $t > t'$ and $t < t'$.

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1.1.1.5 Hamiltonian form of the path integral

Equations (1.20) and (1.21) define the Lagrangian form of the path integral, where the action is expressed in terms of the Lagrangian and the functional integral involves a sum over the paths $q(t)$. Alternatively, one can keep explicitly the variable p appearing in (1.13) and write the propagator as

$$\begin{aligned} U(q_f,q_i;t_f-t_i) &= \lim_{N\to\infty}\int\prod_{k=1}^{N-1}dq_k\int\prod_{k=1}^{N}\frac{dp_k}{2\pi}e^{\sum_{k=1}^{N}\left[ip_k(q_k-q_{k-1})-i\epsilon\frac{p_k^2}{2m}-i\epsilon V(q_{k-1})\right]} \\ &\equiv \int_{q(t_i)=q_i}^{q(t_f)=q_f}\mathcal{D}[p,q]\,e^{iS[p,q]}, \end{aligned} \tag{1.29}$$

where

$$\begin{aligned} \mathcal{D}[p,q] &= \lim_{N\to\infty}\prod_{k=1}^{N-1}dq_k\prod_{k=1}^{N}\frac{dp_k}{2\pi}, \\ S[p,q] &= \lim_{N\to\infty}\sum_{k=1}^{N}\Big[p_k(q_k-q_{k-1})-\epsilon\frac{p_k^2}{2m}-\epsilon V(q_{k-1})\Big] \\ &\equiv \int_{t_i}^{t_f}dt\,[p\dot{q}-H(p,q)]. \end{aligned} \tag{1.30}$$

Equations (1.29) and (1.30) give the Hamiltonian form of the path integral, where the action $S[p,q]$ is a function of the position q as well as the momentum p of the particle. Note that q and p are independent variables in the functional integral (1.29) and only q is fixed at the initial (t_i) and final (t_f) times. When the particle travels from q_{k-1} to q_k, its momentum is given by p_k (which is independent of the velocity $(q_k-q_{k-1})/\epsilon$) so that the path is not a classical trajectory. The interpretation of the propagator as a sum over classical paths is therefore lost in the Hamiltonian form of the path integral. Nevertheless, for $\hbar\to 0$, the path integral is dominated by the trajectories satisfying

$$\frac{\delta S}{\delta p(t)}=\frac{\delta S}{\delta q(t)}=0, \tag{1.31}$$

and one recovers Hamilton's equations of motion.

1.1.1.6 Euclidean path integral

Suppose we want to calculate the partition function

$$Z=\mathrm{Tr}\,e^{-\beta\hat{H}}=\int dq\langle q|e^{-\beta\hat{H}}|q\rangle \tag{1.32}$$

of the quantum particle considered above. Z can be obtained from the matrix elements of the evolution operator $e^{-i\hat{H}t}$ evaluated at the imaginary time $t=-i\beta$ (or $t=-i\beta\hbar$ if we restore $\hbar$). Real-time dynamics and quantum statistical mechanics are thus related by the transformation $t\to-i\tau$ known as a Wick rotation, i.e. a rotation of angle $-\pi/2$ in the complex time plane. The concept of imaginary time plays an essential role in quantum statistical physics. The propagator in imaginary time, $U(q_f,q_i;-i\tau)=\langle q_f|e^{-\hat{H}\tau}|q_i\rangle$ is given

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by (1.11) with the replacement $t \to -i\tau$. It can be represented as a path integral following the same steps as before,[10]

$$U(q_f, q_i; -i\tau) = \lim_{N\to\infty} \left(\frac{mN}{2\pi\tau}\right)^{N/2} \int \prod_{k=1}^{N-1} dq_k \, e^{-\epsilon \sum_{k=1}^{N} \left[\frac{m}{2}\frac{(q_k - q_{k-1})^2}{\epsilon^2} + V(q_{k-1})\right]}$$
$$= \int_{q(0)=q_i}^{q(\tau)=q_f} \mathcal{D}[q] \, e^{-S_E[q]}. \tag{1.33}$$

The imaginary-time path integral is defined by a sum over trajectories from $q(0) = q_i$ to $q(\tau) = q_f$ weighted by the action

$$S_E[q] = \int_0^\tau d\tau' \left[\frac{m}{2}\dot{q}^2 + V(q)\right]. \tag{1.34}$$

$S_E[q]$ is referred to as the Euclidean action since (imaginary) time and space enter the action in a symmetric way.[11,12,13] The kinetic and potential terms have the same sign in the Euclidean Lagrangian, since the kinetic term changes sign under the Wick rotation $\tau = it$. As in the real-time formalism, the kinetic term selects trajectories such that $[q(\tau + \epsilon) - q(\tau)]^2/\epsilon$ remains finite in the limit $\epsilon \to 0$ – paths that do not fulfill this condition are exponentially suppressed – so that the paths contributing to the propagator are continuous (but not necessarily differentiable). The real-time action (1.21) can be recovered from S_E performing a Wick rotation,

$$S_E[q] \xrightarrow[\substack{\text{Wick rotation}\\(\tau = it)}]{} i \int_0^t dt' \left[-\frac{m}{2}\dot{q}^2 + V(q)\right] = -iS[q]. \tag{1.35}$$

Using the preceding results, we can express the partition function of the quantum particle,

$$Z = \int dq \, U(q, q, -i\beta) = \int_{q(\beta)=q(0)} \mathcal{D}[q] \, e^{-S_E[q]}, \tag{1.36}$$

as a path integral in imaginary time over all periodic trajectories of period β. Similarly, we can write the expectation value of the imaginary-time operator $\hat{q}(\tau) = e^{\tau\hat{H}}\hat{q}e^{-\tau\hat{H}}$ as

$$\langle\hat{q}(\tau)\rangle = \frac{1}{Z}\mathrm{Tr}\left[e^{-\beta\hat{H}}\hat{q}(\tau)\right] = \frac{1}{Z}\int_{q(\beta)=q(0)} \mathcal{D}[q] \, q(\tau) e^{-S_E[q]}. \tag{1.37}$$

For the time-ordered product $\langle T_\tau \hat{q}(\tau)\hat{q}(\tau')\rangle$, we obtain

$$\langle T_\tau \hat{q}(\tau)\hat{q}(\tau')\rangle = \frac{1}{Z}\int_{q(\beta)=q(0)} \mathcal{D}[q] \, q(\tau)q(\tau') e^{-S_E[q]}. \tag{1.38}$$

The imaginary-time-ordering operator T_τ is defined analogously to its real-time counterpart [Eq. (1.28)].

[10] Contrary to the real-time propagator, there is no need here to add a small imaginary part to the infinitesimal time ϵ (see the remark after (1.13)). From a mathematical point of view, the Euclidean formalism is preferable to the real-time one.

[11] The Minkowski metric $ds^2 = dt^2 - dx^2$ becomes $-d\tau^2 - dx^2$ in the Euclidean formalism.

[12] The imaginary time is sometimes referred to as the Euclidean time.

[13] The Euclidean action will often be the starting point for the description of quantum many-body systems; in quantum mechanics it is used for the study of tunneling.

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If $\beta \to 0$ (i.e. $T \to \infty$) or $\hbar \to 0$, the propagator $U(q,q;-i\beta\hbar)$ can be calculated with a single time step $N=1$ (we restore $\hbar$ in this paragraph). This leads to

$$\begin{aligned} Z_{\rm cl} &= \frac{1}{\hbar}\sqrt{\frac{m}{2\pi\beta}}\int dq\, e^{-\beta V(q)} \\ &\equiv \int_{-\infty}^{\infty}\frac{dp}{2\pi\hbar}\int_{-\infty}^{\infty} dq\, \exp\biggl\{-\beta\biggl[\frac{p^2}{2m}+V(q)\biggr]\biggr\}, \end{aligned} \tag{1.39}$$

which is nothing but the classical partition function. At finite temperature $T<\infty$, the classical approximation (1.39) is valid if the potential $V(q)$ is slowly varying on a length scale of the order of the de Broglie wavelength $\xi_{\rm th} \sim \hbar/\sqrt{mk_BT}$. In Sec. 1.1.2, we shall further discuss the connection between high-temperature limit and classical behavior of a quantum system.

Equation (1.39) shows a very important property of classical systems: Thermodynamics and dynamics decouple. The particle's position and momentum being independent variables, one can integrate out the momentum (which yields an additive contribution to the free energy) and write the partition function in terms of the position variable only. By contrast, in quantum systems coordinate and momentum variables are non-commuting operators so that statics and dynamics are not independent.[14] This is the reason why the partition function can be related to the evolution operator in imaginary time [Eq. (1.36)].

Example: free particle and harmonic oscillator. Let us first consider the propagator of a free particle,

$$U(q_f,q_i;t) = \lim_{N\to\infty}\left(\frac{mN}{2\pi i t}\right)^{N/2}\int\prod_{k=1}^{N-1} dq_k \exp\biggl\{i\frac{m}{2\epsilon}\sum_{k=1}^{N}(q_k-q_{k-1})^2\biggr\}, \tag{1.40}$$

where $q_0=q_i$ and $q_N=q_f$. One can successively integrate out q_1, q_2, etc. The integral over q_1 gives

$$\begin{aligned} U(q_f,q_i;t) = \frac{1}{\sqrt{2}}\lim_{N\to\infty}&\left(\frac{mN}{2\pi i t}\right)^{(N-1)/2} \\ &\times\int\prod_{k=2}^{N-1} dq_k \exp\biggl\{i\frac{m}{4\epsilon}(q_2-q_i)^2+i\frac{m}{2\epsilon}\sum_{k=3}^{N}(q_k-q_{k-1})^2\biggr\}. \end{aligned} \tag{1.41}$$

Integrating then over q_2 yields

$$\begin{aligned} U(q_f,q_i;t) = \frac{1}{\sqrt{3}}\lim_{N\to\infty}&\left(\frac{mN}{2\pi i t}\right)^{(N-2)/2} \\ &\times\int\prod_{k=3}^{N-1} dq_k \exp\biggl\{i\frac{m}{6\epsilon}(q_3-q_i)^2+i\frac{m}{2\epsilon}\sum_{k=4}^{N}(q_k-q_{k-1})^2\biggr\}, \end{aligned} \tag{1.42}$$

and we easily deduce the final result

$$U(q_f,q_i;t) = \left(\frac{m}{2\pi i t}\right)^{1/2}\exp\left(i\frac{m(q_f-q_i)^2}{2t}\right). \tag{1.43}$$

We recognize in the exponential i times the classical action: The classical path gives the exact propagator for a free particle. The result (1.43) can also be directly obtained

[14] Heuristically, the existence of $\hbar$ implies that any energy scale E that enters thermodynamics necessarily determines a time scale $\hbar/E$.

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from the definition (1.11) of the propagator using the eigenstates $\varphi_p(q) = L^{-1/2}e^{ipq}$ with $\epsilon_p = p^2/2m$.

Let us now consider the harmonic oscillator action

$$S[q] = \frac{m}{2}\int_{t_i}^{t_f} dt\,(\dot{q}^2 - \omega^2 q^2). \tag{1.44}$$

Although it is possible to do the calculation of the evolution operator in discrete time (see e.g. Ref. [5]), we take the continuum-time limit. (The discrete time calculation is carried out below for the partition function.) To compute the propagator, we write $q(t) = q_c(t)+r(t)$. The classical trajectory q_c satisfies the equation of motion $\ddot{q}+\omega^2 q = 0$ and reads

$$q_c(t) = A\cos(\omega t) + B\sin(\omega t) \tag{1.45}$$

with the boundary conditions

$$q_i = A\cos(\omega t_i) + B\sin(\omega t_i) \quad \text{and} \quad q_f = A\cos(\omega t_f) + B\sin(\omega t_f), \tag{1.46}$$

while the classical action is given by

$$S[q_c] = \frac{m\omega}{2\sin(\omega(t_f - t_i))}\left[(q_f^2 + q_i^2)\cos\big(\omega(t_f - t_i)\big) - 2q_i q_f\right]. \tag{1.47}$$

Since

$$S[q] = S[q_c] + \frac{m}{2}\int_{t_i}^{t_f} dt\,\left(\dot{r}^2 - \omega^2 r^2\right) \tag{1.48}$$

with $r(t_i) = r(t_f) = 0$, one obtains

$$U(q_f, q_i; t_f - t_i) = e^{iS[q_c]}\int \mathcal{D}[r]\exp\left\{i\frac{m}{2}\int_{t_i}^{t_f} dt\,\left(\dot{r}^2 - \omega^2 r^2\right)\right\}. \tag{1.49}$$

Taking advantage of the boundary conditions $r(t_i) = r(t_f) = 0$, one can expand $r(t)$ in a Fourier series (assuming that this is possible; see the discussion at the end of Sec. 1.1.1.2)

$$r(t) = \sum_{l=1}^{\infty} a_l \sin\left(\frac{l\pi(t - t_i)}{T}\right) \tag{1.50}$$

with $T = t_f - t_i$. The action then reads

$$S[r] = \frac{mT}{4}\sum_{l=1}^{\infty} a_l^2\left(\frac{l^2\pi^2}{T^2} - \omega^2\right) \tag{1.51}$$

and the integration over $r(t)$ becomes an integration over the coefficients a_l,

$$\mathcal{D}[r] \to J(T)\prod_{l=1}^{\infty}\int_{-\infty}^{\infty} da_l, \tag{1.52}$$

with a Jacobian $J(T)$ that depends on T but not on ω. We then find

$$\begin{aligned}\int \mathcal{D}[r]\, e^{iS[r]} &\propto \prod_{l=1}^{\infty}\int_{-\infty}^{\infty} da_l\, \exp\left[i\frac{mT}{4}\left(\frac{l^2\pi^2}{T^2} - \omega^2\right)a_l^2\right] \\ &\propto \prod_{l=1}^{\infty}\left(\frac{l^2\pi^2}{T^2} - \omega^2\right)^{-1/2} \\ &\propto \left(\frac{\omega T}{\sin(\omega T)}\right)^{1/2},\end{aligned} \tag{1.53}$$

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where at each step the proportionality constant is independent of ω. We have used the identity

$$\prod_{l=1}^{\infty}\left(1-\frac{z^2}{l^2\pi^2}\right)=\frac{\sin z}{z}. \tag{1.54}$$

We then have

$$U(q_f,q_i;T)=\mathcal{N}(T)\left(\frac{\omega T}{\sin(\omega T)}\right)^{1/2}e^{iS[q_c]}. \tag{1.55}$$

For $\omega\to 0$, we must reproduce the free particle result (1.43), which fixes the constant $\mathcal{N}(T)=(m/2i\pi T)^{1/2}$, so that

$$\begin{aligned}U(q_f,q_i;t_f-t_i)=&\left(\frac{m\omega}{2\pi i\sin(\omega(t_f-t_i))}\right)^{1/2}\\&\times\exp\left\{i\frac{m\omega}{2\sin(\omega(t_f-t_i))}\left[(q_f^2+q_i^2)\cos\big(\omega(t_f-t_i)\big)-2q_iq_f\right]\right\}.\end{aligned} \tag{1.56}$$

By setting $t_f-t_i\to -i\beta$, one obtains the imaginary-time propagator

$$\begin{aligned}U(q_f,q_i,-i\beta)=&\left(\frac{m\omega}{2\pi\sinh(\beta\omega)}\right)^{1/2}\\&\times\exp\left\{-\frac{m\omega}{2\sinh(\beta\omega)}\left[(q_f^2+q_i^2)\cosh(\beta\omega)-2q_iq_f\right]\right\}\end{aligned} \tag{1.57}$$

and the partition function

$$Z=\int_{-\infty}^{\infty}dq\,U(q,q;-i\beta)=\frac{e^{-\beta\omega/2}}{1-e^{-\beta\omega}} \tag{1.58}$$

of the harmonic oscillator.

Alternatively, one can calculate the partition function directly in the Euclidean formalism with discrete times by writing

$$q_k=\frac{1}{\sqrt{N}}\sum_{l=0}^{N-1}a_l\,e^{-i\omega_l\tau_k}, \tag{1.59}$$

where $\tau_k=k\epsilon=k\beta/N$, $\omega_l=l2\pi/\beta$, $a_l^*=a_{N-l}$ and $a_0=a_0^*$ (q_k is real). The action reads

$$S_E[a]=\frac{m}{\epsilon}\sum_{l=0}^{N-1}\left\{[1-\cos(\omega_l\epsilon)]+\frac{1}{2}\epsilon^2\omega^2\right\}a_l^*a_l, \tag{1.60}$$

and the partition function is given by a product of Gaussian integrals (Appendix 1.E),

$$\begin{aligned}Z&=\lim_{N\to\infty}\left(\frac{m}{2\pi\epsilon}\right)^{N/2}\int\prod_{l=0}^{N-1}da_l^*\,da_l\,e^{-S_E[a]}\\&=\lim_{N\to\infty}\left(\frac{1}{2\epsilon}\right)^{N/2}\prod_{l=0}^{N-1}\left\{\frac{1}{\epsilon}[1-\cos(\omega_l\epsilon)]+\frac{1}{2}\epsilon\omega^2\right\}^{-1/2}\end{aligned} \tag{1.61}$$

(the exponent 1/2 is due to the constraints $a_l^*=a_{N-l}$ and $a_0^*=a_0$). Writing $\cosh\theta=1+\epsilon^2\omega^2/2$ and using the identity

$$\prod_{l=0}^{N-1}[\cosh\theta-\cos(\omega_l\epsilon)]=2^{1-N}[\cosh(N\theta)-1], \tag{1.62}$$

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one reproduces the result (1.58).

Let us finally try to calculate the partition function in the Euclidean formalism with the continuum-time limit. Assuming that a Fourier series decomposition is justified,

$$q(\tau) = \frac{1}{\sqrt{\beta}} \sum_{l=-\infty}^{\infty} a_l e^{-i\omega_l \tau} \tag{1.63}$$

and $a_l^* = a_{-l}$ ($q(\tau)$ is real), the Euclidean action becomes

$$S_E[a] = \frac{m}{2} \sum_{l=-\infty}^{\infty} \left(\omega_l^2 + \omega^2\right) |a_l|^2, \tag{1.64}$$

which leads to the partition function

$$Z = \int \mathcal{D}[q]\, e^{-S_E[q]} \propto \prod_{l=-\infty}^{\infty} \frac{1}{\sqrt{\omega_l^2 + \omega^2}}, \tag{1.65}$$

using the result for Gaussian integration (Appendix 1.E). The product in the rhs vanishes. To get a finite result for the partition function, one must take into account the infinite factor included in the integration measure $\mathcal{D}[q]$ (that was dropped in (1.65)). This infinite factor being ω independent, the derivative of $\ln Z$ with respect to ω is however well defined,

$$\frac{\partial \ln Z}{\partial \omega} = -\sum_{l=-\infty}^{\infty} \frac{\omega}{\omega_l^2 + \omega^2} = -\frac{\beta}{2\tanh(\beta\omega/2)} \tag{1.66}$$

(the method to compute the sum over ω_l in (1.66) is explained in Appendix 1.F), so that

$$Z = \frac{\mathcal{N}}{\sinh(\beta\omega/2)}, \tag{1.67}$$

where the constant $\mathcal{N}$ is independent of ω. Since $\mathcal{N}$ is dimensionless, it cannot be a function of β alone and is therefore a pure number. It can be determined by requiring $\lim_{\beta\to\infty} Z = e^{-\beta\epsilon_0}$ with $\epsilon_0 = \omega/2$ the energy of the ground state. This yields $\mathcal{N} = 1/2$ and in turn the result (1.58).

1.1.1.7 Path integral with vector potential

To quantize a classical Hamiltonian $H(p, q)$, it is not always sufficient to replace the classical variables p and q by the corresponding quantum operators. When this correspondence rule yields products of the non-commuting operators $\hat{p}$ and $\hat{q}$, the order of the operators must be determined by additional conditions such as the hermiticity of the Hamiltonian. These difficulties also show up in the path integral formulation: The path integral must be carefully defined in order to respect the proper ordering of the operators.

As an example, we consider a free particle moving in a three-dimensional space in the presence of a magnetic field $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}$. Its dynamics is governed by the Lagrangian

$$L(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} m \dot{\mathbf{q}}^2 + e\dot{\mathbf{q}} \cdot \mathbf{A}(\mathbf{q}) \tag{1.68}$$

(e is the charge of the particle). The conjugate momentum is defined by $\mathbf{p} = \partial L/\partial \dot{\mathbf{q}} = m\dot{\mathbf{q}} + e\mathbf{A}(\mathbf{q})$ and the classical Hamiltonian reads

$$H(\mathbf{p}, \mathbf{q}) = \frac{[\mathbf{p} - e\mathbf{A}(\mathbf{q})]^2}{2m}. \tag{1.69}$$

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In order to determine the quantum Hamiltonian, one has to choose the order of the operators in the product $\hat{\mathbf{p}} \cdot \mathbf{A}(\hat{\mathbf{q}})$. By requiring the Hamiltonian to be Hermitian, one obtains

$$\hat{H} = \frac{1}{2m}\left[\hat{\mathbf{p}}^2 - e\hat{\mathbf{p}} \cdot \mathbf{A}(\hat{\mathbf{q}}) - e\mathbf{A}(\hat{\mathbf{q}}) \cdot \hat{\mathbf{p}} + e^2\mathbf{A}(\hat{\mathbf{q}})^2\right] = \frac{[\hat{\mathbf{p}} - e\mathbf{A}(\hat{\mathbf{q}})]^2}{2m}. \tag{1.70}$$

This expression can also be deduced from gauge invariance, which implies that the Hamiltonian can only be a function of the gauge-invariant combination $\hat{\mathbf{p}} - e\mathbf{A}(\hat{\mathbf{q}})$. Any other quantification choice would add to the Hamiltonian a term proportional to the commutator $e[\hat{\mathbf{p}}, \mathbf{A}(\hat{\mathbf{q}})] = -ie\boldsymbol{\nabla} \cdot \mathbf{A}(\hat{\mathbf{q}})$ which would violate both hermiticity and gauge invariance.

Because of the additional term $e\dot{\mathbf{q}} \cdot \mathbf{A}(\mathbf{q})$ in the Lagrangian (1.68), we expect the action $S(\mathbf{q}_k, \mathbf{q}_{k-1}; \epsilon)$ along the infinitesimal trajectory $(\mathbf{q}_k, \mathbf{q}_{k-1})$ to write

$$S(\mathbf{q}_k, \mathbf{q}_{k-1}; \epsilon) = \frac{m}{2}\frac{(\mathbf{q}_k - \mathbf{q}_{k-1})^2}{\epsilon} + e(\mathbf{q}_k - \mathbf{q}_{k-1}) \cdot \mathbf{A}(\mathbf{q}), \tag{1.71}$$

but it is not clear whether $\mathbf{A}(\mathbf{q})$ should be evaluated at $\mathbf{q}_k$, $\mathbf{q}_{k-1}$, or somewhere in between. Since typical paths satisfy $|\mathbf{q}_k - \mathbf{q}_{k-1}| \sim \sqrt{\epsilon}$, different discretizations of $\dot{\mathbf{q}} \cdot \mathbf{A}(\mathbf{q})$ lead to changes in the action of order ϵ which cannot be ignored. The proper choice can be obtained by noting that the hermiticity of the Hamiltonian implies $U(\mathbf{q}_k, \mathbf{q}_{k-1}; \epsilon)^* = U(\mathbf{q}_{k-1}, \mathbf{q}_k; -\epsilon)$ and therefore $S(\mathbf{q}_k, \mathbf{q}_{k-1}; \epsilon) = -S(\mathbf{q}_{k-1}, \mathbf{q}_k; -\epsilon)$. The only possibility is then to evaluate the vector potential at the midpoint $(\mathbf{q}_k + \mathbf{q}_{k-1})/2$. The choice $[\mathbf{A}(\mathbf{q}_k) + \mathbf{A}(\mathbf{q}_{k-1})]/2$ (or any other combination symmetric in $\mathbf{q}_k, \mathbf{q}_{k-1}$) is also possible, since two symmetric combinations differ by a term $\mathcal{O}[(\mathbf{q}_k - \mathbf{q}_{k-1})^2] = \mathcal{O}(\epsilon)$ and the corresponding actions by a negligible term $\mathcal{O}(\epsilon^{3/2})$.

The "midpoint" rule that follows from the hermiticity of the Hamiltonian is the only choice compatible with gauge invariance. In the gauge transformation $\mathbf{A} \to \mathbf{A} + \boldsymbol{\nabla}\Lambda$, the action along the path $(\mathbf{q}_i, t_i) \to (\mathbf{q}_f, t_f)$ changes by a term

$$e\int_{t_i}^{t_f} dt\, \dot{\mathbf{q}} \cdot \boldsymbol{\nabla}\Lambda(\mathbf{q}) = e\int_{\mathbf{q}_i}^{\mathbf{q}_f} d\mathbf{q} \cdot \boldsymbol{\nabla}\Lambda(\mathbf{q}) = e[\Lambda(\mathbf{q}_f) - \Lambda(\mathbf{q}_i)]. \tag{1.72}$$

The corresponding change in the propagator,

$$U(\mathbf{q}_f, \mathbf{q}_i; t_f - t_i) \to e^{ie\Lambda(\mathbf{q}_f)}U(\mathbf{q}_f, \mathbf{q}_i; t_f - t_i)e^{-ie\Lambda(\mathbf{q}_i)}, \tag{1.73}$$

reflects the change in the phase of the wavefunctions

$$\varphi(\mathbf{q}) \to e^{ie\Lambda(\mathbf{q})}\varphi(\mathbf{q}) \tag{1.74}$$

(see Eq. (1.11)). Up to a trivial phase factor, the propagator is independent of the function Λ (gauge invariance). In the path integration formulation, the change in the action is given by

$$e\sum_{k=1}^{N}(\mathbf{q}_k - \mathbf{q}_{k-1}) \cdot \boldsymbol{\nabla}\Lambda(\mathbf{u}_k), \tag{1.75}$$

if we choose to evaluate the vector potential in (1.71) at the point $\mathbf{u}_k$. To see whether (1.72) and (1.75) agree, we assume Λ to be a well-behaved function and rewrite (1.72) as

$$\Lambda(\mathbf{q}_f) - \Lambda(\mathbf{q}_i) = \sum_{k=1}^{N}[\Lambda(\mathbf{q}_k) - \Lambda(\mathbf{q}_{k-1})]$$

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$$= \sum_{k=1}^{N} \Big\{ (\mathbf{q}_k - \mathbf{q}_{k-1}) \cdot \boldsymbol{\nabla} \Lambda(\mathbf{u}_k) + \frac{1}{2}(1-2\theta)[(\mathbf{q}_k - \mathbf{q}_{k-1}) \cdot \boldsymbol{\nabla}]^2 \Lambda(\mathbf{u}_k) + \mathcal{O}[(\mathbf{q}_k - \mathbf{q}_{k-1})^3] \Big\}, \tag{1.76}$$

where $\mathbf{u}_k = \mathbf{q}_{k-1} + \theta(\mathbf{q}_k - \mathbf{q}_{k-1})$ (with $0 \leq \theta \leq 1$). Since $|\mathbf{q}_k - \mathbf{q}_{k-1}|^2$ is of order ϵ for typical paths, the second term in the rhs of the last equation does not vanish in the limit $\epsilon \to 0$ unless $\theta = 1/2$, i.e. $\mathbf{u}_k = (\mathbf{q}_k + \mathbf{q}_{k-1})/2$ (midpoint rule). With this choice, Eqs. (1.72) and (1.75) agree.

We conclude this section by showing that the midpoint rule can also be obtained from an explicit calculation of the propagator $U(\mathbf{q}_k, \mathbf{q}_{k-1}; \epsilon)$. We rewrite the latter as

$$U(\mathbf{q}_k, \mathbf{q}_{k-1}; \epsilon) = \frac{1}{(2i\pi)^{3/2}} \int d^3u \, e^{\frac{i}{2}\mathbf{u}^2} \langle \mathbf{q}_k | e^{-i\bar{\epsilon}\mathbf{u}\cdot[\hat{\mathbf{p}} - eA(\hat{\mathbf{q}})]} | \mathbf{q}_{k-1} \rangle, \tag{1.77}$$

where $\bar{\epsilon} = \sqrt{\epsilon/m}$. Contrary to what was done previously, it is not possible to factorize the exponential. This would introduce an error of order $\bar{\epsilon}^2 = \epsilon/m$, whereas we need the error to be smaller than ϵ to construct the path integral over a finite time interval $t_f - t_i$. This difficulty can be circumvented by using the relation

$$e^{\bar{\epsilon}(\hat{A}+\hat{B})} = e^{\frac{\bar{\epsilon}}{2}\hat{B}} e^{\bar{\epsilon}\hat{A}} e^{\frac{\bar{\epsilon}}{2}\hat{B}} + \mathcal{O}(\bar{\epsilon}^3), \tag{1.78}$$

which can be checked by a direct expansion of the exponentials. We then deduce

$$\langle \mathbf{q}_k | e^{-i\bar{\epsilon}\mathbf{u}\cdot[\hat{\mathbf{p}} - eA(\hat{\mathbf{q}})]} | \mathbf{q}_{k-1} \rangle = e^{\frac{i}{2}e\bar{\epsilon}\mathbf{u}\cdot\mathbf{A}(\mathbf{q}_k)} \langle \mathbf{q}_k | e^{-i\bar{\epsilon}\mathbf{u}\cdot\hat{\mathbf{p}} + \mathcal{O}(\epsilon^{3/2})} | \mathbf{q}_{k-1} \rangle e^{\frac{i}{2}e\bar{\epsilon}\mathbf{u}\cdot\mathbf{A}(\mathbf{q}_{k-1})}, \tag{1.79}$$

where the $\mathcal{O}(\epsilon^{3/2})$ error can now be ignored. The matrix element in (1.79) is calculated by inserting the closure relation $\sum_{\mathbf{p}} |\mathbf{p}\rangle\langle\mathbf{p}| = 1$,

$$\begin{aligned} \langle \mathbf{q}_k | e^{-i\bar{\epsilon}\mathbf{u}\cdot\hat{\mathbf{p}}} | \mathbf{q}_{k-1} \rangle &= \sum_{\mathbf{p},\mathbf{p}'} \langle \mathbf{q}_k | \mathbf{p} \rangle \langle \mathbf{p} | e^{-i\bar{\epsilon}\mathbf{u}\cdot\hat{\mathbf{p}}} | \mathbf{p}' \rangle \langle \mathbf{p}' | \mathbf{q}_{k-1} \rangle \\ &= \int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{q}_k - \mathbf{q}_{k-1} - \bar{\epsilon}\mathbf{u})} = \frac{1}{\bar{\epsilon}^3} \delta\left(\frac{\mathbf{q}_k - \mathbf{q}_{k-1}}{\bar{\epsilon}} - \mathbf{u} \right), \end{aligned} \tag{1.80}$$

which gives

$$U(\mathbf{q}_k, \mathbf{q}_{k-1}; \epsilon) = \left(\frac{m}{2i\pi\epsilon} \right)^{3/2} \exp\left\{ i\epsilon \left[\frac{m}{2} \frac{(\mathbf{q}_k - \mathbf{q}_{k-1})^2}{\epsilon^2} + \frac{e}{2} \frac{\mathbf{q}_k - \mathbf{q}_{k-1}}{\epsilon} \cdot \big(\mathbf{A}(\mathbf{q}_k) + \mathbf{A}(\mathbf{q}_{k-1}) \big) \right] \right\}. \tag{1.81}$$

We recognize in the exponential i times the action (1.71) with the vector potential $\mathbf{A}(\mathbf{q})$ approximated by $[\mathbf{A}(\mathbf{q}_k) + \mathbf{A}(\mathbf{q}_{k-1})]/2$ which, as discussed above, is equivalent to the midpoint rule.[15]

[15] One can also directly verify that $U(\mathbf{q}_k, \mathbf{q}_{k-1}; \epsilon)$ yields the time evolution of the wavefunction as obtained from the Hamiltonian (1.70) [3]. This provides a definitive proof of the validity of the path integral.

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1.1.1.8 Many-particle system

With little more effort, we can now write a path integral for the partition function – or the propagator – of a system with N particles interacting via a two-body potential $v(q_i - q_j)$ (we consider a one-dimensional system for simplicity). The Hilbert space is defined as the subspace of the tensor product $\mathcal{H} \otimes \cdots \otimes \mathcal{H}$ ($\mathcal{H}$ denotes the one-particle Hilbert space) containing all N-particle states properly symmetrized or antisymmetrized according to the quantum statistics of the particles. The partition function can be written as (see Sec. 1.2)

$$Z = \frac{1}{N!} \sum_{P \in S_N} \epsilon_P \int dq_1 \cdots dq_N \, (q_1 \cdots q_N | e^{-\beta \hat{H}} | q_{P(1)} \cdots q_{P(N)}), \tag{1.82}$$

where $|q_1 \cdots q_N) = |q_1\rangle \otimes \cdots \otimes |q_N\rangle$. The sum in (1.82) is over all permutations P of $\{1, \cdots, N\}$. ϵ_P equals one for bosons and the signature of the permutation P for fermions. Proceeding as in the one-particle case and inserting the closure relation

$$\int dq_1 \cdots dq_N \, |q_1 \cdots q_N)(q_1 \cdots q_N| = 1 \tag{1.83}$$

at each time step, we obtain the path integral

$$Z = \frac{1}{N!} \sum_{P \in S_N} \epsilon_P \int_{q_i(\beta) = q_{P(i)}(0)} \mathcal{D}[q] \, e^{-S_E[q]} \tag{1.84}$$

with the Euclidean action

$$S_E[q] = \int_0^\beta d\tau \left[\frac{m}{2} \sum_{i=1}^N \dot{q}_i^2 + \sum_{\substack{i,j=1 \\ (i<j)}}^N v(q_i - q_j) \right]. \tag{1.85}$$

This expression bears an obvious similarity with that obtained for a single particle. It is however not well suited for the study of a many-particle system. In Sec. 1.4 we shall introduce a path integral or, more precisely, a functional integral representation of the partition function that turns out to be much more convenient. This functional integral is based on the coherent states introduced in Sec. 1.3 and involves an integration over all configurations of a field weighted by an appropriate action. This turns out to be an essential aspect of (quantum) statistical physics and field theory: Systems with an infinite number of degrees of freedom are naturally described by fields, rather than the set of coordinates of all particles. Before embarking on the study of quantum many-body systems, we consider in the next section an elementary example of field theory.

1.1.2 Functional integral in statistical physics

In this section, we consider the functional integral approach to the quantum harmonic string. This simple example already exhibits many essential aspects of the field theory of (more complicated, i.e. anharmonic) quantum systems.[16]

[16] In a particle language, an "harmonic" system (i.e. a system whose Lagrangian is quadratic in its variables) corresponds to non-interacting particles, whereas an "anharmonic" system includes interactions between particles. These particles do not necessarily correspond to the bare particles but can refer to elementary excitations (a concept that will be explained in the forthcoming chapters).

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1.1.2.1 The classical harmonic string

We consider a one-dimensional harmonic string of length L and mass per unit length ρ. We denote by $\phi(x,t)$ the displacement of the infinitesimal mass ρdx located between x and $x+dx$ at equilibrium. The dynamics of the chain is governed by the Lagrangian[17]

$$\begin{aligned} L[\phi] &= \int_0^L dx\, \mathcal{L}(\partial_x\phi, \dot{\phi}), \\ \mathcal{L}(\partial_x\phi, \dot{\phi}) &= \frac{1}{2}\rho\dot{\phi}^2 - \frac{1}{2}\kappa(\partial_x\phi)^2, \end{aligned} \tag{1.86}$$

where $\mathcal{L}$ is referred to as the Lagrangian density. The first term in the rhs of (1.86) corresponds to the kinetic energy while the second one gives the energy cost associated with the deformation of the string. The harmonic string can be seen as the low-energy or long-wavelength limit of a one-dimensional "crystal" consisting of masses $m=\rho a$ separated by a distance a (at equilibrium) and connected by springs with stiffness κ/a.[18]

The Euler-Lagrange equation

$$\frac{d}{dt}\frac{\partial\mathcal{L}(\partial_x\phi,\dot{\phi})}{\partial\dot{\phi}} + \frac{d}{dx}\frac{\partial\mathcal{L}(\partial_x\phi,\dot{\phi})}{\partial(\partial_x\phi)} = 0 \tag{1.87}$$

gives the wave equation

$$\rho\ddot{\phi} - \kappa\partial_x^2\phi = 0. \tag{1.88}$$

The solutions are plane waves $\phi(x,t) \propto e^{i(kx-c|k|t)} + \text{c.c.}$ propagating with the velocity $c=\sqrt{\kappa/\rho}$, where $k = p\frac{2\pi}{L}$ (p integer) for periodic boundary conditions $\phi(x+L,t)=\phi(x,t)$.

1.1.2.2 The quantum harmonic string

The momentum conjugate to $\phi(x,t)$ is

$$\Pi(x,t) = \frac{\partial\mathcal{L}(\partial_x\phi,\dot{\phi})}{\partial\dot{\phi}(x,t)} = \rho\dot{\phi}(x,t) \tag{1.89}$$

and the Hamiltonian is defined by

$$H = \int_0^L dx\left[\Pi\dot{\phi} - \mathcal{L}(\partial_x\phi,\dot{\phi})\right] = \int_0^L dx\left[\frac{\Pi^2}{2\rho} + \frac{1}{2}\kappa(\partial_x\phi)^2\right]. \tag{1.90}$$

To quantize the string, we promote ϕ and Π to operators satisfying the commutation relations

$$[\hat{\phi}(x), \hat{\Pi}(x')] = i\delta(x-x'). \tag{1.91}$$

It is convenient to introduce the Fourier transformed operators

$$\begin{aligned} \hat{\phi}(k) &= \frac{1}{\sqrt{L}}\int_0^L dx\, e^{-ikx}\hat{\phi}(x) = \hat{\phi}^\dagger(-k), \\ \hat{\Pi}(k) &= \frac{1}{\sqrt{L}}\int_0^L dx\, e^{-ikx}\hat{\Pi}(x) = \hat{\Pi}^\dagger(-k) \end{aligned} \tag{1.92}$$

[17] In high-energy physics, the analog of the Lagrangian (1.86) in four spacetime dimensions is known as the Klein-Gordon Lagrangian.

[18] The Lagrangian of the one-dimensional crystal reads $L(q_i,\dot{q}_i) = \frac{1}{2}\sum_{i=1}^n[m\dot{q}_i^2 - k_s(q_{i+1}-q_i)^2]$, where q_i denotes the displacement of the ith atom with respect to its equilibrium position. The normal modes (phonons) of this system propagate with frequencies $\omega_k = \sqrt{2k_s/m}(1-\cos ka)^{1/2}$. In the long wavelength limit $|k|a \ll 1$, one recovers the spectrum $\omega = c|k|$ of the harmonic continuous string.

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satisfying the commutation relations

$$[\hat{\phi}(k),\hat{\phi}(k')]=[\hat{\Pi}(k),\hat{\Pi}(k')]=0 \quad \text{and} \quad [\hat{\phi}(k),\hat{\Pi}^\dagger(k')]=i\delta_{k,k'}. \tag{1.93}$$

This allows us to write the Hamiltonian as a sum of harmonic oscillators,

$$\hat{H}=\sum_k\left[\frac{1}{2\rho}\hat{\Pi}^\dagger(k)\hat{\Pi}(k)+\frac{1}{2}\rho\omega_k^2\hat{\phi}^\dagger(k)\hat{\phi}(k)\right], \tag{1.94}$$

where $\omega_k=c|k|$ and the sum is over all vectors $k=p2\pi/L$ ($p\in\mathbb{Z}$) satisfying the boundary conditions $e^{ikL}=1$. $\hat{H}$ is diagonalized by introducing the ladder operators

$$\begin{aligned}\hat{a}(k)&=\sqrt{\frac{\rho\omega_k}{2}}\left[\hat{\phi}(k)+\frac{i}{\rho\omega_k}\hat{\Pi}(k)\right],\\ \hat{a}^\dagger(k)&=\sqrt{\frac{\rho\omega_k}{2}}\left[\hat{\phi}^\dagger(k)-\frac{i}{\rho\omega_k}\hat{\Pi}^\dagger(k)\right].\end{aligned} \tag{1.95}$$

Using the commutation relations

$$[\hat{a}(k),\hat{a}(k')]=[\hat{a}^\dagger(k),\hat{a}^\dagger(k')]=0, \qquad [\hat{a}(k),\hat{a}^\dagger(k')]=\delta_{k,k'}, \tag{1.96}$$

one easily finds

$$\hat{H}=\sum_k\omega_k\left(\hat{a}_k^\dagger\hat{a}_k+\frac{1}{2}\right). \tag{1.97}$$

The eigenstates of the harmonic oscillator indexed by k are defined by

$$\begin{aligned}|n_k\rangle&=\frac{1}{\sqrt{n_k!}}(\hat{a}_k^\dagger)^{n_k}|0\rangle,\\ \omega_k\left(\hat{a}_k^\dagger\hat{a}_k+\frac{1}{2}\right)|n_k\rangle&=\omega_k\left(n_k+\frac{1}{2}\right)|n_k\rangle,\end{aligned} \tag{1.98}$$

where $|0\rangle=|n_k=0\rangle$ is the normalized vacuum state: $\hat{a}_k|0\rangle=0$ and $\langle 0|0\rangle=1$. The ladder operators $\hat{a}_k$ and $\hat{a}_k^\dagger$ satisfy

$$\begin{aligned}\hat{a}_k|n_k\rangle&=\sqrt{n_k}|n_k-1\rangle,\\ \hat{a}_k^\dagger|n_k\rangle&=\sqrt{n_k+1}|n_k+1\rangle,\end{aligned} \tag{1.99}$$

and can be seen as annihilation and creation operators of an elementary excitation (phonon) with momentum k. $n_k=\langle n_k|\hat{a}_k^\dagger\hat{a}_k|n_k\rangle$ denotes the number of phonons in the state $|n_k\rangle$. The eigenstates of the Hamiltonian (1.97) are obtained from the tensor product of the states $|n_k\rangle$,

$$\begin{aligned}|n_{k_1}\cdots n_{k_i}\cdots\rangle&=|n_{k_1}\rangle\otimes\cdots\otimes|n_{k_i}\rangle\otimes\cdots=\prod_i\frac{(\hat{a}_{k_i}^\dagger)^{n_{k_i}}}{\sqrt{n_{k_i}!}}|\text{vac}\rangle,\\ \hat{H}|n_{k_1}\cdots n_{k_i}\cdots\rangle&=\left[\sum_j\left(n_{k_j}+\frac{1}{2}\right)\omega_{k_j}\right]|n_{k_1}\cdots n_{k_i}\cdots\rangle,\end{aligned} \tag{1.100}$$

where $|\text{vac}\rangle$ is the vacuum, i.e. the normalized state defined by $\hat{a}_k|\text{vac}\rangle=0$ (for any k). The eigenstate $|n_{k_1}\cdots n_{k_i}\cdots\rangle$ has a total number of phonons $n=\sum_i n_{k_i}$. The Hilbert space can be written as the direct sum $\mathcal{H}=\mathcal{H}_0\oplus\mathcal{H}_1\oplus\cdots\oplus\mathcal{H}_n\oplus\cdots$ where $\mathcal{H}_n$ is the Hilbert space with n phonons. This type of Hilbert space, generically referred to as a Fock space, is at the basis of the second-quantization formalism described in Sec. 1.2.

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1.1.2.3 Functional integral formulation

A functional integral formulation can be obtained following the same procedure as in the case of the path integral discussed in Sec. 1.1.1. $\hat{\phi}(x)$ and $\hat{\Pi}(x)$ play the role of the position and momentum operator $\hat{q}$ and $\hat{p}$ of the single particle (the only difference being that they are labeled by x). One can introduce states $|\phi\rangle$ and $|\Pi\rangle$ satisfying $\hat{\phi}(x)|\phi\rangle = \phi(x)|\phi\rangle$ and $\hat{\Pi}(x)|\Pi\rangle = \Pi(x)|\Pi\rangle$. Since both sets of states $\{|\phi\rangle\}$ and $\{|\Pi\rangle\}$ form a complete basis of the Hilbert space, we have the closure relations

$$\begin{aligned} \mathcal{N} \lim_{a\to 0} \int \prod_{l=0}^{L/a} d\phi(la)\, |\phi\rangle\langle\phi| = 1, \\ \mathcal{N}' \lim_{a\to 0} \int \prod_{l=0}^{L/a} d\Pi(la)\, |\Pi\rangle\langle\Pi| = 1. \end{aligned} \tag{1.101}$$

Here we have discretized the chain; the continuous variable x becomes a discrete variable $x = la$. The harmonic chain is recovered in the limit $a \to 0$. This step is necessary to properly define the closure relations (1.101) as well as the functional integration measure [Eq. (1.110)]. $\mathcal{N}$ and $\mathcal{N}'$ are unimportant normalization constants that contribute a factor to the partition function but do not affect the expectation values. They will be dropped in the following.

To write the partition function as a functional integral, we use the first closure relation (1.101) to obtain

$$Z = \sum_n \langle n|e^{-\beta\hat{H}}|n\rangle = \int d\phi \sum_n \langle n|e^{-\beta\hat{H}}|\phi\rangle\langle\phi|n\rangle = \int d\phi \langle\phi|e^{-\beta\hat{H}}|\phi\rangle, \tag{1.102}$$

where $d\phi \equiv \prod_{l=0}^{L/a} d\phi(la)$ and $\{|n\rangle\}$ denotes a complete basis of the Hamiltonian $\hat{H}$. We now proceed as in Sec. 1.1.1 and split the imaginary time β into N infinitesimal steps $\epsilon = \beta/N$,[19]

$$Z = \int \prod_{k=1}^{N} d\phi_k \prod_{k=1}^{N} \langle\phi_k|e^{-\epsilon\hat{H}}|\phi_{k-1}\rangle, \tag{1.103}$$

with $\phi_0 = \phi_N$. For $\epsilon \to 0$, we can approximate

$$\langle\phi_k|e^{-\epsilon\hat{H}}|\phi_{k-1}\rangle = \langle\phi_k| \exp\Big\{-\frac{\epsilon}{2\rho}\int dx\, \hat{\Pi}^2\Big\} \exp\Big\{-\frac{\epsilon\kappa}{2}\int dx\, (\partial_x\hat{\phi})^2\Big\}|\phi_{k-1}\rangle. \tag{1.104}$$

Inserting the second closure relation (1.101), we obtain

$$\begin{aligned} &\langle\phi_k|e^{-\epsilon\hat{H}}|\phi_{k-1}\rangle \\ &\quad = \int d\Pi_k \exp\Big\{-\frac{\epsilon}{2\rho}\int dx\, \Pi_k^2 - \frac{\epsilon\kappa}{2}\int dx\, (\partial_x\phi_{k-1})^2\Big\}\langle\phi_k|\Pi_k\rangle\langle\Pi_k|\phi_{k-1}\rangle \\ &\quad = \int d\Pi_k \exp\Big\{-\frac{\epsilon}{2\rho}\int dx\, \Pi_k^2 - \frac{\epsilon\kappa}{2}\int dx\, (\partial_x\phi_{k-1})^2 + i\int dx\, \Pi_k(\phi_k - \phi_{k-1})\Big\}, \end{aligned} \tag{1.105}$$

[19] The discrete time variable k should not be confused with the momentum variable k used in (1.92) or (1.113).

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where we have used

$$\langle \phi_k | \Pi_k \rangle = \exp\Big\{ i \int dx\, \Pi_k \phi_k \Big\} \tag{1.106}$$

(up to an irrelevant multiplicative constant). Inserting (1.105) into (1.103) and taking the continuum-time limit, the partition function

$$Z = \int_{\phi(x,\beta)=\phi(x,0)} \mathcal{D}[\Pi,\phi] \exp\Big\{ - \int_0^\beta d\tau \int_0^L dx \left[\frac{\Pi^2}{2\rho} + \frac{\kappa}{2} (\partial_x \phi)^2 - i \Pi \dot{\phi} \right] \Big\} \tag{1.107}$$

is written as a functional integral in imaginary time over the real fields $\phi(x,\tau)$ and $\Pi(x,\tau)$.[20] Performing the Gaussian functional integral over Π,[21] one obtains

$$Z = \int_{\phi(x,\beta)=\phi(x,0)} \mathcal{D}[\phi]\, e^{-S_E[\phi]} \tag{1.108}$$

with the Euclidean action

$$S_E[\phi] = \frac{1}{2} \int_0^\beta d\tau \int_0^L dx \left[\rho \dot{\phi}^2 + \kappa (\partial_x \phi)^2 \right]. \tag{1.109}$$

The integration measure in (1.108) is defined as

$$\mathcal{D}[\phi] = \lim_{N\to\infty} \lim_{a\to 0} \prod_{k=1}^{N} \prod_{l=0}^{L/a} d\phi(la, k\beta/N) \tag{1.110}$$

(up to a multiplicative factor). Performing a Wick rotation ($\tau = it$), we see that the real-time action is given by the classical action $S[\phi] = \int dt \int_0^L dx\, \mathcal{L}(\partial_x \phi, \dot{\phi})$.

Since the field $\phi(x,\tau)$ is periodic in time, it can be expanded as a Fourier series,

$$\begin{aligned} \phi(x,\tau) &= \frac{1}{\sqrt{\beta}} \sum_{\omega_n} e^{-i\omega_n \tau} \phi(x, i\omega_n), \\ \phi(x, i\omega_n) &= \frac{1}{\sqrt{\beta}} \int_0^\beta d\tau\, e^{i\omega_n \tau} \phi(x,\tau), \end{aligned} \tag{1.111}$$

where the discrete frequency

$$\omega_n = \frac{2\pi}{\beta} n \qquad (n \in \mathbb{Z}) \tag{1.112}$$

is called a Matsubara (imaginary) frequency. The Fourier transformed field $\phi(k, i\omega_n)$ diagonalizes the action,

$$S_E[\phi] = \frac{\rho}{2} \sum_{k,\omega_n} \phi(-k, -i\omega_n) \left(\omega_n^2 + \omega_k^2 \right) \phi(k, i\omega_n), \tag{1.113}$$

so that the partition function (1.108) is given by a product of Gaussian integrals. Note that $\phi(-k,-i\omega_n) = \phi^*(k, i\omega_n)$ since $\phi(x,\tau)$ is real.

The functional integral approach to the quantum harmonic string already exhibits many essential aspects of the field theory of (more complicated) quantum systems:

[20] Equation (1.107) is similar to (1.29).
[21] Gaussian functional integration is discussed in Sec. 1.E.

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- Equation (1.108) is an exact representation of the partition function. Since the action is quadratic in the field, the functional integral can be done exactly:[22]

$$Z = \prod_k \frac{e^{-\frac{\beta}{2}\omega_k}}{1 - e^{-\beta\omega_k}}, \tag{1.114}$$

which is the standard result for a set of decoupled harmonic oscillators. More generally, all results that can be obtained from the Hamiltonian (1.97) can also be derived from the functional integral (1.108). When the action is not quadratic, for example when it contains a term

$$S_{\rm int}[\phi] = \int_0^\beta d\tau \int_0^L dx\, V(\phi) \tag{1.115}$$

with $V(\phi)$ a potential of order 4 or higher, the functional integral cannot be done exactly and one has to resort to approximate strategies such as a perturbative expansion (Sec. 1.5) or a saddle-point approximation (Sec. 1.7).

- Requiring the action to be stationary, $\delta S_E/\delta\phi(x,\tau) = 0$, one obtains the classical equation of motion (1.88) in imaginary time: $\rho\partial_\tau^2\phi + \kappa\partial_x^2\phi = 0$.

- The continuous string model cannot be valid at arbitrarily small length scales. For example, if it corresponds to the low-energy description of a one-dimensional crystal (as discussed at the beginning of this section), it is valid only for momenta k much smaller than $1/a$ where a is the lattice spacing. The dispersion ω_k is therefore bounded by a maximum value $\omega_0 = \omega_{k_{\rm max}} = \omega_{k\sim 1/a}$. At high temperatures, when $\omega_0 \ll T$, the partition function is dominated by the zero Matsubara frequency $\omega_{n=0} = 0$. The action

$$S_E[\phi] \simeq \frac{\rho}{2}\sum_k \phi(-k, -i\omega_{n=0})\omega_k^2\phi(k, i\omega_{n=0}) = \frac{\kappa}{2T}\int_0^L dx(\partial_x\phi)^2 \tag{1.116}$$

(with $\phi(x) \equiv \sqrt{T}\phi(x, i\omega_{n=0})$) then yields the classical partition function

$$Z_{\rm cl} = \int \mathcal{D}[\phi] \exp\left\{-\frac{\kappa}{2T}\int_0^L dx(\partial_x\phi)^2\right\}. \tag{1.117}$$

A quantum system at sufficiently high temperature behaves classically. When $T \sim \omega_0$, the system enters the quantum regime and nonzero Matsubara frequencies cannot be ignored. More precisely, the harmonic oscillator with frequency ω_k behaves classically if $\omega_k \ll 2\pi T$. In that case the nonzero Matsubara frequencies can be ignored when computing the partition function: $Z \simeq T/\omega_k$ (this result is easily obtained from (1.66-1.67) by retaining only the term $\omega_l = 0$), which gives the mean energy $E = T$ in agreement with the equipartition theorem. We recover the well-known result that a harmonic oscillator behaves classically when the mean occupation factor $n_B(\omega_k)$ is much larger than unity. Thus the quantum-mechanical behavior of the system shows up in the imaginary-time dependence of the field.

[22] See the calculation of the partition function of the harmonic oscillator in Sec. 1.1.1 (in particular Eqs. (1.64) and (1.65)).

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- In the limit $T \to 0$ ($\beta \to \infty$), the imaginary time τ varies in the interval $[0, \infty[$ and the discrete Matsubara frequency ω_n becomes a continuous variable ω. By rescaling the time variable, we can rewrite the action as

$$S_E[\phi] = \frac{\kappa}{2c} \int d^2r [\boldsymbol{\nabla}\phi(\mathbf{r})]^2 \tag{1.118}$$

(we assume both $\beta \to \infty$ and $L \to \infty$), where $\mathbf{r} = (x, c\tau)$. The action of the quantum one-dimensional string reduces to the action of a classical two-dimensional string. More generally, zero-temperature d-dimensional quantum systems behave as $(d+1)$-dimensional classical systems (d space dimensions and one (imaginary) time dimension); however the action may not exhibit the ("relativistic") SO$(d+1)$ spacetime symmetry as in (1.118).

- The fact that the excitation energies ω_k can be directly read off from the partition function (1.108,1.113) illustrates the connection between thermodynamics and dynamics in a quantum system. The relevant tool to study the dynamics of the system is the (imaginary-time) propagator of the field ϕ – also called the (imaginary-time) Green function – defined as the correlation function

$$\begin{aligned} G(k,\tau) &= \langle \phi(k,\tau)\phi(-k,0)\rangle \\ &= \frac{1}{Z}\int \mathcal{D}[\phi]\, \phi(k,\tau)\phi(-k,0)\, e^{-S_E[\phi]}. \end{aligned} \tag{1.119}$$

It can also be written as the time-ordered correlation function

$$\begin{aligned} G(k,\tau) &= \langle T_\tau \hat{\phi}(k,\tau)\hat{\phi}(-k,0)\rangle \\ &= \Theta(\tau)\langle \hat{\phi}(k,\tau)\hat{\phi}(-k,0)\rangle + \Theta(-\tau)\langle \hat{\phi}(-k,0)\hat{\phi}(k,\tau)\rangle, \end{aligned} \tag{1.120}$$

where T_τ is the time-ordering operator defined in Sec. 1.1.1 and $\langle \cdots \rangle = Z^{-1}\mathrm{Tr}(e^{-\beta\hat{H}} \cdots)$. The imaginary-time operator $\hat{\phi}(k,\tau)$ is defined by

$$\hat{\phi}(k,\tau) = e^{\tau\hat{H}}\hat{\phi}(k)e^{-\tau\hat{H}}. \tag{1.121}$$

The action (1.113) being quadratic, $G(k, i\omega_n)$ can be calculated exactly,[21]

$$G(k,i\omega_n) = \langle \phi(k,i\omega_n)\phi(-k,-i\omega_n)\rangle = \frac{1}{\rho}\frac{1}{\omega_n^2+\omega_k^2}. \tag{1.122}$$

To obtain the dynamics in real time, one performs an analytic continuation $i\omega_n \to \omega + i\eta$ to real frequencies ω. The infinitesimal $\eta \to 0^+$ is crucial for the analytic continuation to be well defined in the general case (Chap. 3). The resulting correlation function

$$G^R(k,\omega) = G(k,\omega+i\eta) = -\frac{1}{\rho}\frac{1}{(\omega+i\eta)^2-\omega_k^2} \tag{1.123}$$

is called the retarded Green function (Sec. 1.2.3). Its poles $\omega = \pm\omega_k - i\eta$ directly give the excitation energies of the quantum string. The excitation spectrum also shows up in the spectral function

$$A(k,\omega) = \Im[G^R(k,\omega)] = \frac{\pi}{2\rho\omega_k}[\delta(\omega-\omega_k) - \delta(\omega+\omega_k)]. \tag{1.124}$$

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This is a very general result: Green functions – or correlation functions – contain information about the excitation spectrum of the system and are therefore the main quantities of interest besides the free energy $-\frac{1}{\beta}\ln Z$. We shall see in Chap. 3 that correlation functions also determine the linear response of the system to an external field.

1.2 Second quantization and Green functions

In quantum statistical physics one is interested in systems with a large number of particles. The state vectors describing the possible physical states must be symmetric or antisymmetric under the exchange of two particles, depending on the statistics of the particles (bosons or fermions). Furthermore, the number of particles fluctuates when the system can exchange particles with a reservoir – as for instance in the grand canonical ensemble. Second quantization provides a natural formalism to describe many-particle systems (the main results are summarized in Table 1.1). It yields the concept of quantum field and in turn the possibility to express the partition function as a functional integral over suitably defined coherent states (Secs. 1.3.1, 1.3.3 and 1.4).

1.2.1 Fock space

The Fock space is defined as the direct sum

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \cdots = \overset{n=\infty}{\underset{n=0}{\oplus}} \mathcal{H}_n, \tag{1.125}$$

where $\mathcal{H}_n$ is the Hilbert space corresponding to a system of n particles. $\mathcal{H}_0$ contains only the vacuum state $|\text{vac}\rangle$ and $\mathcal{H}_1$ is the Hilbert used when studying the quantum mechanics of a single particle.

Let us first focus on the Hilbert space $\mathcal{H}_n$. $\mathcal{H}_n$ contains all n-particle states properly symmetrized (bosons) or antisymmetrized (fermions). It is a subspace of the direct product $\bar{\mathcal{H}}_n = \mathcal{H}_1 \otimes \mathcal{H}_1 \otimes \cdots \mathcal{H}_1$. If we know an orthonormal basis $\{|\alpha\rangle\}$ of $\mathcal{H}_1$, then we can obtain an orthonormal basis of $\bar{\mathcal{H}}_n$ from the tensor product

$$|\alpha_1 \cdots \alpha_n) = |\alpha_1\rangle \otimes |\alpha_2\rangle \otimes \cdots \otimes |\alpha_n\rangle. \tag{1.126}$$

Note that we use the notation $|\cdots)$ (with a curly bracket) to denote this n-particle state. The overlap of two vectors is simply

$$(\alpha_1' \cdots \alpha_n' | \alpha_1 \cdots \alpha_n) = \langle \alpha_1' | \alpha_1 \rangle \cdots \langle \alpha_n' | \alpha_n \rangle = \delta_{\alpha_1', \alpha_1} \cdots \delta_{\alpha_n', \alpha_n} \tag{1.127}$$

and the completeness of the basis gives the closure relation

$$\sum_{\alpha_1 \cdots \alpha_n} |\alpha_1 \cdots \alpha_n)(\alpha_1 \cdots \alpha_n| = 1, \tag{1.128}$$

where "1" in the rhs means the unit (or identity) operator in $\bar{\mathcal{H}}_n$. A wavefunction in coordinate space is obtained by multiplying the ket $|\alpha_1 \cdots \alpha_n)$ by the bra $(\mathbf{r}_1 \cdots \mathbf{r}_n| = \langle \mathbf{r}_1| \otimes \cdots \otimes \langle \mathbf{r}_n|$,

$$(\mathbf{r}_1 \cdots \mathbf{r}_n | \alpha_1 \cdots \alpha_n) = \varphi_{\alpha_1}(\mathbf{r}_1) \cdots \varphi_{\alpha_n}(\mathbf{r}_n), \tag{1.129}$$

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where $\varphi_\alpha(\mathbf{r}) = \langle \mathbf{r}|\alpha\rangle$ is the coordinate-representation wavefunction of a particle in the quantum state $|\alpha\rangle$.

To obtain a basis of $\mathcal{H}_n$, we must (anti)symmetrize the tensor product (1.126),

$$\begin{aligned} |\alpha_1 \cdots \alpha_n\} &= \sqrt{n!}\mathcal{P}_\zeta |\alpha_1 \cdots \alpha_n) \\ &= \frac{1}{\sqrt{n!}} \sum_{P\in S_n} \zeta^P |\alpha_{P(1)} \cdots \alpha_{P(n)}). \end{aligned} \tag{1.130}$$

$\mathcal{P}_+$ ($\mathcal{P}_-$) is a symmetrization (antisymmetrization) operator and $\zeta = +1$ for bosons and -1 for fermions. The sum in (1.130) is over all permutations P of $\{1\cdots n\}$. ζ^P equals one for bosons and the signature of the permutation P for fermions. One can easily verify that $\mathcal{P}_\zeta$ is a projector, i.e. $\mathcal{P}_\zeta^2 = \mathcal{P}_\zeta$. For fermions the state $|\alpha_1 \cdots \alpha_n\}$ vanishes if two of the α_i's are identical (Pauli principle). The states $|\alpha_1 \cdots \alpha_n\}$ form a basis of $\mathcal{H}_n$. The closure relation (1.128) in $\bar{\mathcal{H}}_n$ becomes a closure relation in $\mathcal{H}_n$,

$$\begin{aligned} 1 &= \sum_{\alpha_1\cdots\alpha_n} \mathcal{P}_\zeta |\alpha_1 \cdots \alpha_n)(\alpha_1 \cdots \alpha_n|\mathcal{P}_\zeta \\ &= \frac{1}{n!} \sum_{\alpha_1\cdots\alpha_n} |\alpha_1 \cdots \alpha_n\}\{\alpha_1 \cdots \alpha_n|, \end{aligned} \tag{1.131}$$

and the overlap between two states is given by

$$\begin{aligned} \{\alpha'_1 \cdots \alpha'_n|\alpha_1 \cdots \alpha_n\} &= n!(\alpha'_1 \cdots \alpha'_n|\mathcal{P}_\zeta^2|\alpha_1 \cdots \alpha_n) \\ &= \sum_P \zeta^P (\alpha'_1 \cdots \alpha'_n|\alpha_{P(1)} \cdots \alpha_{P(n)}). \end{aligned} \tag{1.132}$$

The permutations P contributing to the sum must satisfy $\alpha'_1 = \alpha_{P(1)}, \cdots, \alpha'_n = \alpha_{P(n)}$. Since for fermions all α_i's must be different, there is only one such permutation and the overlap reduces to

$$\{\alpha'_1 \cdots \alpha'_n|\alpha_1 \cdots \alpha_n\} = (-1)^P \qquad \text{(fermions)}. \tag{1.133}$$

In the case of bosons, several particles can occupy the same one-body state, and any permutation which interchanges particles in the same state contributes to the sum in (1.132). The number of these permutations is $n_{\alpha_1}! \cdots n_{\alpha_p}!$ if we denote by n_{α_i} the number of bosons in the one-body state $|\alpha_i\rangle$ (with $\alpha_1, \cdots, \alpha_p$ all different).[23] We therefore find

$$\{\alpha'_1 \cdots \alpha'_n|\alpha_1 \cdots \alpha_n\} = n_{\alpha_1}! \cdots n_{\alpha_p}! \qquad \text{(bosons)}. \tag{1.134}$$

A normalized basis of $\mathcal{H}_n$ is obtained by normalizing the states $|\alpha_1 \cdots \alpha_n\}$,

$$\begin{aligned} |\alpha_1 \cdots \alpha_n\rangle &= \frac{1}{(\prod_{i=1}^p n_{\alpha_i}!)^{1/2}} |\alpha_1 \cdots \alpha_n\} \\ &= \frac{1}{(n! \prod_{i=1}^p n_{\alpha_i}!)^{1/2}} \sum_P \zeta^P |\alpha_{P(1)} \cdots \alpha_{P(n)}). \end{aligned} \tag{1.135}$$

The coordinate-representation wavefunction is given by

$$(\mathbf{r}_1 \cdots \mathbf{r}_n|\alpha_1 \cdots \alpha_n\rangle = \frac{1}{(n! \prod_{i=1}^p n_{\alpha_i}!)^{1/2}} \sum_P \zeta^P \varphi_{\alpha_{P(1)}}(\mathbf{r}_1) \cdots \varphi_{\alpha_{P(n)}}(\mathbf{r}_n)$$

[23] Note that we can always relabel the states so that the different states correspond to $\alpha_1, \ldots, \alpha_p$.

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$$= \begin{cases} \dfrac{1}{\sqrt{n!}} \det\left(\varphi_{\alpha_i}(\mathbf{r}_j)\right) & \text{(fermions)}, \\ \dfrac{1}{(n!\prod_{i=1}^{p} n_{\alpha_i}!)^{1/2}} \operatorname{per}\left(\varphi_{\alpha_i}(\mathbf{r}_j)\right) & \text{(bosons)}. \end{cases} \tag{1.136}$$

We thus obtain a basis of permanents for bosons and determinants for fermions. The latter are known as Slater determinants.

It is also possible to define an (anti)symmetrized many-body state in the coordinate representation,

$$\begin{aligned} |\mathbf{r}_1\sigma_1, \cdots, \mathbf{r}_n\sigma_n\} &= \frac{1}{\sqrt{n!}} \sum_{P\in S_n} \zeta^P |\mathbf{r}_{P(1)}\sigma_{P(1)}, \cdots, \mathbf{r}_{P(n)}\sigma_{P(n)}) \\ &= \frac{1}{\mathcal{V}^{n/2}} \sum_{\mathbf{k}_1\cdots\mathbf{k}_n} e^{i(\mathbf{k}_1\cdot\mathbf{r}_1+\cdots+\mathbf{k}_n\cdot\mathbf{r}_n)} |\mathbf{k}_1\sigma_1, \cdots, \mathbf{k}_n\sigma_n\}, \end{aligned} \tag{1.137}$$

although the states $|\mathbf{r}\sigma\rangle$ are not normalized. σ denotes the spin index (as well as other discrete indices if necessary) and $\mathbf{k}$ the momentum.

1.2.1.1 Creation and annihilation operators

To complete the description of many-particle systems, we need to introduce operators which couple $\mathcal{H}_n$ to $\mathcal{H}_{n\pm1}$, i.e. which add or remove a particle from the system. The creation operator $\hat{\psi}^\dagger_\alpha$ is defined by

$$\begin{aligned} \hat{\psi}^\dagger_\alpha |\alpha_1 \cdots \alpha_n\} &= |\alpha\alpha_1 \cdots \alpha_n\}, \\ \hat{\psi}^\dagger_\alpha |\alpha_1 \cdots \alpha_n\rangle &= \sqrt{n_\alpha + 1} |\alpha\alpha_1 \cdots \alpha_n\rangle. \end{aligned} \tag{1.138}$$

$\hat{\psi}^\dagger_\alpha$ adds a particle in the one-body state α and (anti)symmetrizes the resulting many-body state. When acting on the vacuum state, it gives the one-body state $|\alpha\rangle$: $\hat{\psi}^\dagger_\alpha|\text{vac}\rangle = |\alpha\rangle$. More generally,

$$\begin{aligned} |\alpha_1 \cdots \alpha_n\} &= \hat{\psi}^\dagger_{\alpha_1} \cdots \hat{\psi}^\dagger_{\alpha_n} |\text{vac}\rangle, \\ |\alpha_1 \cdots \alpha_n\rangle &= \frac{1}{(\prod_{i=1}^{p} n_{\alpha_i}!)^{1/2}} \hat{\psi}^\dagger_{\alpha_1} \cdots \hat{\psi}^\dagger_{\alpha_n} |\text{vac}\rangle. \end{aligned} \tag{1.139}$$

The annihilation operator $\hat{\psi}_\alpha$ is defined as the adjoint of the creation operator $\hat{\psi}^\dagger_\alpha$. From (1.138), we find

$$\begin{aligned} \{\alpha_1 \cdots \alpha_n |\hat{\psi}_\alpha| \alpha'_1 \cdots \alpha'_m\} &= \{\alpha'_1 \cdots \alpha'_m |\hat{\psi}^\dagger_\alpha| \alpha_1 \cdots \alpha_n\}^* \\ &= \{\alpha\alpha_1 \cdots \alpha_n |\alpha'_1 \cdots \alpha'_m\}. \end{aligned} \tag{1.140}$$

The result is finite only if $n = m - 1$ so that $\hat{\psi}_\alpha$ removes a particle from the system. This means in particular that $\hat{\psi}_\alpha|\text{vac}\rangle = 0$. Using the closure relation in the Fock space $\mathcal{H}$,

$$\sum_{n=0}^{\infty} \sum_{\alpha_1\cdots\alpha_n} \frac{1}{n!} |\alpha_1 \cdots \alpha_n\}\{\alpha_1 \cdots \alpha_n| = 1, \tag{1.141}$$

we obtain

$$\hat{\psi}_\alpha |\alpha'_1 \cdots \alpha'_m\} = \sum_{n=0}^{\infty} \sum_{\alpha_1\cdots\alpha_n} \frac{1}{n!} \{\alpha_1 \cdots \alpha_n |\hat{\psi}_\alpha| \alpha'_1 \cdots \alpha'_m\} |\alpha_1 \cdots \alpha_n\}$$

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$$= \frac{1}{(m-1)!} \sum_{\alpha_1 \cdots \alpha_{m-1}} \{\alpha\alpha_1 \cdots \alpha_{m-1}|\alpha'_1 \cdots \alpha'_m\}|\alpha_1 \cdots \alpha_{m-1}\}$$
$$= \frac{1}{(m-1)!} \sum_{P\in S_m} \zeta^P \langle\alpha|\alpha'_{P(1)}\rangle|\alpha'_{P(2)} \cdots \alpha'_{P(m)}\}, \tag{1.142}$$

where we have used (1.127) and (1.132). The permutation $(P(2), \cdots, P(m)) \to (1, \cdots, P(1)-1, P(1)+1, \cdots, m)$ has a signature $\zeta^{P+P(1)-1}$,[24] so that we finally obtain[25]

$$\hat{\psi}_\alpha|\alpha'_1 \cdots \alpha'_m\} = \frac{1}{(m-1)!} \sum_{P\in S_m} \zeta^{P(1)-1}\delta_{\alpha,\alpha'_{P(1)}}|\alpha'_1 \cdots \alpha'_{P(1)-1}, \alpha'_{P(1)+1} \cdots \alpha'_m\}$$
$$= \sum_{i=1}^{m} \zeta^{i-1}\delta_{\alpha,\alpha'_i}|\alpha'_1 \cdots \hat{\alpha}'_i \cdots \alpha'_m\}, \tag{1.143}$$

where $\hat{\alpha}'_i$ indicates that α'_i is omitted. The analogous result for the normalized states reads

$$\hat{\psi}_\alpha|\alpha'_1 \cdots \alpha'_m\rangle = \frac{1}{\sqrt{n_\alpha}} \sum_{i=1}^{m} \zeta^{i-1}\delta_{\alpha,\alpha'_i}|\alpha'_1 \cdots \hat{\alpha}'_i \cdots \alpha'_m\rangle. \tag{1.144}$$

For bosons, it is sometimes convenient to introduce the states

$$|n_{\alpha_1} \cdots n_{\alpha_p}\rangle = \frac{1}{(\prod_{i=1}^{p} n_{\alpha_i}!)^{1/2}} \left(\hat{\psi}^\dagger_{\alpha_1}\right)^{n_{\alpha_1}} \cdots \left(\hat{\psi}^\dagger_{\alpha_p}\right)^{n_{\alpha_p}} |\text{vac}\rangle, \tag{1.145}$$

where n_{α_i} denotes the occupation number of the one-body state α_i (with $\alpha_1 \cdots \alpha_p$ all different). These states form an orthonormal basis of the Fock space $\mathcal{H}$ and satisfy the closure relation

$$\sum_{n_{\alpha_1}=0}^{\infty} \sum_{n_{\alpha_2}=0}^{\infty} \cdots |n_{\alpha_1} \cdots n_{\alpha_i} \cdots\rangle\langle n_{\alpha_1} \cdots n_{\alpha_i} \cdots| = 1 \tag{1.146}$$

as well as

$$\begin{array}{l} \hat{\psi}^\dagger_\alpha|n_{\alpha_1} \cdots n_{\alpha_p}\rangle = \sqrt{n_\alpha + 1}|n_{\alpha_1} \cdots (n_\alpha + 1) \cdots n_{\alpha_p}\rangle, \\ \hat{\psi}_\alpha|n_{\alpha_1} \cdots n_{\alpha_p}\rangle = \sum_{i=1}^{p} \delta_{\alpha,\alpha_i}\sqrt{n_\alpha}|n_{\alpha_1} \cdots (n_\alpha - 1) \cdots n_{\alpha_p}\rangle, \end{array} \quad \text{(bosons)}. \tag{1.147}$$

They are therefore eigenstates of the operator $\hat{n}_\alpha = \hat{\psi}^\dagger_\alpha\hat{\psi}_\alpha$ which measures the number of particles in the one-body state $|\alpha\rangle$,

$$\hat{n}_\alpha|n_{\alpha_1} \cdots n_{\alpha_p}\rangle = \sum_{i=1}^{p} \delta_{\alpha,\alpha_i} n_\alpha |n_{\alpha_1} \cdots n_{\alpha_p}\rangle. \tag{1.148}$$

1.2.1.2 (Anti)commutation relations

The symmetry properties of the states $|\alpha_1 \cdots \alpha_n\}$ under the exchange of two particles have important consequences for the operators $\hat{\psi}_\alpha, \hat{\psi}^\dagger_\alpha$. We have

$$\hat{\psi}^\dagger_\alpha\hat{\psi}^\dagger_{\alpha'}|\alpha_1 \cdots \alpha_n\} = |\alpha\alpha'\alpha_1 \cdots \alpha_n\}$$

[24] The permutation $(P(1), P(2), \cdots, P(m)) \to (P(1), 2, \cdots, P(1)-1, P(1)+1, \cdots, m)$ can be written as the permutation P^{-1} $[(P(1), P(2), \cdots, P(m)) \to (1, 2, \cdots, m)$, signature $\zeta^P]$ followed by $P(1)-1$ transpositions $[(1, 2, \cdots, m) \to (P(1), 2, \cdots, P(1)-1, P(1)+1, \cdots, m)$, signature $\zeta^{P(1)-1}]$.

[25] One can easily verify that (1.143) reproduces (1.140).

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$$= \zeta|\alpha'\alpha\alpha_1\cdots\alpha_n\rangle\}$$
$$= \zeta\hat{\psi}^\dagger_{\alpha'}\hat{\psi}^\dagger_\alpha|\alpha_1\cdots\alpha_n\rangle\}. \tag{1.149}$$

Since (1.149) holds for any state $|\alpha_1\cdots\alpha_n\rangle\}$, we deduce

$$[\hat{\psi}^\dagger_\alpha, \hat{\psi}^\dagger_{\alpha'}]_{-\zeta} = [\hat{\psi}_\alpha, \hat{\psi}_{\alpha'}]_{-\zeta} = 0 \tag{1.150}$$

(the second equation is deduced from the first by taking the adjoint). $[\cdot,\cdot]_-$ and $[\cdot,\cdot]_+$ denote the commutator and the anticommutator, respectively. By comparing

$$\hat{\psi}_\alpha\hat{\psi}^\dagger_{\alpha'}|\alpha_1\cdots\alpha_n\rangle\} = \hat{\psi}_\alpha|\alpha'\alpha_1\cdots\alpha_n\rangle\}$$
$$= \delta_{\alpha,\alpha'}|\alpha_1\cdots\alpha_n\rangle\} + \sum_{i=1}^n \zeta^i\delta_{\alpha,\alpha_i}|\alpha'\alpha_1\cdots\hat{\alpha}_i\cdots\alpha_n\rangle\} \tag{1.151}$$

and

$$\hat{\psi}^\dagger_{\alpha'}\hat{\psi}_\alpha|\alpha_1\cdots\alpha_n\rangle\} = \hat{\psi}^\dagger_{\alpha'}\sum_{i=1}^n \zeta^{i-1}\delta_{\alpha,\alpha_i}|\alpha_1\cdots\hat{\alpha}_i\cdots\alpha_n\rangle\}$$
$$= \sum_{i=1}^n \zeta^{i-1}\delta_{\alpha,\alpha_i}|\alpha'\alpha_1\cdots\hat{\alpha}_i\cdots\alpha_n\rangle\}, \tag{1.152}$$

we also obtain

$$[\hat{\psi}_\alpha, \hat{\psi}^\dagger_{\alpha'}]_{-\zeta} = \delta_{\alpha,\alpha'}. \tag{1.153}$$

Equations (1.150) and (1.153) are fundamental relations satisfied by the creation and annihilation operators $\hat{\psi}^\dagger_\alpha, \hat{\psi}_\alpha$. Recall the crucial assumption that $\{|\alpha\rangle\}$ forms an orthonormal basis of the single-particle Hilbert space $\mathcal{H}_1$.

1.2.1.3 Change of basis

Creation and annihilation operators in another basis $\{|\tilde{\alpha}\rangle\}$ can be straightforwardly obtained from the operators $\hat{\psi}^\dagger_\alpha, \hat{\psi}_\alpha$. Using $|\tilde{\alpha}\rangle = \sum_\alpha\langle\alpha|\tilde{\alpha}\rangle|\alpha\rangle$, we deduce

$$\hat{\psi}^\dagger_{\tilde{\alpha}}|\alpha_1\cdots\alpha_n\rangle\} = |\tilde{\alpha}\alpha_1\cdots\alpha_n\rangle\}$$
$$= \sum_\alpha\langle\alpha|\tilde{\alpha}\rangle|\alpha\alpha_1\cdots\alpha_n\rangle\}$$
$$= \sum_\alpha\langle\alpha|\tilde{\alpha}\rangle\hat{\psi}^\dagger_\alpha|\alpha_1\cdots\alpha_n\rangle\}, \tag{1.154}$$

and therefore

$$\hat{\psi}^\dagger_{\tilde{\alpha}} = \sum_\alpha\langle\alpha|\tilde{\alpha}\rangle\hat{\psi}^\dagger_\alpha,$$
$$\hat{\psi}_{\tilde{\alpha}} = \sum_\alpha\langle\tilde{\alpha}|\alpha\rangle\hat{\psi}_\alpha. \tag{1.155}$$

The transformed operators satisfy the (anti)commutation relations

$$[\hat{\psi}^\dagger_{\tilde{\alpha}}, \hat{\psi}^\dagger_{\tilde{\alpha}'}]_{-\zeta} = [\hat{\psi}_{\tilde{\alpha}}, \hat{\psi}_{\tilde{\alpha}'}]_{-\zeta} = 0,$$
$$[\hat{\psi}_{\tilde{\alpha}}, \hat{\psi}^\dagger_{\tilde{\alpha}'}]_{-\zeta} = \langle\tilde{\alpha}|\tilde{\alpha}'\rangle. \tag{1.156}$$

If the new basis is orthonormal, the (anti)commutation relations are preserved; the transformation $\{\hat{\psi}^\dagger_\alpha, \hat{\psi}_\alpha\} \to \{\hat{\psi}^\dagger_{\tilde{\alpha}}, \hat{\psi}_{\tilde{\alpha}}\}$ is unitary.

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1.2.1.4 Field operators

The "field operators" are defined by

$$\begin{aligned}\hat{\psi}_\sigma(\mathbf{r}) &= \sum_\alpha \langle \mathbf{r},\sigma|\alpha\rangle \hat{\psi}_\alpha = \frac{1}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{\psi}_\sigma(\mathbf{k}),\\ \hat{\psi}^\dagger_\sigma(\mathbf{r}) &= \sum_\alpha \langle \alpha|\mathbf{r},\sigma\rangle \hat{\psi}^\dagger_\alpha = \frac{1}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}}\hat{\psi}^\dagger_\sigma(\mathbf{k}),\end{aligned} \tag{1.157}$$

where the sum extends over all states α of the orthonormal basis. The last relations are obtained by choosing the momentum-representation basis $\{|\mathbf{k},\sigma\rangle\}$. The field operators satisfy the (anti)commutation relations

$$\begin{aligned}&[\hat{\psi}^\dagger_\sigma(\mathbf{r}),\hat{\psi}^\dagger_{\sigma'}(\mathbf{r}')]_{-\zeta} = [\hat{\psi}_\sigma(\mathbf{r}),\hat{\psi}_{\sigma'}(\mathbf{r}')]_{-\zeta} = 0,\\ &[\hat{\psi}_\sigma(\mathbf{r}),\hat{\psi}^\dagger_{\sigma'}(\mathbf{r}')]_{-\zeta} = \delta_{\sigma,\sigma'}\delta(\mathbf{r}-\mathbf{r}').\end{aligned} \tag{1.158}$$

From Eqs. (1.138) and (1.143), we deduce that

$$\begin{aligned}&\hat{\psi}^\dagger_\sigma(\mathbf{r})|\mathbf{r}_1\sigma_1,\cdots,\mathbf{r}_n\sigma_n\} = |\mathbf{r}\sigma,\mathbf{r}_1\sigma_1,\cdots\mathbf{r}_n\sigma_n\},\\ &\hat{\psi}_\sigma(\mathbf{r})|\mathbf{r}_1\sigma_1,\cdots,\mathbf{r}_n\sigma_n\} = \sum_{i=1}^n \zeta^{i-1}\delta(\mathbf{r}-\mathbf{r}_i)|\mathbf{r}_1\sigma_1,\cdots,\widehat{\mathbf{r}_i\sigma_i},\cdots,\mathbf{r}_n\sigma_n\},\end{aligned} \tag{1.159}$$

where $\widehat{\mathbf{r}_i\sigma_i}$ means that $\mathbf{r}_i\sigma_i$ is omitted. Thus the field operator $\hat{\psi}^\dagger_\sigma(\mathbf{r})$ adds a spin-σ particle at point $\mathbf{r}$ and (anti)symmetrizes the resulting many-body state.

1.2.2 Operators in second-quantized form

Any operator acting within the Fock space can be written in terms of the creation and annihilation operators. In the following, we discuss in detail one- and two-body operators before generalizing to n-body operators.

An operator $\hat{V}^{(1)}$ is a one-body operator if it acts on each particle separately,[26]

$$\begin{aligned}\hat{V}^{(1)}|\alpha_1\cdots\alpha_n) &= \sum_{i=1}^n \hat{V}_i|\alpha_1\cdots\alpha_n),\\ \hat{V}^{(1)}|\alpha_1\cdots\alpha_n\} &= \frac{1}{\sqrt{n!}}\sum_P \zeta^P \sum_{i=1}^n \hat{V}_i|\alpha_{P(1)}\cdots\alpha_{P(n)}),\end{aligned} \tag{1.160}$$

where $\hat{V}_i$ operates only on the ith particle. $\hat{V}_1|\alpha_1\cdots\alpha_n) = (\hat{V}|\alpha_1\rangle)\otimes|\alpha_2\rangle\otimes\cdots|\alpha_n\rangle$, etc. Let us first choose a basis where the operator $\hat{V}$ is diagonal,

$$\hat{V}|\alpha\rangle = \langle\alpha|\hat{V}|\alpha\rangle|\alpha\rangle \equiv V_\alpha|\alpha\rangle. \tag{1.161}$$

We then have

$$\hat{V}^{(1)}|\alpha_1\cdots\alpha_n\} = \frac{1}{\sqrt{n!}}\sum_P \zeta^P \sum_{i=1}^n V_{\alpha_{P(i)}}|\alpha_{P(1)}\cdots\alpha_{P(n)})$$

[26] Recall that in first-quantization, one would write $\hat{V}^{(1)} = \sum_{i=1}^n V(\hat{\mathbf{r}}_i,\hat{\mathbf{p}}_i,\hat{\sigma}_i)$ where the sum extends over the n particles of the system.

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$$= \sum_{i=1}^{n} V_{\alpha_i} |\alpha_1 \cdots \alpha_n\}$$
$$= \sum_{\alpha} V_\alpha \hat{n}_\alpha |\alpha_1 \cdots \alpha_n\}, \tag{1.162}$$

where the sum extends over the complete set of one-body states α and $\hat{n}_\alpha = \hat{\psi}_\alpha^\dagger \hat{\psi}_\alpha$. We deduce

$$\hat{V}^{(1)} = \sum_{\alpha} \langle \alpha | \hat{V} | \alpha \rangle \hat{n}_\alpha. \tag{1.163}$$

Using (1.155), we obtain the general expression for one-body operators,

$$\hat{V}^{(1)} = \sum_{\alpha,\alpha'} \langle \alpha | \hat{V} | \alpha' \rangle \hat{\psi}_\alpha^\dagger \hat{\psi}_{\alpha'}, \tag{1.164}$$

valid in any complete basis. In particular

$$\hat{V}^{(1)} = \int d^d r d^d r' \sum_{\sigma,\sigma'} \langle \mathbf{r}, \sigma | \hat{V} | \mathbf{r}', \sigma' \rangle \hat{\psi}_\sigma^\dagger(\mathbf{r}) \hat{\psi}_{\sigma'}(\mathbf{r}') \tag{1.165}$$

in terms of the field operators. Note that the expressions (1.164) and (1.165) make no reference to the total number of particles actually present in the system (to be compared with the first-quantized form of a one-body operator acting on a n-particle state[26]).

Examples. The density $\hat{n}(\mathbf{r})$ of particles at point $\mathbf{r}$ reads $\sum_{i=1}^n \delta(\mathbf{r} - \hat{\mathbf{r}}_i)$ in first quantization and

$$\hat{n}(\mathbf{r}) = \int d^d r_1 d^d r_2 \sum_{\sigma,\sigma'} \langle \mathbf{r}_1, \sigma | \delta(\mathbf{r} - \hat{\mathbf{r}}) | \mathbf{r}_2, \sigma' \rangle \hat{\psi}_\sigma^\dagger(\mathbf{r}_1) \hat{\psi}_{\sigma'}(\mathbf{r}_2)$$
$$= \sum_{\sigma} \hat{\psi}_\sigma^\dagger(\mathbf{r}) \hat{\psi}_\sigma(\mathbf{r}) \tag{1.166}$$

in second-quantized form. The number of particles operator is given by

$$\hat{N} = \int d^d r \, \hat{n}(\mathbf{r}) = \sum_{\alpha} \hat{\psi}_\alpha^\dagger \hat{\psi}_\alpha \tag{1.167}$$

for any complete orthonormal basis of one-body states $|\alpha\rangle$. For a one-body potential $\hat{V}^{(1)} = \sum_{i=1}^n V(\hat{\mathbf{r}}_i)$, one finds

$$\hat{V}^{(1)} = \int d^d r_1 d^d r_2 \sum_{\sigma,\sigma'} \langle \mathbf{r}_1, \sigma | V(\hat{\mathbf{r}}) | \mathbf{r}_2, \sigma' \rangle \hat{\psi}_\sigma^\dagger(\mathbf{r}_1) \hat{\psi}_{\sigma'}(\mathbf{r}_2')$$
$$= \int d^d r \, V(\mathbf{r}) \sum_{\sigma} \hat{\psi}_\sigma^\dagger(\mathbf{r}) \hat{\psi}_\sigma(\mathbf{r}) = \int d^d r \, V(\mathbf{r}) \hat{n}(\mathbf{r}). \tag{1.168}$$

The kinetic energy operator $\hat{T} = \sum_{i=1}^n \hat{\mathbf{p}}_i^2 / 2m$ is most easily calculated in the plane-wave basis,

$$\hat{T} = \sum_{\substack{\mathbf{k},\mathbf{k}' \\ \sigma,\sigma'}} \langle \mathbf{k}, \sigma | \frac{\hat{\mathbf{p}}^2}{2m} | \mathbf{k}', \sigma' \rangle \hat{\psi}_\sigma^\dagger(\mathbf{k}) \hat{\psi}_{\sigma'}(\mathbf{k}') = \sum_{\mathbf{k},\sigma} \frac{\mathbf{k}^2}{2m} \hat{\psi}_\sigma^\dagger(\mathbf{k}) \hat{\psi}_\sigma(\mathbf{k}). \tag{1.169}$$

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In direct space, this gives

$$\hat{T} = -\int d^d r \sum_\sigma \hat{\psi}^\dagger_\sigma(\mathbf{r}) \frac{\boldsymbol{\nabla}^2}{2m} \hat{\psi}_\sigma(\mathbf{r}) = \frac{1}{2m} \int d^d r \sum_\sigma \boldsymbol{\nabla}\hat{\psi}^\dagger_\sigma(\mathbf{r}) \cdot \boldsymbol{\nabla}\hat{\psi}_\sigma(\mathbf{r}). \tag{1.170}$$

The last line is obtained from an integration by part assuming that the fields vanish at infinity (or satisfy periodic boundary conditions). For the current operator $\hat{\mathbf{j}}(\mathbf{r}) = \frac{1}{2m}\sum_i [\hat{\mathbf{p}}_i \delta(\mathbf{r}-\hat{\mathbf{r}}_i) + \delta(\mathbf{r}-\hat{\mathbf{r}}_i)\hat{\mathbf{p}}_i]$, one obtains

$$\begin{aligned}\hat{\mathbf{j}}(\mathbf{r}) &= \frac{1}{2m}\int d^d r_1 d^d r_2 \sum_\sigma \langle \mathbf{r}_1 | \hat{\mathbf{p}}\delta(\mathbf{r}-\hat{\mathbf{r}}) + \delta(\mathbf{r}-\hat{\mathbf{r}})\hat{\mathbf{p}} | \mathbf{r}_2\rangle \hat{\psi}^\dagger_\sigma(\mathbf{r}_1)\hat{\psi}_\sigma(\mathbf{r}_2) \\ &= -\frac{i}{2m}\sum_\sigma \left[\hat{\psi}^\dagger_\sigma(\mathbf{r})\boldsymbol{\nabla}\hat{\psi}_\sigma(\mathbf{r}) - \text{H.c.}\right].\end{aligned} \tag{1.171}$$

The most general form of the grand canonical Hamiltonian of non-interacting particles is $\hat{H}_0 = \sum_\alpha \xi_\alpha \hat{\psi}^\dagger_\alpha \hat{\psi}_\alpha$ in the diagonal basis $\{|\alpha\rangle\}$, where $\xi_\alpha = \epsilon_\alpha - \mu$ denotes the energy of the one-body state $|\alpha\rangle$ with respect to the chemical potential μ.

An operator $\hat{V}^{(2)}$ is a two-body operator if it acts only on pairs of particles,

$$\begin{aligned}\hat{V}^{(2)}|\alpha_1 \cdots \alpha_n) &= \sum_{\substack{i,j=1\\(i<j)}}^n \hat{V}_{ij}|\alpha_1 \cdots \alpha_n), \\ \hat{V}^{(2)}|\alpha_1 \cdots \alpha_n\} &= \frac{1}{\sqrt{n!}}\sum_P \zeta^P \sum_{\substack{i,j=1\\(i<j)}}^n \hat{V}_{ij}|\alpha_{P(1)} \cdots \alpha_{P(n)}),\end{aligned} \tag{1.172}$$

where $\hat{V}_{ij}$ acts only on particles i and j. In the basis $\{|\alpha\rangle\}$ where $\hat{V}$ is diagonal,

$$\hat{V}|\alpha\beta) = (\alpha\beta|\hat{V}|\alpha\beta)|\alpha\beta) \equiv V_{\alpha\beta}|\alpha\beta), \tag{1.173}$$

one has

$$\begin{aligned}\hat{V}^{(2)}|\alpha_1 \cdots \alpha_n\} &= \frac{1}{\sqrt{n!}}\sum_P \zeta^P \sum_{\substack{i,i=1\\(i<j)}}^n V_{\alpha_{P(i)}\alpha_{P(j)}}|\alpha_{P(1)} \cdots \alpha_{P(n)}) \\ &= \sum_{\substack{i,j=1\\(i<j)}}^n V_{\alpha_i\alpha_j}|\alpha_1 \cdots \alpha_n\},\end{aligned} \tag{1.174}$$

which leads to

$$\begin{aligned}\hat{V}^{(2)}|\alpha_1 \cdots \alpha_n\} &= \frac{1}{2}\sum_{i,j=1}^n \left(V_{\alpha_i\alpha_j} - \delta_{i,j}V_{\alpha_i\alpha_i}\right)|\alpha_1 \cdots \alpha_n\} \\ &= \frac{1}{2}\sum_{\alpha,\beta} V_{\alpha\beta}\left(\hat{n}_\alpha\hat{n}_\beta - \delta_{\alpha,\beta}\hat{n}_\alpha\right)|\alpha_1 \cdots \alpha_n\},\end{aligned} \tag{1.175}$$

where the sum extends over all states α, β of the complete basis of the one-body states. Using the (anti)commutation relations (1.150,1.153), one finds $\hat{n}_\alpha\hat{n}_\beta - \delta_{\alpha,\beta}\hat{n}_\alpha = \hat{\psi}^\dagger_\alpha\hat{\psi}^\dagger_\beta\hat{\psi}_\beta\hat{\psi}_\alpha$, which leads to

$$\hat{V}^{(2)} = \frac{1}{2}\sum_{\alpha,\beta}(\alpha\beta|\hat{V}|\alpha\beta)\hat{\psi}^\dagger_\alpha\hat{\psi}^\dagger_\beta\hat{\psi}_\beta\hat{\psi}_\alpha. \tag{1.176}$$

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Transforming to an arbitrary basis, we finally find the general expression of a two-body operator,

$$\begin{aligned}\hat{V}^{(2)} &= \frac{1}{2}\sum_{\alpha_1\cdots\alpha_2'}(\alpha_1\alpha_2|\hat{V}|\alpha_1'\alpha_2')\hat{\psi}^\dagger_{\alpha_1}\hat{\psi}^\dagger_{\alpha_2}\hat{\psi}_{\alpha_2'}\hat{\psi}_{\alpha_1'} \\ &= \frac{1}{2}\int d^dr_1\cdots d^dr_2'\sum_{\sigma_1\cdots\sigma_2'}(\mathbf{r}_1\sigma_1,\mathbf{r}_2\sigma_2|\hat{V}|\mathbf{r}_1'\sigma_1',\mathbf{r}_2'\sigma_2')\hat{\psi}^\dagger_{\sigma_1}(\mathbf{r}_1)\hat{\psi}^\dagger_{\sigma_2}(\mathbf{r}_2)\hat{\psi}_{\sigma_2'}(\mathbf{r}_2')\hat{\psi}_{\sigma_1'}(\mathbf{r}_1').\end{aligned} \tag{1.177}$$

Notice the respecting order of α_1' and α_2' in the matrix element and in the product of annihilation operators. It is sometimes convenient to use a variant of (1.177) where the matrix element is (anti)symmetrized,

$$\hat{V}^{(2)} = \frac{1}{4}\sum_{\alpha_1\cdots\alpha_2'}\{\alpha_1\alpha_2|\hat{V}|\alpha_1'\alpha_2'\}\hat{\psi}^\dagger_{\alpha_1}\hat{\psi}^\dagger_{\alpha_2}\hat{\psi}_{\alpha_2'}\hat{\psi}_{\alpha_1'} \tag{1.178}$$

with

$$\{\alpha_1\alpha_2|\hat{V}|\alpha_1'\alpha_2'\} = (\alpha_1\alpha_2|\hat{V}|\alpha_1'\alpha_2') + \zeta(\alpha_1\alpha_2|\hat{V}|\alpha_2'\alpha_1'). \tag{1.179}$$

Example. For a two-body potential $\hat{V}^{(2)} = \sum_{i<j} v(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j)$ (in first-quantized form), one finds

$$\begin{aligned}\hat{V}^{(2)} &= \frac{1}{2}\int d^dr_1\cdots d^dr_2'\sum_{\sigma_1\cdots\sigma_2'}(\mathbf{r}_1\sigma_1,\mathbf{r}_2\sigma_2|v(\hat{\mathbf{r}}-\hat{\mathbf{r}}')|\mathbf{r}_1'\sigma_1',\mathbf{r}_2'\sigma_2') \\ &\qquad\times\hat{\psi}^\dagger_{\sigma_1}(\mathbf{r}_1)\hat{\psi}^\dagger_{\sigma_2}(\mathbf{r}_2)\hat{\psi}_{\sigma_2'}(\mathbf{r}_2')\hat{\psi}_{\sigma_1'}(\mathbf{r}_1') \\ &= \frac{1}{2}\int d^dr_1d^dr_2\sum_{\sigma_1,\sigma_2}v(\mathbf{r}_1-\mathbf{r}_2)\hat{\psi}^\dagger_{\sigma_1}(\mathbf{r}_1)\hat{\psi}^\dagger_{\sigma_2}(\mathbf{r}_2)\hat{\psi}_{\sigma_2}(\mathbf{r}_2)\hat{\psi}_{\sigma_1}(\mathbf{r}_1).\end{aligned} \tag{1.180}$$

Here we have used the fact that $v(\hat{\mathbf{r}}-\hat{\mathbf{r}}')$ is diagonal in the coordinate representation: $v(\hat{\mathbf{r}}-\hat{\mathbf{r}}')|\mathbf{r}_1\sigma_1,\mathbf{r}_2\sigma_2) = v(\mathbf{r}_1-\mathbf{r}_2)|\mathbf{r}_1\sigma_1,\mathbf{r}_2\sigma_2)$. In Fourier space, Eq. (1.180) becomes

$$\hat{V}^{(2)} = \frac{1}{2\mathcal{V}}\sum_{\substack{\mathbf{k},\mathbf{k}',\mathbf{q}\\ \sigma,\sigma'}}v(\mathbf{q})\hat{\psi}^\dagger_\sigma(\mathbf{k}+\mathbf{q})\hat{\psi}^\dagger_{\sigma'}(\mathbf{k}'-\mathbf{q})\hat{\psi}_{\sigma'}(\mathbf{k}')\hat{\psi}_\sigma(\mathbf{k}), \tag{1.181}$$

where $v(\mathbf{q})$ is the Fourier transform of $v(\mathbf{r})$.

To conclude this section, we give without proof the expression of a n-body operator in second-quantized form,

$$\begin{aligned}\hat{V}^{(n)} &= \frac{1}{n!}\sum_{\{\alpha_i,\alpha_i'\}}(\alpha_1\cdots\alpha_n|\hat{V}|\alpha_1'\cdots\alpha_n')\hat{\psi}^\dagger_{\alpha_1}\cdots\hat{\psi}^\dagger_{\alpha_n}\hat{\psi}_{\alpha_n'}\cdots\hat{\psi}_{\alpha_1'} \\ &= \frac{1}{(n!)^2}\sum_{\{\alpha_i,\alpha_i'\}}\{\alpha_1\cdots\alpha_n|\hat{V}|\alpha_1'\cdots\alpha_n'\}\hat{\psi}^\dagger_{\alpha_1}\cdots\hat{\psi}^\dagger_{\alpha_n}\hat{\psi}_{\alpha_n'}\cdots\hat{\psi}_{\alpha_1'}.\end{aligned} \tag{1.182}$$

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Table 1.1: Main results from second quantification.

Many-body states	$\|\alpha_1 \cdots \alpha_n) = \|\alpha_1\rangle \otimes \cdots \otimes \|\alpha_n\rangle$
	$\|\alpha_1 \cdots \alpha_n\} = \frac{1}{\sqrt{n!}} \sum_P \zeta^P \|\alpha_{P(1)} \cdots \alpha_{P(n)})$
	$\|\alpha_1 \cdots \alpha_n\rangle = \frac{1}{(\prod_{i=1}^p n_{\alpha_i}!)^{1/2}} \|\alpha_1 \cdots \alpha_n\}$
Closure relation	$\sum_{n=0}^{\infty} \frac{1}{n!} \sum_{\alpha_1 \cdots \alpha_n} \|\alpha_1 \cdots \alpha_n\}\{\alpha_1 \cdots \alpha_n\| = 1$
Creation and annihilation operators	$\hat{\psi}_\alpha^\dagger \|\alpha_1 \cdots \alpha_n\} = \|\alpha\alpha_1 \cdots \alpha_n\}$
	$\hat{\psi}_\alpha \|\alpha_1 \cdots \alpha_n\} = \sum_{i=1}^n \zeta^{i-1} \delta_{\alpha,\alpha_i} \|\alpha_1 \cdots \hat{\alpha}_i \cdots \alpha_n\}$
	$\hat{\psi}_\alpha^\dagger \|\alpha_1 \cdots \alpha_n\rangle = \sqrt{n_\alpha + 1} \|\alpha\alpha_1 \cdots \alpha_n\rangle$
	$\hat{\psi}_\alpha \|\alpha_1 \cdots \alpha_n\rangle = \frac{1}{\sqrt{n_\alpha}} \sum_{i=1}^n \zeta^{i-1} \delta_{\alpha,\alpha_i} \|\alpha_1 \cdots \hat{\alpha}_i \cdots \alpha_n\rangle$
(Anti)commutation relations	$[\hat{\psi}_\alpha, \hat{\psi}_{\alpha'}]_{-\zeta} = [\hat{\psi}_\alpha^\dagger, \hat{\psi}_{\alpha'}^\dagger]_{-\zeta} = 0$
	$[\hat{\psi}_\alpha, \hat{\psi}_{\alpha'}^\dagger]_{-\zeta} = \delta_{\alpha,\alpha'}$
Field operators	$\hat{\psi}_\sigma(\mathbf{r}) = \sum_\alpha \langle \mathbf{r}, \sigma \| \alpha \rangle \hat{\psi}_\alpha$
	$\hat{\psi}_\sigma^\dagger(\mathbf{r}) = \sum_\alpha \langle \alpha \| \mathbf{r}, \sigma \rangle \hat{\psi}_\alpha^\dagger$
	$[\hat{\psi}_\sigma(\mathbf{r}), \hat{\psi}_{\sigma'}(\mathbf{r}')]_{-\zeta} = [\hat{\psi}_\sigma^\dagger(\mathbf{r}), \hat{\psi}_{\sigma'}^\dagger(\mathbf{r}')]_{-\zeta} = 0$
	$[\hat{\psi}_\sigma(\mathbf{r}), \hat{\psi}_{\sigma'}^\dagger(\mathbf{r}')]_{-\zeta} = \delta_{\sigma,\sigma'} \delta(\mathbf{r} - \mathbf{r}')$
One-body operator	$\hat{V}^{(1)} = \sum_{\alpha,\alpha'} \langle \alpha \| \hat{V} \| \alpha' \rangle \hat{\psi}_\alpha^\dagger \hat{\psi}_{\alpha'}$
	$= \int d^d r d^d r' \sum_{\sigma,\sigma'} \langle \mathbf{r}, \sigma \| \hat{V} \| \mathbf{r}', \sigma' \rangle \hat{\psi}_\sigma^\dagger(\mathbf{r}) \hat{\psi}_{\sigma'}(\mathbf{r}')$
Two-body operator	$\hat{V}^{(2)} = \frac{1}{2} \sum_{\alpha_1 \cdots \alpha_2'} (\alpha_1 \alpha_2 \| \hat{V} \| \alpha_1' \alpha_2') \hat{\psi}_{\alpha_1}^\dagger \hat{\psi}_{\alpha_2}^\dagger \hat{\psi}_{\alpha_2'} \hat{\psi}_{\alpha_1'}$

1.2.3 Green functions

1.2.3.1 Retarded Green function

When studying the quantum mechanics of a single particle in Sec. 1.1.1, we introduced the propagator

$$G^R(\mathbf{r}, \mathbf{r}'; t) = -i\Theta(t) \langle \mathbf{r} | e^{-i\hat{H}t} | \mathbf{r}' \rangle = -i\Theta(t) \sum_n e^{-i\epsilon_n t} \varphi_n(\mathbf{r}) \varphi_n^*(\mathbf{r}') \tag{1.183}$$

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describing the propagation of a particle from $(\mathbf{r}', 0)$ to $(\mathbf{r}, t)$. We ignore for the time being the spin of the particle. To make causality explicit, we have added the factor $\Theta(t)$: The particle can propagate only forward in time. The sum in (1.183) extends over a complete set $\{\varphi_n(\mathbf{r}), \epsilon_n\}$ of one-body eigenstates. G^R is called the retarded propagator or retarded Green function. Its Fourier transform is defined by[27]

$$G^R(\mathbf{r}, \mathbf{r}'; \omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} G^R(\mathbf{r}, \mathbf{r}'; t) = \sum_n \frac{\varphi_n(\mathbf{r})\varphi_n^*(\mathbf{r}')}{\omega + i\eta - \epsilon_n}, \tag{1.184}$$

where $\eta \to 0^+$. Thus the energies ϵ_n of the eigenstates $|n\rangle$ show up as poles of the retarded Green function while the residues give information about the wavefunctions $\varphi_n(\mathbf{r}) = \langle \mathbf{r}|n\rangle$. Since $|\mathbf{r}\rangle = \hat{\psi}^\dagger(\mathbf{r})|\text{vac}\rangle$, G^R can be rewritten as

$$G^R(\mathbf{r}, \mathbf{r}'; t) = -i\Theta(t)\langle \text{vac}|\hat{\psi}(\mathbf{r}) e^{-i\hat{H}t}\hat{\psi}^\dagger(\mathbf{r}')|\text{vac}\rangle. \tag{1.185}$$

In a many-body system, it is tempting to generalize (1.185) as

$$\begin{aligned} G^R(\mathbf{r}, \mathbf{r}'; t) &= -i\Theta(t)\langle 0|\hat{\psi}(\mathbf{r}) e^{-i\hat{H}t}\hat{\psi}^\dagger(\mathbf{r}')|0\rangle \\ &= -i\Theta(t)\langle 0|\hat{\psi}(\mathbf{r}, t)\hat{\psi}^\dagger(\mathbf{r}', 0)|0\rangle, \end{aligned} \tag{1.186}$$

where we take $\epsilon_0 = 0$ for the energy of the many-body ground state $|0\rangle$.

$$\begin{aligned} \hat{\psi}(\mathbf{r}, t) &= e^{i\hat{H}t}\hat{\psi}(\mathbf{r}) e^{-i\hat{H}t}, \\ \hat{\psi}^\dagger(\mathbf{r}, t) &= e^{i\hat{H}t}\hat{\psi}^\dagger(\mathbf{r}) e^{-i\hat{H}t} \end{aligned} \tag{1.187}$$

are the field operators in the Heisenberg picture. In the grand canonical formalism, the term $-\mu\hat{N}$ is included in the Hamiltonian. Contrary to (1.183), the definition (1.186) does not satisfy $G^R(\mathbf{r}, t^+; \mathbf{r}', t) = -i\delta(\mathbf{r} - \mathbf{r}')$ – an expected result for a "propagator" since within an infinitesimal time the particle has no time to propagate. The latter relation is however satisfied if we choose

$$G^R(\mathbf{r}, \mathbf{r}'; t) = -i\Theta(t)\langle 0|\big[\hat{\psi}(\mathbf{r}, t), \hat{\psi}^\dagger(\mathbf{r}', 0)\big]_{-\zeta}|0\rangle. \tag{1.188}$$

This expression reflects the fact that in a many-body system we can not only add but also remove a particle. At finite temperature the ground state expectation value must be replaced by a statistical average and we can define the retarded Green function as

$$\begin{aligned} G^R(\alpha, \alpha'; t) &= -i\Theta(t)\langle\big[\hat{\psi}_\alpha(t), \hat{\psi}^\dagger_{\alpha'}\big]_{-\zeta}\rangle \\ &= -i\Theta(t)\frac{1}{Z}\text{Tr}\left\{e^{-\beta\hat{H}}\big[\hat{\psi}_\alpha(t), \hat{\psi}^\dagger_{\alpha'}\big]_{-\zeta}\right\} \end{aligned} \tag{1.189}$$

for any complete set $\{|\alpha\rangle\}$ of one-body states. α includes a spin index as well as other internal indices if necessary.

For a non-interacting system with grand canonical Hamiltonian $\hat{H}_0 = \sum_\alpha \xi_\alpha \hat{\psi}^\dagger_\alpha \hat{\psi}_\alpha$ ($\xi_\alpha = \epsilon_\alpha - \mu$) in the diagonal basis, the field operators satisfy the equations of motion

$$\begin{aligned} \partial_t \hat{\psi}_\alpha(t) &= i\big[\hat{H}_0, \hat{\psi}_\alpha(t)\big] = -i\xi_\alpha\hat{\psi}_\alpha(t), \\ \partial_t \hat{\psi}^\dagger_\alpha(t) &= i\big[\hat{H}_0, \hat{\psi}^\dagger_\alpha(t)\big] = i\xi_\alpha\hat{\psi}^\dagger_\alpha(t), \end{aligned} \tag{1.190}$$

[27] We use $\int_0^\infty dt\, e^{i\omega t} = i/(\omega + i0^+)$, which follows from the inverse relation $\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{\omega + i0^+} = -i\Theta(t)$ (the latter is easy derived by using the residue theorem).

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and are therefore given by

$$\begin{aligned}\hat{\psi}_\alpha(t) &= e^{-i\xi_\alpha t}\hat{\psi}_\alpha,\\ \hat{\psi}^\dagger_\alpha(t) &= e^{i\xi_\alpha t}\hat{\psi}^\dagger_\alpha.\end{aligned} \tag{1.191}$$

From (1.189), we then deduce

$$G_0^R(\alpha,t) = -i\Theta(t)e^{-i\xi_\alpha t}\langle[\hat{\psi}_\alpha,\hat{\psi}^\dagger_\alpha]_{-\zeta}\rangle = -i\Theta(t)e^{-i\xi_\alpha t}, \tag{1.192}$$

and

$$G_0^R(\alpha,\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t}G^R_{0,\alpha}(t) = \frac{1}{\omega + i\eta - \xi_\alpha}. \tag{1.193}$$

As in the single-particle case [Eq. (1.184)], the poles of the retarded Green function give the single-particle excitation energies, i.e. the energy of the many-body states obtained by adding (or removing) a particle to the equilibrium state. In Chap. 3, we shall study in detail the properties of the retarded Green function in interacting systems and show that its poles are indeed related to the single-particle excitation energies.

1.2.3.2 Matsubara Green function

The retarded Green function is not the one that is calculated in the first place. Instead, one usually focuses on the Matsubara Green function

$$\begin{aligned}G(\alpha,\alpha';\tau) &= -\langle T_\tau\hat{\psi}_\alpha(\tau)\hat{\psi}^\dagger_{\alpha'}\rangle\\ &= -\Theta(\tau)\langle\hat{\psi}_\alpha(\tau)\hat{\psi}^\dagger_{\alpha'}\rangle - \zeta\Theta(-\tau)\langle\hat{\psi}^\dagger_{\alpha'}\hat{\psi}_\alpha(\tau)\rangle,\end{aligned} \tag{1.194}$$

where the time-ordering operator T_τ is defined by the second line. For fermions, the interchange of the two operators when $\tau < 0$ implies a sign change. The imaginary-time operators in (1.194) are defined by

$$\begin{aligned}\hat{\psi}_\alpha(\tau) &= e^{\tau\hat{H}}\hat{\psi}_\alpha e^{-\tau\hat{H}},\\ \hat{\psi}^\dagger_\alpha(\tau) &= e^{\tau\hat{H}}\hat{\psi}^\dagger_\alpha e^{-\tau\hat{H}}.\end{aligned} \tag{1.195}$$

For $\tau = it$, they reduce to the real-time Heisenberg operators (1.187). Note that $\hat{\psi}^\dagger_\alpha(\tau)$ is not the adjoint $[\hat{\psi}_\alpha(\tau)]^\dagger$ of $\hat{\psi}_\alpha(\tau)$. For $\tau > 0$, $G(\alpha,\alpha';\tau)$ describes the creation of a particle and its propagation in the system, while for $\tau < 0$ it describes the annihilation of a particle and the propagation of the corresponding "hole". There is no need to specify here how the equal-time ($\tau = 0$) Green function is defined. As we will see, $G(\alpha,\alpha';0)$ will always be interpreted as either $G(\alpha,\alpha';0^+)$ or $G(\alpha,\alpha';0^-)$.

We shall see later that we need to know $G(\alpha,\alpha';\tau)$ only for $\tau \in [-\beta,\beta]$. For $-\beta < \tau < 0$, one obtains

$$\begin{aligned}G(\alpha,\alpha';\tau) &= -\zeta\langle\hat{\psi}^\dagger_{\alpha'}\hat{\psi}_\alpha(\tau)\rangle\\ &= -\zeta e^{\beta\Omega}\mathrm{Tr}\big[e^{-\beta\hat{H}}\hat{\psi}^\dagger_{\alpha'}e^{\tau\hat{H}}\hat{\psi}_\alpha e^{-\tau\hat{H}}\big]\\ &= -\zeta e^{\beta\Omega}\mathrm{Tr}\big[e^{\tau\hat{H}}\hat{\psi}_\alpha e^{-(\tau+\beta)\hat{H}}\hat{\psi}^\dagger_{\alpha'}\big]\\ &= -\zeta e^{\beta\Omega}\mathrm{Tr}\big[e^{-\beta\hat{H}}\hat{\psi}_\alpha(\tau+\beta)\hat{\psi}^\dagger_{\alpha'}\big]\end{aligned}$$

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$$= \zeta G(\alpha, \alpha'; \tau + \beta) \tag{1.196}$$

(with $\Omega = -\frac{1}{\beta} \ln Z$ the thermodynamic potential) using the cyclic invariance of the trace. A similar reasoning gives

$$G(\alpha, \alpha'; \tau) = \zeta G(\alpha, \alpha'; \tau - \beta) \tag{1.197}$$

for $0 < \tau < \beta$. Since $G(\alpha, \alpha'; \tau)$ is (anti)periodic for $\tau \in [-\beta, \beta]$ with period β, it can be expanded in a Fourier series,

$$\begin{aligned} G(\alpha, \alpha'; \tau) &= \frac{1}{\beta} \sum_{\omega_n} e^{-i\omega_n \tau} G(\alpha, \alpha'; i\omega_n), \\ G(\alpha, \alpha'; i\omega_n) &= \int_0^\beta d\tau \, e^{i\omega_n \tau} G(\alpha, \alpha'; \tau), \end{aligned} \tag{1.198}$$

where $\sum_{\omega_n} = \sum_{n=-\infty}^{\infty}$ and the discrete frequencies

$$\omega_n = \begin{cases} \dfrac{2\pi}{\beta} n & \text{(bosons)}, \\ \dfrac{2\pi}{\beta} \left(n + \frac{1}{2}\right) & \text{(fermions)}, \end{cases} \tag{1.199}$$

are called Matsubara frequencies.

The Matsubara Green function can be easily calculated for a non-interacting system with Hamiltonian $\hat{H}_0 = \sum_\alpha \xi_\alpha \hat{\psi}_\alpha^\dagger \hat{\psi}_\alpha$. Inserting $\hat{\psi}_\alpha(\tau) = e^{-\xi_\alpha \tau} \hat{\psi}_\alpha$ and $\hat{\psi}_\alpha^\dagger(\tau) = e^{-\xi_\alpha \tau} \hat{\psi}_\alpha^\dagger$ in (1.194), we obtain

$$G_0(\alpha, \tau) = -e^{-\xi_\alpha \tau} \big[\Theta(\tau)(1 + \zeta n_\alpha) + \Theta(-\tau) \zeta n_\alpha\big], \tag{1.200}$$

where the occupation number $n_\alpha = \langle \hat{\psi}_\alpha^\dagger \hat{\psi}_\alpha \rangle$ is given by the Fermi-Dirac function $n_F(\xi_\alpha)$ for fermions and the Bose-Einstein function $n_B(\xi_\alpha)$ for bosons. In frequency space

$$G_0(\alpha, i\omega_n) = \frac{1}{i\omega_n - \xi_\alpha}. \tag{1.201}$$

Comparing (1.193) and (1.201), we see that the retarded Green function can be obtained from the Matsubara one by an analytic continuation,

$$G_0^R(\alpha, \omega) = G_0(\alpha, i\omega_n \to \omega + i\eta). \tag{1.202}$$

This result turns out to be valid also in interacting systems. The retarded Green function, which contains physical information about the system, can be deduced from the Matsubara Green function which is the natural quantity to consider in (finite-temperature) interacting systems. We postpone a detailed study of the properties of Green functions to Chap. 3 and in the following sections of this chapter discuss how we can define and compute Green functions in the functional integral formalism.

1.2.3.3 Many-particle Green function

The Green function we have discussed so far is called the one-particle Green function (or propagator) since it describes the propagation of a single particle. One can also define higher-order Green functions

$$G^{(2n)}(\alpha_1 \tau_1, \cdots, \alpha_n \tau_n; \alpha_n' \tau_n', \cdots, \alpha_1' \tau_1')$$

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$$= (-1)^n \langle T_\tau \hat{\psi}_{\alpha_1}(\tau_1) \cdots \hat{\psi}_{\alpha_n}(\tau_n) \hat{\psi}^\dagger_{\alpha'_n}(\tau'_n) \cdots \hat{\psi}^\dagger_{\alpha'_1}(\tau'_1) \rangle. \tag{1.203}$$

The superscript $(2n)$ corresponds to the number of operators rather than the number n of particles propagating in the system. We shall often denote the one-particle Green function $G^{(2)}$ merely by G. As in (1.194), the operator T_τ chronologically orders the operators from left to right with the latest time to the left,

$$T_\tau \hat{A}_{\alpha_1}(\tau_1) \cdots \hat{A}_{\alpha_n}(\tau_n) = \zeta^P \hat{A}_{\alpha_{P(1)}}(\tau_{P(1)}) \cdots \hat{A}_{\alpha_{P(n)}}(\tau_{P(n)}) \tag{1.204}$$

$(\tau_{P(1)} > \tau_{P(2)} > \cdots > \tau_{P(n)})$, where ζ^P equals one for bosons and the signature of the permutation P for fermions.

1.2.3.4 Real-time Green function

Finally, we can define Green functions in real time,

$$\begin{aligned} &G^{(2n)}(\alpha_1 t_1, \cdots, \alpha_n t_n; \alpha'_n t'_n, \cdots, \alpha'_1 t'_1) \\ &\qquad = (-i)^n \langle T \hat{\psi}_{\alpha_1}(t_1) \cdots \hat{\psi}_{\alpha_n}(t_n) \hat{\psi}^\dagger_{\alpha'_n}(t'_n) \cdots \hat{\psi}^\dagger_{\alpha'_1}(t'_1) \rangle, \end{aligned} \tag{1.205}$$

where T is the time-ordering operator (not to be confused with the temperature) defined like its imaginary-time counterpart T_τ, and $\hat{\psi}_\alpha(t), \hat{\psi}^\dagger_\alpha(t)$ the Heisenberg picture operators (1.187). The relation between the various Green functions introduced in this section will be studied in Secs. 3.2.7 and 3.5.

1.3 Coherent states

The quantum mechanical behavior of a single particle is usually described in terms of the position and momentum operators $\hat{\mathbf{q}}$ and $\hat{\mathbf{p}}$. When constructing a path integral representation of the propagator or the partition function, it therefore appears natural to introduce the closure relations $\int d^d q |\mathbf{q}\rangle\langle\mathbf{q}| = 1$ and/or $\sum_{\mathbf{p}} |\mathbf{p}\rangle\langle\mathbf{p}| = 1$ at each time step (Sec. 1.1.1). Since second quantization is the natural formalism for studying many-particle systems, we should try to construct the path integral using a closure relation based on the eigenstates of the creation or annihilation operators.

Now it is clear that an arbitrary state

$$|\psi\rangle = \sum_{\alpha_1 \cdots \alpha_n} C_{\alpha_1 \cdots \alpha_n} |\alpha_1 \cdots \alpha_n\rangle \tag{1.206}$$

cannot be an eigenstate of the creation operator $\hat{\psi}^\dagger_\alpha$. $|\psi\rangle$ has necessarily a component with a minimum number n_0 of particles. Applying $\hat{\psi}^\dagger_\alpha$ to this component will either increase n_0 by 1 or yield zero if the single-particle state α is already occupied and the particles are fermions. In all cases, the minimum numbers of particles in $|\psi\rangle$ and $\hat{\psi}^\dagger_\alpha |\psi\rangle$ differ. On the other hand there is *a priori* nothing that prevents $|\psi\rangle$ to be an eigenstate of $\hat{\psi}_\alpha$. Suppose that $|\psi\rangle$ is an eigenstate of the annihilation operators,

$$\hat{\psi}_\alpha |\psi\rangle = \psi_\alpha |\psi\rangle \quad \forall \alpha, \tag{1.207}$$

with ψ_α a c-number. Then the (anti)commutation relations

$$[\hat{\psi}_\alpha, \hat{\psi}_\beta]_{-\zeta} = 0 \tag{1.208}$$

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imply

$$\begin{aligned}\hat{\psi}_\alpha\hat{\psi}_\beta|\psi\rangle &= \hat{\psi}_\alpha\psi_\beta|\psi\rangle = \psi_\alpha\psi_\beta|\psi\rangle \\ &= \zeta\hat{\psi}_\beta\hat{\psi}_\alpha|\psi\rangle = \zeta\hat{\psi}_\beta\psi_\alpha|\psi\rangle = \zeta\psi_\beta\psi_\alpha|\psi\rangle,\end{aligned} \tag{1.209}$$

i.e.

$$[\psi_\alpha, \psi_\beta]_{-\zeta} = 0. \tag{1.210}$$

Thus, for fermions, the eigenvalues cannot be ordinary numbers. We shall need to introduce anticommuting variables called Grassmann numbers.[28] Postponing the case of fermions to Secs. 1.3.2 and 1.3.3, we now restrict the discussion to the case of bosons.[29]

1.3.1 Boson coherent states

It is straightforward to verify that the so-called boson coherent states

$$\begin{aligned}|\psi\rangle &= \exp\left\{\sum_\alpha \psi_\alpha\hat{\psi}_\alpha^\dagger\right\}|\text{vac}\rangle \\ &= \sum_{n_{\alpha_1}, n_{\alpha_2}, \cdots = 0}^{\infty} \frac{(\psi_{\alpha_1}^{n_{\alpha_1}} \cdots \psi_{\alpha_p}^{n_{\alpha_p}} \cdots)}{(n_{\alpha_1}! \cdots n_{\alpha_p}! \cdots)^{1/2}} |n_{\alpha_1} \cdots n_{\alpha_p} \cdots\rangle\end{aligned} \tag{1.211}$$

(ψ_α is a c-number) are eigenstates of the annihilation operators,[30]

$$\hat{\psi}_\alpha|\psi\rangle = \psi_\alpha|\psi\rangle. \tag{1.212}$$

The action of the creation operator $\hat{\psi}_\alpha^\dagger$ on the coherent states is given by

$$\hat{\psi}_\alpha^\dagger|\psi\rangle = \frac{\partial}{\partial\psi_\alpha}|\psi\rangle. \tag{1.213}$$

By taking the Hermitian conjugate in (1.212) and (1.213), we deduce

$$\begin{aligned}\langle\psi|\hat{\psi}_\alpha^\dagger &= \langle\psi|\psi_\alpha^*, \\ \langle\psi|\hat{\psi}_\alpha &= \frac{\partial}{\partial\psi_\alpha^*}\langle\psi|,\end{aligned} \tag{1.214}$$

where

$$\langle\psi| = \langle\text{vac}|\exp\left\{\sum_\alpha \psi_\alpha^*\hat{\psi}_\alpha\right\} \tag{1.215}$$

is the bra corresponding to the ket $|\psi\rangle$ and ψ_α^* the complex conjugate of ψ_α.

The overlap between two states is given by

$$\langle\psi|\psi'\rangle = \exp\left\{\sum_\alpha \psi_\alpha^*\psi'_\alpha\right\}. \tag{1.216}$$

[28] Note that when ψ_α is an anticommuting variable, (1.209) holds regardless of whether ψ_α commutes or anticommutes with the operator $\hat{\psi}_\beta$.

[29] The properties of coherent states are summarized in Table 1.2 on page 46.

[30] Expand the exponential in (1.211) and use $[\hat{\psi}_\alpha, (\hat{\psi}_\beta^\dagger)^n]_- = \delta_{\alpha,\beta} n(\hat{\psi}_\alpha^\dagger)^{n-1}$.

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Another crucial property of the coherent states is that they form an overcomplete basis in the Fock space,[31] which results in the closure relation

$$\int \prod_\alpha \frac{d\psi_\alpha^* d\psi_\alpha}{2i\pi} e^{-\sum_\alpha |\psi_\alpha|^2} |\psi\rangle\langle\psi| = 1, \tag{1.217}$$

where

$$\frac{d\psi_\alpha^* d\psi_\alpha}{2i\pi} = \frac{1}{\pi} d\Re(\psi_\alpha) d\Im(\psi_\alpha). \tag{1.218}$$

For a single degree of freedom, writing $\psi = \rho e^{i\theta}$, one finds

$$\begin{aligned}\int \frac{d\psi^* d\psi}{2i\pi} e^{-|\psi|^2} |\psi\rangle\langle\psi| &= \frac{1}{\pi}\int_0^\infty \rho d\rho \int_0^{2\pi} d\theta\, e^{-\rho^2} \sum_{n,m=0}^\infty \frac{\rho^{n+m} e^{i(n-m)\theta}}{(n!m!)^{1/2}} |n\rangle\langle m| \\ &= 2\int_0^\infty \rho d\rho\, e^{-\rho^2} \sum_{n=0}^\infty \frac{\rho^{2n}}{n!} |n\rangle\langle n| \\ &= \sum_{n=0}^\infty |n\rangle\langle n| = 1. \end{aligned} \tag{1.219}$$

This result is generalized to many degrees of freedom by writing the coherent state $|\psi\rangle$ in terms of the states $|n_{\alpha_1}\cdots\rangle$ [Eq. (1.211)] and using the closure relation (1.146) in Fock space.

Let us now consider an operator $\hat{A} \equiv A(\hat{\psi}_\alpha^\dagger, \hat{\psi}_\alpha)$ where all creation operators stand to the left of the annihilation operators. Such an operator is said to be normal ordered. Its matrix element between coherent states takes the simple form

$$\langle\psi|A(\hat{\psi}_\alpha^\dagger, \hat{\psi}_\alpha)|\psi'\rangle = e^{\sum_\alpha \psi_\alpha^* \psi'_\alpha} A(\psi_\alpha^*, \psi'_\alpha). \tag{1.220}$$

To conclude this section, we note that the occupation numbers n_α in the coherent state $|\psi\rangle$ are Poisson distributed with mean values $|\psi_\alpha|^2$,

$$|\langle n_{\alpha_1}\cdots n_{\alpha_p}\cdots|\psi\rangle|^2 = \prod_\alpha \frac{|\psi_\alpha|^{2n_\alpha}}{n_\alpha!}. \tag{1.221}$$

The total number of particles has the average value

$$N = \langle\hat{N}\rangle = \frac{\langle\psi|\hat{N}|\psi\rangle}{\langle\psi|\psi\rangle} = \sum_\alpha |\psi_\alpha|^2 \tag{1.222}$$

(recall that $\hat{N} = \sum_\alpha \hat{\psi}_\alpha^\dagger \hat{\psi}_\alpha$) and variance

$$\sigma^2 = \left\langle \left(\hat{N} - \langle\hat{N}\rangle\right)^2 \right\rangle = \langle\hat{N}\rangle. \tag{1.223}$$

In the thermodynamic limit where $N \to \infty$, $\sigma/N = 1/\sqrt{N} \to 0$ and the particle number distribution becomes sharply peaked around N, reflecting the fact that a product of Poisson distributions approaches a normal distribution.

[31] The basis is said to be overcomplete because it contains more vectors than necessary to be complete.

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The physical meaning of the boson coherent states can be understood from the study of a one-dimensional harmonic oscillator,

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \tag{1.224}$$

with $[\hat{x}, \hat{p}] = i$. The ladder operators

$$\hat{a} = \sqrt{\frac{m\omega}{2}}\left(\hat{x} + i\frac{\hat{p}}{m\omega}\right), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2}}\left(\hat{x} - i\frac{\hat{p}}{m\omega}\right) \tag{1.225}$$

satisfy canonical bosonic commutation relations, $[\hat{a}, \hat{a}^\dagger] = 1$, and enable to rewrite the Hamiltonian as

$$\hat{H} = \omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right). \tag{1.226}$$

The states $|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|\mathrm{vac}\rangle$ are eigenstates with eigenvalues $\left(n + \frac{1}{2}\right)\omega$.

Let us now consider the normalized coherent state

$$|z\rangle = e^{-\frac{|z|^2}{2} + z\hat{a}^\dagger}|\mathrm{vac}\rangle. \tag{1.227}$$

A simple calculation yields

$$\begin{aligned} \langle z|\hat{x}|z\rangle &= \sqrt{\frac{2}{m\omega}}\Re(z), \qquad & \langle z|\hat{x}^2|z\rangle &= \frac{1}{2m\omega}[4\Re(z)^2 + 1], \\ \langle z|\hat{p}|z\rangle &= \sqrt{2m\omega}\Im(z), \qquad & \langle z|\hat{p}^2|z\rangle &= \frac{m\omega}{2}[4\Im(z)^2 + 1], \end{aligned} \tag{1.228}$$

and therefore

$$\Delta x^2 = \frac{1}{2m\omega}, \qquad \Delta p^2 = \frac{m\omega}{2}. \tag{1.229}$$

Thus the coherent state $|z\rangle$ yields the minimum position and momentum fluctuations compatible with the uncertainty relation $\Delta x \Delta p \geq \hbar/2$ (we have restored Planck's constant).

In order to further study the relation between coherent states and the classical limit of the harmonic oscillator, we consider the dynamics of the eigenstates $|n\rangle$ in the Schrödinger picture,

$$e^{-i\hat{H}t}|n\rangle = e^{-i\left(n+\frac{1}{2}\right)\omega t}|n\rangle. \tag{1.230}$$

From (1.230) we deduce the time evolution of the coherent states,

$$e^{-i\hat{H}t}|z\rangle = e^{-\frac{|z|^2}{2}}\sum_{n=0}^{\infty}\frac{z^n}{\sqrt{n!}}e^{-i\hat{H}t}|n\rangle = e^{-\frac{i}{2}\omega t}|e^{-i\omega t}z\rangle. \tag{1.231}$$

Apart from the unimportant phase factor $e^{-\frac{i}{2}\omega t}$, the dynamics of the coherent state $|z\rangle$ is simply obtained by replacing z by $z(t) = ze^{-i\omega t}$. One then finds that the expectation values

$$\begin{aligned} \langle\hat{x}\rangle(t) &= \langle z(t)|\hat{x}|z(t)\rangle = \sqrt{\frac{2}{m\omega}}z(0)\cos(\omega t), \\ \langle\hat{p}\rangle(t) &= \langle z(t)|\hat{p}|z(t)\rangle = \sqrt{2m\omega}z(0)\sin(\omega t) \end{aligned} \tag{1.232}$$

(we assume $z(0) = z$ to be real) satisfy the classical equations of motion $\dot{x} = \partial H/\partial p = p/m$ and $\dot{p} = -\partial H/\partial x = -m\omega^2 x$. The amplitude $|z(0)|$ of the oscillations is related to the expectation value of $\hat{N} = \hat{a}^\dagger\hat{a}$,

$$\langle\hat{N}\rangle = \langle z(t)|\hat{a}^\dagger\hat{a}|z(t)\rangle = |z(t)|^2 = |z(0)|^2. \tag{1.233}$$

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The time independence of $\langle\hat{N}\rangle$ follows from $[\hat{H},\hat{N}]=0$ (see Appendix 1.B and Chap. 2). When $\langle\hat{N}\rangle=|z(0)|^2\gg 1$, $|\langle\hat{x}\rangle(t)|\gg\Delta x$ and $|\langle\hat{p}\rangle(t)|\gg\Delta p$ for most times t (with Δx and Δp given by (1.229)). A coherent state with large occupation number $\langle\hat{N}\rangle=|z(0)|^2$ therefore corresponds to a quasi-classical behavior of the harmonic oscillator. Boson coherent states are sometimes referred to as quasi-classical states.[32]

1.3.2 Grassmann variables

A Grassmann algebra is an algebra of anticommuting variables called Grassmann variables (or numbers).[33] It is defined by a set $\{\psi_1,\cdots,\psi_n\}$ of n generators whose product is anticommutative,

$$[\psi_i,\psi_j]_+=\psi_i\psi_j+\psi_j\psi_i=0. \tag{1.234}$$

This implies that the square ψ_i^2 of any generator vanishes. An element of the Grassmann algebra is a polynomial of first order in the generators,

$$f(\psi_1,\cdots,\psi_n)=c_0+\sum_{k=1}^{n}\sum_{i_1\cdots i_k=1}^{n}c_{i_1\cdots i_k}\psi_{i_1}\cdots\psi_{i_k}, \tag{1.235}$$

where the coefficients c_0 and $c_{i_1\cdots i_k}$ are complex and we assume that the indices $i_1<i_2<\cdots<i_k$ are ordered. For example, the elements of a Grassmann algebra with only one or two generators are

$$\begin{aligned} f(\psi)&=c_0+c_1\psi,\\ f(\psi_1,\psi_2)&=c_0+c_1\psi_1+c_2\psi_2+c_{12}\psi_1\psi_2. \end{aligned} \tag{1.236}$$

Differentiation with respect to Grassmann numbers is defined by

$$\frac{\partial}{\partial\psi_i}\psi_j=\delta_{i,j}. \tag{1.237}$$

As with ordinary numbers, this is a linear operation. The anticommutation relations (1.234) imply

$$\left[\partial_{\psi_i},\partial_{\psi_j}\right]_+=0. \tag{1.238}$$

The operator ∂_{ψ_i} is therefore nilpotent, $\partial_{\psi_i}^2=0$, reflecting the fact that the function $f(\psi_1,\cdots,\psi_n)$ is at most of order 1 in each variable. Equation (1.237) defines a left differentiation,

$$\partial_{\psi_i}\psi_j\psi_i=-\partial_{\psi_i}\psi_i\psi_j=-\psi_j\qquad(i\neq j). \tag{1.239}$$

For ∂_{ψ_i} to act on ψ_i, one must first shift ψ_i to the left of the monomial. The chain rule for differentiation then reads

$$\frac{\partial}{\partial\psi_i}f(g)=\frac{\partial g}{\partial\psi_i}\frac{\partial f}{\partial g}. \tag{1.240}$$

Contrary to ordinary variables, the order of the terms in the rhs matters.

For Grassmann numbers, integration and differentiation are identical,

$$\int d\psi_i\, f(\psi_1,\cdots,\psi_n)=\frac{\partial}{\partial\psi_i}f(\psi_1,\cdots,\psi_n). \tag{1.241}$$

[32] Coherent states were introduced in quantum optics by Glauber [29] to describe the electromagnetic field; see the discussion on page 118 in Sec. 1.9.

[33] Recall that an algebra is a vector space endowed with a multiplication rule.

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This property ensures that two fundamental properties of ordinary integrals over functions vanishing at infinity are satisfied. The integral of an exact differential form is zero,

$$\int d\psi_i \frac{\partial}{\partial \psi_i} f(\psi_1, \cdots, \psi_n) = 0, \tag{1.242}$$

and the integral over ψ_i of $f(\psi_1, \cdots, \psi_n)$ does not depend on ψ_i so that its derivative vanishes,

$$\frac{\partial}{\partial \psi_i} \int d\psi_i \, f(\psi_1, \cdots, \psi_n) = 0. \tag{1.243}$$

Both properties (1.242) and (1.243) follow from the definition (1.241) and the nilpotence of the differential operator ∂_{ψ_i}. For a Grassmann algebra with a single generator, one has

$$\begin{aligned} f(\psi) &= c_0 + c_1 \psi, \\ \frac{\partial f}{\partial \psi} &= \int d\psi \, f = c_1. \end{aligned} \tag{1.244}$$

Other properties of Grassmann integration are discussed in Appendix 1.E.

It will often be convenient to consider Grassmann algebras with $2n$ generators $\{\psi_1, \cdots, \psi_n, \psi_1^*, \cdots, \psi_n^*\}$. One can then define a (complex) conjugation for Grassmann variables,

$$\begin{aligned} &(\psi_i)^* = \psi_i^*, \\ &(\psi_i^*)^* = \psi_i, \\ &(c\psi_i)^* = c^* \psi_i^*, \\ &(\psi_i \psi_j)^* = \psi_j^* \psi_i^*, \end{aligned} \qquad (c \in \mathbb{C}). \tag{1.245}$$

The general element of a Grassmann algebra with two generators $\{\psi, \psi^*\}$ reads

$$f(\psi, \psi^*) = c_0 + c_1 \psi^* + c_2 \psi + c_{12} \psi^* \psi. \tag{1.246}$$

Using the differentiation and integration rules, one easily finds

$$\begin{aligned} \frac{\partial f}{\partial \psi} &= \int d\psi \, f = c_2 - c_{12} \psi^*, \\ \frac{\partial f}{\partial \psi^*} &= \int d\psi^* \, f = c_1 + c_{12} \psi, \\ \frac{\partial^2 f}{\partial \psi^* \partial \psi} &= \int d\psi^* d\psi \, f = -c_{12}. \end{aligned} \tag{1.247}$$

1.3.3 Fermion coherent states

Grassmann variables enable us to define coherent states similar to the ones introduced for bosons. Given a set $\{\hat{\psi}_\alpha^\dagger, \hat{\psi}_\alpha\}$ of creation and annihilation operators, we define a Grassmann algebra by associating a generator ψ_α to $\hat{\psi}_\alpha$ and a generator ψ_α^* to $\hat{\psi}_\alpha^\dagger$ for any α. We also require

$$\begin{aligned} &\left[\psi_\alpha^{(*)}, \hat{\psi}_\beta^{(\dagger)}\right]_+ = 0, \\ &(\psi_\alpha \hat{\psi}_\beta)^\dagger = \hat{\psi}_\beta^\dagger \psi_\alpha^*, \quad (\psi_\alpha \hat{\psi}_\beta^\dagger)^\dagger = \hat{\psi}_\beta \psi_\alpha^*, \quad \text{etc.} \end{aligned} \tag{1.248}$$

A fermion coherent state is defined by the ket

$$|\psi\rangle = \exp\left\{-\sum_\alpha \psi_\alpha \hat{\psi}_\alpha^\dagger\right\} |\text{vac}\rangle = \prod_\alpha (1 - \psi_\alpha \hat{\psi}_\alpha^\dagger) |\text{vac}\rangle \tag{1.249}$$

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and the corresponding bra

$$\langle\psi| = \langle\text{vac}|\exp\biggl\{\sum_\alpha \psi^*_\alpha\hat{\psi}_\alpha\biggr\} = \langle\text{vac}|\prod_\alpha(1+\psi^*_\alpha\hat{\psi}_\alpha). \tag{1.250}$$

It is straightforward to show that the properties (1.212-1.214) obtained previously for the boson coherent states can be transposed to the fermionic case,[34]

$$\begin{aligned}
\hat{\psi}_\alpha|\psi\rangle &= \psi_\alpha|\psi\rangle,\\
\hat{\psi}^\dagger_\alpha|\psi\rangle &= -\frac{\partial}{\partial\psi_\alpha}|\psi\rangle,\\
\langle\psi|\hat{\psi}^\dagger_\alpha &= \langle\psi|\psi^*_\alpha,\\
\langle\psi|\hat{\psi}_\alpha &= \frac{\partial}{\partial\psi^*_\alpha}\langle\psi|.
\end{aligned} \tag{1.251}$$

The overlap between two states is given by

$$\langle\psi|\psi'\rangle = \exp\biggl\{\sum_\alpha \psi^*_\alpha\psi'_\alpha\biggr\}, \tag{1.252}$$

and for a normal-ordered operator $\hat{A} = A(\hat{\psi}^\dagger_\alpha,\hat{\psi}_\alpha)$,

$$\langle\psi|A(\hat{\psi}^\dagger_\alpha,\hat{\psi}_\alpha)|\psi'\rangle = \langle\psi|A(\psi^*_\alpha,\psi'_\alpha)|\psi'\rangle = e^{\sum_\alpha \psi^*_\alpha\psi'_\alpha}A(\psi^*_\alpha,\psi'_\alpha). \tag{1.253}$$

The last result is obtained by noting that $|\psi\rangle$ and $\langle\psi|$ commute with any product of Grassmann variables by virtue of (1.248-1.250).

The overcompleteness of the coherent state basis reflects into the closure relation

$$\int\prod_\alpha d\psi^*_\alpha d\psi_\alpha\, e^{-\sum_\alpha \psi^*_\alpha\psi_\alpha}|\psi\rangle\langle\psi| = 1. \tag{1.254}$$

This relation can be proven by considering the matrix element of the lhs of (1.254) between two arbitrary states of the Fock space. From

$$\begin{aligned}
\langle\alpha_1\cdots\alpha_n|\psi\rangle &= \langle\text{vac}|\hat{\psi}_{\alpha_n}\cdots\hat{\psi}_{\alpha_1}|\psi\rangle = \langle\text{vac}|\psi_{\alpha_n}\cdots\psi_{\alpha_1}|\psi\rangle = \psi_{\alpha_n}\cdots\psi_{\alpha_1},\\
\langle\psi|\beta_1\cdots\beta_m\rangle &= \langle\psi|\hat{\psi}^\dagger_{\beta_1}\cdots\hat{\psi}^\dagger_{\beta_m}|\text{vac}\rangle = \langle\psi|\psi^*_{\beta_1}\cdots\psi^*_{\beta_m}|\text{vac}\rangle = \psi^*_{\beta_1}\cdots\psi^*_{\beta_m}
\end{aligned} \tag{1.255}$$

(note that $\langle\text{vac}|\psi\rangle = \langle\psi|\text{vac}\rangle = 1$), we deduce

$$\begin{aligned}
&\int\prod_\alpha d\psi^*_\alpha d\psi_\alpha\, e^{-\sum_\alpha \psi^*_\alpha\psi_\alpha}\langle\alpha_1\cdots\alpha_n|\psi\rangle\langle\psi|\beta_1\cdots\beta_m\rangle\\
&= \int\prod_\alpha d\psi^*_\alpha d\psi_\alpha\prod_\alpha(1-\psi^*_\alpha\psi_\alpha)\psi_{\alpha_n}\cdots\psi_{\alpha_1}\psi^*_{\beta_1}\cdots\psi^*_{\beta_m}.
\end{aligned} \tag{1.256}$$

[34] For instance, the first equation in (1.251) is obtained from $\hat{\psi}_\alpha|\psi\rangle = \hat{\psi}_\alpha\prod_\beta(1-\psi_\beta\hat{\psi}^\dagger_\beta)|\text{vac}\rangle = \prod_{\beta\neq\alpha}(1-\psi_\beta\hat{\psi}^\dagger_\beta)\hat{\psi}_\alpha(1-\psi_\alpha\hat{\psi}^\dagger_\alpha)|\text{vac}\rangle = \prod_{\beta\neq\alpha}(1-\psi_\beta\hat{\psi}^\dagger_\beta)\psi_\alpha|\text{vac}\rangle = \prod_{\beta\neq\alpha}(1-\psi_\beta\hat{\psi}^\dagger_\beta)\psi_\alpha(1-\psi_\alpha\hat{\psi}^\dagger_\alpha)|\text{vac}\rangle = \psi_\alpha|\psi\rangle$.

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Table 1.2: Main properties of coherent states. ψ_α denotes a complex variable for bosons ($\zeta = 1$) and a Grassmann number for fermions ($\zeta = -1$). $\mathcal{N} = 2i\pi$ for bosons and $\mathcal{N} = 1$ for fermions.

Definition	$\|\psi\rangle = e^{\zeta \sum_\alpha \psi_\alpha \hat{\psi}_\alpha^\dagger} \|\text{vac}\rangle$ $\langle\psi\| = \langle\text{vac}\| e^{\sum_\alpha \psi_\alpha^* \hat{\psi}_\alpha}$
	$\hat{\psi}_\alpha \|\psi\rangle = \psi_\alpha \|\psi\rangle, \qquad \langle\psi\| \hat{\psi}_\alpha^\dagger = \langle\psi\| \psi_\alpha^*$
Overlap	$\langle\psi\|\psi'\rangle = e^{\sum_\alpha \psi_\alpha^* \psi'_\alpha}$
Closure relation	$\int \prod_\alpha \frac{d\psi_\alpha^* d\psi_\alpha}{\mathcal{N}} e^{-\sum_\alpha \|\psi_\alpha\|^2} \|\psi\rangle\langle\psi\| = 1$
Trace	$\text{Tr}(\hat{A}) = \int \prod_\alpha \frac{d\psi_\alpha^* d\psi_\alpha}{\mathcal{N}} e^{-\sum_\alpha \|\psi_\alpha\|^2} \langle\zeta\psi\|\hat{A}\|\psi\rangle$
Normal-ordered operator	$\langle\psi\| A(\hat{\psi}_\alpha^\dagger, \hat{\psi}_\alpha) \|\psi'\rangle = e^{\sum_\alpha \psi_\alpha^* \psi'_\alpha} A(\psi_\alpha^*, \psi'_\alpha)$

Four types of integrals appear in (1.256),

$$\int d\psi_\alpha^* d\psi_\alpha (1 - \psi_\alpha^* \psi_\alpha) \begin{pmatrix} \psi_\alpha \psi_\alpha^* \\ \psi_\alpha \\ \psi_\alpha^* \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{1.257}$$

Thus each α must belong to both $\{\alpha_i\}$ and $\{\beta_i\}$ or be absent from both sets for (1.256) to be nonzero. In this case $n = m$ and $\beta_i = \alpha_{P(i)}$ with $P \in S_n$ a permutation, and Eq. (1.256) reduces to

$$\begin{aligned} &\int \prod_\alpha d\psi_\alpha^* d\psi_\alpha \prod_\alpha (1 - \psi_\alpha^* \psi_\alpha) \psi_{\alpha_n} \cdots \psi_{\alpha_1} \psi^*_{\alpha_{P(1)}} \cdots \psi^*_{\alpha_{P(m)}} \\ = &\int \prod_\alpha d\psi_\alpha^* d\psi_\alpha \prod_\alpha (1 - \psi_\alpha^* \psi_\alpha) (-1)^P \psi_{\alpha_n} \cdots \psi_{\alpha_1} \psi^*_{\alpha_1} \cdots \psi^*_{\alpha_m} = (-1)^P. \end{aligned} \tag{1.258}$$

This result should be compared with

$$\langle\alpha_1 \cdots \alpha_n | \beta_1 \cdots \beta_m\rangle = \delta_{n,m} \sum_{P \in S_n} (-1)^P \delta_{\alpha_i, \beta_{P(i)}}, \tag{1.259}$$

which proves the closure relation (1.254).

To conclude this section, we note that fermion coherent states are not physical states of the system; they do not belong to the Fock space. For instance, the expectation value of the particle number operator $\hat{N} = \sum_\alpha \hat{\psi}_\alpha^\dagger \hat{\psi}_\alpha$ in the coherent state $|\psi\rangle$,

$$\frac{\langle\psi|\hat{N}|\psi\rangle}{\langle\psi|\psi\rangle} = \sum_\alpha \psi_\alpha^* \psi_\alpha, \tag{1.260}$$

is not a real number.

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1.4 Functional integral in many-particle systems

We are now in a position to derive a functional integral representation of the partition function of a many-particle system following the general procedure outlined in Sec. 1.1.

1.4.1 Coherent-state functional integral

1.4.1.1 Partition function

Let us first express the partition function Z in the coherent-state basis (the following derivation actually holds for the trace of any operator). Using the closure relation (1.217) or (1.254), we can write

$$Z = \sum_n \langle n|e^{-\beta\hat{H}}|n\rangle = \sum_n \int d(\psi^*,\psi)\, e^{-\sum_\alpha \psi_\alpha^*\psi_\alpha} \langle n|\psi\rangle\langle\psi|e^{-\beta\hat{H}}|n\rangle, \tag{1.261}$$

where $\hat{H}$ is the grand canonical Hamiltonian of the system (including the term $-\mu\hat{N}$), and $\{|n\rangle\}$ a basis in the Fock space. $|\psi\rangle$ is a coherent state, and ψ_α a c-number for bosons and a Grassmann variable for fermions. We use the notation

$$d(\psi^*,\psi) = \begin{cases} \prod_\alpha \dfrac{d\psi_\alpha^* d\psi_\alpha}{2i\pi} & \text{(bosons)}, \\ \prod_\alpha d\psi_\alpha^* d\psi_\alpha & \text{(fermions)}. \end{cases} \tag{1.262}$$

For bosons, we can interchange the two terms $\langle n|\psi\rangle$ and $\langle\psi|e^{-\beta\hat{H}}|n\rangle$ in (1.261) and then use $\sum_n |n\rangle\langle n| = 1$. For fermions, this interchange must be realized with care. Let us write $e^{-\beta\hat{H}}|n\rangle = \sum_m a_m|m\rangle$. Since $\hat{H}$ conserves the particle number, the kets $|n\rangle$ and $|m\rangle$ must have the same number of particles. Writing

$$\begin{aligned} |n\rangle &= \hat{\psi}_{\alpha_1}^\dagger \cdots \hat{\psi}_{\alpha_p}^\dagger |\text{vac}\rangle, \\ |m\rangle &= \hat{\psi}_{\gamma_1}^\dagger \cdots \hat{\psi}_{\gamma_p}^\dagger |\text{vac}\rangle, \end{aligned} \tag{1.263}$$

we obtain

$$\begin{aligned} \langle n|\psi\rangle\langle\psi|m\rangle &= \psi_{\alpha_p}\cdots\psi_{\alpha_1}\psi_{\gamma_1}^*\cdots\psi_{\gamma_p}^* \\ &= \zeta^{p^2}\psi_{\gamma_1}^*\cdots\psi_{\gamma_p}^*\psi_{\alpha_p}\cdots\psi_{\alpha_1} \\ &= \langle\zeta\psi|m\rangle\langle n|\psi\rangle \end{aligned} \tag{1.264}$$

(note that $\zeta^{p^2} = \zeta^p$). We therefore deduce

$$\langle n|\psi\rangle\langle\psi|e^{-\beta\hat{H}}|n\rangle = \sum_m a_m\langle\zeta\psi|m\rangle\langle n|\psi\rangle = \langle\zeta\psi|e^{-\beta\hat{H}}|n\rangle\langle n|\psi\rangle \tag{1.265}$$

and the partition function can be written as

$$Z = \int d(\psi^*,\psi)\, e^{-\sum_\alpha \psi_\alpha^*\psi_\alpha} \langle\zeta\psi|e^{-\beta\hat{H}}|\psi\rangle. \tag{1.266}$$

We are now in a position to write the partition function as a coherent-state functional integral. Dividing the imaginary time β into $N = \beta/\epsilon$ steps and inserting $N-1$ times the

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closure relation (1.217) or (1.254), we obtain

$$Z = \int \prod_{k=1}^{N} d(\psi_k^*, \psi_k)\, e^{-\sum_{k=1}^{N}\sum_\alpha \psi_{k,\alpha}^* \psi_{k,\alpha}} \prod_{k=1}^{N} \langle \psi_k | e^{-\epsilon \hat{H}} | \psi_{k-1} \rangle, \tag{1.267}$$

where we have relabeled $\zeta\psi^{(*)} \equiv \psi_N^{(*)}$ in Eq. (1.266) and introduced the variables (not to be integrated over) $\psi_0^{(*)} = \zeta\psi_N^{(*)} \equiv \psi^{(*)}$. If the Hamiltonian $\hat{H} = H(\hat{\psi}_\alpha^\dagger, \hat{\psi}_\alpha)$ is normal ordered,

$$\langle \psi_k | e^{-\epsilon \hat{H}} | \psi_{k-1} \rangle = \langle \psi_k | \psi_{k-1} \rangle e^{-\epsilon H(\psi_{k,\alpha}^*, \psi_{k-1,\alpha})} + \mathcal{O}(\epsilon^2), \tag{1.268}$$

one finds

$$Z = \int \prod_{k=1}^{N} d(\psi_k^*, \psi_k) \exp\left\{ -\epsilon \sum_{k=1}^{N} \left[\sum_\alpha \psi_{k,\alpha}^* \frac{(\psi_{k,\alpha} - \psi_{k-1,\alpha})}{\epsilon} + H(\psi_{k,\alpha}^*, \psi_{k-1,\alpha}) \right] \right\} \tag{1.269}$$

for $\epsilon \to 0$. As for the Feynman path integral (Sec. 1.1), in the limit $N \to \infty$ it is convenient to represent the set $\{\psi_{0,\alpha}, \cdots, \psi_{N,\alpha}\}$ by a continuous trajectory $\psi_\alpha(\tau)$ with τ an imaginary time varying between 0 and β and the boundary conditions

$$\psi_\alpha(\beta) = \zeta\psi_\alpha(0), \qquad \psi_\alpha^*(\beta) = \zeta\psi_\alpha^*(0). \tag{1.270}$$

We can then (symbolically) write

$$\begin{aligned} \epsilon \sum_{k=1}^{N} &\to \int_0^\beta d\tau, \\ \psi_{k,\alpha}^* \frac{(\psi_{k,\alpha} - \psi_{k-1,\alpha})}{\epsilon} &\to \psi_\alpha^*(\tau)\partial_\tau \psi_\alpha(\tau), \\ H(\psi_{k,\alpha}^*, \psi_{k-1,\alpha}) &\to H\big(\psi_\alpha^*(\tau^+), \psi_\alpha(\tau)\big), \end{aligned} \tag{1.271}$$

so that the partition takes the final form

$$Z = \int_{\substack{\psi(\beta)=\zeta\psi(0) \\ \psi^*(\beta)=\zeta\psi^*(0)}} \mathcal{D}[\psi^*, \psi]\, e^{-S[\psi^*,\psi]} \tag{1.272}$$

with the Euclidean action[35]

$$S[\psi^*, \psi] = \int_0^\beta d\tau \left[\sum_\alpha \psi_\alpha^*(\tau)\partial_\tau \psi_\alpha(\tau) + H\big(\psi_\alpha^*(\tau^+), \psi_\alpha(\tau)\big) \right]. \tag{1.273}$$

The notation τ^+ reminds us that the creation field is always evaluated infinitesimally later than the annihilation field (see (1.271)). For fermions the notation $\partial_\tau \psi_\alpha(\tau)$ is purely symbolic since there is no sense in which $\psi_{k,\alpha} - \psi_{k-1,\alpha}$ is small. The symbol $\partial_\tau \psi_\alpha(\tau)$ must be understood as $\lim_{\epsilon\to 0} \frac{1}{\epsilon}(\psi_{k,\alpha} - \psi_{k-1,\alpha})$. The measure in the functional integral (1.272) is defined as

$$\mathcal{D}[\psi^*, \psi] = \lim_{N\to\infty} \prod_{k=0}^{N} d(\psi_k^*, \psi_k) \tag{1.274}$$

[35] For simplicity we denote the Euclidean action merely by S (and not S_E).

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and must be supplemented with the boundary conditions $\psi(\beta) = \zeta\psi(0)$ and $\psi^*(\beta) = \zeta\psi^*(0)$.

For a system with a non-interacting one-body Hamiltonian $\hat{h}_0$ and a two-body interaction $\hat{v}$, the action reads

$$S[\psi^*,\psi] = \int_0^\beta d\tau \Bigg\{ \sum_{\alpha,\alpha'} \psi^*_\alpha(\tau) \left[\delta_{\alpha,\alpha'}\partial_\tau + \langle\alpha|\hat{h}_0|\alpha'\rangle \right] \psi_{\alpha'}(\tau) + \frac{1}{2} \sum_{\alpha_1,\alpha_2,\alpha'_1,\alpha'_2} (\alpha_1\alpha_2|\hat{v}|\alpha'_1\alpha'_2) \psi^*_{\alpha_1}(\tau^+)\psi^*_{\alpha_2}(\tau^+)\psi_{\alpha'_2}(\tau)\psi_{\alpha'_1}(\tau) \Bigg\}. \tag{1.275}$$

1.4.1.2 Green functions

Let us now consider the Green function $G(\alpha\tau,\alpha'\tau') = -\langle T_\tau \hat{\psi}_\alpha(\tau)\hat{\psi}^\dagger_{\alpha'}(\tau')\rangle$ introduced in Sec. 1.2.3. For $\tau > \tau'$ it reads

$$G(\alpha\tau,\alpha'\tau') = -\frac{1}{Z}\mathrm{Tr}\left[e^{-(\beta-\tau)\hat{H}}\hat{\psi}_\alpha e^{-(\tau-\tau')\hat{H}}\hat{\psi}^\dagger_{\alpha'} e^{-\tau'\hat{H}} \right]. \tag{1.276}$$

We now split the time intervals $\beta-\tau$, $\tau-\tau'$ and τ' in infinitesimal steps $\epsilon = \beta/N$. If we assume that τ corresponds to the discrete time $\tau_{k_1} = k_1\epsilon$ and τ' to $\tau_{k_2} = k_2\epsilon$, we obtain

$$G(\alpha\tau,\alpha'\tau') = -\frac{1}{Z}\int \prod_{k=1}^N d(\psi^*_k,\psi_k)\, e^{-\sum_{k=1}^N \sum_\gamma \psi^*_{k,\gamma}\psi_{k,\gamma}} \langle\psi_N|e^{-\epsilon\hat{H}}|\psi_{N-1}\rangle \cdots \\ \cdots \langle\psi_{k_1+1}|e^{-\epsilon\hat{H}}\hat{\psi}_\alpha|\psi_{k_1}\rangle \cdots \langle\psi_{k_2}|\hat{\psi}^\dagger_{\alpha'}e^{-\epsilon\hat{H}}|\psi_{k_2-1}\rangle \cdots \langle\psi_1|e^{-\epsilon\hat{H}}|\psi_0\rangle, \tag{1.277}$$

with the boundary conditions $\psi_N = \zeta\psi_0$ and $\psi^*_N = \zeta\psi^*_0$. We deduce

$$G(\alpha\tau,\alpha'\tau') = -\frac{1}{Z}\int \prod_{k=1}^N d(\psi^*_k,\psi_k)\, \psi_{k_1,\alpha}\psi^*_{k_2,\alpha'} e^{-S(\{\psi^*_k,\psi_k\})} \\ \equiv -\frac{1}{Z}\int \mathcal{D}[\psi^*,\psi]\, \psi_\alpha(\tau)\psi^*_{\alpha'}(\tau') e^{-S[\psi^*,\psi]}, \tag{1.278}$$

where $S(\{\psi^*_k,\psi_k\})$ is the action defined in (1.269) and $S[\psi^*,\psi]$ its continuum-time limit (1.273). From now on we do not write explicitly the boundary conditions $\psi(\beta) = \zeta\psi(0)$ and $\psi^*(\beta) = \zeta\psi^*(0)$. For $\tau < \tau'$, we can follow a similar reasoning. Starting from

$$G(\alpha\tau,\alpha'\tau') = -\frac{\zeta}{Z}\mathrm{Tr}\left[e^{-(\beta-\tau')\hat{H}}\hat{\psi}^\dagger_{\alpha'} e^{-(\tau'-\tau)\hat{H}}\hat{\psi}_\alpha e^{-\tau\hat{H}} \right], \tag{1.279}$$

one obtains

$$G(\alpha\tau,\alpha'\tau') = -\frac{\zeta}{Z}\int \mathcal{D}[\psi^*,\psi]\, \psi^*_{\alpha'}(\tau')\psi_\alpha(\tau) e^{-S[\psi^*,\psi]} \\ = -\frac{1}{Z}\int \mathcal{D}[\psi^*,\psi]\, \psi_\alpha(\tau)\psi^*_{\alpha'}(\tau') e^{-S[\psi^*,\psi]}, \tag{1.280}$$

which is identical to (1.278). Thus the Green function can be written as a functional integral. This result generalizes to the n-particle Green function (1.203) and any statistical average of a time-ordered product of operators,

$$\langle T_\tau A_1(\hat{\psi}^\dagger_\alpha(\tau_1),\hat{\psi}_\alpha(\tau_1)) \cdots A_n(\hat{\psi}^\dagger_\alpha(\tau_n),\hat{\psi}_\alpha(\tau_n))\rangle$$

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$$= \frac{1}{Z}\mathrm{Tr}\left[e^{-\beta\hat{H}}T_\tau A_1(\hat{\psi}^\dagger_\alpha(\tau_1),\hat{\psi}_\alpha(\tau_1))\cdots A_n(\hat{\psi}^\dagger_\alpha(\tau_n),\hat{\psi}_\alpha(\tau_n))\right]$$
$$= \frac{1}{Z}\int \mathcal{D}[\psi^*,\psi]\, A_1(\psi^*_\alpha(\tau_1),\psi_\alpha(\tau_1))\cdots A_n(\psi^*_\alpha(\tau_n),\psi_\alpha(\tau_n))\, e^{-S[\psi^*,\psi]}. \tag{1.281}$$

1.4.2 Non-interacting particles

For non-interacting particles with grand canonical Hamiltonian $\hat{H}_0 = \sum_\alpha \xi_\alpha \hat{\psi}^\dagger_\alpha \hat{\psi}_\alpha$, the action reads

$$S_0 = \sum_\alpha \sum_{k=1}^{N} \left[\psi^*_{k,\alpha}(\psi_{k,\alpha} - \psi_{k-1,\alpha}) + \epsilon\xi_\alpha \psi^*_{k,\alpha}\psi_{k-1,\alpha}\right]. \tag{1.282}$$

In order to diagonalize S_0, we expand the fields in Fourier series,

$$\psi_{k,\alpha} = \frac{1}{\sqrt{N}}\sum_{n=0}^{N-1} a_{n,\alpha} e^{-i\omega_n\tau_k}, \qquad \psi^*_{k,\alpha} = \frac{1}{\sqrt{N}}\sum_{n=0}^{N-1} a^*_{n,\alpha} e^{i\omega_n\tau_k}, \tag{1.283}$$

where $\tau_k = k\epsilon$ and the Matsubara frequencies ω_n are defined in (1.199). The a_n's are complex variables for bosons and Grassmann numbers for fermions. Since the transformation $\psi^{(*)} \to a^{(*)}$ has unit Jacobian,[36] one obtains

$$Z_0 = \lim_{N\to\infty} \int \prod_{n=0}^{N-1} d(a^*_n, a_n) \exp\left\{-\sum_\alpha \sum_{n=0}^{N-1} |a_{n,\alpha}|^2 \left[1 + (\epsilon\xi_\alpha - 1)e^{i\omega_n\epsilon}\right]\right\}$$
$$= \lim_{N\to\infty} \prod_\alpha \prod_{n=0}^{N-1} \left[1 + (\epsilon\xi_\alpha - 1)e^{i\omega_n\epsilon}\right]^{-\zeta} \tag{1.284}$$

using the results of Appendix 1.E. Writing $1 - \epsilon\xi_\alpha = e^{-\theta_\alpha}$ and using the identity

$$\prod_{n=0}^{N-1}\left(1 - e^{i\omega_n\epsilon - \theta}\right) = 1 - \zeta e^{-N\theta}, \tag{1.285}$$

we finally deduce

$$Z_0 = \lim_{N\to\infty} \prod_\alpha \left(1 - \zeta e^{-N\theta_\alpha}\right)^{-\zeta} = \prod_\alpha \left(1 - \zeta e^{-\beta\xi_\alpha}\right)^{-\zeta}, \tag{1.286}$$

which is the familiar result for non-interacting particles.

The Fourier transform of the one-particle Green function is obtained from

$$-\langle a_{n,\alpha} a^*_{n,\alpha}\rangle = -\lim_{N\to\infty} \frac{1}{Z}\int \prod_{n=0}^{N-1} d(a^*_n, a_n)\, a_{n,\alpha} a^*_{n,\alpha}\, e^{-S_0}$$
$$= \lim_{N\to\infty} \left[(1 - \epsilon\xi_\alpha)e^{i\omega_n\epsilon} - 1\right]^{-1}$$
$$= \frac{1}{\epsilon}\frac{1}{i\omega_n - \xi_\alpha}. \tag{1.287}$$

[36] See Appendix 1.E.3 for change of variables in a Grassmannian integral.

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Up to a factor $1/\epsilon$ we recover the Green function $G_0(\alpha, i\omega_n) = (i\omega_n - \xi_\alpha)^{-1}$ of the non-interacting system [Eq. (1.201)]. Equation (1.287) yields the Green function

$$\begin{aligned} G_0(\alpha, \tau_{k_1} - \tau_{k_2}) &= -\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle a_{n,\alpha} a^*_{n,\alpha} \rangle e^{-i\omega_n(\tau_{k_1} - \tau_{k_2})} \\ &= -\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \sum_{p=0}^{\infty} a^p e^{i\omega_n \epsilon (p - k_1 + k_2)} \\ &= -\frac{1}{2} \lim_{N\to\infty} \sum_{p=0}^{\infty} \sum_{m=-\infty}^{\infty} a^p \delta_{p-k_1+k_2, mN} [1 + \zeta + (1-\zeta)(-1)^m] \end{aligned} \tag{1.288}$$

with $a = 1 - \epsilon\xi_\alpha$ and $\omega_n \epsilon = 2\pi n/N$ (bosons) or $2\pi(n+1/2)/N$ (fermions). We have used the fact that the sum over n vanishes unless $p - k_1 + k_2 = mN$ with m integer. Furthermore, the constraints $p \geq 0$ and $0 \leq k \leq N-1$ imply $m \geq 1$ if $\tau_{k_1} - \tau_{k_2} < 0$ and $m \geq 0$ if $\tau_{k_1} - \tau_{k_2} \geq 0$, so that we finally obtain

$$G_0(\alpha, \tau_{k_1} - \tau_{k_2}) = \begin{cases} -\lim\limits_{N\to\infty} \dfrac{\zeta a^{N+k_1-k_2}}{1-\zeta a^N} = -\zeta n_\alpha e^{-\xi_\alpha(\tau_{k_1} - \tau_{k_2})} & \text{if} \quad \tau_{k_1} < \tau_{k_2}, \\ -\lim\limits_{N\to\infty} \dfrac{a^{k_1-k_2}}{1-\zeta a^N} = -(1+\zeta n_\alpha) e^{-\xi_\alpha(\tau_{k_1} - \tau_{k_2})} & \text{if} \quad \tau_{k_1} \geq \tau_{k_2}, \end{cases} \tag{1.289}$$

where

$$n_\alpha = \frac{1}{e^{\beta\xi_\alpha} - \zeta} \tag{1.290}$$

if the occupation number $n_B(\xi_\alpha)$ for bosons and $n_F(\xi_\alpha)$ for fermions. One may rewrite (1.289) as

$$G_0(\alpha, \tau) = -e^{-\xi_\alpha \tau} \left[\Theta(\tau)(1 + \zeta n_\alpha) + \Theta(-\tau) \zeta n_\alpha \right], \tag{1.291}$$

where $\tau = \tau_{k_1} - \tau_{k_2} \in [-\beta, \beta]$, in agreement with (1.200).

It is also possible to calculate the partition function Z_0 directly from (1.282) without introducing the Fourier transformed variables $a^{(*)}_{n,\alpha}$. To this end, we write the action as

$$S_0 = \sum_\alpha \sum_{k,k'=1}^{N} \psi^*_{k,\alpha} S^{(\alpha)}_{k,k'} \psi_{k',\alpha}, \tag{1.292}$$

where $S^{(\alpha)}$ is the $N \times N$ matrix

$$S^{(\alpha)} = \begin{pmatrix} 1 & 0 & \cdots & \cdots & 0 & -\zeta a \\ -a & 1 & 0 & & & 0 \\ 0 & -a & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & -a & 1 & 0 \\ 0 & \cdots & \cdots & 0 & -a & 1 \end{pmatrix} \tag{1.293}$$

with $a = 1 - \frac{\beta}{N}\xi_\alpha$. Expanding with respect to the first line, we obtain

$$\lim_{N\to\infty} \det S^{(\alpha)} = \lim_{N\to\infty} \left[1 + (-\zeta a)(-1)^{N-1} (-a)^{N-1} \right] = 1 - \zeta e^{-\beta\xi_\alpha}, \tag{1.294}$$

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and finally

$$Z_0 = \lim_{N\to\infty} \prod_\alpha \left[\det S^{(\alpha)}\right]^{-\zeta} = \prod_\alpha \left(1-\zeta e^{-\beta\xi_\alpha}\right)^{-\zeta}. \tag{1.295}$$

Similarly we can calculate the one-particle Green function

$$\begin{aligned} &G_0(\alpha,\tau_{k_1}-\tau_{k_2}) \\ &\quad = -\lim_{N\to\infty}\frac{1}{Z_0}\int\prod_{k=1}^N d(\psi_k^*,\psi_k)\,\psi_{k_1,\alpha}\psi_{k_2,\alpha}^*\exp\Big\{-\sum_{k,k'=1}^N\sum_\alpha \psi_{k,\alpha}^* S_{k,k'}^{(\alpha)}\psi_{k',\alpha}\Big\} \\ &\quad = -\lim_{N\to\infty}\left(S^{(\alpha)-1}\right)_{k_1,k_2}, \end{aligned} \tag{1.296}$$

where $\tau_k = k\beta/N$. The inverse matrix $S^{(\alpha)-1}$ being given by

$$S^{(\alpha)-1} = \frac{1}{1-\zeta a^N}\begin{pmatrix} 1 & \zeta a^{N-1} & \zeta a^{N-2} & \cdots & \cdots & \zeta a \\ a & 1 & \zeta a^{N-1} & \ddots & & \vdots \\ a^2 & a & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \zeta a^{N-1} & \zeta a^{N-2} \\ \vdots & & \ddots & a & 1 & \zeta a^{N-1} \\ a^{N-1} & \cdots & \cdots & a^2 & a & 1 \end{pmatrix}, \tag{1.297}$$

we obtain

$$G_0(\alpha,\tau_{k_1}-\tau_{k_2}) = -\lim_{N\to\infty}\begin{cases} \dfrac{\zeta a^{N+k_1-k_2}}{1-\zeta a^N} & \text{if} \quad \tau_{k_1}<\tau_{k_2}, \\ \dfrac{a^{k_1-k_2}}{1-\zeta a^N} & \text{if} \quad \tau_{k_1}\geq\tau_{k_2}, \end{cases} \tag{1.298}$$

i.e.

$$G_0(\alpha,\tau_{k_1}-\tau_{k_2}) = \begin{cases} -\zeta n_\alpha e^{-\xi_\alpha(\tau_{k_1}-\tau_{k_2})} & \text{if} \quad \tau_{k_1}<\tau_{k_2}, \\ -(1+\zeta n_\alpha)e^{-\xi_\alpha(\tau_{k_1}-\tau_{k_2})} & \text{if} \quad \tau_{k_1}\geq\tau_{k_2}. \end{cases} \tag{1.299}$$

1.4.2.1 Continuum-time limit

In the continuum-time limit, the variables $a_{n,\alpha}^{(*)}$ become Fourier-transformed fields $\psi_\alpha^{(*)}(i\omega_n)$ defined by

$$\begin{aligned} \psi_\alpha(\tau) &= \frac{1}{\sqrt{\beta}}\sum_{\omega_n} e^{-i\omega_n\tau}\psi_\alpha(i\omega_n), \qquad \psi_\alpha(i\omega_n) = \frac{1}{\sqrt{\beta}}\int_0^\beta d\tau\, e^{i\omega_n\tau}\psi_\alpha(\tau), \\ \psi_\alpha^*(\tau) &= \frac{1}{\sqrt{\beta}}\sum_{\omega_n} e^{i\omega_n\tau}\psi_\alpha^*(i\omega_n), \qquad \psi_\alpha^*(i\omega_n) = \frac{1}{\sqrt{\beta}}\int_0^\beta d\tau\, e^{-i\omega_n\tau}\psi_\alpha^*(\tau). \end{aligned} \tag{1.300}$$

The action reads

$$\begin{aligned} S_0[\psi^*,\psi] &= \int_0^\beta d\tau \sum_\alpha \psi_\alpha^*(\tau)\left(\partial_\tau+\xi_\alpha\right)\psi_\alpha(\tau), \\ &= \sum_{\alpha,\omega_n}\psi_\alpha^*(i\omega_n)(-i\omega_n+\xi_\alpha)\psi_\alpha(i\omega_n), \end{aligned} \tag{1.301}$$

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while the functional integral measure is given by

$$\mathcal{D}[\psi^*,\psi] = \prod_{\alpha,\omega_n} \frac{d\psi^*_\alpha(i\omega_n) d\psi_\alpha(i\omega_n)}{\beta^\zeta \mathcal{N}}, \tag{1.302}$$

where the factor $\beta^{-\zeta}$ arises from the Jacobian of the change of variables to Fourier space, and $\mathcal{N} = 2i\pi$ (1) for bosons (fermions). If one naively computes the partition function from the continuum-time limit, one finds

$$\begin{aligned} Z_0 &= \prod_{\alpha,\omega_n} [\beta(-i\omega_n + \xi_\alpha)]^{-\zeta}, \\ \Omega_0 &= -\frac{1}{\beta} \ln Z_0 = \frac{\zeta}{\beta} \sum_{\alpha,\omega_n} \ln[\beta(-i\omega_n + \xi_\alpha)], \end{aligned} \tag{1.303}$$

and the mean particle number

$$\langle \hat{N} \rangle = -\frac{\partial \Omega_0}{\partial \mu} = -\frac{\zeta}{\beta} \sum_{\alpha,\omega_n} \frac{1}{i\omega_n - \xi_\alpha}. \tag{1.304}$$

The Matsubara sums in Eqs. (1.303) and (1.304) are ill-defined. Now compare the expression of $\langle \hat{N} \rangle$ with the result obtained directly from the functional integral,[37]

$$\langle \hat{N} \rangle = \frac{1}{\beta Z_0} \frac{\partial Z_0}{\partial \mu} = \frac{1}{\beta} \int_0^\beta d\tau \sum_\alpha \langle \psi^*_\alpha(\tau^+) \psi_\alpha(\tau) \rangle = \sum_\alpha \langle \psi^*_\alpha(\tau^+) \psi_\alpha(\tau) \rangle, \tag{1.305}$$

where τ^+ enforces the proper equal-time ordering of the chemical-potential term. The last equality follows from time-translation invariance and can also be deduced from the discrete-time action (1.282). From Eq. (1.305), we obtain

$$\langle \hat{N} \rangle = -\zeta \sum_\alpha G_0(\alpha, -\eta) = -\frac{\zeta}{\beta} \sum_{\alpha,\omega_n} G_0(\alpha, i\omega_n)\, e^{i\omega_n \eta} \quad (\eta \to 0^+), \tag{1.306}$$

where the Green function

$$\begin{aligned} G_0(\alpha, i\omega_n) &= -\langle \psi_\alpha(i\omega_n) \psi^*_\alpha(i\omega_n) \rangle \\ &= -\frac{1}{Z_0} \int \mathcal{D}[\psi^*,\psi]\, \psi_\alpha(i\omega_n) \psi^*_\alpha(i\omega_n)\, e^{-S_0[\psi^*,\psi]} \\ &= \frac{1}{i\omega_n - \xi_\alpha} \end{aligned} \tag{1.307}$$

is obtained using standard Gaussian-integration results (Appendix 1.E). The Matsubara sum in (1.306) is convergent and gives $\langle \hat{N} \rangle = \sum_\alpha n_\alpha$,[38] with n_α the Bose-Einstein or Fermi-Dirac occupation number [Eq. (1.290)]. To ensure consistency with $\langle \hat{N} \rangle = -\partial \Omega_0 / \partial \mu$, the thermodynamic potential should be written as

$$\Omega_0 = \frac{\zeta}{\beta} \sum_{\alpha,\omega_n} \ln[\beta(-i\omega_n + \xi_\alpha)] e^{i\omega_n \eta}. \tag{1.308}$$

[37] Equation (1.305) also follows from $\langle \hat{N} \rangle = \sum_\alpha \langle \hat{\psi}^\dagger_\alpha \hat{\psi}_\alpha \rangle = \sum_\alpha \langle T_\tau \hat{\psi}^\dagger_\alpha(\tau^+) \hat{\psi}_\alpha(\tau) \rangle$, which, in the functional-integral representation, becomes $\langle \hat{N} \rangle = \sum_\alpha \langle \psi^*_\alpha(\tau^+) \psi_\alpha(\tau) \rangle$.

[38] See Appendix 1.F.

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The Matsubara sum then yields the standard expression for the thermodynamic potential of non-interacting bosons or fermions.[38]

The difficulties encountered in the continuum-time limit of the coherent-state functional integral are very similar to those that appear in the path integral formulation of a quantum particle in a magnetic field (Sec. 1.1.1). In the formal continuum limit, the ordering of the operators in the Hamiltonian is lost, which results in ambiguities or ill-defined quantities. In practice, however, these difficulties seldom occur and can be fixed by introducing the convergence factor $e^{i\omega_n \eta}$.[39]

1.5 Perturbation theory and Feynman diagrams[40]

Interacting systems are described by non-Gaussian actions so that the functional integral cannot in general be calculated exactly. When the interactions between particles are "weak" (in a sense that is to be explained), they can be perturbatively taken into account with respect to the kinetic energy. Perturbation theory in many-body systems is based on Wick's theorem (Sec. 1.5.1) and formulated in terms of Feynman diagrams (Secs. 1.5.2-1.5.4).

1.5.1 Wick's theorem

We consider the partition function of a non-interacting system,[41]

$$\begin{aligned} Z_0[J^*,J] &= \int \mathcal{D}[\psi^*,\psi] \exp\Big\{ -S_0[\psi^*,\psi] + \int_0^\beta d\tau \sum_\alpha [J^*_\alpha(\tau)\psi_\alpha(\tau) + \text{c.c.}] \Big\}, \\ S_0[\psi^*,\psi] &= -\int_0^\beta d\tau d\tau' \sum_{\alpha\alpha'} \psi^*_\alpha(\tau) G_0^{-1}(\alpha\tau,\alpha'\tau')\psi_{\alpha'}(\tau'). \end{aligned} \tag{1.309}$$

J and J^* are external "sources" that couple linearly to the fields ψ^* and ψ. They are c-numbers for bosons and Grassmann variables for fermions. $Z_0[J^*,J]$ can be calculated exactly using the results (1.E.38) for Gaussian integrals,

$$Z_0[J^*,J] = Z_0 \exp\Big\{ -\int_0^\beta d\tau d\tau' \sum_{\alpha,\alpha'} J^*_\alpha(\tau) G_0(\alpha\tau,\alpha'\tau') J_{\alpha'}(\tau') \Big\}, \tag{1.310}$$

where $Z_0 = Z_0[0,0]$ is the non-interacting partition function for vanishing sources.

Let us now consider the expectation value $\langle \psi_{\alpha_1}(\tau_1)\cdots\psi_{\alpha_n}(\tau_n)\psi^*_{\alpha'_n}(\tau'_n)\cdots\psi^*_{\alpha'_1}(\tau'_1)\rangle_0$ where

$$\langle \cdots \rangle_0 = \frac{1}{Z_0} \int \mathcal{D}[\psi^*,\psi]\, (\cdots) e^{-S_0[\psi^*,\psi]}. \tag{1.311}$$

This average can be expressed as the functional derivative

$$\frac{1}{Z_0} \frac{\zeta^n \delta^{2n} Z_0[J^*,J]}{\delta J^*_{\alpha_1}(\tau_1)\cdots\delta J^*_{\alpha_n}(\tau_n)\delta J_{\alpha'_n}(\tau'_n)\cdots\delta J_{\alpha'_1}(\tau'_1)}\bigg|_{J^*=J=0}. \tag{1.312}$$

[39] There are essentially two instances where the convergence factor $e^{i\omega_n\eta}$ is necessary: when calculating the thermodynamic potential and when computing boson/fermion loops in perturbation theory (Sec. 1.5).

[40] In this section we derive the rules of perturbation theory using the functional integral formalism. The alternative derivation, based on the operator formalism, is briefly reviewed in Appendix 1.G.

[41] We shall see in the following that G_0, which appears in the definition of S_0, is indeed the one-particle Green function.

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This leads to (see (1.E.44) in Appendix 1.E)

$$\begin{aligned}\langle\psi_{\alpha_1}(\tau_1)\cdots\psi_{\alpha_n}(\tau_n)\psi^*_{\alpha'_n}(\tau'_n)\cdots\psi^*_{\alpha'_1}(\tau'_1)\rangle_0 \\ = (-1)^n\sum_{P\in S_n}\zeta^P G_0(\alpha_n\tau_n,\alpha'_{P(n)}\tau'_{P(n)})\cdots G_0(\alpha_1\tau_1,\alpha'_{P(1)}\tau'_{P(1)}),\end{aligned} \tag{1.313}$$

where the sum is over all permutations P of $\{1\cdots n\}$. The average $\langle\psi_\alpha(\tau)\psi^*_{\alpha'}(\tau')\rangle_0$ is often called a contraction and denoted by

$$-G_0(\alpha\tau,\alpha'\tau') = \langle\psi_\alpha(\tau)\psi^*_{\alpha'}(\tau')\rangle_0 = \overbrace{\psi_\alpha(\tau)\psi^*_{\alpha'}}(\tau'). \tag{1.314}$$

Wick's theorem states that the expectation value of a product of fields with the Gaussian action S_0 is given by the sum of all complete contractions [Eq. (1.313)],

$$\langle\psi_{\alpha_1}(\tau_1)\cdots\psi_{\alpha_n}(\tau_n)\psi^*_{\alpha'_n}(\tau'_n)\cdots\psi^*_{\alpha'_1}(\tau'_1)\rangle_0 = \sum \text{all complete contractions}. \tag{1.315}$$

A complete contraction is a configuration in which each ψ is contracted with a ψ^* and the overall sign is specified by ζ^P where P is the permutation such that $\psi_{\alpha_i}(\tau_i)$ is contracted with $\psi^*_{\alpha'_{P(i)}}(\tau'_{P(i)})$. It is easy to show that if all contractions appear consecutively in a complete contraction, one can simply replace $\overbrace{\psi_{\alpha_i}(\tau)\psi^*_{\alpha'_j}}(\tau')$ by $-G_0(\alpha_i\tau,\alpha'_j\tau')$ and $\overbrace{\psi^*_{\alpha_i}(\tau)\psi_{\alpha'_j}}(\tau')$ by $G_0(\alpha'_j\tau',\alpha_i\tau)$.[42]

One can generalize (1.315) by allowing for contractions $\overbrace{\psi\psi}$ and $\overbrace{\psi^*\psi^*}$ since the latter vanish,

$$\begin{aligned}\overbrace{\psi_\alpha(\tau)\psi_{\alpha'}}(\tau') = \langle\psi_\alpha(\tau)\psi_{\alpha'}(\tau')\rangle_0 = \frac{1}{Z_0}\frac{\delta^2 Z_0[J^*,J]}{\delta J^*_\alpha(\tau)J^*_{\alpha'}(\tau')}\bigg|_{J^*=J=0} = 0, \\ \overbrace{\psi^*_\alpha(\tau)\psi^*_{\alpha'}}(\tau') = \langle\psi^*_\alpha(\tau)\psi^*_{\alpha'}(\tau')\rangle_0 = \frac{1}{Z_0}\frac{\delta^2 Z_0[J^*,J]}{\delta J_\alpha(\tau)J_{\alpha'}(\tau')}\bigg|_{J^*=J=0} = 0.\end{aligned} \tag{1.316}$$

The expectation value of a product of fields is nonzero only if there is the same number of fields ψ and ψ^*.

An immediate consequence of Wick's theorem is that once we know the one-particle Green function G_0, we also know the higher-order Green functions in a non-interacting system. For instance, the two-particle Green function reads[43]

$$\begin{aligned}G^{(4)}_0(\alpha_1\tau_1,\alpha_2\tau_2;\alpha'_2\tau'_2,\alpha'_1\tau'_1) &= \langle\psi_{\alpha_1}(\tau_1)\psi_{\alpha_2}(\tau_2)\psi^*_{\alpha'_2}(\tau'_2)\psi^*_{\alpha'_1}(\tau'_1)\rangle_0 \\ &= \overbrace{\psi_{\alpha_1}(\tau_1)\overbrace{\psi_{\alpha_2}(\tau_2)\psi^*_{\alpha'_2}}(\tau'_2)\psi^*_{\alpha'_1}}(\tau'_1) + \overbrace{\psi_{\alpha_1}(\tau_1)\overbrace{\psi_{\alpha_2}(\tau_2)\psi^*_{\alpha'_2}}(\tau'_2)\psi^*_{\alpha'_1}}(\tau'_1) \\ &= G_0(\alpha_1\tau_1,\alpha'_1\tau'_1)G_0(\alpha_2\tau_2,\alpha'_2\tau'_2) + \zeta G_0(\alpha_1\tau_1,\alpha'_2\tau'_2)G_0(\alpha_2\tau_2,\alpha'_1\tau'_1).\end{aligned} \tag{1.317}$$

[42] As an example, the second contraction in (1.317) gives $\overbrace{\psi_{\alpha_1}(\tau_1)\overbrace{\psi_{\alpha_2}(\tau_2)\psi^*_{\alpha'_2}}(\tau'_2)\psi^*_{\alpha'_1}}(\tau'_1) = \zeta\overbrace{\psi_{\alpha_1}(\tau_1)\psi^*_{\alpha'_2}}(\tau'_2)\overbrace{\psi_{\alpha_2}(\tau_2)\psi^*_{\alpha'_1}}(\tau'_1) = \zeta G_0(\alpha_1\tau_1,\alpha'_2\tau'_2)G_0(\alpha_2\tau_2,\alpha'_1\tau'_1)$, where the factor ζ is due to the exchange of $\psi^*_{\alpha'_2}(\tau'_2)$ and $\psi_{\alpha_2}(\tau_2)$.

[43] Recall that the superscript (4) denotes the number of fields appearing in the definition of $G^{(4)}$. It also corresponds to the number of external "legs" in the Feynman diagrams representing $G^{(4)}$.

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The two particles created in the states α_1 and α_2 at times τ_1 and τ_2 propagate independently. The second term in the rhs of (1.317) comes from the indiscernability of the particles with the factor ζ reflecting the bosonic or fermionic statistics. It is convenient to represent the propagator G_0 by a directed line,

$$G_0(\alpha\tau, \alpha'\tau') = \quad \alpha\tau \longleftarrow \alpha'\tau'$$

which starts at (α', τ') (creation of a particle in the one-body state α' at time τ') and ends at (α, τ) (annihilation of a particle in the one-body state α at time τ). The two-particle Green function can be represented by the diagram

$$G_0^{(4)}(\alpha_1\tau_1, \alpha_2\tau_2; \alpha_2'\tau_2', \alpha_1'\tau_1') = \quad \begin{matrix} \alpha_1\tau_1 \longleftarrow \alpha_1'\tau_1' \\ \alpha_2\tau_2 \longleftarrow \alpha_2'\tau_2' \end{matrix} \quad +\zeta \quad \begin{matrix} \alpha_1\tau_1 \searrow\!\!\nearrow \alpha_1'\tau_1' \\ \alpha_2\tau_2 \nearrow\!\!\searrow \alpha_2'\tau_2' \end{matrix}$$

Such a diagram is called a Feynman diagram (or graph). As we shall see in the next sections, Feynman diagrams play an essential role in the perturbation theory of many-body systems.

1.5.2 Partition function and thermodynamic potential

We consider a many-body system with a two-body interaction,

$$\begin{aligned} Z &= \int \mathcal{D}[\psi^*, \psi]\, e^{-S_0[\psi^*,\psi] - S_{\rm int}[\psi^*,\psi]}, \\ S_{\rm int}[\psi^*, \psi] &= \frac{1}{2}\int_0^\beta d\tau \sum_{\alpha_1,\alpha_2,\alpha_1',\alpha_2'} (\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')\psi^*_{\alpha_1}(\tau)\psi^*_{\alpha_2}(\tau)\psi_{\alpha_2'}(\tau)\psi_{\alpha_1'}(\tau). \end{aligned} \tag{1.318}$$

The perturbation expansion of the partition function Z is obtained by expanding $e^{-S_{\rm int}}$ in powers of the two-body interaction v,

$$\frac{Z}{Z_0} = \left\langle e^{-S_{\rm int}} \right\rangle_0 = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \langle S_{\rm int}^n \rangle_0 \tag{1.319}$$

where

$$\begin{aligned} \frac{(-1)^n}{n!}\langle S_{\rm int}^n\rangle_0 = \frac{(-1)^n}{2^n n!} &\sum_{\alpha_1,\beta_1,\gamma_1,\delta_1} \cdots \sum_{\alpha_n,\beta_n,\gamma_n,\delta_n} (\alpha_1\beta_1|\hat{v}|\gamma_1\delta_1)\cdots(\alpha_n\beta_n|\hat{v}|\gamma_n\delta_n) \\ &\times \int_0^\beta d\tau_1 \cdots d\tau_n \langle \psi^*_{\alpha_1}(\tau_1)\psi^*_{\beta_1}(\tau_1)\psi_{\delta_1}(\tau_1)\psi_{\gamma_1}(\tau_1)\cdots \\ &\quad \cdots \psi^*_{\alpha_n}(\tau_n)\psi^*_{\beta_n}(\tau_n)\psi_{\delta_n}(\tau_n)\psi_{\gamma_n}(\tau_n)\rangle_0. \end{aligned} \tag{1.320}$$

The averages $\langle\cdots\rangle_0$ in (1.320) can be evaluated using Wick's theorem. For the $\mathcal{O}(v)$ term, one obtains

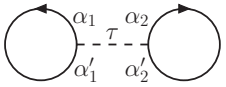
$$\begin{aligned} &-\frac{1}{2}\sum_{\alpha_1\cdots\alpha_2'} (\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2') \int_0^\beta d\tau \langle \psi^*_{\alpha_1}(\tau)\psi^*_{\alpha_2}(\tau)\psi_{\alpha_2'}(\tau)\psi_{\alpha_1'}(\tau)\rangle_0 \\ &= -\frac{1}{2}\sum_{\alpha_1\cdots\alpha_2'} (\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2') \int_0^\beta d\tau [G_0(\alpha_1'\tau, \alpha_1\tau^+)G_0(\alpha_2'\tau, \alpha_2\tau^+) \end{aligned}$$

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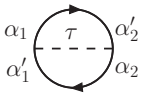
$$+\zeta G_0(\alpha_1'\tau,\alpha_2\tau^+)G_0(\alpha_2'\tau,\alpha_1\tau^+)], \tag{1.321}$$

where the equal-time propagator $G_0(\alpha\tau,\alpha'\tau)$ is interpreted as $G_0(\alpha\tau,\alpha'\tau^+)$ in agreement with the proper time ordering in the interaction term (see the discussion after (1.273)). Equation (1.321) can be represented by the two Feynman diagrams

$$-\frac{1}{2}\sum_{\alpha_1\cdots\alpha_2'}(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')\int_0^\beta d\tau\, G_0(\alpha_1'\tau,\alpha_1\tau^+)G_0(\alpha_2'\tau,\alpha_2\tau^+) =$$

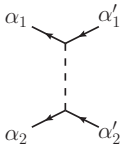


$$-\frac{\zeta}{2}\sum_{\alpha_1\cdots\alpha_2'}(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')\int_0^\beta d\tau\, G_0(\alpha_1'\tau,\alpha_2\tau^+)G_0(\alpha_2'\tau,\alpha_1\tau^+) =$$



where each propagator is represented as a directed line (see preceding section) and each interaction as a vertex with two incoming and two outgoing lines corresponding to $\psi_{\alpha_1'}\psi_{\alpha_2'}$ and $\psi^*_{\alpha_1}\psi^*_{\alpha_2}$, respectively,

$(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2') =$



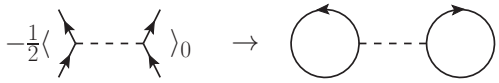
Note that unless otherwise specified we do not write explicitly the sign and prefactor when drawing a Feynman diagram.

All terms appearing in the perturbation expansion of the partition function can be represented by Feynman diagrams. These diagrams are often referred to as vacuum fluctuation graphs since in quantum field theory they correspond to the "vacuum to vacuum transition" amplitude $\langle\text{vac}|e^{-i\hat{H}t}|\text{vac}\rangle$ (Sec. 1.8). The $\mathcal{O}(v^2)$ term

$$\frac{1}{2!2^2}\sum_{\alpha_1\cdots\delta_2}(\alpha_1\beta_1|\hat{v}|\gamma_1\delta_1)(\alpha_2\beta_2|\hat{v}|\gamma_2\delta_2)\int_0^\beta d\tau_1 d\tau_2\langle\psi^*_{\alpha_1}(\tau_1)\cdots\psi_{\gamma_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)\cdots\psi_{\gamma_2}(\tau_2)\rangle_0 \tag{1.322}$$

yields the diagrams shown in Fig. 1.2. What is interesting here is that the $4! = 24$ contractions arising from (1.322) are represented by only 8 diagrams. The number of diagrams even reduces to 5 for the thermodynamic potential (see below). Rather than starting from (1.320) and considering all possible contractions, it is therefore easier to draw all diagrams to a given order in v and compute each of these diagrams. Since we already know that to each line is associated a propagator G_0 and to each vertex a matrix element $(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')$, our remaining task is to determine the combinatorial factor and sign of a diagram.

Let us start with the first of the first-order diagram shown after (1.321). Schematically, this diagram is obtained from



There is only one contraction that leads to the diagram so that the overall factor is 1/2. Consider now the following second-order diagram

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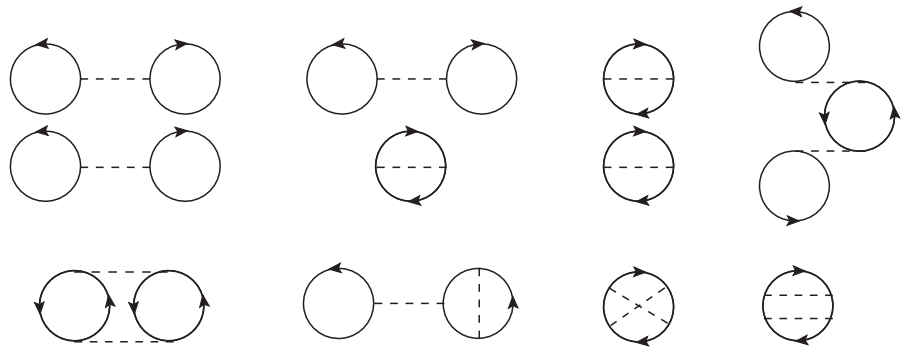
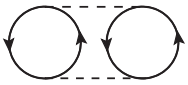
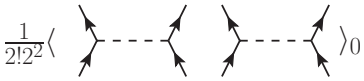


Figure 1.2: Diagrams contributing to Z/Z_0 to second order. The first three diagrams are made of two disconnected pieces.



It is obtained from



where the factor $1/2!2^2$ comes from the expansion of $e^{-S_{\rm int}}$ to second order. There are two contractions consistent with the topology of the diagram. To see this, take one of the outgoing lines of one of the two vertices. It must be contracted with one of the incoming line of the other vertex (hence the 2 contractions). Once this is done, there is no choice in the remaining contractions to obtain the diagram so that the combinatorial factor is[44]

$$\frac{1}{2!2^2}\times 2=\frac{1}{4}. \tag{1.323}$$

To obtain the sign of the diagram, it suffices to consider a particular contraction leading to the diagram, e.g.

$$\begin{aligned}
&\frac{1}{2!2^2}\sum_{\alpha_1\cdots\delta_2}(\alpha_1\beta_1|\hat{v}|\gamma_1\delta_1)(\alpha_2\beta_2|\hat{v}|\gamma_2\delta_2)\int_0^\beta d\tau_1 d\tau_2\\
&\qquad\times \psi^*_{\alpha_1}(\tau_1)\psi^*_{\beta_1}(\tau_1)\psi_{\delta_1}(\tau_1)\psi_{\gamma_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)\psi^*_{\beta_2}(\tau_2)\psi_{\delta_2}(\tau_2)\psi_{\gamma_2}(\tau_2)\\
&=\frac{1}{2!2^2}\sum_{\alpha_1\cdots\delta_2}(\alpha_1\beta_1|\hat{v}|\gamma_1\delta_1)(\alpha_2\beta_2|\hat{v}|\gamma_2\delta_2)\int_0^\beta d\tau_1 d\tau_2\\
&\qquad\times \psi_{\gamma_2}(\tau_2)\psi^*_{\alpha_1}(\tau_1)\psi_{\delta_2}(\tau_2)\psi^*_{\beta_1}(\tau_1)\psi_{\delta_1}(\tau_1)\psi^*_{\beta_2}(\tau_2)\psi_{\gamma_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2),
\end{aligned} \tag{1.324}$$

[44]The combinatorial factor can also be interpreted as $1/S$ where S is the symmetry factor of the graph. S is defined as the number of deformations (obtained by permuting the time labels or exchanging the legs $(\alpha_1\alpha_1')\leftrightarrow(\alpha_2\alpha_2')$ in a vertex $(\alpha_1\alpha_2|\hat{v}|\alpha_2'\alpha_1')$) that leaves the diagram unchanged. See Ref. [9] for a detailed discussion.

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where each contraction $\overbrace{\psi\psi^*}$ can now be replaced by $-G_0$ using (1.314). The sign of the diagram is therefore positive.

It turns out that the sign of a diagram is determined by the number of closed loops of one-particle propagators. Consider a general diagram with a closed loop coming from the contraction

$$(\cdots)\overbrace{\psi^*\psi^*\psi\psi(\cdots)\psi^*\psi^*\psi\psi(\cdots)\psi^*\psi^*\psi\psi}(\cdots), \tag{1.325}$$

where $(\cdots)$ denotes vertices not involved in the closed loop. Each $(\cdots)$ contains an even number of fields and therefore commutes with the fields explicitly written in (1.325). We therefore have to consider

$$\overbrace{\psi^*\psi^*\psi\psi\psi^*\psi^*\psi\psi\psi^*\psi^*\psi\psi}. \tag{1.326}$$

The fields not included in the cycle may be shifted to the left or to the right without changing the sign so that the cycle of contractions has the form

$$\overbrace{\psi^*\psi\psi^*\psi\psi^*\psi} = \zeta\psi\overbrace{\psi\psi^*}\psi\overbrace{\psi\psi^*}\psi\overbrace{\psi\psi^*} = \zeta(-1)^3 G_0 G_0 G_0. \tag{1.327}$$

Thus the contribution of a cycle gives a sign $\zeta(-1)^{n_p}$ where n_p is the number of propagators in the cycle. Since the total number of propagators in a diagram is always even, a diagram with n interaction lines and n_L closed loops has the overall sign $(-1)^n\zeta^{n_L}$ where the factor $(-1)^n$ simply comes from the expansion of $e^{-S_{\rm int}}$ to nth order.

The diagrammatic rules for calculating the nth-order contribution to Z/Z_0 are therefore the following:

1. Draw all distinct diagrams with n vertices connected by directed lines.
2. To each directed line, associate $G_0(\alpha\tau, \alpha'\tau')$. An equal-time propagator, with both ends connected to the same vertex, must be interpreted as $G_0(\alpha\tau, \alpha'\tau^+)$.
3. To each vertex, associate $(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')$.
4. Sum over all indices α and integrate all times $\tau \in [0, \beta]$.
5. Multiply the result by the combinatorial factor times $(-1)^n\zeta^{n_L}$ where n_L is the number of closed loops.

It is possible to write the diagrams directly in terms of the propagators $G_0(\alpha, \alpha'; i\omega_n)$. In frequency space, the interaction action (1.318) becomes

$$S_{\rm int} = \frac{1}{2\beta}\sum_{\substack{\alpha_1,\alpha_2,\alpha_1',\alpha_2' \\ \omega_n,\omega_{n'},\omega_\nu}} (\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')\psi^*_{\alpha_1}(i\omega_n + i\omega_\nu)\psi^*_{\alpha_2}(i\omega_{n'} - i\omega_\nu)\psi_{\alpha_2'}(i\omega_{n'})\psi_{\alpha_1'}(i\omega_n). \tag{1.328}$$

The sum of the frequencies associated with the propagators entering the vertex is conserved in the interaction process (frequency conservation). The preceding diagrammatic rules must therefore be modified as

2'. To each directed line, associate $G_0(\alpha, \alpha'; i\omega_n)$ while satisfying frequency conservation at each vertex. For a propagator with both ends connected to the same vertex (loop), multiply the propagator by $e^{i\omega_n\eta}$.

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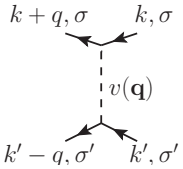
4'. Sum over all indices α and Matsubara frequencies ω_n.

6'. Multiply the result by $\frac{1}{\beta^n}$ where n is the number of vertices.

In translation-invariant systems, it is often convenient to use the momentum basis $\{|\mathbf{k},\sigma\rangle\}$ where σ denotes the spin of the particle (as well as other internal indices if any). For particles interacting *via* a two-body interaction $v(\mathbf{r}-\mathbf{r}')$, one has

$$S_{\rm int} = \frac{1}{2\beta\mathcal{V}} \sum_{\substack{k,k',q \\ \sigma,\sigma'}} v(\mathbf{q})\psi^*_\sigma(k+q)\psi^*_{\sigma'}(k'-q)\psi_{\sigma'}(k')\psi_\sigma(k) \tag{1.329}$$

where $v(\mathbf{q})$ is the Fourier transform of $v(\mathbf{r})$ (see Sec. 1.2.2) and $k=(\mathbf{k},i\omega_n)$, $k'=(\mathbf{k}',i\omega_{n'})$, $q=(\mathbf{q},i\omega_\nu)$; ω_n and $\omega_{n'}$ are bosonic or fermionic Matsubara frequencies in accordance with the statistic of the particles but ω_ν is always a bosonic frequency. The interaction vertex has the diagrammatic representation



The modifications of the previous diagrammatic rules are straightforward:

2". To each directed line, associate $G_{0,\sigma}(\mathbf{k},i\omega_n)$ while satisfying momentum, frequency and spin conservation at each vertex. For a propagator with both ends connected to the same vertex (loop), multiply the propagator by $e^{i\omega_n\eta}$.

3". To each vertex, associate $v(\mathbf{q})$ where $\mathbf{q}$ is the transferred momentum in the interaction process.

4". Sum over all momenta and Matsubara frequencies.

6". Multiply the result by $\frac{1}{(\beta\mathcal{V})^n}$ where n is the number of vertices.

With these rules, the first-order correction to the partition function Z/Z_0 reads

$$\begin{aligned} &-\frac{1}{2\beta\mathcal{V}}v(\mathbf{q}=0)\sum_{k,k',\sigma,\sigma'} G_{0,\sigma}(k)G_{0,\sigma'}(k')e^{i(\omega_n+\omega'_n)\eta} \\ &-\frac{\zeta}{2\beta\mathcal{V}}\sum_{k,k',\sigma} v(\mathbf{k}-\mathbf{k}')G_{0,\sigma}(k)G_{0,\sigma}(k')e^{i(\omega_n+\omega'_n)\eta} \end{aligned} \tag{1.330}$$

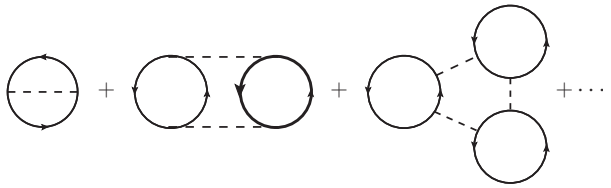
and agrees with the previous result (1.321).

In practice, computing Feynman diagrams beyond the first or second order becomes rapidly cumbersome. However, as will become clear in the following chapters, the diagrammatic representation of the perturbation expansion often brings a physical interpretation which in turn suggests a resummation (to infinite order) of a subclass of diagrams.

> One of the most familiar examples is the random-phase approximation (RPA) which amounts to summing the bubble (or ring) diagrams[45]

[45] Depending on the context, the RPA sums up bubble or ladder diagrams.

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Using the diagrammatic rules, we obtain

$$\left(\frac{Z}{Z_0}\right)_{\text{bubble graphs}} = -\frac{\zeta}{2}\mathrm{Tr}(vG_0G_0) + \frac{1}{4}\mathrm{Tr}(vG_0G_0)^2 - \frac{\zeta}{6}\mathrm{Tr}(vG_0G_0)^3 + \cdots$$
$$= -\frac{1}{2}\mathrm{Tr}\ln(1+\zeta vG_0G_0), \tag{1.331}$$

where Tr stands for summation over spacetime and internal indices. For a translation-invariant system, this gives

$$\left(\frac{Z}{Z_0}\right)_{\text{bubble graphs}} = -\frac{1}{2}\sum_q \ln\left[1-\zeta v(\mathbf{q})\Pi_0(q)\right] \tag{1.332}$$

where

$$\Pi_0(q) = -\frac{1}{\beta\mathcal{V}}\sum_{k,\sigma} G_{0,\sigma}(k)G_{0,\sigma}(k+q) \tag{1.333}$$

Since $\Pi_0(\mathbf{q},\tau) \propto \sum_{\mathbf{k}} G_{0,\sigma}(\mathbf{k},\tau)G_{0,\sigma}(\mathbf{k}+\mathbf{q},-\tau)$, it describes the propagation of a particle and a hole (see Sec. 1.2.3) and is therefore called a particle-hole pair propagator.

1.5.2.1 Linked cluster theorem and thermodynamic potential

To obtain the thermodynamics we need to compute $\ln Z$ rather than Z. The linked cluster theorem states that $\ln(Z/Z_0)$ is given by the sum of the connected graphs contributing to Z/Z_0. A graph is said to be connected if it consists of only one piece.

To derive this theorem, we use

$$\ln\left(\frac{Z}{Z_0}\right) = \lim_{R\to 0}\frac{d}{dR}\exp\left[R\ln\left(\frac{Z}{Z_0}\right)\right] = \lim_{R\to 0}\frac{d}{dR}\left(\frac{Z}{Z_0}\right)^R. \tag{1.334}$$

To compute $(Z/Z_0)^R$, we consider R replicas of the system and introduce a set of fields $\{\psi^*_{r,\alpha},\psi_{r,\alpha}\}$ for the rth copy ($r\in[1,R]$),

$$\left(\frac{Z}{Z_0}\right)^R = \frac{1}{Z_0^R}\int\prod_{r=1}^{R}\mathcal{D}[\psi^*_r,\psi_r]\exp\left\{-\sum_{r=1}^{R}S[\psi^*_r,\psi_r]\right\}. \tag{1.335}$$

The rules for calculating $(Z/Z_0)^R$ are the same as before except that each propagator $G_{0,r}(\alpha\tau,\alpha'\tau')$ now carries a replica index r. Since all fields attached to a given vertex share the same replica index, all propagators joining at this vertex must also share the same replica index. It follows that each connected part of a diagram carries a single index r, which yields a factor R when summed from 1 to R. A graph with n_c connected parts is therefore proportional to R^{n_c} and the graphs proportional to R are fully connected. From (1.319) and (1.334), we then deduce

$$\ln\left(\frac{Z}{Z_0}\right) = \sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\langle S^n_{\text{int}}\rangle_{0,c} = \sum \text{connected graphs} \tag{1.336}$$

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(the subscript c stands for connected diagrams) and

$$\Omega = \Omega_0 - \frac{1}{\beta}\sum \text{connected graphs}, \tag{1.337}$$

where $\Omega_0 = -\frac{1}{\beta}\ln Z_0$ is the thermodynamic potential of the non-interacting system.

1.5.3 Green functions

Green functions can be calculated perturbatively in a similar manner and the various terms in the expansion can be represented by Feynman diagrams. Let us start with the one-particle Green function

$$G(\alpha_1\tau_1,\alpha_2\tau_2) = -\langle\psi_{\alpha_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)\rangle = -\frac{Z_0}{Z}\langle\psi_{\alpha_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)e^{-S_{\rm int}}\rangle_0. \tag{1.338}$$

To compute G to a given order in the interaction, one must expand both Z and the average $\langle\cdots\rangle_0$ in (1.338). To first order, we use the expression (1.321) for Z/Z_0 and

$$-\langle\psi_{\alpha_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)e^{-S_{\rm int}}\rangle_0 = G_0(\alpha_1\tau_1,\alpha_2\tau_2) + \langle\psi_{\alpha_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)S_{\rm int}\rangle_0 \tag{1.339}$$

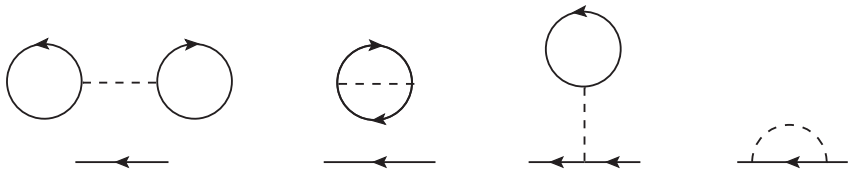
with

$$\begin{aligned}\langle\psi_{\alpha_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)S_{\rm int}\rangle_0 = \frac{1}{2}\sum_{\gamma_1\cdots\gamma_2'}(\gamma_1\gamma_2|\hat v|\gamma_1'\gamma_2')\\ \times\int_0^\beta d\tau\langle\psi_{\alpha_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)\psi^*_{\gamma_1}(\tau)\psi^*_{\gamma_2}(\tau)\psi_{\gamma_2'}(\tau)\psi_{\gamma_1'}(\tau)\rangle_0.\end{aligned} \tag{1.340}$$

Using Wick's theorem, we obtain for the last term

$$\begin{aligned}-\sum_{\gamma_1\cdots\gamma_2'}(\gamma_1\gamma_2|\hat v|\gamma_1'\gamma_2')\int_0^\beta d\tau\Big\{\frac{1}{2}G_0(\alpha_1\tau_1,\alpha_2\tau_2)\big[G_0(\gamma_1'\tau,\gamma_1\tau)G_0(\gamma_2'\tau,\gamma_2\tau)\\ +\zeta G_0(\gamma_1'\tau,\gamma_2\tau)G_0(\gamma_2'\tau,\gamma_1\tau)\big]+G_0(\alpha_1\tau_1,\gamma_1\tau)\big[\zeta G_0(\gamma_1'\tau,\alpha_2\tau_2)G_0(\gamma_2'\tau,\gamma_2\tau)\\ +G_0(\gamma_2'\tau,\alpha_2\tau_2)G_0(\gamma_1'\tau,\gamma_2\tau)\big]\Big\},\end{aligned} \tag{1.341}$$

which can be represented by the following four diagrams:

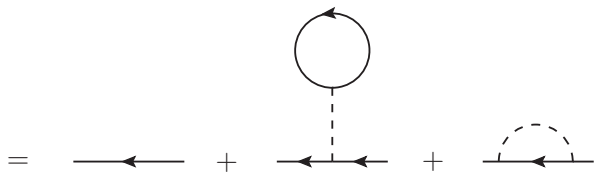


The diagrams for the one-particle Green function have two external lines (or legs) which carry the indices $(\alpha_1\tau_1)$ (outgoing line) and $(\alpha_2\tau_2)$ (incoming line). Only the indices carried by the internal lines are summed over. Combining (1.341) with the $\mathcal{O}(v)$ correction to Z/Z_0, we obtain

$$G(\alpha_1\tau_1,\alpha_2\tau_2) = G_0(\alpha_1\tau_1,\alpha_2\tau_2) - \sum_{\gamma_1\cdots\gamma_2'}(\gamma_1\gamma_2|\hat v|\gamma_1'\gamma_2')\int_0^\beta d\tau\, G_0(\alpha_1\tau_1,\gamma_1\tau)$$

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$$\times \left[\zeta G_0(\gamma_1'\tau, \alpha_2\tau_2) G_0(\gamma_2'\tau, \gamma_2\tau) + G_0(\gamma_2'\tau, \alpha_2\tau_2) G_0(\gamma_1'\tau, \gamma_2\tau)\right] \tag{1.342}$$



The $\mathcal{O}(v)$ correction to G_0 is given by the connected diagrams that contribute to $-\langle \psi_{\alpha_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2) e^{-S_{\rm int}}\rangle_0$; the vacuum fluctuation graphs factorize out in the numerator and denominator of (1.338).

This is a general result: The Green function is given by the sum of all connected graphs. Let us prove this result for the n-particle Green function

$$G^{(2n)}(\alpha_1\tau_1, \cdots, \alpha_n\tau_n; \alpha_n'\tau_n', \cdots, \alpha_1'\tau_1') = (-1)^n \langle \psi_{\alpha_1}(\tau_1) \cdots \psi^*_{\alpha_1'}(\tau_1')\rangle. \tag{1.343}$$

We introduce R replicas of the system and consider the correlation function

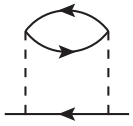
$$\begin{aligned} &G_R^{(2n)}(\alpha_1\tau_1, \cdots, \alpha_n\tau_n; \alpha_n'\tau_n', \cdots, \alpha_1'\tau_1') \\ &\quad = (-1)^n \int \prod_{r=1}^{R} \mathcal{D}[\psi_r^*, \psi_r]\, \psi_{1,\alpha_1}(\tau_1) \cdots \psi^*_{1,\alpha_1'}(\tau_1')\, e^{-\sum_{r=1}^R S[\psi_r^*, \psi_r]} \\ &\quad = (-1)^n Z^{R-1} \int \mathcal{D}[\psi_1^*, \psi_1]\, \psi_{1,\alpha_1}(\tau_1) \cdots \psi^*_{1,\alpha_1'}(\tau_1')\, e^{-S[\psi_1^*, \psi_1]}, \end{aligned} \tag{1.344}$$

where the last line is obtained by integrating out the fields ψ_r^*, ψ_r with $r \geq 2$. Thus $G^{(2n)}$ can be formally obtained from $G_R^{(2n)}$ in the limit $R = 0$. The perturbation expansion for $G_R^{(2n)}$ is obtained as before by expanding $e^{-S_{\rm int}}$. The Feynman diagrams involve propagators carrying a replica index $r \in [1, R]$. All propagators attached to the same vertex share the same value of the replica index. Those corresponding to the external lines $\alpha_1 \cdots \alpha_1'$ have $r = 1$. Since the index r in the interaction vertex is summed over, a diagram with p disconnected parts, i.e. p parts that are not connected to the external legs, is proportional to R^p and vanishes in the limit $R = 0$ when $p \geq 1$. For the connected diagrams ($p = 0$), the integration of the fields $\psi_r^{(*)}$ with $r \in [2, R]$ in (1.344) gives a factor Z_0^{R-1} so that we finally obtain

$$\begin{aligned} &G^{(2n)}(\alpha_1\tau_1, \cdots, \alpha_n\tau_n; \alpha_n'\tau_n', \cdots, \alpha_1'\tau_1') \\ &\quad = \frac{(-1)^n}{Z_0} \int \mathcal{D}[\psi^*, \psi]\, \psi_{\alpha_1}(\tau_1) \cdots \psi^*_{\alpha_1'}(\tau_1')\, e^{-S[\psi^*, \psi]} \Big|_{\text{connected diagrams}} \end{aligned} \tag{1.345}$$

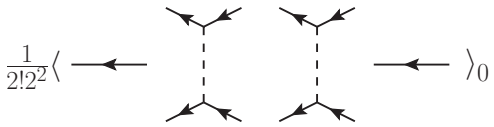
where only the connected diagrams, i.e. with all parts connected to the external lines, must be retained.

The combinatorial factor of a diagram can be computed, as for the partition function, by a direct evaluation of the number of contractions leading to that diagram. Let us consider for instance the following second-order diagram contributing to the one-particle Green function,



which is obtained from

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The combinatorial factor reads

$$\frac{1}{2!2^2} \times \underbrace{2!}_{\substack{\text{interchange}\\ \text{of vertices}}} \times \underbrace{2}_{\substack{\text{connection of}\\ \text{outgoing line}\\ \text{to vertex}}} \times \underbrace{2}_{\substack{\text{connection of}\\ \text{incoming line}\\ \text{to vertex}}} = 1. \tag{1.346}$$

It turns out that the combinatorial factor is always unity for the n-particle Green function. One can easily convince oneself by working out a few examples. A proof can be found in Ref. [9].

The sign of a diagram can be obtained from (1.345) and a particular contraction leading to that diagram. For the one-particle Green function, the sign of a diagram with l vertices is simply given by $(-1)^l\zeta^{n_L}$ where n_L is the number of closed loops. (The proof is similar to that given for the partition function.) In the case of fermions, a diagram for the n-particle Green function takes the additional factor ζ^P where ζ^P is the signature of the permutation such that the incoming line $(\alpha'_i\tau'_i)$ joins the outgoing line $(\alpha_{P(i)}\tau_{P(i)})$.[46]

The diagrammatic rules for calculating the lth-order contribution to the n-particle Green function $G^{(2n)}(\alpha_1\tau_1,\cdots,\alpha_n\tau_n;\alpha'_n\tau'_n,\cdots,\alpha'_1\tau'_1)$ are the following:

1. Draw all distinct connected diagrams with n incoming lines $(\alpha'_n\tau'_n\cdots\alpha'_1\tau'_1)$, n outgoing lines $(\alpha_1\tau_1\cdots\alpha_n\tau_n)$ and l interaction vertices.

2. To each directed line, associate $G_0(\alpha\tau,\alpha'\tau'^+)$.

3. To each vertex, associate $(\alpha\alpha'|\hat{v}|\gamma\gamma')$.

4. Sum over all internal indices α and integrate all internal times $\tau\in[0,\beta]$. The indices $(\alpha'_i\tau'_i)$ and $(\alpha_i\tau_i)$ of incoming and outgoing lines must be held fixed.

5. Multiply the result by the factor $(-1)^l\zeta^{n_L+P}$ where n_L is the number of closed loops and P the permutation such that each incoming line $(\alpha'_i\tau'_i)$ terminates at $(\alpha_{P(i)}\tau_{P(i)})$.

As an example, the second-order diagrams for the one-particle Green function are shown in Fig. 1.3 and the first-order diagrams for the two-particle Green function in Fig. 1.4.

Let us illustrate the various ways to calculate a diagram with the example of a system with the following action,

$$\begin{aligned} S_0 &= -\int_0^\beta d\tau d\tau' \int d^dr d^dr' \sum_\sigma \psi^*_\sigma(\mathbf{r},\tau)G^{-1}_{0,\sigma}(\mathbf{r},\tau;\mathbf{r}',\tau')\psi_\sigma(\mathbf{r}',\tau'),\\ S_{\rm int} &= \frac{1}{2}\int_0^\beta d\tau \int d^dr d^dr' \sum_{\sigma,\sigma'} v(\mathbf{r}-\mathbf{r}')\psi^*_\sigma(\mathbf{r},\tau)\psi^*_{\sigma'}(\mathbf{r}',\tau)\psi_{\sigma'}(\mathbf{r}',\tau)\psi_\sigma(\mathbf{r},\tau). \end{aligned} \tag{1.347}$$

[46] For a simple example, consider the non-interacting two-particle Green function [Eq. (1.317)].

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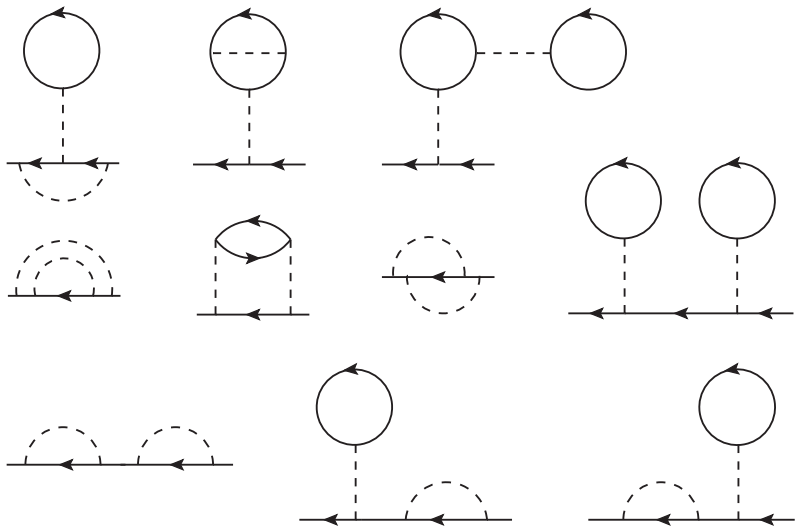
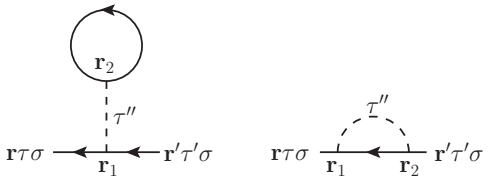


Figure 1.3: Second-order diagrams for the one-particle Green function $G^{(2)} = G$.

The first-order correction to $G_{0,\sigma}(\mathbf{r}, \tau; \mathbf{r}', \tau')$ is obtained from the diagrams



and is given by

$$
\begin{aligned}
-\zeta \int_0^\beta d\tau'' \int d^d r_1 d^d r_2 \sum_{\sigma'} v(\mathbf{r}_1 - \mathbf{r}_2) G_{0,\sigma}(\mathbf{r}, \tau; \mathbf{r}_1, \tau'') G_{0,\sigma}(\mathbf{r}_1, \tau''; \mathbf{r}', \tau') \\
\times G_{0,\sigma'}(\mathbf{r}_2, \tau''; \mathbf{r}_2, \tau''^+) \\
- \int_0^\beta d\tau'' \int d^d r_1 d^d r_2 \, v(\mathbf{r}_1 - \mathbf{r}_2) G_{0,\sigma}(\mathbf{r}, \tau; \mathbf{r}_1, \tau'') \\
\times G_{0,\sigma}(\mathbf{r}_1, \tau''; \mathbf{r}_2, \tau''^+) G_{0,\sigma}(\mathbf{r}_2, \tau''; \mathbf{r}', \tau'). \quad (1.348)
\end{aligned}
$$

When the system is translation invariant, it is convenient to start from

$$
\begin{aligned}
S_0 &= -\sum_{k,\sigma} \psi^*_\sigma(k) G^{-1}_{0,\sigma}(k) \psi_\sigma(k), \\
S_{\text{int}} &= \frac{1}{2\beta\mathcal{V}} \sum_{\substack{k,k',q \\ \sigma,\sigma'}} v(\mathbf{q}) \psi^*_\sigma(k+q) \psi^*_{\sigma'}(k'-q) \psi_{\sigma'}(k') \psi_\sigma(k).
\end{aligned} \quad (1.349)
$$

The first-order correction to $G_{0,\sigma}(k)$ is represented by the two diagrams

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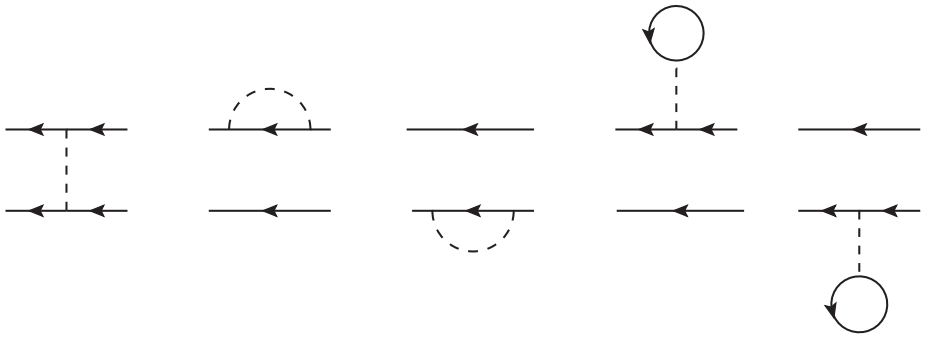
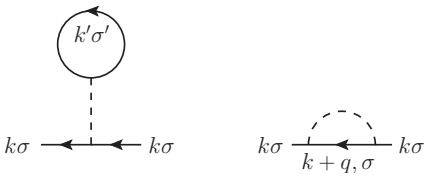


Figure 1.4: First-order diagrams for the two-particle Green function $G^{(4)}$. Diagrams obtained by exchanging the two incoming or outgoing lines are not shown.



and reads

$$-\frac{\zeta}{\beta\mathcal{V}}\sum_{k',\sigma'} v(0)G_{0,\sigma}(k)^2 G_{0,\sigma'}(k')e^{i\omega'_n\eta}$$
$$-\frac{1}{\beta\mathcal{V}}\sum_{q} v(\mathbf{q})G_{0,\sigma}(k)^2 G_{0,\sigma}(k+q)e^{i\omega_{n+\nu}\eta}. \tag{1.350}$$

This result can also be obtained from the Fourier transform of (1.348).

1.5.4 (Anti)symmetrized vertices

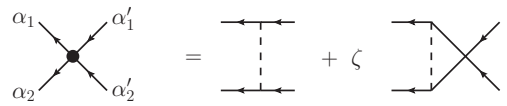
Perturbation theory can be formulated in terms of the (anti)symmetrized vertex introduced in Sec. 1.2 [Eq. (1.179)]. One then starts from the action

$$S_{\rm int}[\psi^*,\psi] = \frac{1}{4}\int_0^\beta d\tau \sum_{\alpha_1,\alpha_2,\alpha'_1,\alpha'_2} \{\alpha_1\alpha_2|\hat{v}|\alpha'_1\alpha'_2\}\psi^*_{\alpha_1}(\tau)\psi^*_{\alpha_2}(\tau)\psi_{\alpha'_2}(\tau)\psi_{\alpha'_1}(\tau) \tag{1.351}$$

with a vertex which is (anti)symmetric under the exchange of the two incoming or outgoing particles,

$$\{\alpha_1\alpha_2|\hat{v}|\alpha'_1\alpha'_2\} = \zeta\{\alpha_1\alpha_2|\hat{v}|\alpha'_2\alpha'_1\} = \zeta\{\alpha_2\alpha_1|\hat{v}|\alpha'_1\alpha'_2\}. \tag{1.352}$$

Since this vertex no longer distinguishes between direct and exchange scatterings of the two incoming particles, it should be graphically represented by a dot,

$\{\alpha_1\alpha_2|\hat{v}|\alpha'_1\alpha'_2\} =$  $=$ $+\ \zeta$

(we write explicitly that the second diagram comes with a sign ζ).

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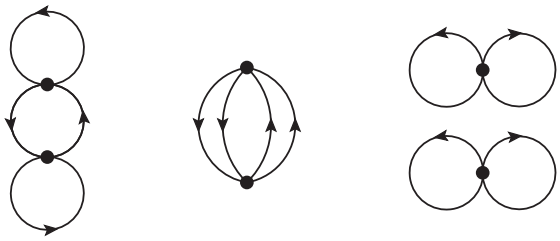
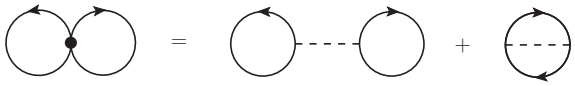


Figure 1.5: Second-order diagrams contributing to the partition function Z/Z_0.

As previously, perturbation theory is obtained by expanding $e^{-S_{\rm int}}$. Using Wick's theorem, we can again express the results by diagrams made of (anti)symmetrized vertices connected by propagators. Because we do not longer distinguish between the two incoming and outgoing lines of a vertex, the number of diagrams is considerably reduced. For instance, the $\mathcal{O}(v)$ correction to Z/Z_0,

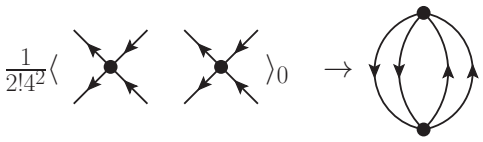
$$-\frac{1}{2}\sum_{\alpha_1,\alpha_2,\alpha_1',\alpha_2'}\{\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2'\}\int_0^\beta d\tau\, G_0(\alpha_1'\tau,\alpha_1\tau^+)G_0(\alpha_2'\tau,\alpha_2\tau^+), \tag{1.353}$$

is represented by a single diagram:



To second order, there are only 3 different diagrams (Fig. 1.5) instead of the 8 to be considered when working with non-(anti)symmetrized vertices (Fig. 1.2).

The combinatorial factor is determined as before. (Note that the action $S_{\rm int}$ is now defined with a factor $\frac{1}{4}$.) For instance, the second of the three diagrams shown in Fig. 1.5 is obtained from



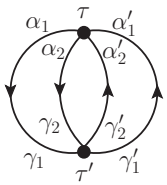
and has a combinatorial factor

$$\frac{1}{2!4^2}\times 4=\frac{1}{8}. \tag{1.354}$$

The factor 4 in the lhs comes from the 4 ways to connect the two vertices in a way consistent with the topology of the diagram.

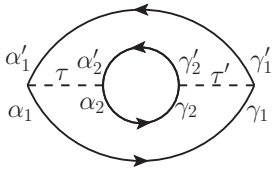
As for the sign of the diagram, it can be determined as before from a particular contraction. One may also use the fact that the diagram should have the same sign as the one obtained *via* the replacement $\{\alpha_i\alpha_j|\hat{v}|\alpha_i'\alpha_j'\}\to(\alpha_i\alpha_j|\hat{v}|\alpha_i'\alpha_j')$. For instance, if we write the preceding diagram as

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$$= \text{sign} \times \frac{1}{8} \sum_{\alpha_1 \cdots \gamma_2'} \{\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2'\}\{\gamma_1'\gamma_2'|\hat{v}|\gamma_1\gamma_2\}$$
$$\times \int_0^\beta d\tau d\tau' G_0(\gamma_1\tau', \alpha_1\tau) G_0(\gamma_2\tau', \alpha_2\tau) G_0(\alpha_2'\tau, \gamma_2'\tau') G_0(\alpha_1'\tau, \gamma_1'\tau'),$$

the replacement $\{\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2'\} \to (\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')$ and $\{\gamma_1'\gamma_2'|\hat{v}|\gamma_1\gamma_2\} \to (\gamma_1'\gamma_2'|\hat{v}|\gamma_1\gamma_2)$ gives



and the sign is therefore $(-1)^2\zeta^2$ since there are two vertices and two closed loops. Note that the sign of the diagram can be unambiguously defined only once we have its expression in terms of the vertices. For instance, we could write $\{\alpha_2\alpha_1|\hat{v}|\alpha_1'\alpha_2'\}$ instead of $\{\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2'\}$ in the preceding equation and the diagram would then have a negative sign.

The diagrammatic rules for calculating a diagram are therefore the following:

1. To each directed line, associate $G_0(\alpha\tau, \alpha'\tau'^+)$.
2. To each vertex, associate $\{\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2'\}$.
3. Sum over all one-particle indices α and integrate all times $\tau \in [0, \beta]$.
4. Multiply the diagram by the combinatorial factor and by $(-1)^l\zeta^{n_L}$ where l is the number of vertices and n_L the number of closed loops in the diagram obtained by replacing $\{\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2'\}$ by $(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')$ in each vertex.

When considering the perturbative expansion of the n-particle Green function $G^{(2n)}(\alpha_1\tau_1, \cdots, \alpha_1'\tau_1')$, the last two rules must be replaced by

3'. Sum over all indices α and integrate all times τ over the interval $[0, \beta]$, while holding fixed the indices $(\alpha_i\tau_i)$ and $(\alpha_i'\tau_i')$ of the external lines.

4'. Multiply by the combinatorial factor and by $(-1)^l\zeta^{n_L}\zeta^P$ where l is the number of vertices, n_L the number of closed loops in the diagram obtained by replacing all vertices $\{\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2'\}$ by $(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')$, and P the permutation of the incoming and outgoing lines.

For example, the first-order contribution to the one-particle Green function $G_0(\alpha\tau, \alpha'\tau')$ is given by the diagram



$$= -\zeta \int_0^\beta d\tau'' \sum_{\alpha_1 \cdots \alpha_4} \{\alpha_1\alpha_2|\hat{v}|\alpha_4\alpha_3\} G_0(\alpha\tau, \alpha_1\tau'') G_0(\alpha_3\tau'', \alpha_2\tau'') G_0(\alpha_4\tau'', \alpha'\tau'). \quad (1.355)$$

The second-order diagrams are shown in Fig. 1.6.

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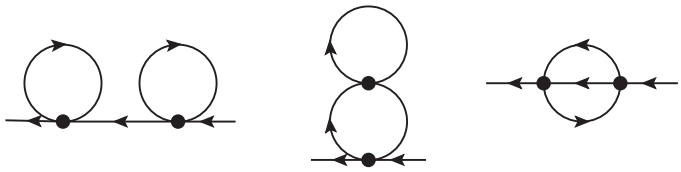


Figure 1.6: Second-order diagrams with (anti)symmetrized vertices contributing to the one-particle Green function.

1.5.4.1 Fully (anti)symmetrized vertices

We define a new field

$$\psi_{\tilde{\alpha}}(\tau) = \begin{cases} \psi_\alpha(\tau) & \text{if} \quad c = -, \\ \psi^*_\alpha(\tau) & \text{if} \quad c = +, \end{cases} \tag{1.356}$$

where $\tilde{\alpha} = (\alpha, c)$ with $c = \pm$ a "charge" index. $\psi_{\alpha+}$ creates a particle whereas $\psi_{\alpha-}$ creates a hole (i.e. removes a particle from the system). The action can be written as

$$\begin{aligned} S_0[\psi] &= \frac{1}{2}\int_0^\beta d\tau d\tau' \sum_{\tilde{\alpha},\tilde{\alpha}'} \psi_{\tilde{\alpha}}(\tau) G_0^{-1}(\tilde{\alpha}\tau, \tilde{\alpha}'\tau')\psi_{\tilde{\alpha}'}(\tau'), \\ S_{\text{int}}[\psi] &= \frac{1}{4!}\int_0^\beta d\tau \sum_{\tilde{\alpha}_1\cdots\tilde{\alpha}_4} v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4}\psi_{\tilde{\alpha}_1}(\tau)\psi_{\tilde{\alpha}_2}(\tau)\psi_{\tilde{\alpha}_3}(\tau)\psi_{\tilde{\alpha}_4}(\tau), \end{aligned} \tag{1.357}$$

where the (anti)symmetric propagator G_0 is defined by

$$G_0(\tilde{\alpha}\tau, \tilde{\alpha}'\tau') = \zeta G_0(\tilde{\alpha}'\tau', \tilde{\alpha}\tau) = \begin{cases} -G_0(\alpha\tau, \alpha'\tau') & \text{if} \quad c = -c' = -, \\ 0 & \text{if} \quad c = c'. \end{cases} \tag{1.358}$$

Note that in (1.357) and (1.358), we use opposite sign conventions for $G_0(\tilde{\alpha}\tau, \tilde{\alpha}'\tau')$ and $G_0(\alpha\tau, \alpha'\tau')$. This choice turns out to be more convenient when dealing with the field $\psi_{\tilde{\alpha}}$ (in particular for the formal manipulations in Sec. 1.6). The vertex $v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4}$ is (anti)symmetric under the exchange of any two of its arguments,

$$v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4} = \zeta v_{\tilde{\alpha}_2\tilde{\alpha}_1\tilde{\alpha}_3\tilde{\alpha}_4} = \zeta v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_4\tilde{\alpha}_3} = \zeta v_{\tilde{\alpha}_3\tilde{\alpha}_2\tilde{\alpha}_1\tilde{\alpha}_4}, \tag{1.359}$$

and satisfies

$$v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4} = \begin{cases} \{\tilde{\alpha}_1\tilde{\alpha}_2|\hat{v}|\tilde{\alpha}_4\tilde{\alpha}_3\} & \text{if} \quad c_1 = c_2 = -c_3 = -c_4 = +, \\ 0 & \text{if} \quad c_1 + c_2 + c_3 + c_4 \neq 0. \end{cases} \tag{1.360}$$

Note the position of $\tilde{\alpha}_3$ and $\tilde{\alpha}_4$ which follows from the definition (1.357) of $v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4}$.

The Wick theorem now reads

$$\langle \psi_{\tilde{\alpha}_1}(\tau_1)\cdots\psi_{\tilde{\alpha}_n}(\tau_n)\rangle = \sum \text{all complete contractions} \tag{1.361}$$

and is a consequence of the results for Gaussian integration over complex or Grassmann variables discussed in Sec. 1.E.4. The various terms appearing in the perturbation expansion of the partition function can be represented by Feynman diagrams. Since we do not distinguish between ψ_α and ψ^*_α (the difference is encoded in the charge index c of $\psi_{\tilde{\alpha}}$), the

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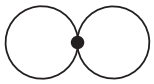
propagator should be represented by a non-directed line

$$G_0(\tilde{\alpha}\tau, \tilde{\alpha}'\tau') = \quad \tilde{\alpha}\tau \underline{\qquad\qquad} \tilde{\alpha}'\tau'$$

and the vertex by a simple dot,

$$v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4} = \quad \begin{matrix} \tilde{\alpha}_1 & & \tilde{\alpha}_4 \\ & \bullet & \\ \tilde{\alpha}_2 & & \tilde{\alpha}_3 \end{matrix}$$

The first-order correction to the partition function reads



$$= -\frac{1}{8}\int_0^\beta d\tau \sum_{\tilde{\alpha}_1\cdots\tilde{\alpha}_4} v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4} G_0(\tilde{\alpha}_1\tau, \tilde{\alpha}_4\tau) G_0(\tilde{\alpha}_2\tau, \tilde{\alpha}_3\tau), \tag{1.362}$$

where the combinatorial factor comes from the $1/4!$ in front of the interaction and the three ways to pair the lines of the vertex. The sign is obtained from a particular contraction, e.g.

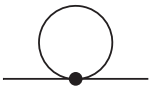
$$-\frac{1}{4!}\int_0^\beta d\tau \sum_{\tilde{\alpha}_1\cdots\tilde{\alpha}_4} v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4} \overbrace{\psi_{\tilde{\alpha}_1}(\tau)\underbrace{\psi_{\tilde{\alpha}_2}(\tau)\psi_{\tilde{\alpha}_3}(\tau)}\psi_{\tilde{\alpha}_4}(\tau)}. \tag{1.363}$$

Equation (1.362) gives

$$\begin{aligned} &-\frac{1}{8}\int_0^\beta d\tau \sum_{\substack{\alpha_1\cdots\alpha_4 \\ c,c'=\pm}} v_{\alpha_1 c,\alpha_2 c',\alpha_3\bar{c}',\alpha_4\bar{c}} G_0(\alpha_1, c, \tau; \alpha_4, \bar{c}, \tau) G_0(\alpha_2, c', \tau; \alpha_3, \bar{c}', \tau) \\ &= -\frac{1}{2}\int_0^\beta d\tau \sum_{\alpha_1\cdots\alpha_4} v_{\alpha_1+,\alpha_2+,\alpha_3-,\alpha_4-} G_0(\alpha_4, -, \tau; \alpha_1, +, \tau^+) G_0(\alpha_3, -, \tau; \alpha_2, +, \tau^+) \end{aligned} \tag{1.364}$$

(with $\bar{c} = -c$) and therefore reproduces (1.353). In (1.364) we have interpreted the equal time propagator $G_0(\alpha\tau, \alpha'\tau)$ as $G_0(\alpha\tau, \alpha'\tau^+)$ for reasons previously discussed. Note that $G_0(\tilde{\alpha}\tau, \tilde{\alpha}'\tau)$ corresponds to $G_0(\tilde{\alpha}\tau, \tilde{\alpha}'\tau^+)$ if $c' = -c = +$ and to $G_0(\tilde{\alpha}\tau^+, \tilde{\alpha}'\tau)$ if $c = -c' = +$.

Similarly, for the first-order correction to the Green function $G(\tilde{\alpha}\tau, \tilde{\alpha}'\tau')$ one obtains



$$= -\frac{1}{2}\int_0^\beta d\tau'' \sum_{\tilde{\alpha}_1\cdots\tilde{\alpha}_4} v_{\tilde{\alpha}_1\tilde{\alpha}_2\tilde{\alpha}_3\tilde{\alpha}_4} G(\tilde{\alpha}\tau, \tilde{\alpha}_1\tau'') G(\tilde{\alpha}_2\tau'', \tilde{\alpha}_3\tau'') G(\tilde{\alpha}_4\tau'', \tilde{\alpha}'\tau'), \tag{1.365}$$

which can be shown to be equivalent to (1.355) when $\tilde{\alpha} = (\alpha, -)$ and $\tilde{\alpha}' = (\alpha', +)$.

One must be careful about the time ordering of the fields when using this formalism. To illustrate this point, let us consider non-interacting particles and express the action

$$S[\Psi^\dagger, \Psi] = \frac{1}{2}\sum_{\omega_n} \Psi^\dagger(i\omega_n)\mathcal{G}^{-1}(i\omega_n)\Psi(i\omega_n) \tag{1.366}$$

in terms of the two-component field

$$\Psi(\tau) = \begin{pmatrix} \psi(\tau) \\ \psi^*(\tau) \end{pmatrix}, \qquad \Psi(i\omega_n) = \begin{pmatrix} \psi(i\omega_n) \\ \psi^*(-i\omega_n) \end{pmatrix}. \tag{1.367}$$

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For simplicity, we consider a single one-body state with energy $\epsilon = \xi + \mu$. In frequency space, the propagator $\mathcal{G}$ is a 2×2 matrix defined by

$$\mathcal{G}^{-1}(i\omega_n) = \begin{pmatrix} -i\omega_n + \xi & 0 \\ 0 & \zeta(i\omega_n + \xi) \end{pmatrix}. \tag{1.368}$$

The functional integral gives

$$Z_0 = \int \mathcal{D}[\psi^*, \psi]\, e^{-S[\Psi^\dagger, \Psi]} = \prod_{\omega_n} [\mathcal{G}_{11}(i\omega_n)\mathcal{G}_{22}(i\omega_n)]^{\zeta/2} \tag{1.369}$$

and

$$\Omega_0 = -\frac{\zeta}{2\beta} \sum_{\omega_n} \ln[\mathcal{G}_{11}(i\omega_n)\mathcal{G}_{22}(i\omega_n)]. \tag{1.370}$$

If now one introduces the convergence factor $e^{i\omega_n \eta}$, one obtains (see Appendix 1.F)[47]

$$\frac{\zeta}{2\beta} \sum_{\omega_n} \ln(\omega_n^2 + \xi^2) e^{i\omega_n \eta} = \frac{\zeta}{\beta} \ln\left(1 - \zeta e^{-\beta\xi}\right) + \zeta \frac{\xi}{2}, \tag{1.371}$$

which differs from the exact result by $\zeta\xi/2$.

The reason for this apparent difficulty is that if we permute creation and annihilation fields, which is what we have done when writing

$$\begin{aligned} \psi^*(\tau) G_0(\tau - \tau') \psi(\tau') &= \zeta \psi(\tau') G_0(-\tau' + \tau) \psi^*(\tau) \\ &\equiv \Psi_2^\dagger(\tau') \mathcal{G}_{22}(\tau' - \tau) \Psi_2(\tau), \end{aligned} \tag{1.372}$$

one must then interpret the equal time propagator $\mathcal{G}_{22}(0)$ as $\mathcal{G}_{22}(0^+)$ rather than $\mathcal{G}_{22}(0^-)$. Accordingly, we must use the convergence factor $e^{-i\omega_n \eta}$ for the (22) component of the propagator. Thus the correct expression of the thermodynamic potential is

$$\begin{aligned} \Omega_0 &= -\frac{\zeta}{2\beta} \sum_{\omega_n} \left\{ \ln[\mathcal{G}_{11}(i\omega_n)] e^{i\omega_n \eta} + \ln[\mathcal{G}_{22}(i\omega_n)] e^{-i\omega_n \eta} \right\} \\ &= -\frac{\zeta}{\beta} \sum_{\omega_n} \ln[\mathcal{G}_{11}(i\omega_n)] e^{i\omega_n \eta}, \end{aligned} \tag{1.373}$$

in agreement with (1.308). The mean particle number $N = -\partial\Omega_0/\partial\mu$ is then given by

$$\begin{aligned} N &= \frac{\zeta}{2\beta} \sum_{\omega_n} \left[\mathcal{G}_{11}(i\omega_n) e^{i\omega_n \eta} + \zeta \mathcal{G}_{22}(i\omega_n) e^{-i\omega_n \eta} \right] \\ &= -\frac{\zeta}{\beta} \sum_{\omega_n} \frac{e^{i\omega_n \eta}}{i\omega_n - \xi}, \end{aligned} \tag{1.374}$$

which coincides with (1.306).

In formal manipulations of the functional integral as well as in some practical calculations, it is sometimes convenient to use the field $\psi_{\tilde{\alpha}}$ rather than ψ_α and ψ_α^*. This is particularly true when one has to deal with superfluid or superconducting systems where

[47] For fermions, the Matsubara frequencies appear in pairs $(\omega_n, \omega_{-n-1} = -\omega_n)$ so that $\det[\mathcal{G}^{-1}(i\omega_n)\mathcal{G}^{-1}(-i\omega_n)] = (\omega_n^2 + \xi^2)^2$ and $\det \mathcal{G}^{-1} = \prod_{\omega_n} (\omega_n^2 + \xi^2)$.

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the (global) U(1) symmetry is broken.[48] In these systems, the one-particle Green function $G(\tilde{\alpha}\tau, \tilde{\alpha}'\tau')$ has both "normal" ($G(\alpha, c, \tau; \alpha', -c, \tau')$) and "anomalous" ($G(\alpha, c, \tau; \alpha', c, \tau')$) components and can be represented as a 2×2 matrix

$$G(\tilde{\alpha}\tau, \tilde{\alpha}'\tau') = \begin{pmatrix} G(\alpha, +, \tau; \alpha', +, \tau') & G(\alpha, +, \tau; \alpha', -, \tau') \\ G(\alpha, -, \tau; \alpha', +, \tau') & G(\alpha, -, \tau; \alpha', -, \tau') \end{pmatrix} \tag{1.375}$$

$$= \left(\right.$$  $$\left. \right)$$

Moreover, in bosonic superfluid systems, the Green functions $G^{(2n+1)}$ with an odd number of legs do not vanish, and $G^{(1)}(\tilde{\alpha}, \tau) = \langle \psi_{\tilde{\alpha}}(\tau) \rangle$ is the order parameter of the superfluid phase (see Chap. 7).[49] Superfluid or superconducting systems cannot be described within a perturbation expansion starting from the non-interacting limit. As for any system where a symmetry is spontaneously broken, it is necessary to reorganize the perturbation theory about a broken-symmetry state. The perturbation expansion in bosonic superfluid systems is explained in detail in Sec. 1.7. Superconductors (i.e. superfluid fermion systems) are discussed in Chap. 7.

1.6 Generating functionals

In this section, we define connected Green functions, one-particle irreducible (1PI) and two-particle irreducible (2PI) vertices. 1PI and 2PI vertices allow one to reformulate and simplify the perturbation expansion discussed in Sec. 1.5. Furthermore, we shall see in the forthcoming chapters that the corresponding generating functionals enable to set up non-perturbative approaches to strongly correlated quantum systems (Chap. 9 and 10).

1.6.1 Connected Green functions

In this section we define the connected Green functions and their generating functional. We first consider normal systems where global U(1) invariance is not broken[48] before discussing the general case where we allow the system to be superfluid.

1.6.1.1 Normal systems

We consider the partition function in the presence of external sources,

$$Z[J^*, J] = \int \mathcal{D}[\psi^*, \psi] \exp\Big\{ -S[\psi^*, \psi] + \int_0^\beta d\tau \sum_\alpha [J_\alpha^*(\tau)\psi_\alpha(\tau) + \text{c.c.}] \Big\}, \tag{1.376}$$

where J^*, J are c-numbers for bosons and anticommuting variables for fermions. The Green functions

$$G^{(2n)}(\alpha_1\tau_1, \cdots, \alpha_n\tau_n; \alpha_n'\tau_n', \cdots, \alpha_1'\tau_1') = (-1)^n \langle \psi_{\alpha_1}(\tau_1) \cdots \psi_{\alpha_n}(\tau_n) \psi_{\alpha_n'}^*(\tau_n') \cdots \psi_{\alpha_1'}^*(\tau_1') \rangle$$

[48] The (global) U(1) invariance, i.e. the invariance of the action in the transformation $\psi_\sigma(\mathbf{r}, \tau) \to \psi_\sigma(\mathbf{r}, \tau) e^{i\Lambda}$ and $\psi_\sigma^*(\mathbf{r}, \tau) \to \psi_\sigma^*(\mathbf{r}, \tau) e^{-i\Lambda}$, is related to the conservation of the number of particles (Sec. 2.2.7). Broken U(1) symmetry in superfluid and superconducting systems is discussed in Chap. 7, while the concept of spontaneous symmetry breaking is introduced in Chaps. 2 and 3.

[49] In fermion systems, $\langle \psi_{\tilde{\alpha}}(\tau) \rangle$ always vanishes. The spontaneously broken U(1) symmetry in a superfluid fermion system manifests itself by nonzero expectation values $\langle \psi_\alpha(\tau) \psi_{\alpha'}(\tau') \rangle$ and $\langle \psi_\alpha^*(\tau) \psi_{\alpha'}^*(\tau') \rangle$ (Chap. 7).

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$$= (-\zeta)^n \frac{\delta^{2n}\mathcal{G}[J^*,J]}{\delta J^*_{\alpha_1}(\tau_1)\cdots\delta J^*_{\alpha_n}(\tau_n)\delta J_{\alpha'_n}(\tau'_n)\cdots\delta J_{\alpha'_1}(\tau'_1)}\bigg|_{J^*=J=0} \tag{1.377}$$

can be obtained from the generating functional

$$\mathcal{G}[J^*,J] = \frac{Z[J^*,J]}{Z[0,0]}, \tag{1.378}$$

where $Z[0,0] = Z$ is the partition function in the absence of external sources. In a normal system, the only non-vanishing Green functions are of the type (1.377) with n incoming particles and n outgoing particles.

In Sec. 1.5, we showed that the Green function $G^{(2n)}$ can be obtained from the sum of all Feynman diagrams that are connected to the external legs. However, these diagrams need not be fully connected.[50] As the disconnected diagrams can be expressed in terms of lower-order Green functions ($G^{(2n-2)}$, $G^{(2n-4)}$, etc.), it is useful to deal with the "connected" Green functions $G_c^{(2n)}$ defined as the sum of all fully connected diagrams. The corresponding generating functional is $W[J^*,J] = \ln Z[J^*,J]$. This can be shown using the replica method as in the derivation of the linked cluster theorem (Sec. 1.5.2). The generating functional $\mathcal{G}[J^*,J]^R$ of the Green functions of R replicas of the system can be written as a functional integral over R distinct fields. The connected diagrams are proportional to R (Sec. 1.5.2), while a diagram with n disconnected pieces is of order R^n. The connected diagrams can therefore be obtained from the functional

$$\lim_{R\to 0}\frac{\partial}{\partial_R}\mathcal{G}[J^*,J]^R = \ln\mathcal{G}[J^*,J] = \ln Z[J^*,J] - \ln Z[0,0], \tag{1.379}$$

or more simply $W[J^*,J] = \ln Z[J^*,J]$ since $\ln Z[0,0]$ is independent of the external sources.

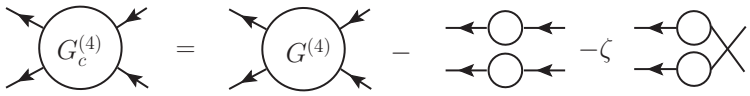
A direct calculation gives

$$G_c(\alpha_1\tau_1,\alpha'_1\tau'_1) = -\zeta\frac{\delta^2 W[J^*,J]}{\delta J^*_{\alpha_1}(\tau_1)\delta J_{\alpha'_1}(\tau'_1)}\bigg|_{J^*=J=0} = G(\alpha_1\tau_1,\alpha'_1\tau'_1) \tag{1.380}$$

and

$$\begin{aligned} G_c^{(4)}(\alpha_1\tau_1,\alpha_2\tau_2;\alpha'_2\tau'_2,\alpha'_1\tau'_1) &= \frac{\delta^4 W[J^*,J]}{\delta J^*_{\alpha_1}(\tau_1)\delta J^*_{\alpha_2}(\tau_2)\delta J_{\alpha'_2}(\tau'_2)\delta J_{\alpha'_1}(\tau'_1)}\bigg|_{J^*=J=0} \\ &= G^{(4)}(\alpha_1\tau_1,\alpha_2\tau_2;\alpha'_2\tau'_2,\alpha'_1\tau'_1) \\ &\quad - G(\alpha_1\tau_1,\alpha'_1\tau'_1)G(\alpha_2\tau_2,\alpha'_2\tau'_2) \\ &\quad - \zeta G(\alpha_1\tau_1,\alpha'_2\tau'_2)G(\alpha_2\tau_2,\alpha'_1\tau'_1) \end{aligned} \tag{1.381}$$

in a normal system where $\langle\psi_\alpha(\tau)\rangle = \langle\psi^*_\alpha(\tau)\rangle = 0$. The last equation can be graphically represented as



(writing explicitly the sign of the diagrams), where the empty circles stand for the one-particle Green function $G \equiv G^{(2)}$. Clearly, $G_c^{(4)}$ is given by the sum of all fully connected diagrams.

[50] For instance, in a non-interacting system, none of the diagrams contributing to $G^{(4)}$ are fully connected (see Eq. (1.317) and the corresponding figure).

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The connected Green functions are similar to the cumulants of a probability distribution $P(x)$ with moment-generating function

$$Z(j) = \int_{-\infty}^{\infty} dx\, P(x) e^{jx} = \sum_{n=0}^{\infty} \frac{j^n}{n!} \langle x^n \rangle. \tag{1.382}$$

The nth-order cumulant $\langle x^n \rangle_c$ is defined by

$$\langle x^n \rangle_c = \frac{\partial^n}{\partial j^n} W(j)\Big|_{j=0}, \qquad W(j) = \ln Z(j) = \sum_{n=1}^{\infty} \frac{j^n}{n!} \langle x^n \rangle_c. \tag{1.383}$$

One easily finds

$$\begin{aligned} \langle x \rangle_c &= \langle x \rangle, \\ \langle x^2 \rangle_c &= \langle x^2 \rangle - \langle x \rangle^2 = \langle (x - \langle x \rangle)^2 \rangle, \\ \langle x^3 \rangle_c &= \langle x^3 \rangle - 3\langle x^2 \rangle \langle x \rangle + 2\langle x \rangle^3 = \langle (x - \langle x \rangle)^3 \rangle, \\ \langle x^4 \rangle_c &= \langle (x - \langle x \rangle)^4 \rangle - 3\langle (x - \langle x \rangle)^2 \rangle^2. \end{aligned} \tag{1.384}$$

For a Gaussian probability distribution $P(x) = (a/2\pi)^{1/2} e^{-(x-x_0)^2/2a}$,

$$W(j) = \ln\left\{ \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{\infty} dx\, e^{-\frac{1}{2a}(x-x_0)^2 + jx} \right\} = \frac{1}{2} a j^2 + j x_0, \tag{1.385}$$

so that $\langle x \rangle_c = x_0$, $\langle x^2 \rangle_c = a$, and $\langle x^n \rangle_c = 0$ for $n \geq 3$. By inverting (1.384), one can express all averages $\langle x^n \rangle$ in terms of the average value x_0 and the second-order cumulant $\langle x^2 \rangle_c$ (which is nothing but Wick's theorem). The analogous result for a many-particle system is that when the action $S[\psi^*, \psi]$ is Gaussian (non-interacting particles), the only non-vanishing connected Green function is $G_c^{(2)}$, and all higher-order Green functions $G^{(n \geq 4)}$ are entirely determined by $G_c^{(2)}$.

1.6.1.2 General case

As pointed out in Sec. 1.5.4, in systems where the U(1) symmetry is broken the Green functions take a more general form where the number of outgoing particles may differ from that of incoming particles. The total number of fields can even be odd in bosonic (superfluid) systems. It is then convenient to use the field $\psi_{\tilde{\alpha}}(\tau)$ (with $\tilde{\alpha} = (\alpha, c)$ and $c = \pm$ a charge index) introduced in Sec. 1.5.4 and define the Green functions as[51]

$$G^{(n)}(\tilde{\alpha}_1 \tau_1, \cdots, \tilde{\alpha}_n \tau_n) = \langle \psi_{\tilde{\alpha}_1}(\tau_1) \cdots \psi_{\tilde{\alpha}_n}(\tau_n) \rangle = \frac{\delta^n \mathcal{G}[J]}{\delta J_{\tilde{\alpha}_1}(\tau_1) \cdots \delta J_{\tilde{\alpha}_n}(\tau_n)}\bigg|_{J=0}. \tag{1.386}$$

This definition will be used from now on whenever we work with the field $\psi_{\tilde{\alpha}}$. The generating functional is given by

$$\begin{aligned} \mathcal{G}[J] &= \frac{1}{Z[0]} \int \mathcal{D}[\psi] \exp\left\{ -S[\psi] + \int_0^\beta d\tau \sum_{\tilde{\alpha}} J_{\tilde{\alpha}}(\tau) \psi_{\tilde{\alpha}}(\tau) \right\} \\ &= 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \sum_{\tilde{\alpha}_1 \cdots \tilde{\alpha}_n} G^{(n)}(\tilde{\alpha}_1 \tau_1, \cdots, \tilde{\alpha}_n \tau_n) J_{\tilde{\alpha}_n}(\tau_n) \cdots J_{\tilde{\alpha}_1}(\tau_1). \end{aligned} \tag{1.387}$$

[51] The sign convention in (1.386) agrees with that of (1.358).

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Similarly, the connected Green functions and their generating functional $W[J] = \ln Z[J]$ are defined by

$$W[J] = \ln Z[0] + \sum_{n=1}^{\infty} \frac{1}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \sum_{\tilde{\alpha}_1 \cdots \tilde{\alpha}_n} G_c^{(n)}(\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_n\tau_n) J_{\tilde{\alpha}_n}(\tau_n) \cdots J_{\tilde{\alpha}_1}(\tau_1) \tag{1.388}$$

and

$$G_c^{(n)}(\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_n\tau_n) = \left. \frac{\delta^n W[J]}{\delta J_{\tilde{\alpha}_1}(\tau_1) \cdots \delta J_{\tilde{\alpha}_n}(\tau_n)} \right|_{J=0}. \tag{1.389}$$

For the lowest-order Green functions one finds

$$\begin{aligned} G_c^{(1)}(\tilde{\alpha}\tau) &= G^{(1)}(\tilde{\alpha}\tau) = \langle \psi_{\tilde{\alpha}}(\tau) \rangle, \\ G_c^{(2)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2\tau_2) &= G^{(2)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2\tau_2) - \langle \psi_{\tilde{\alpha}_1}(\tau_1) \rangle \langle \psi_{\tilde{\alpha}_2}(\tau_2) \rangle \end{aligned} \tag{1.390}$$

and

$$\begin{aligned} G_c^{(3)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2\tau_2, \tilde{\alpha}_3\tau_3) = G^{(3)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2\tau_2, \tilde{\alpha}_3\tau_3) &- \langle \psi_{\tilde{\alpha}_1}(\tau_1) \rangle G_c^{(2)}(\tilde{\alpha}_2\tau_2, \tilde{\alpha}_3\tau_3) \\ &- \zeta \langle \psi_{\tilde{\alpha}_2}(\tau_2) \rangle G_c^{(2)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_3\tau_3) - \langle \psi_{\tilde{\alpha}_3}(\tau_3) \rangle G_c^{(2)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2\tau_2) \\ &- \langle \psi_{\tilde{\alpha}_1}(\tau_1) \rangle \langle \psi_{\tilde{\alpha}_2}(\tau_2) \rangle \langle \psi_{\tilde{\alpha}_3}(\tau_3) \rangle. \end{aligned} \tag{1.391}$$

For a fermion system, all Green functions with an odd number of legs vanish.

1.6.1.3 Equations of motion: Schwinger-Dyson equations

The Green functions $G^{(n)}$ are not independent but are related by "equations of motion".[52] Let us consider the generating functional $\mathcal{G}[J]$ [Eq. (1.387)]. The change of variables $\psi_{\tilde{\alpha}}(\tau) \to \psi_{\tilde{\alpha}}(\tau) + \epsilon f_{\tilde{\alpha}}(\tau)$, with $\epsilon \to 0$ and $f_{\tilde{\alpha}}(\tau)$ a complex or Grassmannian field, has a unit Jacobian and gives

$$\mathcal{G}[J] = \frac{1}{Z[0]} \int \mathcal{D}[\psi]\, e^{-S[\psi+\epsilon f] + \int_0^\beta d\tau \sum_{\tilde{\alpha}} J_{\tilde{\alpha}}(\tau)[\psi_{\tilde{\alpha}}(\tau) + \epsilon f_{\tilde{\alpha}}(\tau)]}. \tag{1.392}$$

Expanding to first order in ϵ, one finds

$$\frac{1}{Z[0]} \int \mathcal{D}[\psi]\, e^{-S[\psi] + \int_0^\beta d\tau \sum_{\tilde{\alpha}} J_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}}(\tau)} \int_0^\beta d\tau \sum_{\tilde{\alpha}} f_{\tilde{\alpha}}(\tau) \left[\zeta \frac{\delta S}{\delta \psi_{\tilde{\alpha}}(\tau)} - J_{\tilde{\alpha}}(\tau) \right] = 0 \tag{1.393}$$

(for Grassmann variables we use the chain rule (1.240) for derivation) and, since the function $f_{\tilde{\alpha}}(\tau)$ is arbitrary,

$$\frac{1}{Z[0]} \int \mathcal{D}[\psi]\, e^{-S[\psi] + \int_0^\beta d\tau \sum_{\tilde{\alpha}} J_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}}(\tau)} \left[\zeta \frac{\delta S}{\delta \psi_{\tilde{\alpha}}(\tau)} - J_{\tilde{\alpha}}(\tau) \right] = 0. \tag{1.394}$$

If we write the action as in (1.357), Eq. (1.394) gives

$$\frac{1}{Z[0]} \int \mathcal{D}[\psi]\, e^{-S[\psi] + \int_0^\beta d\tau \sum_{\tilde{\alpha}} J_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}}(\tau)} \left[\int_0^\beta d\tau_1' \sum_{\tilde{\alpha}_1'} G_0^{-1}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_1'\tau_1') \psi_{\tilde{\alpha}_1'}(\tau_1') \right.$$

[52] In the operator formalism, the equations of motion are obtained by calculating $\partial_{\tau_1} G^{(n)}(\tilde{\alpha}_1\tau_1 \cdots \tilde{\alpha}_n\tau_n) = \partial_{\tau_1} \langle T_\tau \hat{\psi}_{\tilde{\alpha}_1}(\tau_1) \cdots \hat{\psi}_{\tilde{\alpha}_n}(\tau_n) \rangle$ using $\partial_\tau \hat{\psi}_{\tilde{\alpha}}(\tau) = [\hat{H}, \hat{\psi}_{\tilde{\alpha}}(\tau)]$.

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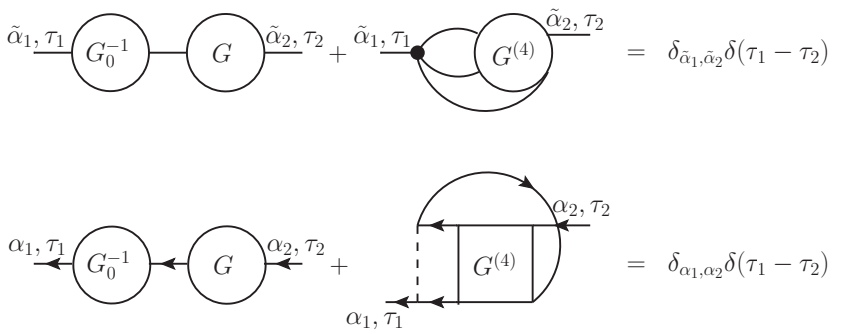


Figure 1.7: Diagrammatic representation of the equation of motion for $G = G^{(2)}$ in the general case [Eq. (1.397)] and for a normal system [Eq. (1.398)].

$$+\frac{1}{3!}\sum_{\tilde{\alpha}_2'\tilde{\alpha}_3'\tilde{\alpha}_4'} v_{\tilde{\alpha}_1\tilde{\alpha}_2'\tilde{\alpha}_3'\tilde{\alpha}_4'}\psi_{\tilde{\alpha}_2'}(\tau_1)\psi_{\tilde{\alpha}_3'}(\tau_1)\psi_{\tilde{\alpha}_4'}(\tau_1) - \zeta J_{\tilde{\alpha}_1}(\tau_1)\Big] = 0. \tag{1.395}$$

The equation of motion for the Green function $G^{(n)}$ is obtained by taking the $(n-1)$th-order functional derivative with respect to J and setting $J = 0$ in the end,

$$\begin{aligned}\int_0^\beta d\tau_1' \sum_{\tilde{\alpha}_1'} & G_0^{-1}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_1'\tau_1')G^{(n)}(\tilde{\alpha}_n\tau_n, \cdots, \tilde{\alpha}_2\tau_2, \tilde{\alpha}_1'\tau_1') \\ &+ \frac{1}{3!}\sum_{\tilde{\alpha}_2',\tilde{\alpha}_3',\tilde{\alpha}_4'} v_{\tilde{\alpha}_1\tilde{\alpha}_2'\tilde{\alpha}_3'\tilde{\alpha}_4'}G^{(n+2)}(\tilde{\alpha}_n\tau_n, \cdots, \tilde{\alpha}_2\tau_2, \tilde{\alpha}_2'\tau_1, \tilde{\alpha}_3'\tau_1, \tilde{\alpha}_4'\tau_1) \\ &= \zeta\sum_{i=2}^{n}\delta_{\tilde{\alpha}_i,\tilde{\alpha}_1}\delta(\tau_i-\tau_1)G^{(n-2)}(\tilde{\alpha}_n\tau_n, \cdots, \tilde{\alpha}_{i+1}\tau_{i+1}, \tilde{\alpha}_{i-1}\tau_{i-1}, \cdots, \tilde{\alpha}_2\tau_2).\end{aligned} \tag{1.396}$$

The equation of motion for $G^{(2)} = G$,

$$\begin{aligned}&\int_0^\beta d\tau_1' \sum_{\tilde{\alpha}_1'} G_0^{-1}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_1'\tau_1')G(\tilde{\alpha}_1'\tau_1', \tilde{\alpha}_2\tau_2) \\ &+\frac{1}{3!}\sum_{\tilde{\alpha}_2',\tilde{\alpha}_3',\tilde{\alpha}_4'} v_{\tilde{\alpha}_1\tilde{\alpha}_2'\tilde{\alpha}_3'\tilde{\alpha}_4'}G^{(4)}(\tilde{\alpha}_2'\tau_1, \tilde{\alpha}_3'\tau_1, \tilde{\alpha}_4'\tau_1, \tilde{\alpha}_2\tau_2) = \delta_{\tilde{\alpha}_1,\tilde{\alpha}_2}\delta(\tau_1-\tau_2)\end{aligned} \tag{1.397}$$

is represented diagrammatically in Fig. 1.7. For a normal system, one finds (with $\tilde{\alpha}_1 = (\alpha_1, +)$ and $\tilde{\alpha}_2 = (\alpha_2, +)$)

$$\begin{aligned}&\int_0^\beta d\tau_1' \sum_{\alpha_1'} G_0^{-1}(\alpha_1\tau_1, \alpha_1'\tau_1')G(\alpha_1'\tau_1', \alpha_2\tau_2) \\ &+ \sum_{\alpha_2',\alpha_3',\alpha_4'} (\alpha_1\alpha_2'|\hat{v}|\alpha_4'\alpha_3')G^{(4)}(\alpha_3'\tau_1, \alpha_4'\tau_1, \alpha_2'\tau_1, \alpha_2\tau_2) = \delta_{\alpha_1,\alpha_2}\delta(\tau_1-\tau_2)\end{aligned} \tag{1.398}$$

(see Fig. 1.7), where (α_3', τ_1) and (α_4', τ_1) must be understood as (α_3', τ_1^-) and (α_4', τ_1^-) (with $\tau_1^- = \tau_1 - 0^+$) because of the time ordering associated with the matrix element $(\alpha_1\alpha_2'|\hat{v}|\alpha_4'\alpha_3')$.

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1.6.2 Effective action and 1PI vertices

In this section, we introduce the effective action and the 1PI vertices considering again first normal systems before discussing the general case.[53] The effective action is further discussed in Sec. 1.7 and plays a fundamental role in the non-perturbative functional RG (Chap. 10).

1.6.2.1 Normal systems

In presence of external sources, the fields $\psi_\alpha, \psi_\alpha^*$ acquire nonzero expectation values,[54]

$$\begin{aligned} \phi_\alpha[\tau; J^*, J] &= \langle \psi_\alpha(\tau) \rangle = \frac{\delta W[J^*, J]}{\delta J_\alpha^*(\tau)}, \\ \phi_\alpha^*[\tau; J, J^*] &= \langle \psi_\alpha^*(\tau) \rangle = \zeta \frac{\delta W[J^*, J]}{\delta J_\alpha(\tau)}, \end{aligned} \tag{1.399}$$

where $\langle \cdots \rangle$ denotes an expectation value in the presence of the external sources. ϕ_α and ϕ_α^* are c-numbers for bosonic fields and Grassmann variables for fermionic fields.

The (1PI) effective action $\Gamma[\phi^*, \phi]$ is defined as the Legendre transform of $\ln Z[J^*, J]$,

$$\Gamma[\phi^*, \phi] = -\ln Z[J^*, J] + \int_0^\beta d\tau \sum_\alpha \left[J_\alpha^*(\tau) \phi_\alpha(\tau) + \text{c.c.} \right]. \tag{1.400}$$

In (1.400), $J_\alpha(\tau)$ and $J_\alpha^*(\tau)$ must be considered as functionals of $\phi_\alpha(\tau)$ and $\phi_\alpha^*(\tau)$ obtained by inverting (1.399). $\Gamma[\phi^*, \phi]$ satisfies[55]

$$\begin{aligned} \frac{\delta \Gamma[\phi^*, \phi]}{\delta \phi_\alpha(\tau)} = & -\int_0^\beta d\tau' \sum_{\alpha'} \left(\frac{\delta J_{\alpha'}(\tau')}{\delta \phi_\alpha(\tau)} \frac{\delta W[J^*, J]}{\delta J_{\alpha'}(\tau')} + \frac{\delta J_{\alpha'}^*(\tau')}{\delta \phi_\alpha(\tau)} \frac{\delta W[J^*, J]}{\delta J_{\alpha'}^*(\tau')} \right) \\ & + \int_0^\beta d\tau' \sum_{\alpha'} \left(\frac{\delta J_{\alpha'}^*(\tau')}{\delta \phi_\alpha(\tau)} \phi_{\alpha'}(\tau') + \zeta \frac{\delta J_{\alpha'}(\tau')}{\delta \phi_\alpha(\tau)} \phi_{\alpha'}^*(\tau') \right) + \zeta J_\alpha^*(\tau) \\ = & \, \zeta J_\alpha^*(\tau) \end{aligned} \tag{1.401}$$

and the companion equation

$$\frac{\delta \Gamma[\phi^*, \phi]}{\delta \phi_\alpha^*(\tau)} = J_\alpha(\tau). \tag{1.402}$$

The effective action is similar to the Gibbs free energy introduced in statistical mechanics. Let us consider for instance a classical spin system with partition function

$$Z(\mathbf{h}) = e^{-\beta F(\mathbf{h})} = \int \mathcal{D}[\mathbf{S}] \, e^{-\beta \int d^d r [\mathcal{H}(\mathbf{S}(\mathbf{r})) - \mathbf{h} \cdot \mathbf{S}(\mathbf{r})]}, \tag{1.403}$$

where $\mathcal{H}(\mathbf{S}(\mathbf{r}))$ is the Hamiltonian density and $\mathbf{h}$ an external field. The magnetization density of the system is given by

$$\mathbf{m} = \frac{1}{\mathcal{V}} \int d^d r \langle \mathbf{S}(\mathbf{r}) \rangle = \frac{1}{\beta \mathcal{V}} \frac{\partial}{\partial \mathbf{h}} \ln Z(\mathbf{h}) = -\frac{1}{\mathcal{V}} \frac{\partial F(\mathbf{h})}{\partial \mathbf{h}}. \tag{1.404}$$

[53] In this section the effective action is defined by the Legendre transform of $\ln Z[J^*, J]$. The same terminology is used for an action $S[\psi^*, \psi]$ providing an effective description of the system, e.g. at low energies or for a subset of degrees of freedom.

[54] The notation emphasizes that $\phi_\alpha^{(*)}[\tau; J^*, J]$ is a functional of the external sources.

[55] In the case of Grassmann variables, we use the chain rule (1.240) for derivation.

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The Gibbs free energy is defined as the Legendre transform of the Helmholtz free energy $F(\mathbf{h})$,

$$G(\mathbf{m}) = F(\mathbf{h}) + \mathcal{V}\mathbf{m}\cdot\mathbf{h}, \tag{1.405}$$

and satisfies

$$\frac{\partial G(\mathbf{m})}{\partial \mathbf{m}} = \mathcal{V}\mathbf{h}. \tag{1.406}$$

$F(\mathbf{h})$ and $G(\mathbf{m})$ contain the same physical information. In the absence of an external field, the magnetization of the system at thermodynamic equilibrium is obtained by requiring the Gibbs free energy to be stationary. Note that $G(\mathbf{m})$ (if calculated exactly) includes all effects of thermal fluctuations.

More generally, one can consider a spatially varying field $\mathbf{h}(\mathbf{r})$ giving rise to a non uniform magnetization density $\mathbf{m}(\mathbf{r}) = \langle \mathbf{S}(\mathbf{r})\rangle$. $F[\mathbf{h}]$ and $G[\mathbf{m}]$ are then functionals of $\mathbf{h}(\mathbf{r})$ and $\mathbf{m}(\mathbf{r})$, respectively. $F[\mathbf{h}]$ is the generating functional of the connected correlation functions of the spin density field $\mathbf{S}(\mathbf{r})$ whereas $G[\mathbf{m}]$ is the generating functional of the 1PI vertices (see below).

1.6.2.2 Self-energy and Dyson's equation

The 1PI vertices[56] are defined as the functional derivatives of the effective action,

$$\Gamma^{(2n)}(\alpha_1\tau_1,\cdots,\alpha_n\tau_n;\alpha'_n\tau'_n,\cdots,\alpha'_1\tau'_1) = \frac{\delta^{2n}\Gamma[\phi^*,\phi]}{\delta\phi_{\alpha'_1}(\tau'_1)\cdots\delta\phi_{\alpha'_n}(\tau'_n)\delta\phi^*_{\alpha_n}(\tau_n)\cdots\delta\phi^*_{\alpha_1}(\tau_1)}\bigg|_{\phi^*=\phi=0}. \tag{1.407}$$

Inverting (1.407), one obtains

$$\Gamma[\phi^*,\phi] = \Gamma[0,0] + \sum_{n=1}^{\infty}\frac{1}{(n!)^2}\int_0^\beta d\tau_1\cdots d\tau'_1 \sum_{\alpha_1\cdots\alpha'_n}\Gamma^{(2n)}(\alpha_1\tau_1,\cdots,\alpha_n\tau_n;\alpha'_n\tau'_n,\cdots,\alpha'_1\tau'_1) \times \phi^*_{\alpha_1}(\tau_1)\cdots\phi^*_{\alpha_n}(\tau_n)\phi_{\alpha'_n}(\tau'_n)\cdots\phi_{\alpha'_1}(\tau'_1), \tag{1.408}$$

where $\Gamma[0,0] = -W[0,0] = \beta\Omega$.

Taking the functional derivatives of the first of Eqs. (1.399) with respect to $\phi_{\alpha'}(\tau')$, one obtains

$$\begin{aligned}\delta_{1,2} &= \frac{\delta\phi_1}{\delta\phi_2}\\ &= \sum_3\left(\frac{\delta J_3}{\delta\phi_2}\frac{\delta^2 W[J^*,J]}{\delta J_3\delta J_1^*} + \frac{\delta J_3^*}{\delta\phi_2}\frac{\delta^2 W[J^*,J]}{\delta J_3^*\delta J_1^*}\right)\\ &= \sum_3\left(\frac{\delta^2\Gamma[\phi^*,\phi]}{\delta\phi_2\delta\phi_3^*}\frac{\delta^2 W[J^*,J]}{\delta J_3\delta J_1^*} + \zeta\frac{\delta^2\Gamma[\phi^*,\phi]}{\delta\phi_2\delta\phi_3}\frac{\delta^2 W[J^*,J]}{\delta J_3^*\delta J_1^*}\right),\end{aligned} \tag{1.409}$$

where we use the short-hand notation $i \equiv (\alpha_i,\tau_i)$, $\delta_{i,j} = \delta_{\alpha_i,\alpha_j}\delta(\tau_i-\tau_j)$ and $\sum_i = \int_0^\beta d\tau_i\sum_{\alpha_i}$. By considering the functional derivatives of (1.399) with respect to both $\phi_{\alpha'}(\tau')$ and $\phi^*_{\alpha'}(\tau')$, one obtains four equations that can be cast in the form

$$\sum_3\begin{pmatrix}\dfrac{\delta^2\Gamma}{\delta\phi_2\delta\phi_3^*} & \zeta\dfrac{\delta^2\Gamma}{\delta\phi_2\delta\phi_3}\\ \zeta\dfrac{\delta^2\Gamma}{\delta\phi_2^*\delta\phi_3^*} & \dfrac{\delta^2\Gamma}{\delta\phi_2^*\delta\phi_3}\end{pmatrix}\begin{pmatrix}\dfrac{\delta^2 W}{\delta J_3\delta J_1^*} & \dfrac{\delta^2 W}{\delta J_3\delta J_1}\\ \dfrac{\delta^2 W}{\delta J_3^*\delta J_1^*} & \dfrac{\delta^2 W}{\delta J_3^*\delta J_1}\end{pmatrix} = \delta_{1,2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}. \tag{1.410}$$

[56] The reason for this terminology is explained below.

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In a normal system, the matrices in (1.410) become diagonal for $J^* = J = 0$ and one obtains

$$-\int_0^\beta d\tau_3 \sum_{\alpha_3} G_c^{(2)}(\alpha_1\tau_1, \alpha_3\tau_3)\Gamma^{(2)}(\alpha_3\tau_3, \alpha_2\tau_2) = \delta_{\alpha_1,\alpha_2}\delta(\tau_1 - \tau_2), \tag{1.411}$$

i.e.

$$\Gamma^{(2)} = -G_c^{(2)-1} = -G^{(2)-1} \tag{1.412}$$

in a matrix sense (recall that $G_c^{(2)} = G^{(2)}$ when $\langle\psi_\alpha(\tau)\rangle = \langle\psi_\alpha^*(\tau)\rangle = 0$).

It is conventional to express $\Gamma^{(2)}$ in terms of the self-energy Σ defined by $\Gamma^{(2)} = -G_0^{-1} + \Sigma$ or $G^{-1} = G_0^{-1} - \Sigma$ (with $G \equiv G^{(2)}$). This equation can be rewritten as

$$G = G_0 + G_0 \Sigma G \tag{1.413}$$

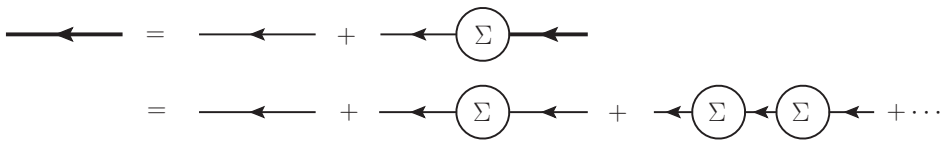
or, more explicitly,

$$\begin{aligned} G(\alpha\tau, \alpha'\tau') &= G_0(\alpha\tau, \alpha'\tau') \\ &\quad + \int_0^\beta d\tau_1 d\tau_2 \sum_{\alpha_1,\alpha_2} G_0(\alpha\tau, \alpha_1\tau_1)\Sigma(\alpha_1\tau_1, \alpha_2\tau_2)G(\alpha_2\tau_2, \alpha'\tau'). \end{aligned} \tag{1.414}$$

Equation (1.413) is known as Dyson's equation. For a translation-invariant system, we obtain the simple expression

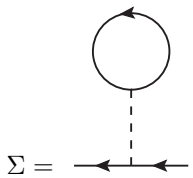
$$G(\mathbf{k}, i\omega_n) = \frac{1}{G_0^{-1}(\mathbf{k}, i\omega_n) - \Sigma(\mathbf{k}, i\omega_n)} = \frac{1}{i\omega_n - \xi_\mathbf{k} - \Sigma(\mathbf{k}, i\omega_n)} \tag{1.415}$$

(neglecting for simplicity internal degrees of freedom such as the spin of the particle), where $\Sigma(\mathbf{k}, i\omega_n)$ is the Fourier transform of $\Sigma(\mathbf{r}, \tau; \mathbf{r}', \tau') = \Sigma(\mathbf{r} - \mathbf{r}', \tau - \tau')$. The Dyson equation can be graphically represented as



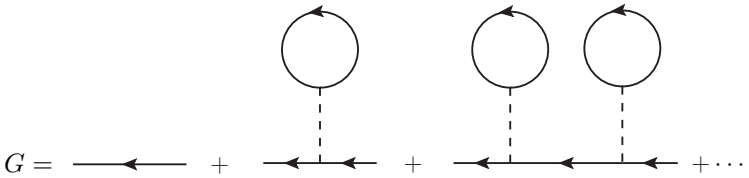
where the thick line stands for G and the thin one for G_0.

Let us introduce two definitions. i) A diagram is n-particle irreducible (nPI) if it cannot be separated into two or more disconnected pieces by cutting n internal lines. ii) An amputated diagram has no propagator attached to the external legs $(\alpha_i\tau_i)$; the latter connect directly to an interaction vertex. With these definitions, it is easy to see that the self-energy is given by the sum of 1PI amputated diagrams. One easily verifies that all diagrams for the Green function G are indeed obtained from the Dyson equation. For instance, the self-energy diagram



generates an infinite number of diagrams for the Green function,

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It should be understood that the external legs in the self-energy diagrams are not part of the self-energy even though they are often drawn for clarity.

The diagrammatic rules for the self-energy follow directly from those corresponding to the one-particle Green function. To calculate the nth order contribution to the self-energy:

1. Draw all distinct 1PI amputated connected (two-leg) diagrams with n interaction vertices.

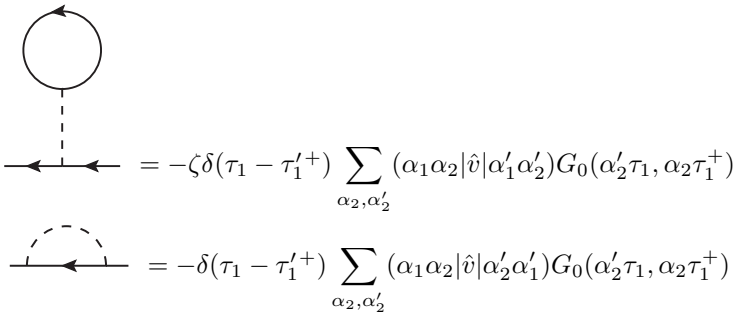
2. To each directed line, associate $G_0(\alpha\tau, \alpha', \tau'^+)$.

3. To each vertex, associate $(\gamma_1\gamma_2|\hat{v}|\gamma_1'\gamma_2')$.

4. Sum over all internal indices α and integrate all internal times $\tau \in [0, \beta]$.

5. Multiply the result by the factor $(-1)^n \zeta^{n_L}$ where n_L is the number of closed loops. (The combinatorial factor is unity since we use non-(anti)symmetrized vertices.)

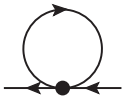
The two first-order diagrams contributing to the self-energy $\Sigma(\alpha_1\tau_1, \alpha_1'\tau_1')$ are



$$= -\zeta\delta(\tau_1 - \tau_1'^+) \sum_{\alpha_2,\alpha_2'} (\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2') G_0(\alpha_2'\tau_1, \alpha_2\tau_1^+)$$

$$= -\delta(\tau_1 - \tau_1'^+) \sum_{\alpha_2,\alpha_2'} (\alpha_1\alpha_2|\hat{v}|\alpha_2'\alpha_1') G_0(\alpha_2'\tau_1, \alpha_2\tau_1^+)$$

The first-order self-energy correction can also be represented by the diagram



The second-order diagrams are given by those of Fig. 1.3 (with amputated legs) except the last four that are not 1PI. They are also given by the last two diagrams (with amputated legs) of Fig. 1.6.

In perturbation theory, one always computes the self-energy rather than directly the one-particle Green function. Any finite order calculation of Σ leads (*via* the Dyson equation) to the summation of an infinite number of diagrams for the Green function. Moreover, a perturbative calculation of G to finite order is likely to violate basic properties of the Green function such as causality, etc. (see Sec. 3.5).

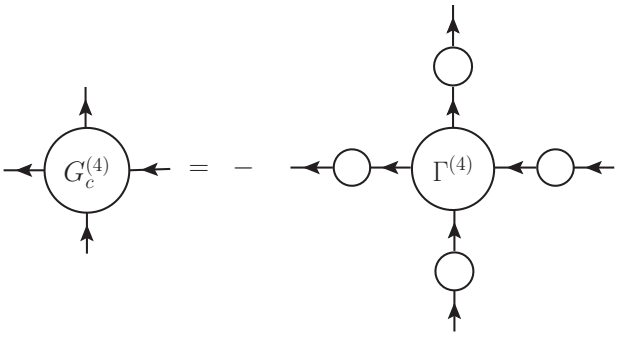
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1.6.2.3 Higher-order vertices

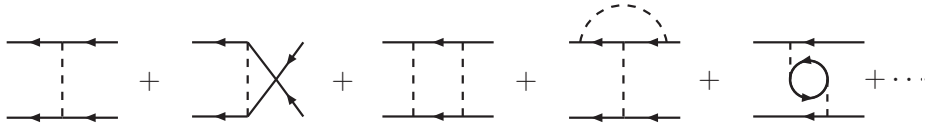
By taking functional derivatives of (1.410) with respect to J^* and J, one can relate the higher-order connected Green functions $G_c^{(n)}$ ($n \geq 4$) to the 1PI vertices. After a little algebra, one finds

$$G_c^{(4)}(\alpha_1\tau_1, \alpha_2\tau_2; \alpha_2'\tau_2', \alpha_1'\tau_1') = -\int_0^\beta du_1 du_2 du_2' du_1' \sum_{\gamma_1,\gamma_2,\gamma_2',\gamma_1'} G(\alpha_1\tau_1, \gamma_1 u_1) \times G(\alpha_2\tau_2, \gamma_2 u_2)\Gamma^{(4)}(\gamma_1 u_1, \gamma_2 u_2; \gamma_2' u_2', \gamma_1' u_1')G(\gamma_2' u_2', \alpha_2'\tau_2')G(\gamma_1' u_1', \alpha_1'\tau_1'). \quad (1.416)$$

This expression can be graphically represented as



(writing explicitly the sign of the diagrams), where the empty circle stands for the one-particle Green function G, and identifies $\Gamma^{(4)}(\alpha_1\tau_1, \alpha_2\tau_2; \alpha_2'\tau_2', \alpha_1'\tau_1')$ as the four-leg (or two-particle) 1PI vertex. $\Gamma^{(4)}$ can be seen as the effective interaction vertex between two particles. To lowest order, it is given by the bare vertex $\{\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2'\}$, and its perturbation expansion can be represented by diagrams,



or, equivalently,

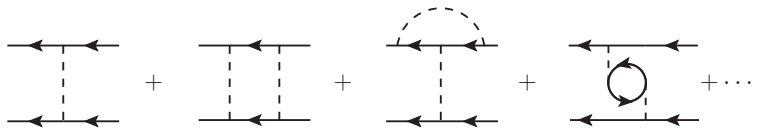


where the lines may contain any self-energy corrections. The diagrammatic rules follow from those corresponding to the two-particle Green function $G_c^{(4)}$. Since $\Gamma^{(4)}$ shares the same symmetry properties as $G_c^{(4)}$ [Eq. (1.416)], it is (anti)symmetric under the exchange of the two incoming or outgoing particles,

$$\begin{aligned}\Gamma^{(4)}(\alpha_1\tau_1, \alpha_2\tau_2; \alpha_2'\tau_2', \alpha_1'\tau_1') &= \zeta\Gamma^{(4)}(\alpha_2\tau_2, \alpha_1\tau_1; \alpha_2'\tau_2', \alpha_1'\tau_1') \\ &= \zeta\Gamma^{(4)}(\alpha_1\tau_1, \alpha_2\tau_2; \alpha_1'\tau_1', \alpha_2'\tau_2'). \end{aligned} \quad (1.417)$$

One can also write down a perturbation expansion for the (non-(anti)symmetrized) vertex, equal to $(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')$ to lowest order,

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1.6.2.4 General case

It is possible to define fully (anti)symmetric 1PI vertices starting from the partition function

$$Z[J] = \int \mathcal{D}[\psi] \exp\Big\{-S[\psi] + \int_0^\beta d\tau \sum_{\tilde{\alpha}} J_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}}(\tau)\Big\} \tag{1.418}$$

and the effective action $\Gamma[\phi]$ defined as the Legendre transform

$$\begin{aligned} \phi_{\tilde{\alpha}}[\tau; J] &= \langle \psi_{\tilde{\alpha}}(\tau)\rangle = \frac{\delta W[J]}{\delta J_{\tilde{\alpha}}(\tau)}, \\ \Gamma[\phi] &= -W[J] + \int_0^\beta d\tau \sum_{\tilde{\alpha}} J_{\tilde{\alpha}}(\tau)\phi_{\tilde{\alpha}}(\tau), \end{aligned} \tag{1.419}$$

where $W[J] = \ln Z[J]$. $\Gamma[\phi]$ satisfies

$$\frac{\delta\Gamma[\phi]}{\delta\phi_{\tilde{\alpha}}(\tau)} = \zeta J_{\tilde{\alpha}}(\tau). \tag{1.420}$$

In the absence of external sources ($J = 0$), the effective action is stationary. If the stationary value $\bar{\phi}$ is nonzero, the U(1) symmetry is spontaneously broken.[57]

The 1PI vertices in the state $\bar{\phi}$ are defined by[58]

$$\Gamma^{(n)}(\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_n\tau_n) = \frac{\delta^n\Gamma[\phi]}{\delta\phi_{\tilde{\alpha}_n}(\tau_n)\cdots\delta\phi_{\tilde{\alpha}_1}(\tau_1)}\bigg|_{\phi=\bar{\phi}} \tag{1.421}$$

or the inverse relation

$$\begin{aligned} \Gamma[\phi] = \Gamma[\bar{\phi}] + \sum_{n=1}^\infty \frac{1}{n!}\int_0^\beta d\tau_1\cdots d\tau_n \sum_{\tilde{\alpha}_1\cdots\tilde{\alpha}_n} \Gamma^{(n)}(\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_n\tau_n) \\ \times [\phi_{\tilde{\alpha}_1}(\tau_1) - \bar{\phi}_{\tilde{\alpha}_1}(\tau_1)]\cdots[\phi_{\tilde{\alpha}_n}(\tau_n) - \bar{\phi}_{\tilde{\alpha}_n}(\tau_n)]. \end{aligned} \tag{1.422}$$

They are (anti)symmetric under the exchange of two particles,

$$\begin{aligned} &\Gamma^{(n)}(\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_i\tau_i, \cdots, \tilde{\alpha}_j\tau_j, \cdots, \tilde{\alpha}_n\tau_n) \\ &\qquad = \zeta\Gamma^{(n)}(\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_j\tau_j, \cdots, \tilde{\alpha}_i\tau_i, \cdots, \tilde{\alpha}_n\tau_n). \end{aligned} \tag{1.423}$$

From the first of Eqs. (1.419), one easily deduces[59]

$$\frac{\delta^2\Gamma[\phi]}{\delta\phi\delta\phi} = \zeta\left(\frac{\delta^2 W[J]}{\delta J\delta J}\right)^{-1} \tag{1.424}$$

[57] See footnote 49 on page 71.

[58] The 1PI vertices can also be defined with respect to the state $\phi = 0$, i.e. by expanding the effective action $\Gamma[\phi]$ about $\phi = 0$. In fermion systems, this is the only possibility since $\bar{\phi} = 0$ in the physical state.

[59] The sign difference between (1.412) and (1.425) comes from the sign convention for the Green function $G^{(2)}(\tilde{\alpha}\tau, \tilde{\alpha}'\tau') = +\langle\psi_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}'}(\tau')\rangle$ (see Sec. 1.6.1).

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and, for $J = 0$ and $\phi = \bar{\phi}$,

$$\Gamma^{(2)} = G_c^{(2)-1}. \tag{1.425}$$

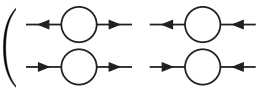
The self-energy is defined as $\Gamma^{(2)} = G_0^{-1} + \Sigma$,[60] i.e.

$$G_c^{-1} = G_0^{-1} + \Sigma, \tag{1.426}$$

and has a 2×2 matrix structure with respect to the charge index c,

$$\Sigma(\alpha\tau, \alpha'\tau') = \begin{pmatrix} \Sigma(\alpha, +, \tau; \alpha', +, \tau') & \Sigma(\alpha, +, \tau; \alpha', -, \tau') \\ \Sigma(\alpha, -, \tau; \alpha', +, \tau') & \Sigma(\alpha, -, \tau; \alpha', -, \tau') \end{pmatrix} \tag{1.427}$$

$$= \left(\quad \right).$$



Equation (1.426) is a generalization of the Dyson equation to systems with spontaneous U(1) symmetry breaking. It is known as the Dyson-Beliaev equation. The perturbative expansion of $\Sigma(\tilde{\alpha}\tau, \tilde{\alpha}'\tau')$ in boson systems will be discussed in Sec. 1.7.

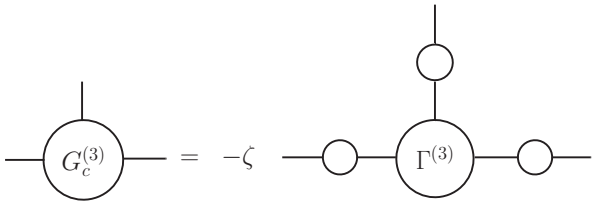
From (1.424) and using the relations[61]

$$\begin{aligned} \frac{\delta}{\delta J_{\tilde{\alpha}_{n+1}}(\tau_{n+1})} G_c^{(n)}[\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_n\tau_n; J] &= \zeta^n G_c^{(n+1)}[\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_{n+1}\tau_{n+1}; J], \\ \frac{\delta}{\delta J_{\tilde{\alpha}_{n+1}}(\tau_{n+1})} \Gamma^{(n)}[\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_n\tau_n; J] &= \int_0^\beta d\tau \sum_{\tilde{\alpha}} G_c^{(2)}[\tilde{\alpha}_{n+1}\tau_{n+1}, \tilde{\alpha}\tau; J] \\ &\quad \times \Gamma^{(n+1)}[\tilde{\alpha}_1\tau_1, \cdots, \tilde{\alpha}_n\tau_n, \tilde{\alpha}\tau; J], \end{aligned} \tag{1.428}$$

one can express higher-order Green functions $G_c^{(n)}$ in terms of $\Gamma^{(m\leq n)}$ and $G_c^{(m<n)}$. For instance, one finds

$$\begin{aligned} G_c^{(3)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2\tau_2, \tilde{\alpha}_3\tau_3) = &-\zeta \int_0^\beta du_1 du_2 du_3 \sum_{\tilde{\gamma}_1, \tilde{\gamma}_2, \tilde{\gamma}_3} G_c(\tilde{\alpha}_1\tau_1, \tilde{\gamma}_1 u_1) \\ &\times G_c(\tilde{\alpha}_2\tau_2, \tilde{\gamma}_2 u_2) G_c(\tilde{\alpha}_3\tau_3, \tilde{\gamma}_3 u_3) \Gamma^{(3)}(\tilde{\gamma}_1 u_1, \tilde{\gamma}_2 u_2, \tilde{\gamma}_3 u_3). \end{aligned} \tag{1.429}$$

This relation can be graphically represented as



(the empty circles stand for the connected one-particle Green function) and identifies $\Gamma^{(3)}$ as the three-leg 1PI vertex. To the next order, we obtain the equation

$$G_c^{(4)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2\tau_2, \tilde{\alpha}_3\tau_3, \tilde{\alpha}_4\tau_4)$$

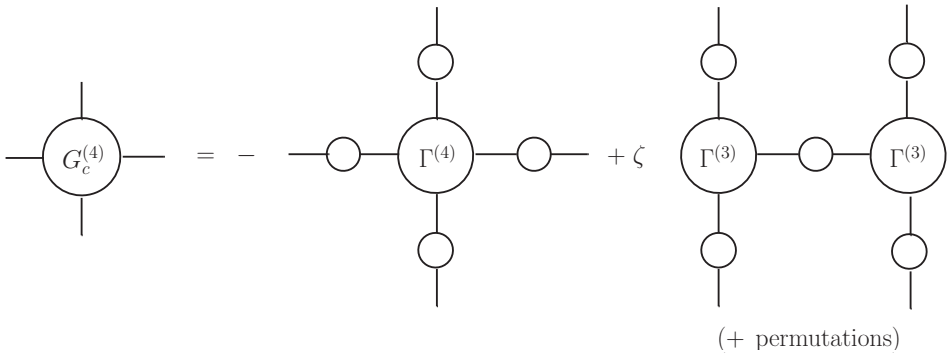
[60] Given the definition of the Green function $G^{(2)}$,[59] the self-energy Σ in (1.426) is defined with the same sign convention as the one introduced for normal systems.

[61] The notation in (1.428) emphasizes that the Green functions $G_c^{(n)}[J]$ and the vertices $\Gamma^{(n)}[J]$, computed for arbitrary external sources, are functional of J.

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$$
\begin{aligned}
= & -\int_0^\beta du_1 du_2 du_3 du_4 \sum_{\tilde{\gamma}_1 \cdots \tilde{\gamma}_4} G_c(\tilde{\alpha}_1\tau_1, \tilde{\gamma}_1 u_1) G_c(\tilde{\alpha}_2\tau_2, \tilde{\gamma}_2 u_2) \\
& \times G_c(\tilde{\alpha}_3\tau_3, \tilde{\gamma}_3 u_3) G_c(\tilde{\alpha}_4\tau_4, \tilde{\gamma}_4 u_4) \Gamma^{(4)}(\tilde{\gamma}_1 u_1, \tilde{\gamma}_2 u_2, \tilde{\gamma}_3 u_3, \tilde{\gamma}_4 u_4) \\
& -\int_0^\beta du_1 du_2 du_3 \sum_{\tilde{\gamma}_1, \tilde{\gamma}_2, \tilde{\gamma}_3} \Gamma^{(3)}(\tilde{\gamma}_1 u_1, \tilde{\gamma}_2 u_2, \tilde{\gamma}_3 u_3) \\
& \times \Big[G_c^{(3)}(\tilde{\alpha}_1\tau_1, \tilde{\gamma}_1 u_1, \tilde{\alpha}_4\tau_4) G_c(\tilde{\alpha}_2\tau_2, \tilde{\gamma}_2 u_2) G_c(\tilde{\alpha}_3\tau_3, \tilde{\gamma}_3 u_3) \\
& + G_c(\tilde{\alpha}_1\tau_1, \tilde{\gamma}_1 u_1) G_c^{(3)}(\tilde{\alpha}_2\tau_2, \tilde{\gamma}_2 u_2, \tilde{\alpha}_4\tau_4) G_c(\tilde{\alpha}_3\tau_3, \tilde{\gamma}_3 u_3) \\
& + G_c(\tilde{\alpha}_1\tau_1, \tilde{\gamma}_1 u_1) G_c(\tilde{\alpha}_2\tau_2, \tilde{\gamma}_2 u_2) G_c^{(3)}(\tilde{\alpha}_3\tau_3, \tilde{\gamma}_3 u_3, \tilde{\alpha}_4\tau_4) \Big]. \qquad (1.430)
\end{aligned}
$$

Combining (1.429) and (1.430), we finally obtain the following diagrammatic representation of $G_c^{(4)}$



The last diagram represents the contribution to $G_c^{(4)}$ which is one-particle reducible. It vanishes in a normal system where $\Gamma^{(3)} = 0$.

1.6.2.5 Equations of motion

The equations of motion of the Green functions imply relations between the 1PI vertices. Equation (1.395) can be rewritten as[62]

$$
\sum_{1'} G_0^{-1}(1,1')\phi_{1'} + \frac{1}{3!} \sum_{2',3',4'} v_{12'3'4'} \mathcal{G}^{(3)}(2',3',4') - \frac{\delta\Gamma}{\delta\phi_1} = 0, \qquad (1.431)
$$

where we use the shorthand notation $i \equiv (\tilde{\alpha}_i, \tau_i)$ and $\sum_i \equiv \int_0^\beta d\tau_i \sum_{\tilde{\alpha}_i}$, and

$$
\mathcal{G}^{(3)}(2',3',4') = \frac{\delta^3 \mathcal{G}[J]}{\delta J_{2'} \delta J_{3'} \delta J_{4'}}. \qquad (1.432)
$$

$\mathcal{G}[J] = Z[J]/Z[0]$ is the generating functional of the Green functions [Eq. (1.392)]. Using

$$
\begin{aligned}
\mathcal{G}^{(3)}(2',3',4') = & W^{(3)}(2',3',4') + \phi_{2'} W^{(2)}(3',4') + \zeta \phi_{3'} W^{(2)}(2',4') \\
& + \phi_{4'} W^{(2)}(2',3') + \phi_{2'}\phi_{3'}\phi_{4'} \qquad (1.433)
\end{aligned}
$$

[62] To alleviate the notations, we do not indicate explicitly that $\mathcal{G}^{(3)}[2',3',4';J]$ in (1.431), $W^{(3)}[2',3',4';J]$ in (1.433), etc., are actually functionals of J.

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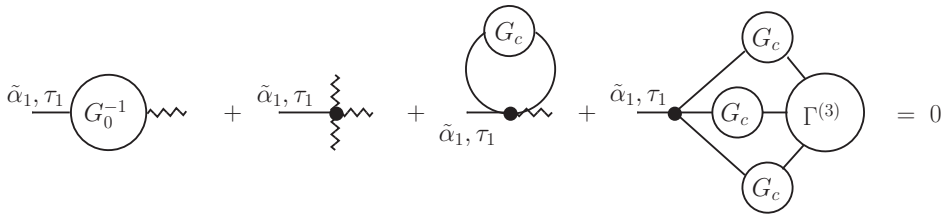


Figure 1.8: Diagrammatic representation of Eq. (1.435) with $\delta\Gamma/\delta\phi = \zeta J = 0$. The zigzag line stands for the order parameter $\bar{\phi}_{\tilde{\alpha}}(\tau)$.

(see Eq. (1.391)) and

$$W^{(3)}(2',3',4') = -\zeta \sum_{2'',3'',4''} W^{(2)}(2',2'')W^{(2)}(3',3'')W^{(2)}(4',4'')\Gamma^{(3)}(2'',3'',4''), \quad (1.434)$$

one finds

$$\begin{aligned}
&\sum_{1'} G_0^{-1}(1,1')\phi_{1'} - \frac{\delta\Gamma}{\delta\phi_1} \\
&+\frac{1}{3!}\sum_{2',3',4'} v_{1,2',3',4'}\big[-\zeta \sum_{2'',3'',4''} W^{(2)}(2',2'')W^{(2)}(3',3'')W^{(2)}(4',4'')\Gamma^{(3)}(2'',3'',4'') \\
&+\phi_{2'}W^{(2)}(3',4') + \zeta\phi_{3'}W^{(2)}(2',4') + \phi_{4'}W^{(2)}(2',3') + \phi_{2'}\phi_{3'}\phi_{4'}\big] = 0. \qquad (1.435)
\end{aligned}$$

If we consider this equation in the physical state, where $\delta\Gamma/\delta\phi = \zeta J = 0$ and $W^{(2)} = G_c$, we obtain an equation for the order parameter $\bar{\phi}_{\tilde{\alpha}}(\tau) = \langle\psi_{\tilde{\alpha}}(\tau)\rangle$ which is diagrammatically represented in Fig. 1.8.

Equation (1.435) is also the starting point for deriving the equations of motion satisfied by the 1PI vertices. If we take the functional derivative with respect to ϕ_2, we obtain

$$\begin{aligned}
\Sigma(1,2) = &-\frac{\zeta}{3!}\sum_{2'\cdots4''} v_{1,2',3',4'}G_c(2',2'')G_c(3',3'')G_c(4',4'')\Gamma^{(4)}(2'',3'',4'',2) \\
&+\frac{\zeta}{2}\sum_{2'\cdots4''} v_{1,2',3',4'}\big(G_c\Gamma^{(3)}(2)G_c\big)(2',2'')G_c(3',3'')G_c(4',4'')\Gamma^{(3)}(2'',3'',4'') \\
&-\frac{\zeta}{2}\sum_{2',3',4'} v_{1,2',3',4'}\phi_{2'}\big(G_c\Gamma^{(3)}(2)G_c\big)(3',4') \\
&+\frac{1}{2}\sum_{3',4'} v_{1,2,3',4'}[G_c(3',4') + \phi_{3'}\phi_{4'}]. \qquad (1.436)
\end{aligned}$$

For a normal system ($\phi = 0$ when $J = 0$), the self-energy reads

$$\begin{aligned}
\Sigma(1,2) = &\sum_{3',4'}\{1,3'|\hat{v}|4',2\}G(4',3') \\
&-\frac{1}{2}\sum_{2'\cdots4''}\{1,2'|\hat{v}|4',3'\}G(2'',2')G(3',3'')G(4',4'')\Gamma^{(4)}(3'',4'';2'',2) \qquad (1.437)
\end{aligned}$$

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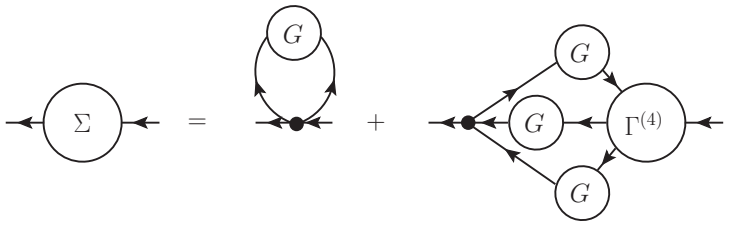


Figure 1.9: Diagrammatic representation of Eq. (1.437) in a normal system.

and is shown diagrammatically in Fig. 1.9. Equation (1.437) can be used to derive a relation between the self-energy and the expectation value of the interacting Hamiltonian. Multiplying $\Sigma(1,3)$ by $G(3,2)$ and summing over 3 gives[63]

$$\sum_3 \Sigma(1,3)G(3,2) = \sum_{2',3',4'} (1,2'|\hat{v}|4',3')G^{(4)}(3',4';2',2), \tag{1.438}$$

where $G^{(4)}$ is the two-particle Green function. Taking $2 = 1^+$ (i.e. $\alpha_2 = \alpha_1$ and $\tau_2 = \tau_1^+ = \tau_1 + 0^+$) and summing over 1, we obtain[64]

$$\begin{aligned} &\int_0^\beta d\tau_1 d\tau_3 \sum_{\alpha_1,\alpha_3} \Sigma(\alpha_1\tau_1,\alpha_3\tau_3)G(\alpha_3\tau_3,\alpha_1\tau_1^+) \\ &= \int_0^\beta d\tau_1 \sum_{\alpha_1,\alpha_2',\alpha_3',\alpha_4'} (\alpha_1\alpha_2'|\hat{v}|\alpha_4'\alpha_3')G^{(4)}(\alpha_3'\tau_1^-,\alpha_4'\tau_1^-;\alpha_2'\tau_1,\alpha_1\tau_1^+), \end{aligned} \tag{1.439}$$

i.e.

$$\frac{1}{\beta}\sum_{\alpha,\alpha',\omega_n} \Sigma(\alpha,\alpha';i\omega_n)G(\alpha',\alpha;i\omega_n)e^{i\omega_n 0^+} = -2\langle \hat{H}_{\rm int}\rangle. \tag{1.440}$$

Equations (1.438) and (1.440) relate the single-particle properties, *via* the self-energy, to a two-particle quantity. Note that the propagator in (1.437-1.440) is defined by $G(1,2) = \langle \psi_{\alpha_1}(\tau_1)\psi^*_{\alpha_2}(\tau_2)\rangle$ (with a + sign).

1.6.3 Luttinger-Ward functional and 2PI vertices

In this section, we introduce the Luttinger-Ward functional and the 2PI vertices. The Luttinger-Ward functional is used to set up "conserving" approximations, i.e. approximations that satisfy conservation laws, as well as non-perturbative approaches (Chap. 9).

1.6.3.1 Normal systems

We consider the partition function in the presence of a bilinear source which couples to a particle-hole pair operator,

$$Z[J] = \int \mathcal{D}[\psi^*,\psi] \exp\Big\{-S[\psi^*,\psi] + \int_0^\beta d\tau d\tau' \sum_{\alpha,\alpha'} J_{\alpha\alpha'}(\tau,\tau')\psi^*_\alpha(\tau)\psi_{\alpha'}(\tau')\Big\}. \tag{1.441}$$

[63] Equation (1.438) is derived by using the relation between $G^{(4)}$ and $\Gamma^{(4)}$.
[64] We use $(1,2'|\hat{v}|4',3') = \delta(\tau_1-\tau_2')\delta(\tau_3'-\tau_4')\delta(\tau_1^- - \tau_3')(\alpha_1\alpha_2'|\hat{v}|\alpha_4'\alpha_3')$.

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By taking the functional derivative with respect to the source, we obtain the one-particle Green function

$$G[\alpha\tau,\alpha'\tau';J]=-\langle\psi_\alpha(\tau)\psi^*_{\alpha'}(\tau')\rangle=-\zeta\frac{\delta\ln Z[J]}{\delta J_{\alpha'\alpha}(\tau',\tau)}. \tag{1.442}$$

The Legendre transform of $-\ln Z[J]$, the 2PI effective action, is a functional of the propagator G,

$$\Gamma[G]=-\ln Z[J]-\zeta\int_0^\beta d\tau d\tau'\sum_{\alpha,\alpha'}J_{\alpha\alpha'}(\tau,\tau')G(\alpha'\tau',\alpha\tau) \tag{1.443}$$

(with $J\equiv J[G]$ a functional of G obtained from (1.442), assuming that $G[J]$ is invertible), which satisfies

$$\frac{\delta\Gamma[G]}{\delta G(\alpha\tau,\alpha'\tau')}=-\zeta J_{\alpha'\alpha}(\tau',\tau). \tag{1.444}$$

The actual Green function $\bar{G}$, obtained for vanishing external sources, corresponds to the stationary value of the functional $\Gamma[G]$.

For a non-interacting system, $\Gamma[G]$ can be calculated exactly since the action is Gaussian. The partition function reads[65]

$$Z[J]=\det\left(-G_0^{-1}-J\right)^{-\zeta}=e^{-\zeta\mathrm{Tr}\ln(-G_0^{-1}-J)}, \tag{1.445}$$

and (1.442) then gives $G[J]=(G_0^{-1}+J)^{-1}$. We deduce

$$\Gamma[G]=-\zeta\mathrm{Tr}\ln(-G)+\zeta\mathrm{Tr}(G_0^{-1}G-1). \tag{1.446}$$

For an interacting system, we write the functional Γ as

$$\Gamma[G]=-\zeta\mathrm{Tr}\ln(-G)+\zeta\mathrm{Tr}(G_0^{-1}G-1)+\Phi[G], \tag{1.447}$$

where $\Phi[G]$ is referred to as the Luttinger-Ward functional. $\Gamma[G]$ and $\Phi[G]$ are well defined only if $G[J]$ is invertible, i.e. if there is a one-to-one correspondence between J and G. This is not necessarily the case and the Luttinger-Ward functional can be multivalued: See the discussion in Sec. 9.1.4. In the following $\Phi[G]$ should be understood as the physical branch of the Luttinger-Ward functional.

From Eq. (1.444) we deduce

$$-\zeta G^{-1}(\alpha'\tau',\alpha\tau)+\zeta G_0^{-1}(\alpha'\tau',\alpha\tau)+\frac{\delta\Phi[G]}{\delta G(\alpha\tau,\alpha'\tau')}=-\zeta J_{\alpha'\alpha}(\tau',\tau), \tag{1.448}$$

which is nothing but the Dyson equation $G^{-1}=G_0^{-1}-\Sigma$ [Eq. (1.413)]. For $J=0$, the self-energy is given by the first-order functional derivative of the Luttinger-Ward functional,

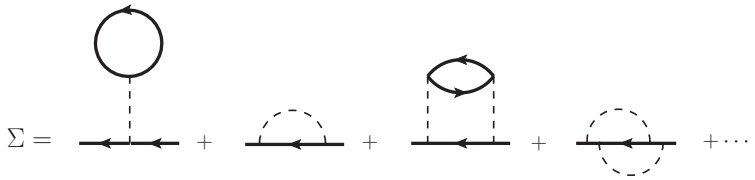
$$\Sigma(\alpha\tau,\alpha'\tau')=-\zeta\frac{\delta\Phi[G]}{\delta G(\alpha'\tau',\alpha\tau)}\bigg|_{G=\bar{G}}, \tag{1.449}$$

where $\bar{G}=(G_0^{-1}-\Sigma)^{-1}$ is the solution of the equation $\delta\Gamma/\delta G=0$. Equation (1.449) can be used to find a diagrammatic interpretation of $\Phi[G]$. The perturbation expansion of Σ with respect to G_0 (Sec. 1.6.2) can be rewritten as an expansion with respect to the full

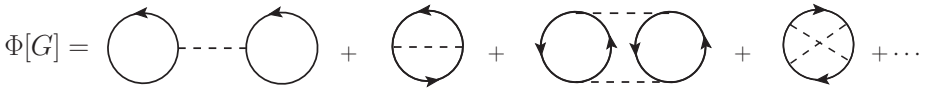
[65] We use Eq. (1.E.41).

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propagator G (represented here by a thick solid line),[66]



One can verify that this diagrammatic resummation makes sense by expanding G to recover the Feynman diagrams in terms of G_0 with proper factors and signs. Σ being defined as the functional derivative of the Luttinger-Ward functional $\Phi[G]$ [Eq. (1.449)], we deduce that $\Phi[G]$ is given by the sum of the 2PI Feynman diagrams defined as the graphs that cannot be separated into two disconnected pieces by cutting one or two lines:[67]



Since the functional derivation with respect to G amounts to removing a propagator G from the diagrams contributing to the Luttinger-Ward functional, the diagrammatic definition of $\Phi[G]$ reproduces the expansion of Σ in terms of G. Note that a two-particle reducible diagram for $\Phi[G]$ would give a contribution to the self-energy which is not 1PI. The diagrammatic rules for $\Phi[G]$ are the same as those for $-\ln Z$. One can check that they are consistent with the diagrammatic rules for the self-energy. For instance, to first-order $\Phi[G]$ is given by

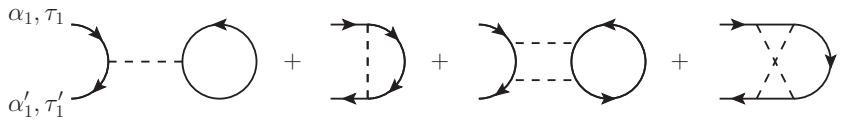
$$\Phi[G] = \frac{1}{2}\sum_{\alpha_1\cdots\alpha_2'}(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')\int_0^\beta d\tau\, G(\alpha_1'\tau,\alpha_1\tau^+)G(\alpha_2'\tau,\alpha_2\tau^+) + \frac{\zeta}{2}\sum_{\alpha_1\cdots\alpha_2'}(\alpha_1\alpha_2|\hat{v}|\alpha_1'\alpha_2')\int_0^\beta d\tau\, G(\alpha_1'\tau,\alpha_2\tau^+)G(\alpha_2'\tau,\alpha_1\tau^+), \tag{1.450}$$

and the first-order functional derivative $\Phi^{(1)}$ reproduces the lowest-order contribution to the self-energy with G_0 replaced by G (see page 80).

The 2PI vertices $\Phi^{(n)}$ are defined by

$$\Phi^{(n)}(\alpha_1\tau_1,\alpha_1'\tau_1';\cdots;\alpha_n\tau_n,\alpha_n'\tau_n') = \left.\frac{(-\zeta)^n\delta^n\Phi[G]}{\delta G(\alpha_1\tau_1,\alpha_1'\tau_1')\cdots\delta G(\alpha_n\tau_n,\alpha_n'\tau_n')}\right|_{G=\bar{G}}. \tag{1.451}$$

$\Phi^{(n)}$ is a vertex with n external "bosonic" legs $(\alpha_i\tau_i,\alpha_i'\tau_i')$. For example, to second-order $\Phi[G]$ is defined by the four diagrams of the preceding figure, so that $\Phi^{(1)}$ is given by the (amputated) diagrams



where the external legs $(\alpha_i\tau_i)$ and $(\alpha_i'\tau_i')$ are drawn nearby so that they can be seen as a

[66] We assume that the particles interact *via* a two-body interaction $\hat{v}$.

[67] The 2PI diagrams are also called skeleton diagrams or bold-line diagrams as they do not contain self-energy insertions.

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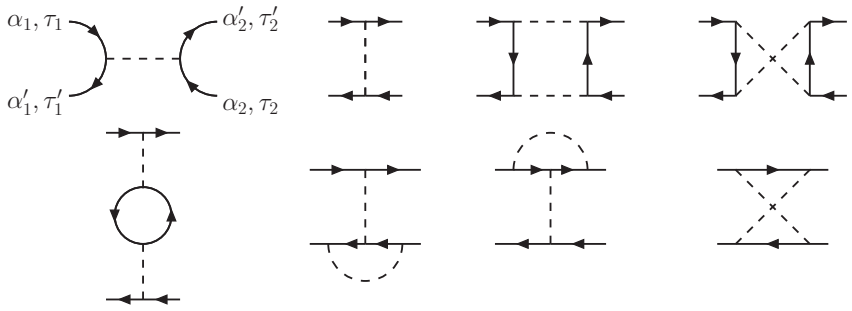


Figure 1.10: Diagrams contributing to the 2PI vertex $\Phi^{(2)}$ to second order in the interaction.

single bosonic (particle-hole) leg $(\alpha_i\tau_i, \alpha_i'\tau_i')$. The diagrams contributing to $\Phi^{(2)}$ to $\mathcal{O}(v^2)$ are shown in Fig. 1.10. It is clear that $\Phi^{(1)}$, $\Phi^{(2)}$ and more generally all vertices obtained from $\Phi[G]$ are 2PI: They cannot be separated into two disconnected pieces by cutting only two internal lines if one considers each external bosonic leg as a single piece. Since the leg $(\alpha_i\tau_i, \alpha_i'\tau_i')$ corresponds to a particle-hole pair, the vertices $\Phi^{(n)}$ are said to be irreducible in the particle-hole channel. This, of course, follows from the choice of $J_{\alpha\alpha'}(\tau, \tau')$ as an external source that couples to a particle-hole pair. We shall see in the next section that it is also possible to define irreducible vertices in the particle-particle channel.

In the 2PI formalism, the Green functions are naturally defined by

$$W^{(n)}(\alpha_1\tau_1, \alpha_1'\tau_1'; \cdots; \alpha_n\tau_n, \alpha_n'\tau_n') = \frac{(-\zeta)^n \delta^n \ln Z[J]}{\delta J_{\alpha_1'\alpha_1}(\tau_1', \tau_1) \cdots \delta J_{\alpha_n'\alpha_n}(\tau_n', \tau_n)}\bigg|_{J=0}. \tag{1.452}$$

One easily verifies that

$$\begin{aligned} W^{(1)}(\alpha_1\tau_1, \alpha_1'\tau_1') &= G^{(2)}(\alpha_1\tau_1, \alpha_1'\tau_1') \equiv \bar{G}(\alpha_1\tau_1, \alpha_1'\tau_1'), \\ W^{(2)}(\alpha_1\tau_1, \alpha_1'\tau_1'; \alpha_2\tau_2, \alpha_2'\tau_2') &= G^{(4)}(\alpha_1\tau_1, \alpha_2\tau_2; \alpha_2'\tau_2', \alpha_1'\tau_1') \\ &\quad - \bar{G}(\alpha_1\tau_1, \alpha_1'\tau_1')\bar{G}(\alpha_2\tau_2, \alpha_2'\tau_2'). \end{aligned} \tag{1.453}$$

$W^{(2)}$ is a "connected" Green function with respect to the external bosonic legs $(\alpha_1\tau_1, \alpha_1'\tau_1')$ and $(\alpha_2\tau_2, \alpha_2'\tau_2')$. It plays a role similar to that of the connected Green function $G_c^{(4)}$ introduced in the 1PI formalism (Sec. 1.6.2).[68] By taking a functional derivative with respect to J in (1.444), one obtains the relation

$$\frac{\delta^2 W[J]}{\delta J \delta J} = \left(\frac{\delta\Gamma[G]}{\delta G \delta G}\right)^{-1}, \tag{1.454}$$

where the inverse should be understood in a bosonic matrix sense, the multiplication being defined by

$$(AB)_{\gamma_1\gamma_2} = \sum_{\gamma_3} A_{\gamma_1\gamma_3} B_{\gamma_3\gamma_2}, \tag{1.455}$$

[68] Note that for a non-interacting system, one has $W^{(2)}(\alpha_1\tau_1, \alpha_1'\tau_1'; \alpha_2\tau_2, \alpha_2'\tau_2') = \zeta G_0(\alpha_1\tau_1, \alpha_2'\tau_2') \times G_0(\alpha_2\tau_2, \alpha_1'\tau_1')$ while $G_c^{(4)}(\alpha_1\tau_1, \alpha_1'\tau_1'; \alpha_2\tau_2, \alpha_2'\tau_2') = 0$.

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with the short-hand notation

$$\gamma = (\alpha\tau, \alpha'\tau'), \qquad \sum_{\gamma} = \int_0^\beta d\tau d\tau' \sum_{\alpha,\alpha'}. \tag{1.456}$$

Using

$$\left.\frac{\delta^2\Gamma[G]}{\delta G_{\gamma_1}\delta G_{\gamma_2}}\right|_{G=\bar{G}} = \Pi^{-1}_{\gamma_1\gamma_2} + \Phi^{(2)}_{\gamma_1\gamma_2} \tag{1.457}$$

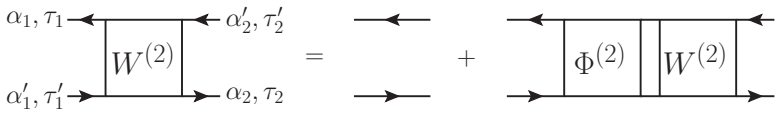
(which follows from (1.447)), where

$$\Pi_{\gamma_1\gamma_2} = \zeta\bar{G}(\alpha_1\tau_1, \alpha_2'\tau_2')\bar{G}(\alpha_2\tau_2, \alpha_1'\tau_1') \tag{1.458}$$

is a particle-hole pair propagator, we can rewrite (1.454) as

$$W^{(2)} = \left(\Pi^{-1} + \Phi^{(2)}\right)^{-1} = \Pi - \Pi\Phi^{(2)}W^{(2)}. \tag{1.459}$$

Equation (1.459) is known as the Bethe-Salpeter equation and relates the two-particle Green function $W^{(2)}$ to the 2PI vertex $\Phi^{(2)}$:



It is similar to the Dyson equation (1.413) with $\Phi^{(2)}$ and Π playing the role of the self-energy Σ and the non-interacting propagator G_0, respectively.

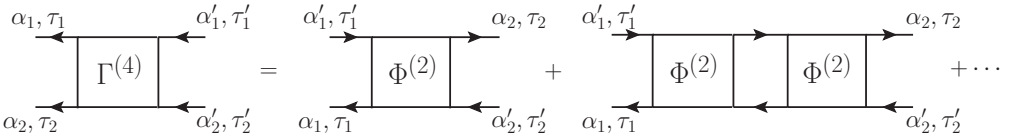
Using the relation between $W^{(2)}$ and $G_c^{(4)}$,

$$W^{(2)}_{\gamma_1\gamma_2} = G_c^{(4)}(\alpha_1\tau_1, \alpha_2\tau_2; \alpha_2'\tau_2', \alpha_1'\tau_1') + \zeta\bar{G}(\alpha_1\tau_1, \alpha_2'\tau_2')\bar{G}(\alpha_2\tau_2, \alpha_1'\tau_1') \tag{1.460}$$

(which follows from (1.453)) and the relation (1.416) between $G_c^{(4)}$ and the 1PI vertex $\Gamma^{(4)}$, we also obtain a relation between $\Gamma^{(4)}$ and $\Phi^{(2)}$,

$$\Gamma^{(4)}(\alpha_1\tau_1, \alpha_2\tau_2; \alpha_2'\tau_2', \alpha_1'\tau_1') = \left[\Phi^{(2)}\left(1 + \Pi\Phi^{(2)}\right)^{-1}\right](\alpha_1'\tau_1', \alpha_1\tau_1; \alpha_2'\tau_2', \alpha_2\tau_2), \tag{1.461}$$

which takes the diagrammatic form



1.6.3.2 General case

We can generalize the preceding analysis by considering the action

$$S[\psi] - \frac{1}{2}\int_0^\beta d\tau d\tau' \sum_{\tilde{\alpha},\tilde{\alpha}'} J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau')\psi_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}'}(\tau'), \tag{1.462}$$

where $J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau') = \zeta J_{\tilde{\alpha}'\tilde{\alpha}}(\tau',\tau)$ is an (anti)symmetric source. With our sign convention for the Green function (Sec. 1.6.1),

$$G[\tilde{\alpha}\tau, \tilde{\alpha}'\tau'; J] = \langle\psi_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}'}(\tau')\rangle = \frac{\delta \ln Z[J]}{\delta J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau')}. \tag{1.463}$$

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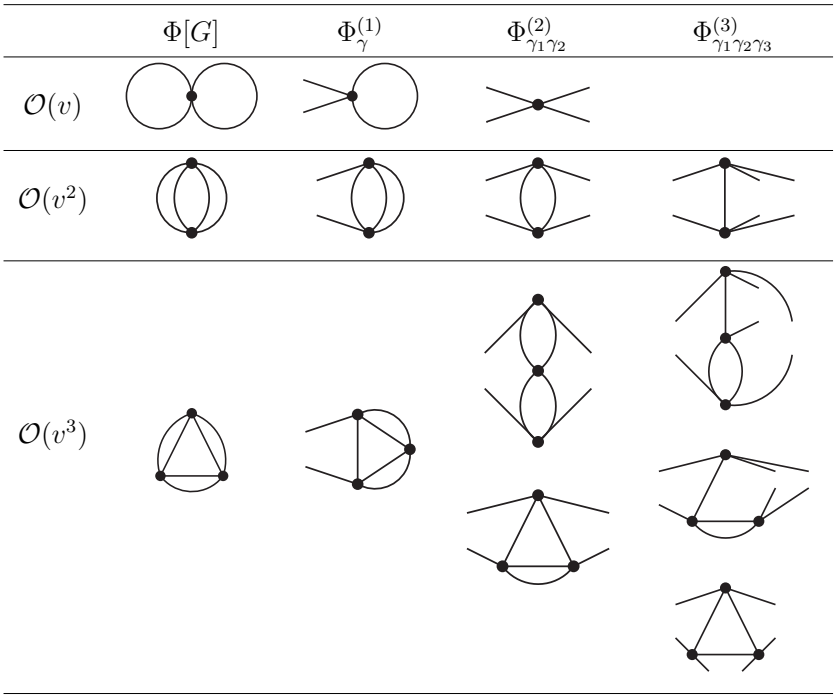


Figure 1.11: Diagrammatic representation of the Luttinger-Ward functional $\Phi[G]$ and the 2PI vertices $\Phi^{(1)}_{\gamma}$, $\Phi^{(2)}_{\gamma_1\gamma_2}$ and $\Phi^{(3)}_{\gamma_1\gamma_2\gamma_3}$ to third order in the interaction.

The 2PI effective action is now defined as

$$\begin{aligned}\Gamma[G] &= -\ln Z[J] + \frac{1}{2}\int_0^\beta d\tau d\tau' \sum_{\tilde{\alpha},\tilde{\alpha}'} J_{\tilde{\alpha},\tilde{\alpha}'}(\tau,\tau')G(\tilde{\alpha}\tau,\tilde{\alpha}'\tau') \\ &= -\frac{\zeta}{2}\mathrm{Tr}\ln(G) + \frac{\zeta}{2}\mathrm{Tr}(G_0^{-1}G - 1) + \Phi[G] \end{aligned} \tag{1.464}$$

and satisfies

$$\frac{\delta\Gamma[G]}{\delta G_\gamma} = -G_\gamma^{-1} + G_{0,\gamma}^{-1} + \frac{\delta\Phi[G]}{\delta G_\gamma} = J_\gamma \tag{1.465}$$

with the short-hand notation $\gamma = (\tilde{\alpha}\tau, \tilde{\alpha}'\tau')$. Equation (1.465) implies that for $J = 0$ the self-energy is defined by

$$\Sigma_\gamma = \left.\frac{\delta\Phi[G]}{\delta G_\gamma}\right|_{G=\bar{G}}, \tag{1.466}$$

where $\bar{G} = (G_0^{-1} + \Sigma)^{-1}$ is solution of $\delta\Gamma/\delta G = 0$, and the Luttinger-Ward functional given by the sum of 2PI diagrams (Fig. 1.11).

Because of the (anti)symmetry of the source J_γ and the Green function G_γ in the exchange $(\tilde{\alpha}\tau) \leftrightarrow (\tilde{\alpha}'\tau')$ (with $\gamma = (\tilde{\alpha}\tau, \tilde{\alpha}'\tau')$), we define a "bosonic" unit matrix by

$$I_{\gamma_1\gamma_2} = \delta_{\tilde{\alpha}_1,\tilde{\alpha}_2}\delta_{\tilde{\alpha}'_1,\tilde{\alpha}'_2}\delta(\tau_1-\tau_2)\delta(\tau'_1-\tau'_2) + \zeta\delta_{\tilde{\alpha}_1,\tilde{\alpha}'_2}\delta_{\tilde{\alpha}'_1,\tilde{\alpha}_2}\delta(\tau_1-\tau'_2)\delta(\tau'_1-\tau_2), \tag{1.467}$$

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so that $\delta J_{\gamma_1}/\delta J_{\gamma_2} = I_{\gamma_1\gamma_2}$. The variation of a functional, e.g. $W[J]$, involves an additional factor 1/2,

$$\delta W[J] = \frac{1}{2}\sum_{\gamma}\frac{\delta W[J]}{\delta J_\gamma}\delta J_\gamma \tag{1.468}$$

and the chain rule for derivation reads

$$\frac{\delta W[J]}{\delta G_\gamma} = \frac{1}{2}\sum_{\gamma'}\frac{\delta W[J]}{\delta J_{\gamma'}}\frac{\delta J_{\gamma'}}{\delta G_\gamma}. \tag{1.469}$$

The 2PI vertices are defined by

$$\Phi^{(n)}_{\gamma_1\cdots\gamma_n} = \left.\frac{\delta^n\Phi[G]}{\delta G_{\gamma_1}\cdots G_{\gamma_n}}\right|_{G=\bar{G}}. \tag{1.470}$$

$\Phi^{(n)}$ is a vertex with n external bosonic legs (Fig. 1.11) and is 2PI in the sense that it cannot be separated into two disconnected pieces by cutting only two internal lines. An external leg $\gamma = (\tilde{\alpha}\tau, \tilde{\alpha}'\tau')$ corresponds to a particle-hole pair if $c = -c'$ and to a particle-particle pair if $c = c'$. Since G_γ is (anti)symmetric in the exchange $(\tilde{\alpha}\tau) \leftrightarrow (\tilde{\alpha}'\tau')$, $\Phi^{(n)}$ satisfies

$$\begin{aligned}\Phi^{(n)}_{\gamma_1\cdots\gamma_i\cdots\gamma_j\cdots\gamma_n} &= \Phi^{(n)}_{\gamma_1\cdots\gamma_j\cdots\gamma_i\cdots\gamma_n},\\ \Phi^{(n)}_{\gamma_1\cdots\gamma_i\cdots\gamma_n} &= \zeta\Phi^{(n)}_{\gamma_1\cdots\bar{\gamma}_i\cdots\gamma_n},\end{aligned} \tag{1.471}$$

where $\bar{\gamma}_i = (\tilde{\alpha}_i'\tau_i', \tilde{\alpha}_i\tau_i)$.

The Green functions are defined by

$$W^{(n)}_{\gamma_1\cdots\gamma_n} = \left.\frac{\delta^n \ln Z[J]}{\delta J_{\gamma_1}\cdots\delta J_{\gamma_n}}\right|_{J=0}. \tag{1.472}$$

To lowest-order they read

$$\begin{aligned}W^{(1)}_\gamma &= \bar{G}_\gamma,\\ W^{(2)}_{\gamma_1\gamma_2} &= G^{(4)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_1'\tau_1', \tilde{\alpha}_2\tau_2, \tilde{\alpha}_2'\tau_2') - \bar{G}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_1'\tau_1')\bar{G}(\tilde{\alpha}_2\tau_2, \tilde{\alpha}_2'\tau_2').\end{aligned} \tag{1.473}$$

$W^{(2)}_{\gamma_1\gamma_2}$ is given by the sum of diagrams that are fully connected to the external bosonic legs γ_1 and γ_2. To relate $W^{(2)}$ to $\Phi^{(2)}$, we start from

$$\frac{\delta W}{\delta J\delta J} = \left(\frac{\delta^2\Gamma[G]}{\delta G\delta G}\right)^{-1}, \tag{1.474}$$

where the inverse should be understood in a bosonic matrix sense with the multiplication defined by

$$(AB)_{\gamma_1\gamma_2} = \frac{1}{2}\sum_{\gamma_3} A_{\gamma_1\gamma_3}B_{\gamma_3\gamma_2} \tag{1.475}$$

and the identity matrix by (1.467). These definitions are consistent with the (anti)symmetry properties of the Green functions and the vertices under the exchange $(\tilde{\alpha}_i\tau_i, \tilde{\alpha}_i'\tau_i') \leftrightarrow (\tilde{\alpha}_i'\tau_i', \tilde{\alpha}_i\tau_i)$ of the two particles in the bosonic external leg γ_i. Defining the particle-particle pair propagator

$$\Pi_{\gamma_1\gamma_2} = \bar{G}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2'\tau_2')\bar{G}(\tilde{\alpha}_1'\tau_1', \tilde{\alpha}_2\tau_2) + \zeta\bar{G}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}_2\tau_2)\bar{G}(\tilde{\alpha}_1'\tau_1', \tilde{\alpha}_2'\tau_2'), \tag{1.476}$$

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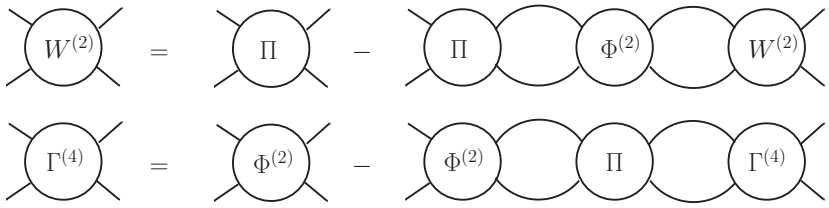


Figure 1.12: Diagrammatic representation of the Bethe-Salpeter equations relating $W^{(2)}$ to $\Phi^{(2)}$ and the 1PI vertex $\Gamma^{(4)}$ to $\Phi^{(2)}$ [Eqs. (1.477,1.478)].

we obtain the Bethe-Salpeter equation relating $W^{(2)}$ to $\Phi^{(2)}$ (Fig. 1.12),

$$W^{(2)} = \left(\Pi^{-1} + \Phi^{(2)}\right)^{-1} = \Pi - \Pi\Phi^{(2)}W^{(2)}. \tag{1.477}$$

Similarly, we can derive a relation between $\Phi^{(2)}$ and the 1PI vertex $\Gamma^{(4)}$,

$$\Gamma^{(4)} = \left(\Phi^{(2)-1} + \Pi\right)^{-1} = \Phi^{(2)} - \Phi^{(2)}\Pi\Gamma^{(4)}, \tag{1.478}$$

with the definition $\Gamma^{(4)}_{\gamma_1\gamma_2} = \Gamma^{(4)}(\tilde{\alpha}_1\tau_1, \tilde{\alpha}'_1\tau'_1, \tilde{\alpha}_2\tau_2, \tilde{\alpha}'_2\tau'_2)$ (Fig. 1.12). In a normal system, $c_1 + c'_1 + c_2 + c'_2 = 0$, and the Bethe-Salpeter equation can be projected onto the particle-hole ($c_1 + c'_1 = c_2 + c'_2 = 0$) and particle-particle ($c_1 = c'_1 = -c_2 = -c'_2$) channels. The corresponding vertices, $\Phi^{(2)}_{\rm ph}$ and $\Phi^{(2)}_{\rm pp}$, are 2PI in the particle-hole and particle-particle channels, respectively.

When dealing with bosons, it appears natural to work with a functional $\Phi[G_c, \phi]$ of both the connected Green function G_c and the classical field $\phi = \langle\psi\rangle$. Let us consider the partition function with both linear and bilinear external sources,

$$Z[J,h] = \int \mathcal{D}[\psi] \exp\Big\{-S[\psi] + \frac{1}{2}\int_0^\beta d\tau d\tau' \sum_{\tilde{\alpha},\tilde{\alpha}'} \psi_{\tilde{\alpha}}(\tau) J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau')\psi_{\tilde{\alpha}'}(\tau') + \int_0^\beta d\tau \sum_{\tilde{\alpha}} h_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}}(\tau)\Big\}, \tag{1.479}$$

where $J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau') = J_{\tilde{\alpha}'\tilde{\alpha}}(\tau',\tau)$ is symmetric. We have

$$\begin{aligned} \phi_{\tilde{\alpha}}[\tau; J, h] &= \langle\psi_{\tilde{\alpha}}(\tau)\rangle = \frac{\delta \ln Z[J,h]}{\delta h_{\tilde{\alpha}}(\tau)}, \\ G_c[\tilde{\alpha}\tau, \tilde{\alpha}'\tau'; J, h] &= \langle\psi_{\tilde{\alpha}}(\tau)\psi_{\tilde{\alpha}'}(\tau')\rangle - \langle\psi_{\tilde{\alpha}}(\tau)\rangle\langle\psi_{\tilde{\alpha}'}(\tau')\rangle \\ &= \frac{\delta \ln Z[J,h]}{\delta J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau')} - \phi_{\tilde{\alpha}}(\tau)\phi_{\tilde{\alpha}'}(\tau'). \end{aligned} \tag{1.480}$$

The effective action is defined as a double Legendre transform,

$$\Gamma[G_c,\phi] = -\ln Z[J,h] + \int_0^\beta d\tau \sum_{\tilde{\alpha}} h_{\tilde{\alpha}}(\tau)\phi_{\tilde{\alpha}}(\tau) + \frac{1}{2}\int_0^\beta d\tau d\tau' \sum_{\tilde{\alpha},\tilde{\alpha}'} J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau')\left[G_c(\tilde{\alpha}\tau,\tilde{\alpha}'\tau') + \phi_{\tilde{\alpha}}(\tau)\phi_{\tilde{\alpha}'}(\tau')\right], \tag{1.481}$$

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and satisfies

$$\begin{aligned}\frac{\delta\Gamma[G_c,\phi]}{\delta\phi_{\tilde{\alpha}}(\tau)} &= h_{\tilde{\alpha}}(\tau)+\int_0^\beta d\tau'\sum_{\tilde{\alpha}'}J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau')\phi_{\tilde{\alpha}'}(\tau'),\\ \frac{\delta\Gamma[G_c,\phi]}{\delta G_c(\tilde{\alpha}\tau,\tilde{\alpha}'\tau')} &= J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau').\end{aligned} \tag{1.482}$$

It can be written as

$$\begin{aligned}\Gamma[G_c,\phi] = &-\frac{1}{2}\mathrm{Tr}\ln(G_c)+\frac{1}{2}\int_0^\beta d\tau d\tau'\phi_{\tilde{\alpha}}(\tau)G_0^{-1}(\tilde{\alpha}\tau,\tilde{\alpha}'\tau')\phi_{\tilde{\alpha}'}(\tau')\\ &+\frac{1}{2}\mathrm{Tr}(G_0^{-1}G_c-1)+\Phi[G_c,\phi],\end{aligned} \tag{1.483}$$

where $\Phi[G_c,\phi]$ is the Luttinger-Ward functional. The equations of state (1.482) give

$$\begin{aligned}&\int_0^\beta d\tau'\sum_{\tilde{\alpha}'}G_0^{-1}(\tilde{\alpha}\tau,\tilde{\alpha}'\tau')\phi_{\tilde{\alpha}'}(\tau')+\frac{\delta\Phi[G_c,\phi]}{\delta\phi_{\tilde{\alpha}}(\tau)}\\ &=h_{\tilde{\alpha}}(\tau)+\int_0^\beta d\tau'\sum_{\tilde{\alpha}'}J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau')\phi_{\tilde{\alpha}'}(\tau')\end{aligned} \tag{1.484}$$

and

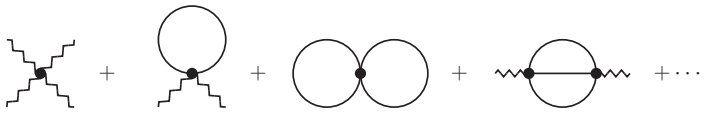
$$-G_c^{-1}(\tilde{\alpha}\tau,\tilde{\alpha}'\tau')+G_0^{-1}(\tilde{\alpha}\tau,\tilde{\alpha}'\tau')+\frac{\delta\Phi[G_c,\phi]}{\delta G_c(\tilde{\alpha}\tau,\tilde{\alpha}'\tau')}=J_{\tilde{\alpha}\tilde{\alpha}'}(\tau,\tau'). \tag{1.485}$$

The first equation is an equation for the order parameter $\phi_{\tilde{\alpha}}(\tau)$. The second one is Dyson's equation $G_c^{-1}=G_0^{-1}+\Sigma$ with the self-energy

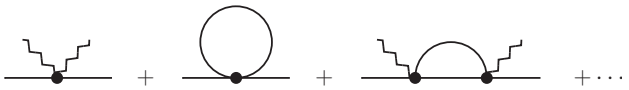
$$\Sigma(\tilde{\alpha}\tau,\tilde{\alpha}'\tau')=\frac{\delta\Phi[G_c,\phi]}{\delta G_c(\tilde{\alpha}\tau,\tilde{\alpha}'\tau')} \tag{1.486}$$

for vanishing external sources ($J=h=0$).

The Luttinger-Ward functional $\Phi[G_c,\phi]$ is given by the sum of the 2PI diagrams,



and the corresponding self-energy has the following diagrammatic expansion



The zigzag line in these diagrams stands for the order parameter $\phi_{\tilde{\alpha}}(\tau)$. The perturbation expansion (in the 1PI formalism) about a broken symmetry state ($\phi_{\tilde{\alpha}}(\tau)\neq 0$) in boson systems is discussed in detail in Sec. 1.7.2.

1.7 Saddle-point approximation and loop expansion

In Sec. 1.5.4, we have pointed out that the perturbation theory of interacting bosons about the non-interacting limit breaks down when the field takes a finite expectation value $\langle\psi_\alpha\rangle$ which breaks the U(1) symmetry. In this section, we show how the stationary-phase approximation allows us to reorganize the perturbation expansion as a "loop expansion" about a broken-symmetry state. We start by introducing the principle of the method on the simple example of a one-dimensional integral.

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1.7.1 Steepest-descent and stationary-phase approximations

We consider the integral

$$I(l) = \int_{-\infty}^{\infty} dt\, e^{-lf(t)}, \tag{1.487}$$

where $f(t)$ is a real function with an absolute minimum at t_0 and look for an expansion in $1/l$. For large l, the integral is dominated by the vicinity of t_0. Expanding $f(t)$ about t_0 and using $f_0' \equiv f'(t_0) = 0$, we obtain

$$\begin{aligned} I(l) &= e^{-lf_0} \int_{-\infty}^{\infty} dt\, \exp\left\{ -\frac{l}{2} f_0''(t-t_0)^2 - l \sum_{n=3}^{\infty} \frac{f_0^{(n)}}{n!} (t-t_0)^n \right\} \\ &= e^{-lf_0} \sqrt{\frac{2\pi}{f_0'' l}} \int_{-\infty}^{\infty} \frac{d\tau}{\sqrt{2\pi}} \exp\left\{ -\frac{\tau^2}{2} - \sum_{n=3}^{\infty} \frac{\tau^n}{n!} \frac{f_0^{(n)}}{(f_0'')^{n/2} l^{n/2-1}} \right\}, \end{aligned} \tag{1.488}$$

where $\tau = \sqrt{f_0'' l}(t - t_0)$. The integral can be calculated in powers of $1/l$ using Wick's theorem with the bare propagator

$$\overset{\sqcap}{\tau\tau} = \int_{-\infty}^{\infty} \frac{d\tau}{\sqrt{2\pi}} \tau^2 e^{-\tau^2/2} = 1 \tag{1.489}$$

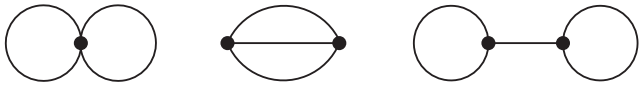
and the vertices

$$V_n = \frac{f_0^{(n)}}{(f_0'')^{n/2} l^{n/2-1}}. \tag{1.490}$$

By the linked cluster theorem (Sec. 1.5.2)

$$I(l) = e^{-lf_0} \sqrt{\frac{2\pi}{f_0'' l}} \exp\Big\{ \sum \text{connected diagrams} \Big\}. \tag{1.491}$$

Since $V_n = \mathcal{O}(l^{-n/2+1})$, the $\mathcal{O}(1/l)$ contribution is given by the three diagrams



,

with respective combinatorial factors $-1/8$, $1/12$ and $1/8$. We deduce

$$\ln I(l) = -lf_0 + \frac{1}{2} \ln\left(\frac{2\pi}{f_0'' l} \right) - \frac{1}{8l} \frac{f_0^{(4)}}{f_0''^2} + \frac{5}{24l} \frac{\left(f_0^{(3)}\right)^2}{f_0''^3} + \mathcal{O}\left(l^{-2}\right). \tag{1.492}$$

A similar analysis can be made when the function $f(t)$ is purely imaginary. In this case, for large l the integral is dominated by the vicinity of the point t_0 where the phase of the integrand is stationary ($f'(t_0) = 0$). The expansion (1.491) is then known as the stationary-phase method.

More generally, let us consider the integral

$$I(l) = \int_{\mathcal{C}} dz\, e^{-lf(z)} \tag{1.493}$$

along a path $\mathcal{C}$ in the complex plane. The complex function $f(z)$ is analytic along $\mathcal{C}$ and vanishes at the end points. Since we can deform the integration contour in the domain of analyticity of $f(z)$, we deform $\mathcal{C}$ into a new contour $\mathcal{C}'$ which satisfies the three conditions:

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1. The end points of $\mathcal{C}'$ are the same as those of $\mathcal{C}$.
2. $\Re[f(z)]$ has a minimum at the point z_0 of $\mathcal{C}'$.
3. $\Im[f(z)]$ is constant along $\mathcal{C}'$ near z_0.

The last two conditions imply that z_0 is a stationary point of $f(z)$ ($f'(z_0)=0$) and therefore a saddle point of $\Re[f(z)]$.[69] For large l, we expect the vicinity of z_0 to give the dominant contribution to $I(l)$ (condition (2)). The condition (3) ensures that $\Im[f(z)]$ does not give rise to rapid oscillations in the integrand as $l\to\infty$ which would suppress the contribution of the vicinity of z_0. To calculate $I(l)$, we then write $z=z_0+\rho e^{i\theta}$ and expand $f(z)$ about z_0,

$$f(z)=f(z_0)+\frac{1}{2}\rho^2|f_0''|e^{i(2\theta+\theta_0)}+\sum_{n=3}^{\infty}\frac{\rho^n e^{in\theta}}{n!}f_0^{(n)}, \tag{1.494}$$

where $f_0''\equiv f''(z_0)=|f_0''|e^{i\theta_0}$. Condition (3) implies $2\theta+\theta_0=0$ or π, and the existence of a minimum of $\Re[f(z)]$ at z_0 selects the solution $2\theta+\theta_0=0$.[70] It is easy to see that $\mathcal{C}'$ is then the path of steepest descents of $e^{-l\Re[f(z)]}$. The integral over ρ can be calculated as in the previous case using the linked cluster theorem.

1.7.2 Saddle-point approximation and loop expansion in field theory

We consider the partition function

$$Z[J^*,J]=\int\mathcal{D}[\psi^*,\psi]\,e^{-l\left(S[\psi^*,\psi]-\int_0^\beta d\tau\int d^dr[J^*(x)\psi(x)+\text{c.c.}]\right)} \tag{1.495}$$

($x=(\mathbf{r},\tau)$) describing a system of (spin-zero) bosons (ψ is a complex field). The factor l is introduced to organize the perturbation expansion and will be set to unity in the end. We assume that the bosons interact *via* a contact interaction,

$$S[\psi^*,\psi]=\int_0^\beta d\tau\int d^dr\left\{\psi^*(x)\left(\partial_\tau-\mu-\frac{\boldsymbol{\nabla}^2}{2m}\right)\psi(x)+\frac{g}{2}|\psi(x)|^4\right\} \tag{1.496}$$

($g>0$).

1.7.2.1 Partition function and thermodynamic potential

Within a saddle-point approximation, the action reads

$$S_c=\beta\mathcal{V}\left(-\mu|\psi_c|^2+\frac{g}{2}|\psi_c|^4\right) \tag{1.497}$$

where the "classical" field ψ_c (assumed to be time and space independent) is determined from the condition

$$\frac{\partial S_c}{\partial\psi_c}=\frac{\partial S_c}{\partial\psi_c^*}=0 \tag{1.498}$$

[69] Recall that an analytic complex function $f(z)$ satisfies the Cauchy-Riemann conditions $\partial_x\Re[f(z)]=\partial_y\Im[f(z)]$ and $\partial_x\Im[f(z)]=-\partial_y\Re[f(z)]$ (obtained by calculating $f'(z)=\lim_{dz\to0}[f(z+dz)-f(z)]/dz$ with $dz=dx$ and $dz=idy$). The Cauchy-Riemann conditions imply $\boldsymbol{\nabla}^2\Re[f(z)]=\boldsymbol{\nabla}^2\Im[f(z)]=0$, which prevents $\Re[f(z)]$ and $\Im[f(z)]$ to have an absolute minimum or maximum.

[70] The fact that the path of steepest descents of $e^{-l\Re[f(z)]}$ keeps $\Im[f(z)]$ constant near z_0 (stationary phase) is a consequence of the Cauchy-Riemann conditions.[69]

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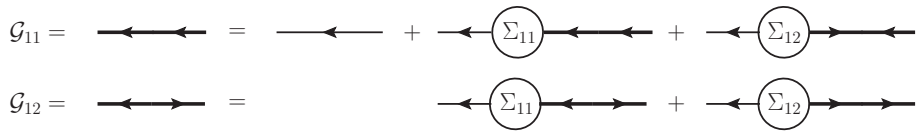


Figure 1.13: Diagrammatic representation of the Beliaev-Dyson equations (1.504). The thin line stands for G_0 and the thick one for $\mathcal{G}$.

in the absence of external sources ($J = J^* = 0$). When $\mu > 0$, the solution which minimizes the action is $|\psi_c| = \sqrt{\mu/g}$. As will be discussed in Chap. 7, this solution corresponds to a superfluid system and $|\psi_c|^2$ determines the density of the Bose-Einstein condensate. Expanding about the saddle-point solution, $\psi = \psi_c + \chi/\sqrt{l}$, we obtain

$$lS[\chi^*, \chi] = lS_c + S^{(2)} + \mathcal{O}\big(l^{-1/2}\big), \tag{1.499}$$

where the Gaussian part of the action

$$S^{(2)} = -\frac{1}{2}\int_0^\beta d\tau d\tau' \int d^d r d^d r' \, (\chi^*(x), \chi(x)) \, \mathcal{G}_c^{-1}(x, x') \begin{pmatrix} \chi(x') \\ \chi^*(x') \end{pmatrix} \tag{1.500}$$

is defined by the inverse "classical" propagator

$$\begin{aligned} \mathcal{G}_c^{-1}(x, x') = & -\delta(\tau - \tau')\delta(\mathbf{r} - \mathbf{r}') \\ & \times \begin{pmatrix} \partial_{\tau'} - \mu - \frac{\nabla_{\mathbf{r}'}^2}{2m} + 2g|\psi_c|^2 & g\psi_c^2 \\ g\psi_c^{*2} & -\partial_{\tau'} - \mu - \frac{\nabla_{\mathbf{r}'}^2}{2m} + 2g|\psi_c|^2 \end{pmatrix}. \end{aligned} \tag{1.501}$$

In the superfluid phase ($|\psi_c| > 0$), the bosonic (connected) propagator $\mathcal{G}$ becomes a 2×2 matrix,[71]

$$\mathcal{G}(x, x') = -\begin{pmatrix} \langle\psi(x)\psi^*(x')\rangle_c & \langle\psi(x)\psi(x')\rangle_c \\ \langle\psi^*(x)\psi^*(x')\rangle_c & \langle\psi^*(x)\psi(x')\rangle_c, \end{pmatrix}. \tag{1.502}$$

where $\langle\psi(x)\psi^*(x')\rangle_c = \langle\psi(x)\psi^*(x')\rangle - \langle\psi(x)\rangle\langle\psi^*(x')\rangle$, etc. It satisfies the Beliaev-Dyson equation $\mathcal{G}^{-1}(k) = \mathcal{G}_0^{-1}(k) - \Sigma(k)$ where

$$\mathcal{G}_0(k) = \begin{pmatrix} G_0(k) & 0 \\ 0 & G_0(-k) \end{pmatrix} \tag{1.503}$$

is the non-interacting propagator and $\Sigma(k)$ the self-energy. More explicitly, the Beliaev-Dyson equations read

$$\begin{aligned} \mathcal{G}_{11}(k) &= G_0(k) + G_0(k)\Sigma_{11}(k)\mathcal{G}_{11}(k) + G_0(k)\Sigma_{12}(k)\mathcal{G}_{21}(k), \\ \mathcal{G}_{12}(k) &= G_0(k)\Sigma_{11}(k)\mathcal{G}_{12}(k) + G_0(k)\Sigma_{12}(k)\mathcal{G}_{22}(k), \end{aligned} \tag{1.504}$$

with $\mathcal{G}_{11}(k) = \mathcal{G}_{22}(-k) = -\langle\psi(k)\psi^*(k)\rangle$ and $\mathcal{G}_{12}(k) = \mathcal{G}_{21}^*(k) = -\langle\psi(k)\psi(-k)\rangle$ (Fig. 1.13). The classical propagator (1.501) corresponds to the self-energy

$$\begin{aligned} \Sigma_{11}(k) &= \Sigma_{22}(k) = 2g|\psi_c|^2, \\ \Sigma_{12}(k) &= \Sigma_{21}^*(k) = g\psi_c^2 \end{aligned} \tag{1.505}$$

[71] In this section, $\mathcal{G}$ denotes the connected propagator and the subscript "c" of $\mathcal{G}_c$ stands for "classical". Note the sign convention in the definition of $\mathcal{G}$ [Eq. (1.500)].

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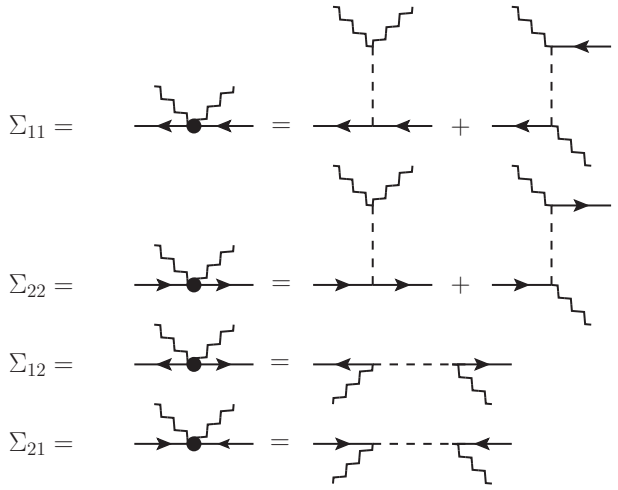


Figure 1.14: Diagrammatic representation of the self-energies (1.505). The zigzag line stands for ψ_c or ψ_c^*. One sometimes introduces a directed zigzag line to distinguish between ψ_c (incoming line) and ψ_c^* (outgoing line).

(Fig. 1.14).

Since the action (1.499) is Gaussian (ignoring the $\mathcal{O}(l^{-1/2})$ terms), we can carry out the functional integral (see Appendix 1.E.4),[72]

$$Z = e^{-lS_c} \left(\det \mathcal{G}_c^{-1}\right)^{-1/2}, \tag{1.506}$$

and obtain the thermodynamic potential[73]

$$\Omega = \frac{l}{\beta} S_c - \frac{1}{2\beta} \mathrm{Tr} \ln \mathcal{G}_c + \mathcal{O}(l^{-1}). \tag{1.507}$$

Let us now consider the corrections to the Gaussian action (1.499),

$$lS[\chi^*, \chi] = lS_c + S^{(2)}[\chi^*, \chi] + \frac{1}{\sqrt{l}} S^{(3)}[\chi^*, \chi] + \frac{1}{l} S^{(4)}[\chi^*, \chi], \tag{1.508}$$

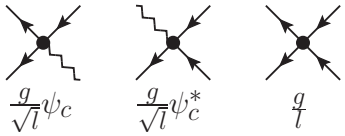
with

$$\begin{aligned} S^{(3)} &= g \int_0^\beta d\tau \int d^d r \left[\psi_c \chi^*(x) |\chi(x)|^2 + \text{c.c.}\right], \\ S^{(4)} &= \frac{g}{2} \int_0^\beta d\tau \int d^d r \, |\chi(x)|^4. \end{aligned} \tag{1.509}$$

These corrections give rise to three- and four-leg vertices which can be represented by the following diagrams

[72] Because of the 2×2 matrix structure of $\mathcal{G}_c$, $\det(-\mathcal{G}_c) = \det(\mathcal{G}_c)$.
[73] We use $\ln \det \mathcal{G} = \mathrm{Tr} \ln \mathcal{G}$ (see Appendix 1.E.4).

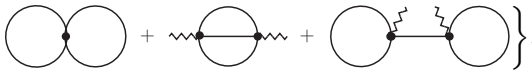
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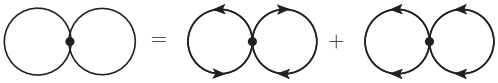
The partition function can be calculated treating $S^{(3)}$ and $S^{(4)}$ in perturbation and using the linked cluster theorem,

$$\begin{aligned} Z &= e^{-lS_c} \int \mathcal{D}[\chi^*,\chi]\, e^{-S^{(2)}-\frac{1}{\sqrt{l}}S^{(3)}-\frac{1}{l}S^{(4)}} \\ &= e^{-lS_c} \left(\det \mathcal{G}_c^{-1}\right)^{-1/2} \exp\left\{\sum \text{connected diagrams}\right\}. \end{aligned} \tag{1.510}$$

To order $1/l$, one finds that the partition function is given by the following diagrams,

$$\ln Z = -lS_c + \frac{1}{2}\mathrm{Tr}\ln \mathcal{G}_c + \frac{1}{l}\Big\{$$

$$\Big\},$$

where the lines in the diagrams must be drawn in all possible ways compatible with $S^{(3)}$, $S^{(4)}$, and the classical propagator $\mathcal{G}_c$. For example, the first diagram means



where the lines stand for the classical propagator $\mathcal{G}_c$.

The expansion in $1/l$ turns out to be an expansion in the number of loops. Consider a diagram with $n_V = n_V^{(3)} + n_V^{(4)}$ vertices, where $n_V^{(3)}$ ($n_V^{(4)}$) is the number of three-leg (four-leg) vertices. The propagator $\mathcal{G}_c$ being independent of l, the diagram is of order $l^{-n_V^{(4)}-n_V^{(3)}/2} = l^{-n_V+n_V^{(3)}/2}$. The number n_I of internal lines is given by

$$2n_I = 4n_V^{(4)} + 3n_V^{(3)} = 4n_V - n_V^{(3)} \tag{1.511}$$

(obtained by counting the total number of lines attached to the n_V vertices). Since each vertex satisfies momentum conservation, a diagram with n_V vertices has $n_V - 1$ momentum conservation constraints, so that the number of independent momentum loops is

$$n_M = n_I - (n_V - 1) = n_V - \frac{n_V^{(3)}}{2} + 1. \tag{1.512}$$

The order of the diagram, $l^{-n_V+n_V^{(3)}/2} = l^{-(n_M-1)}$, is therefore given by the number of loops. In the absence of a condensate ($\psi_c = 0$), $n_V^{(3)} = 0$ and $\mathcal{G}_c = \mathcal{G}_0$, a diagram with n_M loops is of order $g^{n_V} = g^{n_M-1}$. The loop expansion then corresponds to the perturbation expansion about the non-interacting limit discussed in Sec. 1.5. On the other hand, when $|\psi_c| > 0$, the classical propagator $\mathcal{G}_c$ contains an infinite resummation of interaction terms and a diagram with n_M loops contains from $n_M - 1$ to $2(n_M - 1)$ vertices. To any order, the loop expansion then corresponds to an infinite resummation of perturbation theory.

1.7.2.2 Effective action and 1PI vertices

We now consider the effective action defined in Sec. 1.6.2,

$$\Gamma[\phi^*,\phi] = -\frac{1}{l}\ln Z[J^*,J] + \int_0^\beta d\tau \int d^d r [J^*(x)\phi(x) + \text{c.c.}], \tag{1.513}$$

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where

$$\phi(x) = \frac{1}{l}\frac{\delta \ln Z[J^*,J]}{\delta J^*(x)}, \qquad \phi^*(x) = \frac{1}{l}\frac{\delta \ln Z[J^*,J]}{\delta J(x)} \tag{1.514}$$

(with $Z[J^*,J]$ given by (1.495)). For arbitrary external sources, the saddle point equations read

$$\left.\frac{\delta S[\psi^*,\psi]}{\delta\psi(x)}\right|_{\psi^{(*)}=\psi_c^{(*)}} = J^*(x), \qquad \left.\frac{\delta S[\psi^*,\psi]}{\delta\psi^*(x)}\right|_{\psi^{(*)}=\psi_c^{(*)}} = J(x). \tag{1.515}$$

We expand about the classical state, $\psi(x) = \psi_c(x) + \frac{1}{\sqrt{l}}\chi(x)$, using

$$\begin{aligned} &lS[\psi^*,\psi] - l\int_0^\beta d\tau \int d^dr[J^*(x)\psi(x) + \text{c.c.}] \\ &\quad = lS_c - l\int_0^\beta d\tau \int d^dr[J^*(x)\psi_c(x) + \text{c.c.}] + S^{(2)}[\chi^*,\chi] + \mathcal{O}(l^{-1/2}), \end{aligned} \tag{1.516}$$

where $S_c = S[\psi_c^*,\psi_c]$ and the Gaussian part $S^{(2)}$ of the action is defined by (1.500) and (1.501) with a time- and space-dependent classical field $\psi_c(x)$. From

$$\ln Z[J^*,J] = -lS_c + l\int_0^\beta d\tau \int d^dr[J^*(x)\psi_c(x) + \text{c.c.}] + \frac{1}{2}\text{Tr}\ln\mathcal{G}_c + \mathcal{O}(l^{-1}), \tag{1.517}$$

and (1.514,1.515), we deduce

$$\begin{aligned} \phi(x) &= -\frac{\delta S_c}{\delta J^*(x)} + \frac{\delta}{\delta J^*(x)}\int_0^\beta d\tau' \int d^dr'[J^*(x')\psi_c(x') + \text{c.c.}] + \mathcal{O}(l^{-1}) \\ &= \psi_c(x) + \mathcal{O}(l^{-1}) \end{aligned} \tag{1.518}$$

and $\phi^*(x) = \psi_c^*(x) + \mathcal{O}(l^{-1})$. This yields

$$\Gamma[\phi^*,\phi] = S_c + \int_0^\beta d\tau \int d^dr\big\{J^*(x)[\phi(x) - \psi_c(x)] + \text{c.c.}\big\} - \frac{1}{2l}\text{Tr}\ln\mathcal{G}_c + \mathcal{O}(l^{-2}), \tag{1.519}$$

where the $\mathcal{O}(l^{-2})$ contribution comes from the two-loop and higher-order diagrams. Noting that

$$\begin{aligned} &S_c + \int_0^\beta d\tau \int d^dr\big\{J^*(x)[\phi(x) - \psi_c(x)] + \text{c.c.}\big\} \\ &\quad = S[\psi_c^*,\psi_c] + \int_0^\beta d\tau \int d^dr\left\{\left.\frac{\delta S}{\delta\psi(x)}\right|_{\psi^{(*)}=\psi_c^{(*)}}[\phi(x) - \psi_c(x)] + \text{c.c.}\right\} \\ &\quad = S[\phi^*,\phi] + \mathcal{O}(l^{-2}), \end{aligned} \tag{1.520}$$

we finally deduce

$$\Gamma[\phi^*,\phi] = S[\phi^*,\phi] - \frac{1}{2l}\text{Tr}\ln(\mathcal{G}_c[\phi^*,\phi]) + \mathcal{O}(l^{-2}), \tag{1.521}$$

where $\mathcal{G}_c[\phi^*,\phi]$ is obtained from (1.501) by replacing $\psi_c(x)$ by $\phi(x)$, i.e.

$$\mathcal{G}_c^{-1}[x,x';\phi^*,\phi] = -\begin{pmatrix} \dfrac{\delta^2 S[\phi^*,\phi]}{\delta\phi^*(x)\delta\phi(x')} & \dfrac{\delta^2 S[\phi^*,\phi]}{\delta\phi^*(x)\delta\phi^*(x')} \\ \dfrac{\delta^2 S[\phi^*,\phi]}{\delta\phi(x)\delta\phi(x')} & \dfrac{\delta^2 S[\phi^*,\phi]}{\delta\phi(x)\delta\phi^*(x')} \end{pmatrix}. \tag{1.522}$$

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In (1.520) and (1.521), we have used the fact that $\phi(x) - \psi_c(x)$ is of order $1/l$.

To order $1/l$ and for vanishing external sources, the equations of state (1.401,1.402) read

$$\begin{aligned} 0 = \frac{\delta\Gamma}{\delta\phi^*(x)} &= \frac{\delta S}{\delta\phi^*(x)} - \frac{1}{2l}\mathrm{Tr}\left(\mathcal{G}_c \frac{\delta\Sigma}{\delta\phi^*(x)}\right), \\ 0 = \frac{\delta\Gamma}{\delta\phi(x)} &= \frac{\delta S}{\delta\phi(x)} - \frac{1}{2l}\mathrm{Tr}\left(\mathcal{G}_c \frac{\delta\Sigma}{\delta\phi(x)}\right), \end{aligned} \tag{1.523}$$

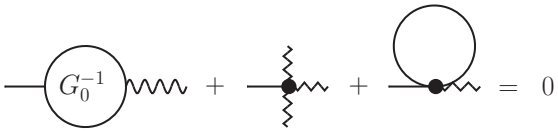
where we have used $\mathcal{G}_c^{-1} = \mathcal{G}_0^{-1} - \Sigma$, with

$$\Sigma[x,x';\phi^*,\phi] = g\delta(\mathbf{r}-\mathbf{r}')\delta(\tau-\tau') \begin{pmatrix} 2|\phi(x)|^2 & \phi(x)^2 \\ \phi^*(x)^2 & 2|\phi(x)|^2 \end{pmatrix}. \tag{1.524}$$

We therefore obtain

$$\begin{aligned} 0 = &-\int_0^\beta d\tau' \int d^d r'\, G_0^{-1}(x,x')\phi(x') + g\phi^*(x)\phi(x)^2 \\ &- \frac{g}{l}\Big\{\big(\mathcal{G}_{c,11}[x,x;\phi^*,\phi] + \mathcal{G}_{c,22}[x,x;\phi^*,\phi]\big)\phi(x) + \mathcal{G}_{c,12}[x,x;\phi^*,\phi]\phi^*(x)\Big\} \end{aligned} \tag{1.525}$$

and a similar equation with $\phi(x) \leftrightarrow \phi^*(x)$. Equation (1.525) can be diagrammatically represented as



and corresponds to a one-loop approximation to the exact result (1.435) (Fig. 1.8). Here the zigzag line stands for $\phi(x)$ or $\phi^*(x)$ and the solid one (in the loop) for the classical propagator $\mathcal{G}_c \equiv \mathcal{G}_c[\phi^*,\phi]$. The one-loop term in (1.525) gives the $\mathcal{O}(1/l)$ correction to the classical (saddle-point) result $|\phi(x)| = \sqrt{\mu/g}$. From (1.521), one can also obtain the two-point vertices

$$\begin{aligned} \Gamma_{11}^{(2)}(x,x') &= \frac{\delta^2\Gamma}{\delta\phi^*(x)\delta\phi(x')}, \\ \Gamma_{12}^{(2)}(x,x') &= \frac{\delta^2\Gamma}{\delta\phi^*(x)\delta\phi^*(x')}, \\ \Gamma_{22}^{(2)}(x,x') &= \frac{\delta^2\Gamma}{\delta\phi(x)\delta\phi^*(x')} \end{aligned} \tag{1.526}$$

to order $1/l$. For instance, one finds

$$\begin{aligned} \Gamma_{11}^{(2)}(x,x') = &-\mathcal{G}_{c,11}^{-1}(x,x') \\ &- \frac{1}{2l}\mathrm{Tr}\left(\mathcal{G}_c \frac{\delta^2\Sigma}{\delta\phi^*(x)\delta\phi(x')}\right) - \frac{1}{2l}\mathrm{Tr}\left(\mathcal{G}_c \frac{\delta\Sigma}{\delta\phi^*(x)}\mathcal{G}_c \frac{\delta\Sigma}{\delta\phi(x')}\right) \end{aligned} \tag{1.527}$$

with Σ the classical self-energy (1.524). The $\mathcal{O}(1/l)$ correction to $-\mathcal{G}_{c,ij}^{-1}$ gives the one-loop self-energy. It can be diagrammatically represented as

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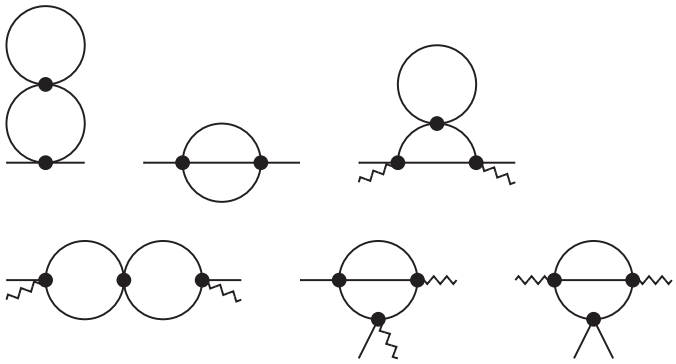
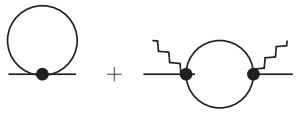
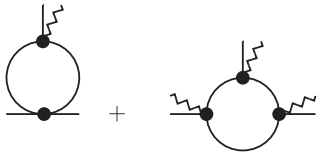


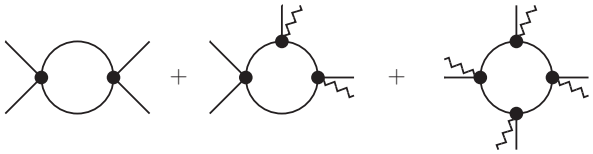
Figure 1.15: Two-loop corrections to the self-energy.



Higher-order vertices are obtained by taking the functional derivative of $\Gamma^{(2)}$ with respect to $\phi(x)$ or $\phi^*(x)$. The derivative of the (bare) three-leg vertex $g\phi^{(*)}(x)$ gives the (bare) four-leg vertex g. On the other hand, when acting on the classical propagator $\mathcal{G}_c = (\mathcal{G}_0^{-1} - \Sigma)^{-1}$, the functional derivative pulls out an external line attached to a three-leg vertex. Thus we obtain diagrams of the type



(diagrams obtained by exchanging external lines are not shown) for the one-loop correction to the three-leg vertex $\Gamma^{(3)}$, and



for the four-leg vertex.

A similar analysis can be carried out at order $1/l^2$. One finds that the effective action

$$\Gamma[\phi^*,\phi] = S[\phi^*,\phi] - \frac{1}{2l}\mathrm{Tr}\ln\mathcal{G}_c[\phi^*,\phi] + \frac{1}{l^2}\Big\{$$  $$\Big\} \qquad (1.528)$$

to this order is given by the 1PI two-loop diagrams (see the calculation below). The two-loop order vertices are obtained from the functional derivatives of (1.528). For instance, the two-loop correction to the self-energy is given by the diagrams of Fig. 1.15. Note that the absence of one-particle reducible diagrams in the effective action is a necessary condition for the self-energy to be 1PI. The one- and two-loop results easily generalize to all orders in perturbation theory: The effective action $\Gamma[\phi^*,\phi]$ may be expressed as the sum of all

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1PI diagrams obtained from the classical propagator $\mathcal{G}_c[\phi^*,\phi]$ and the vertices $S^{(n)}[\phi^*,\phi]$ derived from the action $S[\phi^*,\phi]$ [5].

The effective action to two-loop order. Let us compute the effective action to order $1/l^2$. For simplicity, we consider a real field. The partition function reads

$$Z[J]=\int\mathcal{D}[\psi]\exp\Big\{-lS[\psi]+l\int dx\,J(x)\psi(x)\Big\} \tag{1.529}$$

with the action

$$S[\psi]=-\frac{1}{2}\int dxdx'\,\psi(x)G_0^{-1}(x,x')\psi(x')+S_{\rm int}[\psi]. \tag{1.530}$$

The saddle-point approximation gives

$$\left.\frac{\delta S[\psi]}{\delta\psi(x)}\right|_{\psi=\psi_c}=J(x). \tag{1.531}$$

Expanding about the classical field ψ_c, $\psi=\psi_c+\chi/\sqrt{l}$, we rewrite the action as

$$\begin{aligned}
&lS[\psi]-l\int dx\,J(x)\psi(x)\\
&\quad=lS[\psi_c]-l\int dx\,J(x)\psi_c(x)-\frac{1}{2}\int dx_1dx_2\,\chi(x_1)G_c^{-1}(x_1,x_2)\chi(x_2)\\
&\qquad+\frac{1}{3!\sqrt{l}}\int dx_1dx_2dx_3\,\chi(x_1)\chi(x_2)\chi(x_3)S_c^{(3)}(x_1,x_2,x_3)\\
&\qquad+\frac{1}{4!l}\int dx_1dx_2dx_3dx_4\,\chi(x_1)\chi(x_2)\chi(x_3)\chi(x_4)S_c^{(4)}(x_1,x_2,x_3,x_4)+\mathcal{O}(l^{-3/2}),
\end{aligned} \tag{1.532}$$

where

$$S_c^{(n)}(x_1,\cdots,x_n)=\left.\frac{\delta^nS[\psi]}{\delta\psi(x_1)\cdots\delta\psi(x_n)}\right|_{\psi=\psi_c} \tag{1.533}$$

and $G_c^{-1}(x_1,x_2)=-S^{(2)}(x_1,x_2)$ is the inverse classical propagator. $Z[J]$ is calculated using the linked cluster theorem,

$$Z[J]=e^{-lS_c+l\int dx\,J(x)\psi_c(x)}\left[\det(-G_c^{-1})\right]^{-1/2}\exp\Big\{\sum\text{connected graphs}\Big\}. \tag{1.534}$$

The $\mathcal{O}(1/l)$ contribution is given by the three two-loop diagrams shown after (1.491), i.e.

$$\begin{aligned}
\ln Z[J]=&-lS_c+l\int dx\,J(x)\psi_c(x)-\frac{1}{2}\mathrm{Tr}\ln(-G_c^{-1})\\
&-\frac{1}{8l}\int dx_1dx_2dx_3dx_4\,G_c(x_1,x_2)G_c(x_3,x_4)S_c^{(4)}(x_1,x_2,x_3,x_4)\\
&-\frac{1}{l}\int dx_1dx_2dx_3dy_1dy_2dy_3\,S_c^{(3)}(x_1,x_2,x_3)S_c^{(3)}(y_1,y_2,y_3)\\
&\qquad\times\left[\frac{1}{12}G_c(x_1,y_1)G_c(x_2,y_2)G_c(x_3,y_3)\right.\\
&\qquad\quad\left.+\frac{1}{8}G_c(x_1,x_2)G_c(y_1,y_2)G_c(x_3,y_3)\right]+\mathcal{O}(l^{-2}).
\end{aligned} \tag{1.535}$$

Equation (1.535) gives the thermodynamic potential to two-loop order.

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We can now compute the effective action $\Gamma[\phi]$. We have

$$\begin{aligned}\phi(x) &= \frac{1}{l}\frac{\delta \ln Z[J]}{\delta J(x)} \\ &= \psi_c(x) - \frac{\delta S_c}{\delta J(x)} + \int dx'\, J(x')\frac{\delta \psi_c(x')}{\delta J(x)} - \frac{1}{2l}\frac{\delta}{\delta J(x)}\mathrm{Tr}\ln(-G_c^{-1}) + \mathcal{O}(l^{-2}) \\ &= \psi_c(x) - \frac{1}{2l}\frac{\delta}{\delta J(x)}\mathrm{Tr}\ln(-G_c^{-1}) + \mathcal{O}(l^{-2}). \end{aligned} \tag{1.536}$$

Using

$$\frac{\delta}{\delta J(x)}\mathrm{Tr}\ln(-G_c^{-1}) = \frac{\delta}{\delta J(x)}\mathrm{Tr}\ln S_c^{(2)} = \int dy \frac{\delta \psi_c(y)}{\delta J(x)}\mathrm{Tr}\left[\frac{\delta S_c^{(2)}}{\delta \psi_c(y)} S_c^{(2)-1}\right], \tag{1.537}$$

with $\delta J(x)/\delta\psi_c(y) = -G_c^{-1}(x,y)$, we obtain

$$\begin{aligned}\phi(x) &= \psi_c(x) - \frac{1}{2l}\int dy dy_1 dy_2\, G_c(x,y) S_c^{(3)}(y,y_1,y_2) G_c(y_2,y_1) + \mathcal{O}(l^{-2}). \\ &= \psi_c(x) - \frac{1}{2l}\int dy dy_1 dy_2\, G_c(x,y;\phi) S_c^{(3)}(y,y_1,y_2;\phi) G_c(y_2,y_1;\phi) + \mathcal{O}(l^{-2}), \end{aligned} \tag{1.538}$$

where the last line is obtained by replacing G_c by $G_c(\phi) \equiv G_c[\phi]$ (and similarly $S_c^{(3)} \to S_c^{(3)}(\phi) \equiv S_c^{(3)}[\phi]$), which does not change the result to $\mathcal{O}(1/l)$. To obtain $\Gamma[\phi]$, we also need $J(x)$ to $\mathcal{O}(1/l)$,

$$\begin{aligned} J(x) &= \left.\frac{\delta S}{\delta \psi(x)}\right|_{\psi=\phi} + \int dy \left.\frac{\delta^2 S}{\delta\psi(x)\delta\psi(y)}\right|_{\psi=\phi}[\psi_c(y) - \phi(y)] + \mathcal{O}(l^{-2}) \\ &= \frac{\delta S[\phi]}{\delta\phi(x)} - \frac{1}{2l}\int dy\, G_c^{-1}(x,y;\phi)\int dy_1 dy_2 dy_3\, G_c(y,y_3;\phi) \\ &\qquad \times S_c^{(3)}(y_1,y_2,y_3;\phi) G_c(y_2,y_1;\phi) + \mathcal{O}(l^{-2}) \\ &= \frac{\delta S[\phi]}{\delta\phi(x)} - \frac{1}{2l}\int dy_1 dy_2\, S_c^{(3)}(x,y_1,y_2;\phi) G_c(y_2,y_1;\phi) + \mathcal{O}(l^{-2}). \end{aligned} \tag{1.539}$$

We deduce

$$\begin{aligned}\Gamma[\phi] &= -\frac{1}{l}\ln Z[J] + \int dx\, J(x)\phi(x) \\ &= S_c + \frac{1}{2l}\mathrm{Tr}\ln\big(-G_c^{-1}(\phi)\big) + \int dx J(x)[\phi(x) - \psi_c(x)] \\ &\quad - (\text{two-loop diagrams}) + \mathcal{O}(l^{-3}), \end{aligned} \tag{1.540}$$

where the contribution of the three two-loop diagrams is given in (1.535) (up to a factor $1/l$). Using (1.538) we then find

$$\begin{aligned}\Gamma[\phi] = S[\phi] &+ \frac{1}{2l}\mathrm{Tr}\ln\big(-G_c^{-1}(\phi)\big) - (\text{two-loop diagrams}) \\ &- \frac{1}{8l^2}\int dx_1 dx_2 dx_3 dy_1 dy_2 dy_3\, G_c(x_3,y_3;\phi) G_c(x_1,x_2;\phi) G_c(y_1,y_2;\phi) \\ &\times S_c^{(3)}(x_1,x_2,x_3;\phi) S_c^{(3)}(y_1,y_2,y_3;\phi) + \mathcal{O}(l^{-3}). \end{aligned} \tag{1.541}$$

The last term cancels the two-loop diagram which is one-particle reducible and we finally obtain

$$\Gamma[\phi] = S[\phi] + \frac{1}{2l}\mathrm{Tr}\ln\big(-G_c^{-1}(\phi)\big)$$

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$$+\frac{1}{8l^2}\int dx_1dx_2dx_3dx_4\, S_c^{(4)}(x_1,x_2,x_3,x_4;\phi)G_c(x_1,x_2;\phi)G_c(x_3,x_4;\phi)$$
$$+\frac{1}{12l^2}\int dx_1dx_2dx_3dy_1dy_2dy_3\, S_c^{(3)}(x_1,x_2,x_3;\phi)S_c^{(3)}(y_1,y_2,y_3;\phi)$$
$$\times G_c(x_1,y_1;\phi)G_c(x_2,y_2;\phi)G_c(x_3,y_3;\phi)+\mathcal{O}(l^{-3}), \qquad (1.542)$$

in agreement with (1.528).

1.8 Perturbation theory at zero temperature

In this section, we briefly discuss perturbation theory at zero temperature for systems having a fixed number of particles.[74] The advantage of this approach is that one works directly in real time and no analytic continuation is necessary, contrary to the finite-temperature formalism in the grand canonical ensemble described in the previous sections. On the other hand, we shall see that the zero-temperature approach suffers from some drawbacks and the finite-temperature formalism is often preferable in practice.

We write the Hamiltonian of the N-particle system as $\hat{H}=\hat{H}_0+\hat{H}_{\rm int}$, where $\hat{H}_0$ is the non-interacting Hamiltonian whose ground state $|\phi_0\rangle$ is known. If we let $|\phi_0\rangle$ evolve during a time T, we obtain the state

$$e^{-i\hat{H}T}|\phi_0\rangle=e^{-i\hat{H}T}\sum_n|n\rangle\langle n|\phi_0\rangle=\sum_n e^{-iE_nT}|n\rangle\langle n|\phi_0\rangle, \qquad (1.543)$$

where $\{|n\rangle,E_n\}$ denotes an orthonormal basis of $\hat{H}$. Sending T to infinity with a (small) negative imaginary part, we project out the ground state $|0\rangle$ of $\hat{H}$,

$$\lim_{T\to\infty(1-i\eta)} e^{-i\hat{H}T}|\phi_0\rangle=e^{-iE_0T}|0\rangle\langle 0|\phi_0\rangle \qquad (1.544)$$

($\eta=0^+$), provided that $|0\rangle$ is non-degenerate and has a finite overlap with the non-interacting ground state $|\phi_0\rangle$. We deduce that the ground state energy of the interacting system can be obtained from

$$E_0=\lim_{T\to\infty(1-i\eta)}\frac{i}{T}\ln\langle\phi_0|e^{-i\hat{H}T}|\phi_0\rangle. \qquad (1.545)$$

Let us now consider the one-particle Green function[75]

$$G(\alpha t,\alpha' t')=-i\langle 0|T\hat{\psi}_\alpha(t)\hat{\psi}^\dagger_{\alpha'}(t')|0\rangle \qquad (1.546)$$

defined in Sec. 1.2.3 [Eq. (1.205)]. $\hat{\psi}_\alpha(t)$ and $\hat{\psi}^\dagger_\alpha(t)$ are operators in the Heisenberg picture and the index α labels the one-body states. Using

$$\lim_{T\to\infty(1-i\eta)} e^{-i\hat{H}\frac{T}{2}}|\phi_0\rangle=e^{-iE_0\frac{T}{2}}|0\rangle\langle 0|\phi_0\rangle,$$
$$\lim_{T\to\infty(1-i\eta)}\langle\phi_0|e^{-i\hat{H}\frac{T}{2}}=e^{-iE_0\frac{T}{2}}\langle\phi_0|0\rangle\langle 0|, \qquad (1.547)$$

[74] A thorough discussion of perturbation theory at zero temperature can be found in Chap. 3 of Ref. [9].

[75] The time-ordering operator T in (1.546) should not be confused with the time T appearing e.g. in (1.545).

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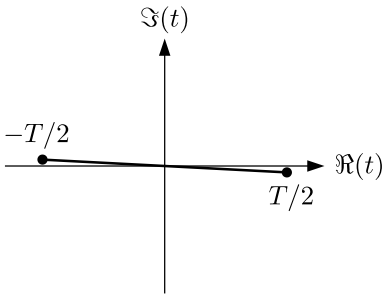


Figure 1.16: Contour in the complex t plane used in the definition of the functional integral at zero temperature.

we obtain

$$iG(\alpha t, \alpha' t') = \lim_{T\to\infty(1-i\eta)} \frac{\langle\phi_0|e^{-i\hat{H}\frac{T}{2}}\left(T\hat{\psi}_\alpha(t)\hat{\psi}^\dagger_{\alpha'}(t')\right)e^{-i\hat{H}\frac{T}{2}}|\phi_0\rangle}{\langle\phi_0|e^{-i\hat{H}T}|\phi_0\rangle}. \tag{1.548}$$

We show below that both $\langle\phi_0|e^{-i\hat{H}T}|\phi_0\rangle$ and the Green function (1.548) can be expressed as a coherent-state functional integral. We shall consider a contour in the complex t plane which is slightly tilted with respect to the real axis and extends from $-T/2$ and $T/2$ (Fig. 1.16). A time $t = \tilde{t}(1-i\eta)$ on this contour can be parameterized by a real variable $\tilde{t}$. In practice, we will distinguish between t and $\tilde{t}$ only when necessary.

1.8.1 Fermions

In the case of fermions, the non-interacting ground state is a Slater determinant,

$$|\phi_0\rangle = \prod_{\alpha=1}^{N} \hat{\psi}^\dagger_\alpha|\text{vac}\rangle, \tag{1.549}$$

with the N lowest one-particle states occupied. Here we assume the one-body states $|\alpha\rangle$ to be labeled in order of increasing eigenvalues ϵ_α. We require $|\phi_0\rangle$ to be non-degenerate, which implies that there is a finite energy gap $\epsilon_{N+1} - \epsilon_N$ between the last occupied state and the first unoccupied state. We shift the energies ϵ_α such that

$$\begin{cases} \epsilon_\alpha < 0 & \text{for} \quad \alpha \leq N, \\ \epsilon_\alpha > 0 & \text{for} \quad \alpha > N, \end{cases} \tag{1.550}$$

and define the operators

$$\hat{\chi}^\dagger_\alpha = \begin{cases} \hat{\psi}^\dagger_\alpha & (\alpha > N), \\ \hat{\psi}_\alpha & (\alpha \leq N), \end{cases} \qquad \hat{\chi}_\alpha = \begin{cases} \hat{\psi}_\alpha & (\alpha > N), \\ \hat{\psi}^\dagger_\alpha & (\alpha \leq N). \end{cases} \tag{1.551}$$

$\hat{\chi}^\dagger_\alpha$ creates an elementary excitation with respect to $|\phi_0\rangle$, i.e. a particle for $\alpha > N$ and a hole for $\alpha \leq N$. Since $\hat{\chi}_\alpha|\phi_0\rangle = 0$ for any α, $|\phi_0\rangle$ is the vacuum state with respect to the excitations of the operator $\hat{\chi}$. The Green function (1.546) is expressed as

$$G(\alpha t, \alpha' t') = \begin{cases} -i\langle 0|T\hat{\chi}_\alpha(t)\hat{\chi}^\dagger_{\alpha'}(t')|0\rangle & \text{if} \quad \alpha > N, \\ i\langle 0|T\hat{\chi}_{\alpha'}(t')\hat{\chi}^\dagger_\alpha(t)|0\rangle & \text{if} \quad \alpha \leq N. \end{cases} \tag{1.552}$$

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1.8.1.1 Coherent-state functional integral

We can now construct coherent states by associating with $\hat{\chi}_\alpha$ and $\hat{\chi}_\alpha^\dagger$ a pair of complex conjugate Grassmann variables ψ_α and ψ_α^* (Sec. 1.3.3). Since $|\phi_0\rangle$ is the vacuum state with respect to the field ψ, it can be seen formally as the coherent state $|\psi\rangle$ with $\psi_\alpha = 0$. Dividing the time T into $M = \beta/\epsilon$ steps and inserting $M-1$ times the closure relation (1.217) or (1.254), one obtains

$$\begin{aligned}\langle\phi_0|e^{-i\hat{H}T}|\phi_0\rangle &= \int \prod_{k=1}^{M-1} d(\psi_k^*,\psi_k)\, e^{-\sum_{k=1}^{M-1}\sum_\alpha |\psi_{k,\alpha}|^2} \prod_{k=1}^{M} \langle\psi_k|e^{-i\epsilon\hat{H}}|\psi_{k-1}\rangle \\ &= \int \prod_{k=1}^{M-1} d(\psi_k^*,\psi_k)\, e^{-\sum_{k=1}^{M}\left[\sum_\alpha \psi_{k,\alpha}^*(\psi_{k,\alpha}-\psi_{k-1,\alpha})+i\epsilon H(\psi_{k,\alpha}^*,\psi_{k,-1\alpha})\right]}, \quad (1.553)\end{aligned}$$

with $\psi_{0,\alpha} = \psi_{M,\alpha}^* = 0$. The integration measure $d(\psi_\alpha^*,\psi_\alpha)$ is defined by (1.262) and we assume the Hamiltonian $\hat{H} = H(\hat{\chi}_\alpha^\dagger,\hat{\chi}_\alpha)$ to be normal ordered. In the continuum-time limit ($M\to\infty$), we may write

$$\langle\phi_0|e^{-i\hat{H}T}|\phi_0\rangle = \int \mathcal{D}[\psi^*,\psi]\, e^{iS[\psi^*,\psi]} \quad (1.554)$$

with the action

$$S[\psi^*,\psi] = \int_{-T/2}^{T/2} dt \left[\sum_\alpha \psi_\alpha^*(t) i\partial_t \psi_\alpha(t) - H\big(\psi_\alpha^*(t),\psi_\alpha(t)\big)\right], \quad (1.555)$$

and the boundary conditions $\psi_\alpha(-T/2) = \psi_\alpha^*(T/2) = 0$. Except for the boundary conditions at $\pm T/2$, the real-time action (1.555) can be deduced from the Euclidean action (1.273) by a Wick rotation $\tau \to it$ and $-S_E \to iS$.

Similarly, we find

$$\langle\phi_0|e^{-i\hat{H}\frac{T}{2}}\big(T\hat{\chi}_\alpha(t)\hat{\chi}_{\alpha'}^\dagger(t')\big)e^{-i\hat{H}\frac{T}{2}}|\phi_0\rangle = \int \prod_{k=1}^{M-1} d(\psi_k^*,\psi_k)\, \psi_{k_1,\alpha}\psi_{k_2,\alpha'}^*\, e^{iS(\{\psi_k^*,\psi_k\})}, \quad (1.556)$$

where the action $S(\{\psi_k^*,\psi_k\})$ is defined by (1.553). $t_{k_1} = -T/2 + k_1\epsilon$ and $t_{k_2} = -T/2 + k_2\epsilon$ are the discrete times corresponding to t and t', respectively. Equation (1.556) is similar to (1.278) and can be derived in the same way. In the continuum-time limit, we therefore obtain

$$\langle 0|T\hat{\chi}_\alpha(t)\hat{\chi}_{\alpha'}^\dagger(t')|0\rangle = \lim_{T\to\infty(1-i\eta)} \frac{\int \mathcal{D}[\psi^*,\psi]\,\psi_\alpha(t)\psi_{\alpha'}^*(t')\,e^{iS[\psi^*,\psi]}}{\int \mathcal{D}[\psi^*,\psi]\,e^{iS[\psi^*,\psi]}} \quad (1.557)$$

with the boundary conditions $\psi_\alpha(-T/2) = \psi_\alpha^*(T/2) = 0$. The Green function $G(\alpha t,\alpha' t')$ can be deduced from (1.552) and (1.557).

1.8.1.2 Non-interacting fermions

Assuming that the non-interacting Hamiltonian is diagonal in the one-body state basis $\{|\alpha\rangle\}$,

$$\hat{H}_0 = \sum_{\alpha=1}^{N} \epsilon_\alpha \hat{\chi}_\alpha\hat{\chi}_\alpha^\dagger + \sum_{\alpha>N} \epsilon_\alpha \hat{\chi}_\alpha^\dagger\hat{\chi}_\alpha = \sum_{\alpha=1}^{N} \epsilon_\alpha(1-\hat{\chi}_\alpha^\dagger\hat{\chi}_\alpha) + \sum_{\alpha>N} \epsilon_\alpha\hat{\chi}_\alpha^\dagger\hat{\chi}_\alpha, \quad (1.558)$$

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which gives the action

$$\begin{aligned} iS_0 &= -iT\sum_{\alpha=1}^{N}\epsilon_\alpha - \sum_{k=1}^{M-1}\sum_{\alpha}\left[\psi^*_{k,\alpha}(\psi_{k,\alpha}-\psi_{k-1,\alpha}) + i\frac{T}{M}|\epsilon_\alpha|\psi^*_{k,\alpha}\psi_{k-1,\alpha}\right] \\ &\equiv -iT\sum_{\alpha=1}^{N}\epsilon_\alpha - \sum_{\alpha}\sum_{k,k'=1}^{M-1}\psi^*_{k,\alpha}S^{(\alpha)}_{k,k'}\psi_{k',\alpha}, \end{aligned} \tag{1.559}$$

where $S^{(\alpha)}$ is the $(M-1)\times(M-1)$ matrix

$$S^{(\alpha)} = \begin{pmatrix} 1 & 0 & \cdots & \cdots & 0 & 0 \\ -a & 1 & 0 & & & 0 \\ 0 & -a & 1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & 1 & \ddots & \vdots \\ \vdots & & \ddots & -a & 1 & 0 \\ 0 & \cdots & \cdots & 0 & -a & 1 \end{pmatrix} \tag{1.560}$$

with $a = 1 - i\frac{T}{M}|\epsilon_\alpha|$. Since $\det S^{(\alpha)} = 1$, we find

$$\langle\phi_0|e^{-i\hat{H}_0 T}|\phi_0\rangle = e^{-iT\sum_{\alpha=1}^{N}\epsilon_\alpha}\prod_{\alpha}\det S^{(\alpha)} = e^{-iT\sum_{\alpha=1}^{N}\epsilon_\alpha} \tag{1.561}$$

so that the non-interacting ground state energy is $E_0^{(0)} = \sum_{\alpha=1}^{N}\epsilon_\alpha$ [Eq. (1.545)]. From the inverse matrix

$$S^{(\alpha)-1} = \begin{pmatrix} 1 & 0 & 0 & \cdots & \cdots & 0 \\ a & 1 & 0 & \ddots & & \vdots \\ a^2 & a & 1 & 0 & \ddots & \vdots \\ \vdots & \ddots & a & 1 & 0 & 0 \\ \vdots & & \ddots & a & 1 & 0 \\ a^{M-2} & \cdots & \cdots & a^2 & a & 1 \end{pmatrix}, \tag{1.562}$$

we deduce

$$\langle\phi_0|T\hat{\chi}_\alpha(t)\hat{\chi}^\dagger_{\alpha'}(t')|\phi_0\rangle = \lim_{\substack{M\to\infty \\ T\to\infty(1-i\eta)}} S^{(\alpha)-1}_{k,k'} = \lim_{\substack{M\to\infty \\ T\to\infty(1-i\eta)}} \begin{cases} 0 & \text{if} \quad k<k', \\ a^{k-k'} & \text{if} \quad k\geq k', \end{cases} \tag{1.563}$$

where $t_k = -T/2 + kT/M$ and $t_{k'} = -T/2 + k'T/M$ are the discrete times corresponding to t and t', respectively. Using (1.552) and

$$\lim_{M\to\infty} a^{k-k'} = \lim_{M\to\infty}\left(1 - i\frac{T}{M}|\epsilon_\alpha|\right)^{(t_k-t_{k'})\frac{M}{T}} = e^{-i|\epsilon_\alpha|(t-t')}, \tag{1.564}$$

we finally obtain

$$iG_0(\alpha,t) = e^{-i\epsilon_\alpha t}\left[(1-n_\alpha)\Theta(t) - n_\alpha\Theta(-t)\right], \tag{1.565}$$

where

$$n_\alpha = \begin{cases} 1 & \text{if} \quad \alpha\leq N, \\ 0 & \text{if} \quad \alpha > N, \end{cases} \tag{1.566}$$

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is the occupation of the one-body state $|\alpha\rangle$ in the non-interacting ground state $|\phi_0\rangle$.

So far we have ignored the fact that the time $t = \tilde{t}(1 - i\eta)$ ($\tilde{t}$ real) contains a small imaginary part (Fig. 1.16). A more correct expression of the Green function would then be

$$iG_0(\alpha, \tilde{t}) = e^{-i\epsilon_\alpha \tilde{t}(1-i\eta)} \left[(1 - n_\alpha)\Theta(\tilde{t}) - n_\alpha \Theta(-\tilde{t})\right]. \tag{1.567}$$

In Fourier space, this yields

$$\begin{aligned} G_0(\alpha, \omega) &= \int_{-\infty}^{\infty} d\tilde{t}\, e^{i\omega\tilde{t}} G_0(\alpha, \tilde{t}) \\ &= \frac{1 - n_\alpha}{\omega - \epsilon_\alpha(1 - i\eta)} e^{i\tilde{t}[\omega - \epsilon_\alpha(1-i\eta)]}\Big|_0^\infty - \frac{n_\alpha}{\omega - \epsilon_\alpha(1 - i\eta)} e^{i\tilde{t}[\omega - \epsilon_\alpha(1-i\eta)]}\Big|_{-\infty}^0. \end{aligned} \tag{1.568}$$

Since $\epsilon_\alpha < 0$ for $n_\alpha = 1$ and $\epsilon_\alpha > 0$ for $n_\alpha = 0$, the infinitesimal $\eta = 0^+$ ensures the convergence of the time integral,

$$G_0(\alpha, \omega) = \frac{n_\alpha}{\omega - \epsilon_\alpha - i\eta} + \frac{1 - n_\alpha}{\omega - \epsilon_\alpha + i\eta}. \tag{1.569}$$

1.8.1.3 Perturbation theory

Perturbation theory is derived by expanding e^{iS} in powers of the interacting action

$$S_{\rm int}[\psi^*, \psi] = -\int_{-T/2}^{T/2} dt\, H_{\rm int}\big(\psi_\alpha^*(t), \psi_\alpha(t)\big) \tag{1.570}$$

For example, to obtain the ground state energy using (1.545), one uses

$$\begin{aligned} \langle\phi_0|e^{-i\hat{H}T}|\phi_0\rangle &= \int \mathcal{D}[\psi^*, \psi]\, e^{iS_0[\psi^*,\psi] + iS_{\rm int}[\psi^*,\psi]} \\ &= \int \mathcal{D}[\psi^*, \psi]\, e^{iS_0[\psi^*,\psi]} \sum_{n=0}^{\infty} \frac{i^n}{n!} (S_{\rm int}[\psi^*, \psi])^n. \end{aligned} \tag{1.571}$$

The average value

$$\langle (S_{\rm int}[\psi^*, \psi])^n \rangle_0 = \frac{\int \mathcal{D}[\psi^*, \psi]\, (S_{\rm int}[\psi^*, \psi])^n\, e^{iS_0[\psi^*,\psi]}}{\int \mathcal{D}[\psi^*, \psi]\, e^{iS_0[\psi^*,\psi]}} \tag{1.572}$$

can be evaluated using Wick's theorem and represented by Feynman diagrams. The diagrammatic rules follow straightforwardly from those of the finite-temperature formalism. The linked cluster theorem states that the ground state energy

$$\begin{aligned} E_0 &= \lim_{T\to\infty(1-i\eta)} \frac{i}{T} \ln\langle\phi_0|e^{-i\hat{H}T}|\phi_0\rangle \\ &= E_0^{(0)} + \lim_{T\to\infty(1-i\eta)} \frac{i}{T} \sum \text{connected graphs} \end{aligned} \tag{1.573}$$

is given by the sum of the connected diagrams (vacuum fluctuation diagrams). More generally, most results derived in Secs. 1.4-1.6 can be transposed to the zero-temperature formalism.

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1.8.2 Bosons

The ground state of a non-interacting system of N bosons is a Bose-Einstein condensate,

$$|\phi_0(N)\rangle = \frac{1}{\sqrt{N!}}\left[\hat{\psi}^\dagger(\mathbf{k}=0)\right]^N|\text{vac}\rangle, \tag{1.574}$$

where all the particles occupy the $\mathbf{k}=0$ one-body state. For simplicity, we consider spin-zero bosons and assume the system to be translation invariant. Since

$$\begin{aligned}\hat{\psi}(\mathbf{k}=0)|\phi_0(N)\rangle &= \sqrt{N}|\phi_0(N-1)\rangle,\\ \hat{\psi}^\dagger(\mathbf{k}=0)|\phi_0(N)\rangle &= \sqrt{N+1}|\phi_0(N+1)\rangle,\end{aligned} \tag{1.575}$$

neither $\hat{\psi}(\mathbf{k}=0)$ nor $\hat{\psi}^\dagger(\mathbf{k}=0)$ annihilates the ground state $|\phi_0(N)\rangle$, and it is not possible to separate the operators between creation and annihilation parts as in the fermion case [Eq. (1.551)].

Let us consider the rescaled $\mathbf{k}=0$ operators

$$\begin{aligned}\hat{\xi}_0 &= \frac{1}{\sqrt{\mathcal{V}}}\hat{\psi}(\mathbf{k}=0) = \frac{1}{\mathcal{V}}\int d^dr\,\hat{\psi}(\mathbf{r}),\\ \hat{\xi}_0^\dagger &= \frac{1}{\sqrt{\mathcal{V}}}\hat{\psi}^\dagger(\mathbf{k}=0) = \frac{1}{\mathcal{V}}\int d^dr\,\hat{\psi}^\dagger(\mathbf{r}).\end{aligned} \tag{1.576}$$

Their commutator

$$[\hat{\xi}_0,\hat{\xi}_0^\dagger] = \frac{1}{\mathcal{V}} \tag{1.577}$$

vanishes in the thermodynamic limit, while they yield a finite factor when acting on $|\phi_0(N)\rangle$,

$$\begin{aligned}\hat{\xi}_0|\phi_0(N)\rangle &= \left(\frac{N}{\mathcal{V}}\right)^{1/2}|\phi_0(N-1)\rangle \simeq \left(\frac{N}{\mathcal{V}}\right)^{1/2}|\phi_0(N)\rangle,\\ \hat{\xi}_0^\dagger|\phi_0(N)\rangle &= \left(\frac{N+1}{\mathcal{V}}\right)^{1/2}|\phi_0(N+1)\rangle \simeq \left(\frac{N}{\mathcal{V}}\right)^{1/2}|\phi_0(N)\rangle.\end{aligned} \tag{1.578}$$

They can therefore be replaced by c-numbers as long as we consider only states where a finite fraction of the particles occupies the $\mathbf{k}=0$ one-body state. This leads us to rewrite the boson operator as

$$\begin{aligned}\hat{\psi}(\mathbf{r}) &= \xi_0 + \frac{1}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k}\neq 0} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{\psi}(\mathbf{k}) \equiv \xi_0 + \hat{\chi}(\mathbf{r}),\\ \hat{\psi}^\dagger(\mathbf{r}) &= \xi_0^* + \frac{1}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k}\neq 0} e^{-i\mathbf{k}\cdot\mathbf{r}}\hat{\psi}^\dagger(\mathbf{k}) \equiv \xi_0^* + \hat{\chi}^\dagger(\mathbf{r}).\end{aligned} \tag{1.579}$$

Since $\hat{\chi}(\mathbf{r})$ annihilates the non-interacting ground state $|\phi_0(N)\rangle$, we can proceed as in the fermion case and write $\langle\phi_0(N)|e^{-i\hat{H}T}|\phi_0(N)\rangle$ as a functional integral over a complex bosonic field with the action

$$S[\psi^*,\psi] = \int_{-T/2}^{T/2} dt\left[\int d^dr\,\psi^*(\mathbf{r},t)i\partial_t\psi(\mathbf{r},t) - H\big(\psi^*(\mathbf{r},t),\psi(\mathbf{r},t)\big)\right], \tag{1.580}$$

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where $\psi(\mathbf{r}, -T/2) = \psi^*(\mathbf{r}, T/2) = 0$. Again, we see that the real-time action (1.580) can be deduced from the Euclidean action (1.273) by a Wick rotation $\tau = it$. As in Sec. 1.7.2, the interaction vertices coming from the interacting Hamiltonian are of four different types depending on the number of condensate lines attached to them.

Considering $\hat{\xi}_0$ and $\hat{\xi}_0^\dagger$ as c-numbers breaks the (global) U(1) invariance, i.e. the invariance of the Hamiltonian under the transformation $\hat{\psi}(\mathbf{r}) \to e^{i\Lambda}\hat{\psi}(\mathbf{r})$ and $\hat{\psi}^\dagger(\mathbf{r}) \to e^{-i\Lambda}\hat{\psi}^\dagger(\mathbf{r})$. As a result the total number of particles is not conserved, $[\hat{N}, \hat{H}] \neq 0$, and the canonical ensemble cannot be used. Instead, one must work in the grand canonical ensemble where the Hamiltonian is replaced by $\hat{H} - \mu\hat{N}$, with the chemical potential μ chosen such that $\langle 0|\hat{N}|0\rangle = N$. In the interacting ground state, not all the particles are in the $\mathbf{k} = 0$ one-body state, and the average value

$$\langle 0|\hat{\xi}_0^\dagger\hat{\xi}_0|0\rangle = \frac{N_0}{\mathcal{V}} = n_0 \tag{1.581}$$

is less than the total density $N/\mathcal{V}$. As long as the condensate density n_0 remains finite, the replacement of $\hat{\xi}_0$ and $\hat{\xi}_0^\dagger$ by c-numbers ξ_0 and ξ_0^* (with $|\xi_0| = \sqrt{n_0}$) is permissible. To determine the condensate density n_0, we require that

$$\frac{\partial}{\partial n_0}\langle 0|\hat{H} - \mu\hat{N}|0\rangle\Big|_\mu = 0, \tag{1.582}$$

which is an implicit equation for $n_0(\mu)$.

1.8.3 The zero-temperature limit

We have now two different ways to compute perturbatively the ground state energy or a correlation function. One can either use the $T = 0$ perturbation theory or consider the $T \to 0$ limit of the finite-temperature formalism discussed in Secs. 1.4-1.6. While one would naively expect these two approaches to give always the same result, this is in fact not the case. In the zero-temperature formalism, the interacting ground state $|0\rangle$ is obtained from (1.545) provided that $|0\rangle$ and $|\phi_0\rangle$ are not orthogonal. In some cases, $e^{-i\hat{H}T}|\phi_0\rangle$ is trapped in a given symmetry space selected by $|\phi_0\rangle$ and produces the lowest energy eigenstate in that space. In contrast, the finite-temperature theory obtains the state of the system from a statistical average (over a certain ensemble) which has access to all possible symmetries. In principle, it is always possible to use the zero-temperature theory, provided that one chooses a Hamiltonian $\hat{H}_0$ with a ground state $|\phi_0\rangle$ which has the right symmetries. In practice however, it is often preferable to use the (Euclidean) finite-temperature even when one is interested only in the $T = 0$ limit.

From a technical point of view, there are two differences between the zero- and finite-temperature formalisms:

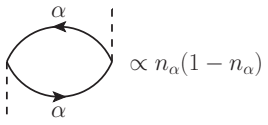
1. In the finite-temperature formalism, the chemical potential μ must be adjusted so that $N = -\partial\Omega/\partial\mu$. In general, μ differs from the chemical potential μ_0 of the non-interacting system defined by $N = -\partial\Omega_0/\partial\mu_0$. Thus the correction

$$E_0 - E_0^{(0)} = \lim_{T\to 0}[\Omega - \Omega_0 + (\mu - \mu_0)N] \tag{1.583}$$

to the ground state energy is determined not only by $\Omega - \Omega_0$ (i.e. the vacuum fluctuation diagrams) but also by the variation $\mu - \mu_0$ of the chemical potential. In the zero-temperature theory, $E_0 - E_0^{(0)}$ is directly given by the perturbation expansion of $\langle\phi_0(N)|e^{-i\hat{H}T}|\phi_0(N)\rangle$, i.e. by the vacuum fluctuation diagrams [Eq. (1.573)].

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2. The finite-temperature perturbation theory yields "anomalous" diagrams which include parts such as



where a particle and a hole propagate in the same one-body state α. Since the one-body state α is either empty ($n_\alpha = 0$) or occupied ($n_\alpha = 1$) in the zero-temperature non-interacting ground state, the anomalous diagrams vanish at $T = 0$. On the contrary, $\lim_{T\to 0} n_\alpha(1 - n_\alpha)$ does not necessarily vanish in the finite-temperature formalism.[76]

For the zero- and finite-temperature perturbation theories to agree, the differences (1) and (2) must cancel.

1.9 Quantization of the electromagnetic field

In this section, we discuss the functional integral describing charged particles coupled to the electromagnetic field. We start by briefly reviewing the Lagrangian and Hamiltonian formalism of classical electromagnetism. The difficulties due to the redundancy of degrees of freedom are resolved by using the Coulomb gauge. We then discuss the canonical quantization in this gauge and express the partition function of the coupled system matter-field as a functional integral.

1.9.1 Lagrangian and Hamiltonian formalism

The standard Lagrangian of classical (three-dimensional) electrodynamics reads

$$\begin{aligned} L = L'_m + \frac{\epsilon_0}{2}\int d^3r \left\{ [\boldsymbol{\nabla}\phi(\mathbf{r},t) + \partial_t \mathbf{A}(\mathbf{r},t)]^2 - c_l^2 \left[\boldsymbol{\nabla}\times\mathbf{A}(\mathbf{r},t)\right]^2 \right\} \\ - \int d^3r \left[\phi(\mathbf{r},t)\rho(\mathbf{r},t) - \mathbf{J}(\mathbf{r},t)\cdot\mathbf{A}(\mathbf{r},t)\right] \end{aligned} \tag{1.584}$$

(c_l denotes the velocity of light), where $L'_m = (m/2)\sum_i \dot{\mathbf{r}}_i^2$ is the Lagrangian of the matter in the absence of electromagnetic interactions with $\mathbf{r}_i$ and $\dot{\mathbf{r}}_i$ the position and velocity of the ith particle. The electric and magnetic fields can be expressed as a function of the electromagnetic potentials,

$$\begin{aligned} \mathbf{E}(\mathbf{r},t) &= -\boldsymbol{\nabla}\phi(\mathbf{r},t) - \partial_t \mathbf{A}(\mathbf{r},t), \\ \mathbf{B}(\mathbf{r},t) &= \boldsymbol{\nabla}\times\mathbf{A}(\mathbf{r},t). \end{aligned} \tag{1.585}$$

The field-matter coupling involves the charge and current densities

$$\begin{aligned} \rho(\mathbf{r},t) &= e\sum_i \delta\big(\mathbf{r} - \mathbf{r}_i(t)\big), \\ \mathbf{J}(\mathbf{r},t) &= e\sum_i \dot{\mathbf{r}}_i \delta\big(\mathbf{r} - \mathbf{r}_i(t)\big) \end{aligned} \tag{1.586}$$

[76] For example, $\beta n_F(\xi_\alpha)[1 - n_F(\xi_\alpha)] = -\frac{\partial n_F(\xi_\alpha)}{\partial \xi_\alpha} \to \delta(\xi_\alpha)$ for $T \to 0$.

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(e denotes the charge of the particles). The Euler-Lagrange equations deduced from (1.584) reproduce Maxwell equations (Appendix 1.A).

It is tempting to quantize the theory in the usual way, i.e. by introducing the momenta $\mathbf{p}_i$ of the particles and the canonical conjugates $(\Pi_0, \mathbf{\Pi})$ to the electromagnetic potential $(\phi, \mathbf{A})$, and assuming the usual commutation rules between conjugate variables. However, since $\dot{\phi} = \partial_t \phi$ does not appear in the Lagrangian, $\Pi_0 = 0$. One must therefore first eliminate the scalar potential by expressing it as a function of the other dynamical variables to obtain a reduced Lagrangian which is a function of the vector potential $\mathbf{A}$ and its time derivative $\dot{\mathbf{A}}$, as well as the positions and velocities of the particles. Before quantizing, one should also get rid of the arbitrariness of $\mathbf{A}$ which follows form gauge invariance. A gauge transformation

$$\begin{aligned} \mathbf{A}'(\mathbf{r},t) &= \mathbf{A}(\mathbf{r},t) + \boldsymbol{\nabla}\Lambda(\mathbf{r},t), \\ \phi'(\mathbf{r},t) &= \phi(\mathbf{r},t) - \partial_t \Lambda(\mathbf{r},t) \end{aligned} \tag{1.587}$$

does not change the electromagnetic field and therefore leaves the Maxwell equations invariant. By choosing a particular gauge, we fix the longitudinal component $\mathbf{A}_\parallel$ of the vector potential, which is otherwise arbitrary. Once this is done, we are left with four degrees of freedom of the field at each position in space, the two components of the transverse vector potential $\mathbf{A}_\perp$ and their time derivatives. Since the longitudinal components $\mathbf{E}_\parallel$ and $\mathbf{B}_\parallel$ of the field are fixed by the Maxwell equations $\boldsymbol{\nabla}\cdot\mathbf{E} = \rho/\epsilon_0$ and $\boldsymbol{\nabla}\cdot\mathbf{B} = 0$, they are not true dynamical variables and we indeed expect any description of the electromagnetic field to contain four independent variables.

We can eliminate the scalar potential using $\boldsymbol{\nabla}\cdot\mathbf{E} = \rho/\epsilon_0$ (which follows from the Euler-Lagrange equation for ϕ), i.e. $\boldsymbol{\nabla}^2\phi + \boldsymbol{\nabla}\cdot\dot{\mathbf{A}}_\parallel = -\rho/\epsilon_0$ or

$$\phi(\mathbf{k},t) = \frac{1}{\mathbf{k}^2}\Big[i\mathbf{k}\cdot\dot{\mathbf{A}}_\parallel(\mathbf{k},t) + \frac{1}{\epsilon_0}\rho(\mathbf{k},t)\Big]. \tag{1.588}$$

The electric field is then given by

$$\begin{aligned} \mathbf{E}_\parallel(\mathbf{k},t) &= -i\mathbf{k}\frac{\rho(\mathbf{k},t)}{\epsilon_0\mathbf{k}^2}, \\ \mathbf{E}_\perp(\mathbf{k},t) &= -\dot{\mathbf{A}}_\perp(\mathbf{k},t). \end{aligned} \tag{1.589}$$

In the following, we shall work in the Coulomb gauge where $\mathbf{A}_\parallel$ vanishes,

$$\boldsymbol{\nabla}\cdot\mathbf{A}(\mathbf{r},t) = 0 \quad \text{(Coulomb gauge)}. \tag{1.590}$$

Using (1.588-1.590) we obtain the Lagrangian

$$\begin{aligned} L = L'_m + L_c + \int d^3r \Big\{ \frac{\epsilon_0}{2}\Big[\big(\partial_t\mathbf{A}_\perp(\mathbf{r},t)\big)^2 - c_l^2\big(\boldsymbol{\nabla}\times\mathbf{A}_\perp(\mathbf{r},t)\big)^2\Big] \\ + \mathbf{J}_\perp(\mathbf{r},t)\cdot\mathbf{A}_\perp(\mathbf{r},t)\Big\}, \end{aligned} \tag{1.591}$$

where[77]

$$L_c = -\frac{1}{2}\sum_{\mathbf{k}}\frac{\rho(-\mathbf{k},t)\rho(\mathbf{k},t)}{\epsilon_0\mathbf{k}^2} = -\frac{1}{2}\int d^3r d^3r' \frac{\rho(\mathbf{r},t)\rho(\mathbf{r}',t)}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|} \tag{1.592}$$

is (minus) the Coulomb energy of the charge distribution $\rho(\mathbf{r},t)$.

[77] The Fourier transformed variables are defined by $\rho(\mathbf{r},t) = \frac{1}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}\rho(\mathbf{k},t)$, etc.

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Let us discuss in more detail the role of the longitudinal component of the vector potential. Equation (1.588) enables to eliminate ϕ and rewrite the Lagrangian as

$$L = L'_m + L_c + \frac{\epsilon_0}{2}\sum_{\mathbf{k}}\left[\dot{\mathbf{A}}_\perp(-\mathbf{k},t)\cdot\dot{\mathbf{A}}_\perp(\mathbf{k},t) - c_l^2\mathbf{k}^2\mathbf{A}_\perp(-\mathbf{k},t)\cdot\mathbf{A}_\perp(\mathbf{k},t)\right]$$
$$+\sum_{\mathbf{k}}\Big[\mathbf{J}_\perp(-\mathbf{k},t)\cdot\mathbf{A}_\perp(\mathbf{k},t) + \mathbf{J}_\parallel(-\mathbf{k},t)\cdot\mathbf{A}_\parallel(\mathbf{k},t) - i\rho(-\mathbf{k},t)\frac{\mathbf{k}}{\mathbf{k}^2}\cdot\dot{\mathbf{A}}_\parallel(\mathbf{k},t)\Big]. \quad (1.593)$$

The Lagrange equation for $\mathbf{A}_\parallel$,

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\mathbf{A}}_\parallel(\mathbf{k},t)} = \frac{\partial L}{\partial\mathbf{A}_\parallel(\mathbf{k},t)}, \quad (1.594)$$

gives

$$i\frac{\mathbf{k}}{\mathbf{k}^2}\dot{\rho}(\mathbf{k},t) = \mathbf{J}_\parallel(\mathbf{k},t), \quad \text{i.e.} \quad \partial_t\rho(\mathbf{r},t) + \boldsymbol{\nabla}\cdot\mathbf{J}_\parallel(\mathbf{r},t) = 0. \quad (1.595)$$

This is a continuity equation, not an equation of motion for $\mathbf{A}_\parallel$. Using (1.595), we find

$$\sum_{\mathbf{k}}\Big[\mathbf{J}_\parallel(-\mathbf{k},t)\cdot\mathbf{A}_\parallel(\mathbf{k},t) - i\rho(-\mathbf{k},t)\frac{\mathbf{k}}{\mathbf{k}^2}\cdot\dot{\mathbf{A}}_\parallel(\mathbf{k},t)\Big]$$
$$= \sum_{\mathbf{k}}\frac{1}{\mathbf{k}^2}\left[\mathbf{k}\cdot\mathbf{J}_\parallel(-\mathbf{k},t)\mathbf{k}\cdot\mathbf{A}_\parallel(\mathbf{k},t) - i\rho(-\mathbf{k},t)\mathbf{k}\cdot\dot{\mathbf{A}}_\parallel(\mathbf{k},t)\right]$$
$$= \sum_{\mathbf{k}}\frac{1}{\mathbf{k}^2}\left[-\dot{\rho}(-\mathbf{k},t)i\mathbf{k}\cdot\mathbf{A}_\parallel(\mathbf{k},t) - \rho(-\mathbf{k},t)i\mathbf{k}\cdot\dot{\mathbf{A}}_\parallel(\mathbf{k},t)\right]$$
$$= -i\frac{d}{dt}\sum_{\mathbf{k}}\frac{1}{\mathbf{k}^2}\rho(-\mathbf{k},t)\mathbf{k}\cdot\mathbf{A}_\parallel(\mathbf{k},t), \quad (1.596)$$

and the Lagrangian becomes

$$L = L'_m + L_c + \frac{\epsilon_0}{2}\sum_{\mathbf{k}}\left[\dot{\mathbf{A}}_\perp(-\mathbf{k},t)\cdot\dot{\mathbf{A}}_\perp(\mathbf{k},t) - c_l^2\mathbf{A}_\perp(-\mathbf{k},t)\cdot\mathbf{A}_\perp(\mathbf{k},t)\right]$$
$$+\sum_{\mathbf{k}}\mathbf{J}_\perp(-\mathbf{k},t)\cdot\mathbf{A}_\perp(\mathbf{k},t) - i\frac{d}{dt}\sum_{\mathbf{k}}\frac{1}{\mathbf{k}^2}\rho(-\mathbf{k},t)\mathbf{k}\cdot\mathbf{A}_\parallel(\mathbf{k},t). \quad (1.597)$$

The last term being a total derivative, it does not affect the equations of motion and can therefore be removed from the Lagrangian which then becomes independent of $\mathbf{A}_\parallel$. Thus the longitudinal component of the vector potential does not affect the dynamics of the system and is not a true dynamical variable. Of course, the possibility to arbitrarily choose $\mathbf{A}_\parallel$ is nothing but a consequence of gauge invariance.

The Hamiltonian is obtained by introducing the momenta of the particles,

$$\mathbf{p}_i = \frac{\partial L}{\partial\dot{\mathbf{r}}_i} = m\dot{\mathbf{r}}_i + e\mathbf{A}_\perp(\mathbf{r}_i,t), \quad (1.598)$$

and the momentum conjugate to $\mathbf{A}_\perp$,

$$\boldsymbol{\Pi}_\perp(\mathbf{r},t) = \frac{\delta L}{\delta\dot{\mathbf{A}}_\perp(\mathbf{r},t)}. \quad (1.599)$$

Since $\mathbf{A}_\perp(\mathbf{r},t)$ is transverse, we cannot vary its components independently. We must therefore obtain $\boldsymbol{\Pi}_\perp$ in reciprocal space, using

$$\frac{\epsilon_0}{2}\int d^3r\,\dot{\mathbf{A}}_\perp(\mathbf{r},t)^2 = \frac{\epsilon_0}{2}\sum_{\mathbf{k}}\dot{\mathbf{A}}_\perp(-\mathbf{k},t)\cdot\dot{\mathbf{A}}_\perp(\mathbf{k},t)$$

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$$= \frac{\epsilon_0}{2} \sum_{\substack{\mathbf{k} \\ \epsilon=\epsilon_1,\epsilon_2}} \dot{A}_{\perp\epsilon}(-\mathbf{k},t)\dot{A}_{\perp\epsilon}(\mathbf{k},t), \tag{1.600}$$

where $A_{\perp\epsilon_1}$ and $A_{\perp\epsilon_2}$ denote the two independent components of $\mathbf{A}_\perp$,

$$\mathbf{A}_\perp(\mathbf{k},t) = A_{\perp\epsilon_1}(\mathbf{k},t)\boldsymbol{\epsilon}_1 + A_{\perp\epsilon_2}(\mathbf{k},t)\boldsymbol{\epsilon}_2, \tag{1.601}$$

with $\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2, \mathbf{k}$ mutually orthogonal and $\boldsymbol{\epsilon}_1^2 = \boldsymbol{\epsilon}_2^2 = 1$. The conjugate momentum is then given by

$$\Pi_{\perp\epsilon}(\mathbf{k},t) = \frac{\partial L}{\partial \dot{A}_{\perp\epsilon}(-\mathbf{k},t)} = \epsilon_0 \dot{A}_{\perp\epsilon}(\mathbf{k},t) \tag{1.602}$$

(see Eq. (1.A.46) in Appendix 1.A), so that

$$\boldsymbol{\Pi}_\perp(\mathbf{r},t) = \epsilon_0 \dot{\mathbf{A}}_\perp(\mathbf{r},t). \tag{1.603}$$

$\boldsymbol{\Pi}_\perp$ is transverse: $\boldsymbol{\nabla}\cdot\boldsymbol{\Pi}_\perp(\mathbf{r},t) = 0$. In the Coulomb gauge, the Hamiltonian is therefore defined by

$$\begin{aligned} H &= \sum_i \mathbf{p}_i(t)\cdot\dot{\mathbf{r}}_i(t) + \int d^3r\, \boldsymbol{\Pi}_\perp(\mathbf{r},t)\cdot\dot{\mathbf{A}}_\perp(\mathbf{r},t) - L, \\ &= \sum_i \frac{[\mathbf{p}_i(t) - e\mathbf{A}_\perp(\mathbf{r}_i,t)]^2}{2m} + H_c + \frac{1}{2\epsilon_0}\int d^3r \left[\boldsymbol{\Pi}_\perp(\mathbf{r},t)^2 + \epsilon_0^2 c_l^2 \big(\boldsymbol{\nabla}\times\mathbf{A}_\perp(\mathbf{r},t)\big)^2\right], \end{aligned} \tag{1.604}$$

where $H_c = -L_c$ is the Coulomb energy. The electromagnetic field is now given by

$$\begin{array}{lll} \mathbf{E}_\parallel(\mathbf{k},t) = -i\mathbf{k}\dfrac{\rho(\mathbf{k},t)}{\epsilon_0\mathbf{k}^2}, & & \mathbf{E}_\parallel(\mathbf{r},t) = \int d^3r' \dfrac{\rho(\mathbf{r}',t)(\mathbf{r}-\mathbf{r}')}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|^3}, \\ \mathbf{E}_\perp(\mathbf{k},t) = -\dfrac{1}{\epsilon_0}\boldsymbol{\Pi}_\perp(\mathbf{k},t), & \text{i.e.} & \mathbf{E}_\perp(\mathbf{r},t) = -\dfrac{1}{\epsilon_0}\boldsymbol{\Pi}_\perp(\mathbf{r},t), \\ \mathbf{B}(\mathbf{k},t) = i\mathbf{k}\times\mathbf{A}_\perp(\mathbf{k},t), & & \mathbf{B}(\mathbf{r},t) = \boldsymbol{\nabla}\times\mathbf{A}_\perp(\mathbf{r},t). \end{array} \tag{1.605}$$

The Hamiltonian of the particles (i.e. without the Hamiltonian $\hat{H}_{\rm em}$ of the free field given by the last line of (1.604)) can be written as

$$H = H'_m + H_c - \int d^3r\, \mathbf{j}(\mathbf{r},t)\cdot\mathbf{A}_\perp(\mathbf{r},t) + \frac{e}{2m}\int d^3r\, \rho(\mathbf{r},t)\mathbf{A}_\perp^2(\mathbf{r},t), \tag{1.606}$$

where

$$\mathbf{j}(\mathbf{r},t) = \frac{e}{m}\sum_i \mathbf{p}_i\delta(\mathbf{r}-\mathbf{r}_i) \tag{1.607}$$

is the paramagnetic part of the current

$$\mathbf{J}(\mathbf{r},t) = e\sum_i \dot{\mathbf{r}}_i\delta(\mathbf{r}-\mathbf{r}_i) = \frac{e}{m}\sum_i [\mathbf{p}_i - e\mathbf{A}_\perp(\mathbf{r}_i,t)]\delta(\mathbf{r}-\mathbf{r}_i) \tag{1.608}$$

and $H'_m = \sum_i \mathbf{p}_i^2/2m$ is the Hamiltonian of the matter in the absence of electromagnetic interactions.

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1.9.2 Canonical quantization in the Coulomb gauge

We are now in a position to quantize the theory by promoting the variables $(\mathbf{r}_i, \mathbf{p}_i)$ and $(\mathbf{A}_\perp, \mathbf{\Pi}_\perp)$ to operators. The position and momentum operators $\hat{\mathbf{r}}_i$ and $\hat{\mathbf{p}}_i$ satisfy the usual commutations relations (Appendix 1.B). As for the electromagnetic field, there are two independent dynamical variables $A_{\perp\epsilon_1}$ and $A_{\perp\epsilon_2}$, and their conjugate momenta $\Pi_{\perp\epsilon_1}$ and $\Pi_{\perp\epsilon_2}$, for each wavevector $\mathbf{k}$. The reality of $\mathbf{A}_\perp(\mathbf{r},t)$ and $\mathbf{\Pi}_\perp(\mathbf{r},t)$ implies $A^*_{\perp\epsilon}(\mathbf{k},t) = A_{\perp\epsilon}(-\mathbf{k},t)$ and $\Pi^*_{\perp\epsilon}(\mathbf{k},t) = \Pi_{\perp\epsilon}(-\mathbf{k},t)$. The corresponding operators satisfy

$$\begin{aligned} \left[\hat{A}_{\perp\epsilon}(\mathbf{k}), \hat{A}_{\perp\epsilon'}(\mathbf{k}')\right] &= \left[\hat{\Pi}_{\perp\epsilon}(\mathbf{k}), \hat{\Pi}_{\perp\epsilon'}(\mathbf{k}')\right] = 0, \\ \left[\hat{A}_{\perp\epsilon}(\mathbf{k}), \hat{\Pi}^\dagger_{\perp\epsilon'}(\mathbf{k}')\right] &= i\delta_{\epsilon,\epsilon'}\delta_{\mathbf{k},\mathbf{k}'}, \end{aligned} \tag{1.609}$$

where $\hat{\Pi}^\dagger_{\perp\epsilon}(\mathbf{k}) = \hat{\Pi}_{\perp\epsilon}(-\mathbf{k})$ and $\hat{A}^\dagger_{\perp\epsilon}(\mathbf{k}) = \hat{A}_{\perp\epsilon}(-\mathbf{k})$. Note that we consider here the Schrödinger picture where the operators do not depend on time (Appendix 1.B). The quantum Hamiltonian $\hat{H}$ is obtained from (1.604) by replacing the classical variables by the corresponding operators. Using

$$\begin{aligned} \hat{A}_{\perp\mu}(\mathbf{k}) &= \sum_\epsilon \hat{A}_{\perp\epsilon}(\mathbf{k})\epsilon_\mu, \\ \hat{\Pi}_{\perp\mu}(\mathbf{k}) &= \sum_\epsilon \hat{\Pi}_{\perp\epsilon}(\mathbf{k})\epsilon_\mu \end{aligned} \tag{1.610}$$

($\mu = x, y, z$), we obtain[78]

$$\left[\hat{A}_{\perp\mu}(\mathbf{k}), \hat{\Pi}^\dagger_{\perp\mu'}(\mathbf{k}')\right] = i\delta_{\mathbf{k},\mathbf{k}'}\sum_\epsilon \epsilon_\mu\epsilon_{\mu'} = i\delta_{\mathbf{k},\mathbf{k}'}\left(\delta_{\mu,\mu'} - k_\mu k_{\mu'}/\mathbf{k}^2\right). \tag{1.611}$$

In direct space, the commutation relations (1.609) become

$$\begin{aligned} \left[\hat{A}_{\perp\mu}(\mathbf{r}), \hat{A}_{\perp\mu'}(\mathbf{r}')\right] &= \left[\hat{\Pi}_{\perp\mu}(\mathbf{r}), \hat{\Pi}_{\perp\mu'}(\mathbf{r}')\right] = 0, \\ \left[\hat{A}_{\perp\mu}(\mathbf{r}), \hat{\Pi}_{\perp\mu'}(\mathbf{r}')\right] &= i\delta^\perp_{\mu,\mu'}(\mathbf{r}-\mathbf{r}'). \end{aligned} \tag{1.612}$$

The "transverse delta function"

$$\delta^\perp_{\mu,\mu'}(\mathbf{r}) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}}\left(\delta_{\mu,\mu'} - \frac{k_\mu k_{\mu'}}{\mathbf{k}^2}\right) = \delta_{\mu,\mu'}\delta(\mathbf{r}) - \frac{\partial}{\partial r_\mu}\frac{\partial}{\partial r_{\mu'}}\frac{1}{4\pi|\mathbf{r}|} \tag{1.613}$$

appearing in (1.612) (in place of $\delta_{\mu,\mu'}\delta(\mathbf{r}-\mathbf{r}')$) reflects the fact that the three components of $\hat{\mathbf{A}}_\perp(\mathbf{r})$ and $\hat{\mathbf{\Pi}}_\perp(\mathbf{r})$ are not independent but constrained by the transversality conditions $\boldsymbol{\nabla}\cdot\hat{\mathbf{A}}_\perp(\mathbf{r}) = \boldsymbol{\nabla}\cdot\hat{\mathbf{\Pi}}_\perp(\mathbf{r}) = 0$.

1.9.2.1 Second quantization and photons

Let us consider the Hamiltonian of the free field (no coupling to the matter)

$$\hat{H}_{\text{em}} = \frac{1}{2\epsilon_0}\sum_{\mathbf{k},\epsilon}\left[\hat{\Pi}^\dagger_{\perp\epsilon}(\mathbf{k})\hat{\Pi}_{\perp\epsilon}(\mathbf{k}) + \epsilon_0^2 c_l^2 \mathbf{k}^2 \hat{A}^\dagger_{\perp\epsilon}(\mathbf{k})\hat{A}_{\perp\epsilon}(\mathbf{k})\right]. \tag{1.614}$$

[78] Since $(\mathbf{k}, \epsilon_1, \epsilon_2)$ form an orthonormal basis in reciprocal space, $\epsilon_{1\mu}\epsilon_{1\mu'} + \epsilon_{2\mu}\epsilon_{2\mu'} + k_\mu k_{\mu'}/\mathbf{k}^2 = \delta_{\mu,\mu'}$.

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Given the commutation relations (1.609) between $\hat{A}_{\perp\epsilon}$ and $\hat{\Pi}_{\perp\epsilon}$, this Hamiltonian describes a sum of independent harmonic oscillators, one for each mode $(\mathbf{k},\epsilon)$. It can be diagonalized by introducing the ladder operators

$$\begin{aligned}\hat{a}_\epsilon(\mathbf{k}) &= \sqrt{\frac{\epsilon_0\omega_\mathbf{k}}{2}}\Big[\hat{A}_{\perp\epsilon}(\mathbf{k}) + \frac{i}{\epsilon_0\omega_\mathbf{k}}\hat{\Pi}_{\perp\epsilon}(\mathbf{k})\Big],\\ \hat{a}^\dagger_\epsilon(\mathbf{k}) &= \sqrt{\frac{\epsilon_0\omega_\mathbf{k}}{2}}\Big[\hat{A}^\dagger_{\perp\epsilon}(\mathbf{k}) - \frac{i}{\epsilon_0\omega_\mathbf{k}}\hat{\Pi}^\dagger_{\perp\epsilon}(\mathbf{k})\Big],\end{aligned} \tag{1.615}$$

with $\omega_\mathbf{k} = c_l|\mathbf{k}|$. $\hat{a}$ and $\hat{a}^\dagger$ satisfy the commutation relations

$$\begin{aligned}\big[\hat{a}_\epsilon(\mathbf{k}),\hat{a}_{\epsilon'}(\mathbf{k}')\big] &= \big[\hat{a}^\dagger_\epsilon(\mathbf{k}),\hat{a}^\dagger_{\epsilon'}(\mathbf{k}')\big] = 0,\\ \big[\hat{a}_\epsilon(\mathbf{k}),\hat{a}^\dagger_{\epsilon'}(\mathbf{k}')\big] &= \delta_{\epsilon,\epsilon'}\delta_{\mathbf{k},\mathbf{k}'},\end{aligned} \tag{1.616}$$

and $\hat{H}_{\rm em}$ takes the form

$$\hat{H}_{\rm em} = \sum_{\mathbf{k},\epsilon}\omega_\mathbf{k}\left(\hat{a}^\dagger_\epsilon(\mathbf{k})\hat{a}_\epsilon(\mathbf{k}) + \frac{1}{2}\right). \tag{1.617}$$

The Hamiltonian of the free field is similar to the one obtained for the quantum harmonic string in Sec. 1.1.2. The elementary excitations of the quantized field (photons) are created and annihilated by the operators $\hat{a}^\dagger$ and $\hat{a}$. The eigenstates of $\hat{H}_{\rm em}$ can be written as

$$\begin{aligned}|n_1\cdots n_i\cdots\rangle &= \prod_i \frac{(\hat{a}^\dagger_i)^{n_i}}{\sqrt{n_i!}}|\text{vac}\rangle,\\ \hat{H}_{\rm em}|n_1\cdots n_i\cdots\rangle &= \left[\sum_j \omega_j\left(n_j+\frac{1}{2}\right)\right]|n_1\cdots n_i\cdots\rangle,\end{aligned} \tag{1.618}$$

where n_i denotes the number of photons in the mode $i\equiv(\mathbf{k}_i,\epsilon_i)$ and $|\text{vac}\rangle$ the (normalized) vacuum state: $\hat{a}_i|\text{vac}\rangle = 0$. From (1.605) and (1.615), we can express the transverse field in terms of the photon operators,

$$\begin{aligned}\hat{\mathbf{E}}_\perp(\mathbf{r}) &= \frac{i}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k},\epsilon}\sqrt{\frac{\omega_\mathbf{k}}{2\epsilon_0}}\Big[\hat{a}_\epsilon(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{r}} - \hat{a}^\dagger_\epsilon(\mathbf{k})e^{-i\mathbf{k}\cdot\mathbf{r}}\Big]\boldsymbol{\epsilon},\\ \hat{\mathbf{B}}(\mathbf{r}) &= \frac{i}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k},\epsilon}\frac{1}{\sqrt{2\epsilon_0\omega_\mathbf{k}}}\Big[\hat{a}_\epsilon(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{r}} - \hat{a}^\dagger_\epsilon(\mathbf{k})e^{-i\mathbf{k}\cdot\mathbf{r}}\Big]\mathbf{k}\times\boldsymbol{\epsilon}.\end{aligned} \tag{1.619}$$

Coherent states. As with any bosonic operator, we can define the (normalized) coherent state

$$|z\rangle = \exp\Big\{-\sum_i\left(\frac{|z_i|^2}{2} - z_i\hat{a}^\dagger_i\right)\Big\}|\text{vac}\rangle, \tag{1.620}$$

where z_i is a c-number. One easily finds[79]

$$\begin{aligned}\langle z|\hat{A}_i|z\rangle &= \sqrt{\frac{2}{\epsilon_0\omega_i}}\Re(z_i), \qquad \langle z|\hat{A}_i^2|z\rangle = \frac{1}{2\epsilon_0\omega_i}[4\Re(z_i)^2+1],\\ \langle z|\hat{\Pi}_i|z\rangle &= \sqrt{2\epsilon_0\omega_i}\Im(z_i), \qquad \langle z|\hat{\Pi}_i^2|z\rangle = \frac{\epsilon_0\omega_i}{2}[4\Im(z_i)^2+1],\end{aligned} \tag{1.621}$$

[79] These relations can be deduced from (1.228) and (1.229) by the correspondence $\hat{p}\to\hat{\Pi}_i$, $\hat{x}\to\hat{A}_i$, $m\to\epsilon_0$ and $\omega\to\omega_i$.

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and therefore

$$\Delta A_i^2 = \frac{1}{2\epsilon_0\omega_i}, \qquad \Delta\Pi_i^2 = \frac{\epsilon_0\omega_i}{2}. \tag{1.622}$$

The coherent state minimizes the uncertainty relation $\Delta A_i \Delta\Pi_i \geq \hbar/2$ (we have restored Planck's constant), and is therefore the quantum state which is closest to the classical electromagnetic field. Since the coherent state evolves in time as

$$e^{-i\hat{H}t}|z\rangle = e^{-\frac{i}{2}\sum_i \omega_i t}|z(t)\rangle \tag{1.623}$$

(see Eq. (1.231)) with $z_i(t) = z_i e^{-i\omega_i t}$, the expectation value of the transverse field reads

$$\begin{aligned}
\langle z(t)|\hat{\mathbf{E}}_\perp(\mathbf{r})|z(t)\rangle &= \frac{i}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k},\epsilon}\sqrt{\frac{\omega_\mathbf{k}}{2\epsilon_0}}\Big[z_\epsilon(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} - \text{c.c.}\Big]\boldsymbol{\epsilon}, \\
\langle z(t)|\hat{\mathbf{B}}(\mathbf{r})|z(t)\rangle &= \frac{i}{\sqrt{\mathcal{V}}}\sum_{\mathbf{k},\epsilon}\frac{1}{\sqrt{2\epsilon_0\omega_\mathbf{k}}}\Big[z_\epsilon(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} - \text{c.c.}\Big]\mathbf{k}\times\boldsymbol{\epsilon}.
\end{aligned} \tag{1.624}$$

One easily verifies that these mean values are solutions of the Maxwell equations. By virtue of (1.622), the fluctuations $\Delta\mathbf{E}_\perp(\mathbf{r})^2$ and $\Delta\mathbf{B}(\mathbf{r})^2$ are the smallest ones compatible with the Heisenberg uncertainty relations. When the average number of photons $n_i = \langle\hat{a}_i^\dagger\hat{a}_i\rangle = |z_i|^2$ is large, $|\langle\hat{A}_i\rangle(t)| \gg \Delta A_i$ and $|\langle\Pi_i\rangle(t)| \gg \Delta\Pi_i$ for most times t and the electromagnetic field in the mode $i = (\mathbf{k},\epsilon)$ behaves classically.

1.9.3 Functional integral in the Coulomb gauge

We are now in a position to write the partition function $Z = \operatorname{Tr} e^{-\beta\hat{H}}$ as a functional integral. In the free field case, the Hamiltonian (1.614) reduces to a sum of independent harmonic oscillators. Following the general method introduced in Sec. 1.1.1, and in particular the derivation of the functional integral of the quantum harmonic string in Sec. 1.1, one finds

$$Z_{\text{em}} = \int \mathcal{D}[\mathbf{A}_\perp]\, e^{-S_{\text{em}}[\mathbf{A}_\perp]}, \tag{1.625}$$

with the boundary condition $\mathbf{A}_\perp(\mathbf{r},\beta) = \mathbf{A}_\perp(\mathbf{r},0)$ and the Euclidean action

$$\begin{aligned}
S_{\text{em}}[\mathbf{A}_\perp] &= \int_0^\beta d\tau \sum_{\mathbf{k},\epsilon}\frac{\epsilon_0}{2}\Big[|\partial_\tau A_{\perp\epsilon}(\mathbf{k},\tau)|^2 + c_l^2\mathbf{k}^2|A_{\perp\epsilon}(\mathbf{k},\tau)|^2\Big] \\
&= \int_0^\beta d\tau \int d^3r\, \frac{\epsilon_0}{2}\Big[\big(\partial_\tau\mathbf{A}_\perp(x)\big)^2 + c_l^2\big(\boldsymbol{\nabla}\times\mathbf{A}_\perp(x)\big)^2\Big].
\end{aligned} \tag{1.626}$$

The measure in the functional integral is defined by

$$\mathcal{D}[\mathbf{A}_\perp] = \prod_{\mathbf{k},\epsilon,\tau}{}' dA^*_{\perp\epsilon}(\mathbf{k},\tau)dA_{\perp\epsilon}(\mathbf{k},\tau) = \mathcal{D}[\mathbf{A}]\prod_{\mathbf{r},\tau}\delta\big(\boldsymbol{\nabla}\cdot\mathbf{A}(\mathbf{r},\tau)\big) \tag{1.627}$$

(up to an irrelevant multiplicative constant), where the prime means that the product over $\mathbf{k}$ is restricted to half the reciprocal space in order to take into account the constraint $A^*_{\perp\epsilon}(\mathbf{k}) = A_{\perp\epsilon}(-\mathbf{k})$. The discretization of space (necessary to properly define the integration measure) introduces an irrelevant multiplicative constant in the continuum limit.

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When the coupling to the matter is introduced, the Hamiltonian is given by

$$\hat{H} = \hat{H}'_m + \hat{H}_c - \int d^3r\, \hat{\mathbf{j}}(\mathbf{r},t)\cdot\hat{\mathbf{A}}_\perp(\mathbf{r},t) + \frac{e}{2m}\int d^3r\, \hat{\rho}(\mathbf{r},t)\hat{\mathbf{A}}_\perp^2(\mathbf{r},t) + \hat{H}_{\rm em} \tag{1.628}$$

(see Eq. (1.606)), where

$$\hat{H}_c = \frac{1}{2}\int d^3r d^3r' \frac{\hat{\rho}(\mathbf{r})\hat{\rho}(\mathbf{r}')}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|}. \tag{1.629}$$

The degrees of freedom of the particles can be described by field operators (Sec. 1.2.2). The charge and current density operators are then defined by

$$\begin{aligned}
\hat{\rho}(\mathbf{r}) &= e\sum_\sigma \hat{\psi}^\dagger_\sigma(\mathbf{r})\hat{\psi}_\sigma(\mathbf{r}),\\
\hat{\mathbf{J}}(\mathbf{r}) &= -\frac{ie}{2m}\sum_\sigma\left[\hat{\psi}^\dagger_\sigma(\mathbf{r})\boldsymbol{\nabla}\hat{\psi}_\sigma(\mathbf{r}) - \text{H.c.}\right] - \frac{e}{m}\hat{\rho}(\mathbf{r})\hat{\mathbf{A}}_\perp(\mathbf{r})\\
&= \hat{\mathbf{j}}(\mathbf{r}) - \frac{e}{m}\hat{\rho}(\mathbf{r})\hat{\mathbf{A}}_\perp(\mathbf{r}),
\end{aligned} \tag{1.630}$$

for spin-σ particles (Sec. 1.2.2). $\hat{\mathbf{j}}$ is the paramagnetic part of the current and $\hat{H}'_m = \hat{H}_m - \hat{H}_c$ the Hamiltonian of the particles in the absence of electromagnetic interactions. Proceeding as in Sec. 1.4, we derive from (1.628) the coherent-state functional integral

$$Z = \int \mathcal{D}[\psi^*,\psi,\mathbf{A}_\perp]\, e^{-S[\psi^*,\psi,\mathbf{A}_\perp]} \tag{1.631}$$

with the action

$$\begin{aligned}
S[\psi^*,\psi,\mathbf{A}_\perp] &= S_{\rm em}[\mathbf{A}_\perp] + S_m[\psi^*,\psi]\\
&\quad + \int_0^\beta d\tau \int d^3r \left[\frac{e}{2m}\rho(x)\mathbf{A}_\perp^2(x) - \hat{\mathbf{j}}(x)\cdot\mathbf{A}_\perp(x)\right]
\end{aligned} \tag{1.632}$$

and the boundary conditions $\mathbf{A}_\perp(\mathbf{r},\beta) = \mathbf{A}_\perp(\mathbf{r},0)$, $\psi^{(*)}_\sigma(\mathbf{r},\beta) = \zeta\psi^{(*)}_\sigma(\mathbf{r},0)$.

It is now possible to reintroduce the scalar potential by rewriting the Coulomb interaction as a functional integral over an auxiliary real bosonic field $\phi(x)$,[80]

$$\begin{aligned}
&\exp\left\{-\frac{1}{2}\sum_k \frac{\rho(-k)\rho(k)}{\epsilon_0\mathbf{k}^2}\right\}\\
&\qquad = \int \mathcal{D}[\phi]\exp\left\{-\frac{1}{2}\sum_k\left[\epsilon_0\mathbf{k}^2\phi(-k)\phi(k) + 2i\phi(-k)\rho(k)\right]\right\}\\
&\qquad = \int \mathcal{D}[\phi]\exp\left\{-\frac{1}{2}\int_0^\beta d\tau\int d^3r\left[\epsilon_0\big(\boldsymbol{\nabla}\phi(x)\big)^2 + 2i\phi(x)\rho(x)\right]\right.
\end{aligned} \tag{1.633}$$

(again we neglect an irrelevant multiplicative term). Equation (1.633) is obtained using the basic results of Gaussian integration derived in Appendix 1.E. The action becomes

$$S[\psi^*,\psi,\phi,\mathbf{A}_\perp] = S_{\rm em}[\phi,\mathbf{A}_\perp] + S'_m[\psi^*,\psi]$$

[80] The transformation (1.633) is known as a Hubbard-Stratonovich transformation. It will be encountered many times in the following chapters (see, e.g., Sec. 5.2.3.1 or 6.3.3.1).

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$$+\int_0^\beta d\tau \int d^3r \left[\frac{e}{2m}\rho(x)\mathbf{A}_\perp^2(x) + i\phi(x)\rho(x) - \mathbf{j}(x)\cdot\mathbf{A}_\perp(x)\right], \quad (1.634)$$

with the action of the electromagnetic potentials given by

$$S_{\rm em}[\phi,\mathbf{A}_\perp] = \int_0^\beta d\tau \int d^3r\, \frac{\epsilon_0}{2}\Big[\big(\boldsymbol{\nabla}\phi(x)\big)^2 + \big(\partial_\tau \mathbf{A}_\perp(x)\big)^2 + c_l^2\big(\boldsymbol{\nabla}\times\mathbf{A}_\perp(x)\big)^2\Big]. \quad (1.635)$$

$S'_m = S_m - S_c$ denotes the action of the particles with no electromagnetic interactions. According to (1.634), the field $i\phi(x)$ plays the role of the scalar potential, since it couples to the particle density. The factor i cannot be eliminated by a redefinition of the ϕ field, $i\phi \to \phi$, since this would change the sign of $(\boldsymbol{\nabla}\phi)^2$ in $S_{\rm em}$ and makes the functional integral ill-defined. Thus, in the Euclidean formalism, the mean values of the electric and magnetic fields are

$$\begin{aligned}\langle\mathbf{E}(x)\rangle &= -\boldsymbol{\nabla}\langle i\phi(x)\rangle - i\partial_\tau\langle\mathbf{A}_\perp(x)\rangle,\\ \langle\mathbf{B}(x)\rangle &= \boldsymbol{\nabla}\times\langle\mathbf{A}_\perp(x)\rangle,\end{aligned} \quad (1.636)$$

or, in Fourier space,

$$\begin{aligned}\langle\mathbf{E}(k)\rangle &= \mathbf{k}\langle\phi(k)\rangle - \omega_n\langle\mathbf{A}_\perp(k)\rangle,\\ \langle\mathbf{B}(k)\rangle &= i\mathbf{k}\times\langle\mathbf{A}_\perp(k)\rangle.\end{aligned} \quad (1.637)$$

According to (1.605), the mean value of the longitudinal electric field is given by

$$\langle\mathbf{E}_\parallel(k)\rangle = -i\mathbf{k}\frac{\langle\rho(k)\rangle}{\epsilon_0\mathbf{k}^2}. \quad (1.638)$$

The mean particle density can be calculated from

$$\langle\rho(k)\rangle = \frac{\partial \ln Z[h]}{\partial h(-k)}\bigg|_{h=0}, \quad (1.639)$$

where $Z[h]$ is the partition function obtained by adding to the action the source term $-\sum_k h(-k)\rho(k)$. After the Hubbard-Stratonovich transformation (1.633), the action reads

$$\begin{aligned}S &= \frac{1}{2}\sum_k \big[\epsilon_0\mathbf{k}^2\phi(-k)\phi(k) + 2i\phi(-k)\rho(k) - 2h(-k)\rho(k)\big] + \cdots\\ &= \frac{1}{2}\sum_k \big[\epsilon_0\mathbf{k}^2\big(\phi(-k) - ih(-k)\big)\big(\phi(k) - ih(k)\big) + 2i\phi(-k)\rho(k)\big]\cdots\end{aligned} \quad (1.640)$$

(we do not write the part of the action which is independent of h). The second line of (1.640) is obtained by shifting the field $\phi \to \phi - ih$ in the functional integral. From (1.639) and (1.640), we then deduce

$$\langle\rho(k)\rangle = \epsilon_0\mathbf{k}^2\langle i\phi(k)\rangle, \qquad \langle\mathbf{E}_\parallel(k)\rangle = \mathbf{k}\langle\phi(k)\rangle \quad (1.641)$$

in agreement with (1.637).

More generally, one can define the propagators of the electromagnetic potentials,

$$\begin{aligned}D_{00}(k) &= \langle\phi(k)\phi(-k)\rangle,\\ D_{0\mu}(k) &= \langle\phi(k)A_\mu(-k)\rangle,\\ D_{\mu\nu}(k) &= \langle A_\mu(k)A_\nu(-k)\rangle = \frac{k_\mu k_\nu}{\mathbf{k}^2}D_\parallel(k) + \left(\delta_{\mu,\nu} - \frac{k_\mu k_\nu}{\mathbf{k}^2}\right)D_\perp(k)\end{aligned} \quad (1.642)$$

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($\mu = x, y, z$), where $D_\parallel(k)$ and $D_\perp(k)$ are functions of $(|\mathbf{k}|, i\omega_n)$. In (1.642), we have taken advantage of the homogeneity and isotropy of the system to write the propagator $D_{\mu\nu}$ in terms of its longitudinal and transverse parts. In the absence of coupling to the particles, one finds

$$\begin{aligned} D_{00}^{(0)}(k) &= \langle \phi(k)\phi(-k)\rangle = \frac{1}{\epsilon_0 \mathbf{k}^2}, \\ D_\perp^{(0)}(k) &= \frac{1}{\epsilon_0(\omega_n^2 + c_l^2\mathbf{k}^2)}. \end{aligned} \tag{1.643}$$

In the Coulomb gauge, the propagator of the vector potential is transverse: $D_\parallel^{(0)}(k) = D_{0\mu}^{(0)}(k) = 0$. $D_{00}^{(0)}(k)$ is the Fourier transform of the Coulomb interaction, whereas $D_\perp^{(0)}(k)$ is the propagator of the (transverse) electromagnetic field in free space. We shall see in Sec. 3.4 that the propagators D_{00} and $D_\perp$ contain important information about the collective excitations of the particles, besides describing the propagation of the electromagnetic field in the matter.

In the Coulomb gauge, the electric field is split into a Coulomb field and a transverse field. Although the Coulomb interaction is instantaneous [Eq. (1.629)], the interaction between two charged particles takes place with a retardation due to the propagation of the field with velocity c_l. This retardation results from an exact compensation of the instantaneous contributions of the Coulomb and transverse fields [18]. In non-relativistic systems, the separation between an instantaneous Coulomb field and a transverse field is very natural since the magnetic interactions mediated by the vector potential are negligible when the characteristic velocity of the particles is small with respect to the velocity of light. To see this, let us integrate out the vector potential in (1.632). If we approximate $\rho(x)$ in the diamagnetic term $\rho(x)\mathbf{A}_\perp^2(x)$ by its mean value en, we are left with a simple Gaussian functional integral (Appendix 1.E.4) which gives a contribution[81]

$$-\frac{1}{2\epsilon_0}\sum_k \frac{\mathbf{j}_\perp(-k)\cdot\mathbf{j}_\perp(k)}{\omega_n^2 + c_l^2\mathbf{k}^2 + \omega_p^2} \tag{1.644}$$

to the action of the particles. $\mathbf{j}_\perp$ is the transverse part of the paramagnetic current and $\omega_p = (ne^2/\epsilon_0 m)^{1/2}$ is called the plasma frequency for reasons given in Chap. 3. For particles with characteristic velocity v, $j_\perp \sim v\rho$ so that the magnetic interaction (1.644) is of order v^2/c_l^2 with respect to to the Coulomb interaction $\rho(k)\rho(-k)/\epsilon_0\mathbf{k}^2$. If $v \ll c_l$, the magnetic interaction can be omitted, which amounts to neglecting the retardation effects in the electromagnetic interactions between particles. When studying systems of charged particles, we will therefore in general consider the action $S_m = S'_m + S_c$ including the Coulomb interaction but not the coupling to the vector potential. Only when considering the response to an external transverse field (i.e. induced by a current distribution external to the system) or the propagation of the electromagnetic field in the matter should we include the transverse vector potential in the action (see e.g. Sec. 3.4.2).

Appendix 1.A Review of classical mechanics[82]

In the following sections, we review the Lagrangian and Hamiltonian formalisms for both a finite number and a continuum of degrees of freedom. In particular we discuss symmetries

[81] The functional integral must be done in Fourier space where the transverse vector potential is defined by its two independent components $A_{\perp\epsilon_1}(\mathbf{k},\tau)$ and $A_{\perp\epsilon_2}(\mathbf{k},\tau)$.

[82] A thorough discussion of classical mechanics can be found in Refs. [22, 23].

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of their consequences.

1.A.1 Lagrangian formalism

Let us consider a system with n degrees of freedom described by n generalized coordinates $q_1 \cdots q_n$. The latter can be represented by a point in a n-dimensional configuration space. The trajectory of the system is determined by the time dependence of the coordinates. The coordinates q_i and the velocities $\dot{q}_i = \partial_t q_i$ form the dynamical variables of the system. The accelerations $\ddot{q}_i$ can be expressed at any time as a function of these variables. The resulting equations of motion are then second-order differential equations in time.

In the Lagrangian approach, the dynamics of the system is determined by the Lagrangian $L(q_i, \dot{q}_i, t)$, a function of the dynamical variables which can also explicitly depend on time. The principle of least action stipulates that the trajectory of the system between times t_1 and t_2, with initial and final coordinates assumed to be known, corresponds to an extremum of the action

$$S = \int_{t_1}^{t_2} dt\, L(q_i, \dot{q}_i, t). \tag{1.A.1}$$

For an arbitrary variation δq_i of the coordinate q_i, the variation of the action is

$$\begin{aligned} \delta S &= \int_{t_1}^{t_2} dt \sum_i \left[\frac{\partial L}{\partial q_i} \delta q_i + \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i \right] \\ &= \int_{t_1}^{t_2} dt \sum_i \left[\frac{\partial L}{\partial q_i} \delta q_i - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \delta q_i \right] + \sum_i \frac{\partial L}{\partial \dot{q}_i} \delta q_i \Big|_{t_1}^{t_2}, \end{aligned} \tag{1.A.2}$$

where we have integrated by parts. The last term in (1.A.2) vanishes since $\delta q_i(t_1) = \delta q_i(t_2) = 0$ (the initial and final coordinates $q_i(t_1)$ and $q_i(t_2)$ are fixed), and we deduce the Euler-Lagrange equations

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0. \tag{1.A.3}$$

The resolution of these n second-order differential equations requires the knowledge of $2n$ parameters: $q_i(t_1)$ and $q_i(t_2)$.

The Lagrangian is not unique. In particular, adding to the Lagrangian a total time derivative,

$$L(q_i, \dot{q}_i, t) \to L(q_i, \dot{q}_i, t) + \frac{d}{dt} F(q_i, t) \tag{1.A.4}$$

($F(q_i, t)$ is an arbitrary function) modifies the action only by the surface term

$$F(q_i(t_2), t_2) - F(q_i(t_1), t_1) \tag{1.A.5}$$

and therefore does not affect the equations of motion ($q_i(t_1)$ and $q_i(t_2)$ being fixed). For the transformed Lagrangian to be a function of the dynamical variables q_i and $\dot{q}_i$, the function $F(q_i, t)$ should not depend on the velocities.

1.A.1.1 Symmetries and invariance

Suppose we describe the system by two sets of coordinates, $\{q_i\}$ and $\{q'_i\}$, corresponding to Lagrangians $L(q_i, \dot{q}_i, t)$ and $L'(q'_i, \dot{q}'_i, t)$, respectively. The transformation $q_i \to q'_i$ is called

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a symmetry transformation.[83] Since both sets $\{q_i\}$ and $\{q'_i\}$ refer to the same system, we must have $L(q_i, \dot{q}_i, t) = L'(q'_i, \dot{q}'_i, t)$ (up to a total time derivative). The equations of motion for the new set of coordinates read

$$\frac{d}{dt}\frac{\partial L'}{\partial \dot{q}'_i} = \frac{\partial L'}{\partial q'_i} \tag{1.A.6}$$

and follow from the principle of least action applied to the action $S'[q'] = \int dt\, L'(q'_i, \dot{q}'_i, t)$. In general, L and L' are different functions.[84] If, however, the action $S'[q']$ and $S[q']$ coincide – i.e. if $L'(q'_i, \dot{q}'_i, t)$ and $L(q'_i, \dot{q}'_i, t)$ differ only by a total derivative – then the equation of motion of q'_i [Eq. (1.A.6)] is identical to that of q_i [Eq. (1.A.3)]; the system is invariant in the symmetry transformation $q_i \to q'_i$.

Let us now consider an infinitesimal (continuous) transformation $q_i \to q_i + \epsilon f_i$ ($\epsilon \to 0$) which leaves the system invariant. To order ϵ, the variation of the Lagrangian can be written as

$$\delta L = \epsilon \sum_i \left(\frac{\partial L}{\partial q_i} f_i + \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i \right) = \epsilon \frac{d}{dt} F(q_i, t), \tag{1.A.7}$$

with $F(q_i, t)$ some function, i.e.

$$\sum_i \left[\frac{\partial L}{\partial q_i} f_i + \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} f_i \right) - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} \right) f_i \right] = \frac{d}{dt} F(q_i, t). \tag{1.A.8}$$

If q_i satisfies the equation of motion (1.A.3), Eq. (1.A.8) implies that

$$\frac{d}{dt}\left(\sum_i \frac{\partial L}{\partial \dot{q}_i} f_i - F(q_i, t) \right) = 0. \tag{1.A.9}$$

Thus, to any transformation which leaves the action invariant, there corresponds a constant of motion (Noether's theorem).

Let us illustrate Noether's theorem with a few examples. We consider a three-dimensional system.

For a system which is translation invariant $L(\mathbf{q}_i + \boldsymbol{\epsilon}, \dot{\mathbf{q}}_i, t) = L(\mathbf{q}_i, \dot{\mathbf{q}}_i, t)$, we can apply (1.A.9) with $F = 0$ and $f_{i\mu} = \delta_{\mu,x}$ if $\boldsymbol{\epsilon}$ is parallel to the x axis ($f_{i\mu} = \delta_{\mu,y}$ if $\boldsymbol{\epsilon}$ is parallel to the y axis, etc.). One then finds that the total momentum of the system, defined as

$$\mathbf{P} = \sum_i \mathbf{p}_i = \sum_i \frac{\partial L}{\partial \dot{\mathbf{q}}_i}, \tag{1.A.10}$$

is conserved. $\mathbf{p}_i$ denotes the momentum of the ith particle (see Sec. 1.A.2).

The Lagrangian of a rotation-invariant system is unchanged by the transformation $\mathbf{q}_i \to \mathbf{q}_i + \theta \mathbf{n} \times \mathbf{q}_i$, i.e. $q_{i\mu} \to q_{i\mu} + \theta \sum_{\nu,\delta} \epsilon_{\mu\nu\delta} n_\nu q_{i\delta}$ (rotation of angle θ about the axis $\mathbf{n}$); $\epsilon_{\mu\nu\delta}$ denotes the antisymmetric tensor. One then finds that the quantity

$$\sum_{i,\mu,\nu,\delta} \frac{\partial L}{\partial \dot{q}_{i\mu}} \epsilon_{\mu\nu\delta} n_\nu q_{i\delta} = \sum_{i,\mu,\nu,\delta} n_\nu \epsilon_{\nu\delta\mu} q_{i\delta} \frac{\partial L}{\partial \dot{q}_{i\mu}} = \sum_i \mathbf{n} \cdot \left(\mathbf{q}_i \times \frac{\partial L}{\partial \dot{\mathbf{q}}_i} \right) \tag{1.A.11}$$

[83] This is the so-called passive point of view where two observers, $\mathcal{O}$ (coordinates $\{q_i\}$) and $\mathcal{O}'$ (coordinates $\{q'_i\}$), look at the same system. Alternatively, one can adopt the active point of view where a symmetry transformation is defined with a single observer $\mathcal{O}$ and two systems $\mathcal{S}$ and $\mathcal{S}'$.

[84] It is crucial here to distinguish between L and L'. In general, however, we shall follow the usual convention to denote a given dynamical variable (such as the Lagrangian or the Hamiltonian) by the same symbol in all coordinate systems.

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is conserved. Since (1.A.11) holds for any unit vector $\mathbf{n}$, we conclude that the total angular momentum

$$\mathbf{J} = \sum_i \mathbf{q}_i \times \frac{\partial L}{\partial \dot{\mathbf{q}}_i} = \sum_i \mathbf{q}_i \times \mathbf{p}_i \tag{1.A.12}$$

is a constant of motion.

If the Lagrangian $L(\mathbf{q}_i, \dot{\mathbf{q}}_i)$ does not depend explicitly on time,

$$L\big(\mathbf{q}_i(t+\epsilon), \dot{\mathbf{q}}_i(t+\epsilon)\big) = L(\mathbf{q}_i + \epsilon\dot{\mathbf{q}}_i, \dot{\mathbf{q}}_i + \epsilon\ddot{\mathbf{q}}_i) = L + \epsilon\frac{dL}{dt} \tag{1.A.13}$$

to first order in ϵ. From (1.A.9) with $\mathbf{f}_i = \dot{\mathbf{q}}_i$ and $F = L$, we deduce that the Hamiltonian of the system

$$H = \sum_i \frac{\partial L}{\partial \dot{\mathbf{q}}_i} \cdot \dot{\mathbf{q}}_i - L = \sum_i \mathbf{p}_i \cdot \dot{\mathbf{q}}_i - L \tag{1.A.14}$$

(which is usually identified with the total energy) is conserved.

Let us finally consider a Galilean transformation $\mathbf{q}_i \to \mathbf{q}_i + \mathbf{v}t$ ($\mathbf{f}_i = \mathbf{v}t$) for a Lagrangian of the form

$$L = \sum_i \frac{1}{2} m\dot{\mathbf{q}}_i^2 - V, \tag{1.A.15}$$

where V depends only on the differences $\mathbf{q}_i - \mathbf{q}_j$. To $\mathcal{O}(\mathbf{v})$, the Lagrangian changes by

$$\delta L = \sum_i \left(\frac{\partial L}{\partial \mathbf{q}_i} \cdot \mathbf{v}t + \frac{\partial L}{\partial \dot{\mathbf{q}}_i} \cdot \mathbf{v} \right) = \sum_i m\dot{\mathbf{q}}_i \cdot \mathbf{v} = \mathbf{v} \cdot \frac{d}{dt} M\mathbf{R}, \tag{1.A.16}$$

where $\mathbf{R}$ is the position of the center of mass and M the total mass. We conclude that

$$\sum_i \frac{\partial L}{\partial \dot{\mathbf{q}}_i} t - M\mathbf{R} = \mathbf{P}t - M\mathbf{R} \tag{1.A.17}$$

is conserved. Since the total momentum $\mathbf{P}$ is conserved for a translation-invariant system, we deduce that the center of mass moves with a constant velocity,

$$M\frac{d\mathbf{R}}{dt} = \mathbf{P}. \tag{1.A.18}$$

1.A.2 Hamiltonian formalism

In the Hamiltonian approach, the state of the system is specified by n generalized coordinates q_i and n conjugate momenta p_i defined by

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \tag{1.A.19}$$

Coordinates and momenta can be represented by a point in a $2n$-dimensional phase space. The Hamiltonian $H(p_i, q_i, t)$, which is a function of coordinates and momenta, is defined as the Legendre transform of the Lagrangian,

$$H(p_i, q_i, t) = \sum_i p_i\dot{q}_i - L(q_i, \dot{q}_i, t). \tag{1.A.20}$$

Using the Euler-Lagrange equations (1.A.3), one obtains Hamilton's equations of motion,

$$\begin{aligned} \dot{q}_i &= \frac{\partial H}{\partial p_i}, \\ \dot{p}_i &= -\frac{\partial H}{\partial q_i}. \end{aligned} \tag{1.A.21}$$

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The dynamics of the system is described by $2n$ first-order differential equations.

The Poisson bracket of two functions $F(p_i, q_i, t)$ and $G(p_i, q_i, t)$ is defined by

$$[F, G] = \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial G}{\partial q_i} \frac{\partial F}{\partial p_i} \right). \tag{1.A.22}$$

Hamilton's equations of motion can be rewritten as

$$\begin{aligned} \dot{q}_i &= [q_i, H], \\ \dot{p}_i &= [p_i, H]. \end{aligned} \tag{1.A.23}$$

The time evolution of an arbitrary function $F(p_i, q_i, t)$,

$$\begin{aligned} \frac{dF}{dt} &= \frac{\partial F}{\partial t} + \sum_i \left(\frac{\partial F}{\partial p_i} \dot{p}_i + \frac{\partial F}{\partial q_i} \dot{q}_i \right) \\ &= \frac{\partial F}{\partial t} + \sum_i \left(-\frac{\partial F}{\partial p_i} \frac{\partial H}{\partial q_i} + \frac{\partial F}{\partial q_i} \frac{\partial H}{\partial p_i} \right) \\ &= \frac{\partial F}{\partial t} + [F, H], \end{aligned} \tag{1.A.24}$$

can also be expressed by means of its Poisson bracket with the Hamiltonian.

It is possible to describe the system in terms of new coordinates and momenta,

$$\begin{aligned} Q_i &\equiv Q_i(p_j, q_j, t), \\ P_i &\equiv P_i(p_j, q_j, t). \end{aligned} \tag{1.A.25}$$

The transformation $(q_i, p_i) \to (Q_i, P_i)$ is canonical if P_i and Q_i are canonically conjugate, i.e. if they satisfy

$$[Q_i, Q_j] = [P_i, P_j] = 0 \quad \text{and} \quad [Q_i, P_j] = \delta_{i,j}. \tag{1.A.26}$$

In this case, the Poisson bracket (1.A.22) of two functions F and G keeps the same form

$$[F, G] = \sum_i \left(\frac{\partial F}{\partial Q_i} \frac{\partial G}{\partial P_i} - \frac{\partial G}{\partial Q_i} \frac{\partial F}{\partial P_i} \right). \tag{1.A.27}$$

When the transformation (1.A.25) does not depend explicitly on time, Hamilton's equations of motion (1.A.23) are invariant, i.e. $\dot{Q}_i = [Q_i, H]$ and $\dot{P}_i = [P_i, H]$, where the Poisson bracket is calculated with the new coordinates Q_i and P_i.

1.A.2.1 Symmetries and invariance

For any function $G(p_i, q_i, t)$, we can define the infinitesimal (canonical) continuous transformation ($\epsilon \to 0$)

$$\begin{aligned} q_i &\to q_i - \epsilon[G, q_i] = q_i + \epsilon \frac{\partial G}{\partial p_i}, \\ p_i &\to p_i - \epsilon[G, p_i] = p_i - \epsilon \frac{\partial G}{\partial q_i}. \end{aligned} \tag{1.A.28}$$

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G is called the generator of the transformation.[85] For a given transformation, the generator is unique, apart from an arbitrary additive function of t, independent of p_i and q_i. If the transformation (1.A.28) does not involve the time explicitly, G can be chosen time independent.

The system is invariant in the transformation if the equations of motion are preserved. If G is time independent, this implies that the Hamiltonian is unchanged (see the remark after (1.A.27)), i.e.

$$\delta H = \epsilon \sum_i \left(-\frac{\partial H}{\partial p_i}\frac{\partial G}{\partial q_i} + \frac{\partial H}{\partial q_i}\frac{\partial G}{\partial p_i} \right) = \epsilon [H, G] = 0. \tag{1.A.29}$$

to order ϵ. Since $\partial G/\partial t = 0$, Eqs. (1.A.24) and (1.A.29) imply

$$\frac{dG}{dt} = 0. \tag{1.A.30}$$

The generator of the transformation is a constant of motion. When G has an explicit dependence on t, the Hamiltonian cannot be invariant in the transformation (1.A.28), since the energy of a system does depend on the choice of the (moving) reference frame. Nevertheless, the condition for the equations of motion to be unchanged is still given by (1.A.30). For the first of Eqs. (1.A.21) to be unchanged, we need[86]

$$\delta \dot{q}_i = \delta \frac{\partial H}{\partial p_i} \quad \text{i.e.} \quad \frac{d}{dt}\frac{\partial G}{\partial p_i} = -\left[G, \frac{\partial H}{\partial p_i} \right]. \tag{1.A.31}$$

Using (1.A.24) for the time evolution of $\partial G/\partial p_i$ and G, one finds that (1.A.31) is equivalent to

$$\frac{\partial}{\partial p_i}\frac{dG}{dt} = 0. \tag{1.A.32}$$

Similarly, from the invariance of the second of Eqs. (1.A.21), one deduces

$$\frac{\partial}{\partial q_i}\frac{dG}{dt} = 0. \tag{1.A.33}$$

It follows that dG/dt must be a function of t alone. But since we can always add to G any function of t without affecting the transformation it generates, we can choose it such that $dG/dt = 0$.

It is straightforward to verify that the constants of motion obtained from the Lagrangian formalism (Sec. 1.A.1) are the generators of the corresponding symmetry transformations. The total momentum $\mathbf{P} = \sum_i \mathbf{p}_i$ satisfies

$$\begin{aligned} \mathbf{q}_i - [\mathbf{P}\cdot\delta\mathbf{a}, \mathbf{q}_i] &= \mathbf{q}_i + \delta\mathbf{a}, \\ \mathbf{p}_i - [\mathbf{P}\cdot\delta\mathbf{a}, \mathbf{p}_i] &= \mathbf{p}_i, \end{aligned} \tag{1.A.34}$$

[85] There is a close analogy between symmetries in the (classical) Hamiltonian formalism and symmetries in quantum mechanics. In particular, to the canonical transformation generated by $G(p_i, q_i, t)$ corresponds, in quantum mechanics, a unitary transformation generated by $G(\hat{p}_i, \hat{q}_i, t)$ (Sec. 2.1).

[86] We use the fact that any function $F(p_i, q_i, t)$ changes by $\delta F = \sum_i \left(\frac{\partial F}{\partial q_i}\delta q_i + \frac{\partial F}{\partial p_i}\delta p_i \right) = -\epsilon[G, F]$ in the transformation (1.A.28).

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and is therefore the generator of translations. Similarly, one finds that the total angular momentum $\mathbf{J} = \sum_i \mathbf{q}_i \times \mathbf{p}_i$ is the generator of rotations,

$$\begin{aligned} \mathbf{q}_i - [\mathbf{J}\cdot\delta\mathbf{a}, \mathbf{q}_i] &= \mathbf{q}_i + \delta\mathbf{a}\times\mathbf{q}_i, \\ \mathbf{p}_i - [\mathbf{J}\cdot\delta\mathbf{a}, \mathbf{p}_i] &= \mathbf{p}_i + \delta\mathbf{a}\times\mathbf{p}_i \end{aligned} \tag{1.A.35}$$

($\delta\mathbf{a} = \delta\theta\mathbf{n}$ for a rotation of angle $\delta\theta$ about the $\mathbf{n}$ axis), and the Hamiltonian the generator of time translations [Eq. (1.A.23)]. The generator of a Galilean transformation is given by $\mathbf{P}t - M\mathbf{R}$,

$$\begin{aligned} \mathbf{q}_i - [(\mathbf{P}t - M\mathbf{R})\cdot\delta\mathbf{v}, \mathbf{q}_i] &= \mathbf{q}_i + \delta\mathbf{v}t, \\ \mathbf{p}_i - [(\mathbf{P}t - M\mathbf{R})\cdot\delta\mathbf{v}, \mathbf{p}_i] &= \mathbf{p}_i + m\delta\mathbf{v}. \end{aligned} \tag{1.A.36}$$

1.A.3 Classical field theory

The Lagrangian and Hamiltonian formalisms also apply to systems with a continuum of degrees of freedom described by a field $\psi_j(\mathbf{r},t)$ which depends on a continuous variable $\mathbf{r}$ (which is generally the space coordinate) and a discrete index j. $\psi_j(\mathbf{r},t)$ is usually assumed to be real.[87] The Lagrangian is a functional of the dynamical variables $\psi_j(\mathbf{r},t)$ and $\dot{\psi}_j(\mathbf{r},t) = \partial_t\psi_j(\mathbf{r},t)$,[88] which can be written as

$$L[\psi(t), \dot{\psi}(t)] = \int d^d r\, \mathcal{L}(\psi_j, \dot{\psi}_j, \boldsymbol{\nabla}\psi_j), \tag{1.A.37}$$

where $\mathcal{L}$ is called the Lagrangian density. For simplicity we assume that $\mathcal{L}$ has no explicit dependence on $\mathbf{r}$ and t. The notation $L[\psi(t), \dot{\psi}(t)]$ emphasizes that the Lagrangian is a functional of $\psi_j(\mathbf{r},t)$ and $\dot{\psi}_j(\mathbf{r},t)$ at a given time t. It should be noticed that $\boldsymbol{\nabla}\psi_j(\mathbf{r},t)$ is not an independent dynamical variable but a mere function of the field $\psi_j(\mathbf{r},t)$ at different positions $\mathbf{r}$. Although the Lagrangian (1.A.37) describes a large number of physical systems, the Lagrangian density $\mathcal{L}$ may depend on spatial derivatives to all orders.

To obtain the Euler-Lagrange equations of motion, we require the action[89]

$$S[\psi] = \int_{t_1}^{t_2} dt\, L[\psi(t), \dot{\psi}(t)] = \int_{t_1}^{t_2} dt \int d^d r\, \mathcal{L}(\psi_j, \dot{\psi}_j, \boldsymbol{\nabla}\psi_j) \tag{1.A.38}$$

to be stationary under infinitesimal variations $\delta\psi_j(\mathbf{r},t)$ with the condition $\delta\psi_j(\mathbf{r},t_1) = \delta\psi_j(\mathbf{r},t_2) = 0$. Applying the rules of functional derivatives, one finds[90]

$$\begin{aligned} 0 = \frac{\delta S}{\delta\psi_j(\mathbf{r},t)} &= \frac{\delta L}{\delta\psi_j(\mathbf{r},t)} - \frac{d}{dt}\frac{\delta L}{\delta\dot{\psi}_j(\mathbf{r},t)} \\ &= \frac{\partial\mathcal{L}}{\partial\psi_j(\mathbf{r},t)} - \boldsymbol{\nabla}\cdot\frac{\partial\mathcal{L}}{\partial(\boldsymbol{\nabla}\psi_j(\mathbf{r},t))} - \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{\psi}_j(\mathbf{r},t)} \end{aligned} \tag{1.A.39}$$

[87] If the theory is naturally expressed in terms of a complex field ψ and its complex conjugate ψ^*, we write $\psi = \psi_1 + i\psi_2$ and $\psi^* = \psi_1 - i\psi_2$ where ψ_1 and ψ_2 are two real fields.

[88] Functionals are defined and discussed in Appendix 1.D.

[89] The knowledge of $\psi_j(\mathbf{r},t)$ over the time interval $[t_1,t_2]$ fixes the value of $\dot{\psi}_j(\mathbf{r},t)$ over the same time interval. The action $S[\psi]$ is therefore a functional of $\psi_j(\mathbf{r},t)$, whereas the Lagrangian $L[\psi(t),\dot{\psi}(t)]$ at time t is a functional of both $\psi_j(\mathbf{r},t)$ and $\dot{\psi}_j(\mathbf{r},t)$.

[90] Equation (1.A.39) is a direct consequence of Eq. (1.D.5) in Appendix 1.D.

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under the assumption that the field vanishes at infinity ($|\mathbf{r}| \to \infty$) or satisfies periodic boundary conditions. Without changing the equations of motion, we can add to the Lagrangian density a total time or space derivative,[91]

$$\mathcal{L} \to \mathcal{L} + \frac{d}{dt}\mathcal{F}_0(\psi_j(\mathbf{r},t)) + \boldsymbol{\nabla}\cdot\boldsymbol{\mathcal{F}}(\psi_j(\mathbf{r},t)). \tag{1.A.40}$$

The space integral of $\boldsymbol{\nabla}\cdot\boldsymbol{\mathcal{F}}$ gives a surface integral at infinity which vanishes. The transformed Lagrangian then differs from the original one by a total time derivative which does not affect the equations of motion.

The conjugate momentum and the Hamiltonian are defined by

$$\begin{aligned}
\Pi_j(\mathbf{r},t) &= \frac{\delta L}{\delta\dot{\psi}_j(\mathbf{r},t)} = \frac{\partial\mathcal{L}}{\partial\dot{\psi}_j(\mathbf{r},t)},\\
H[\Pi(t),\psi(t)] &= \int d^dr \sum_j \Pi_j(\mathbf{r},t)\dot{\psi}_j(\mathbf{r},t) - L[\psi(t),\dot{\psi}(t)]\\
&= \int d^dr\, \mathcal{H}(\Pi_j,\psi_j,\boldsymbol{\nabla}\psi_j),
\end{aligned} \tag{1.A.41}$$

where $\mathcal{H}$ is the Hamiltonian density. From (1.A.39) and (1.A.41), we obtain Hamilton's equations of motion

$$\begin{aligned}
\dot{\psi}_j(\mathbf{r},t) &= \frac{\delta H}{\delta\Pi_j(\mathbf{r},t)} = \frac{\partial\mathcal{H}}{\partial\Pi_j(\mathbf{r},t)},\\
\dot{\Pi}_j(\mathbf{r},t) &= -\frac{\delta H}{\delta\psi_j(\mathbf{r},t)} = -\frac{\partial\mathcal{H}}{\partial\psi_j(\mathbf{r},t)} + \boldsymbol{\nabla}\cdot\frac{\partial\mathcal{H}}{\partial(\boldsymbol{\nabla}\psi_j(\mathbf{r},t))}.
\end{aligned} \tag{1.A.42}$$

It is often convenient to work with the Fourier transformed variables

$$\begin{aligned}
\psi_j(\mathbf{k},t) &= \frac{1}{\sqrt{\mathcal{V}}}\int d^dr\, e^{-i\mathbf{k}\cdot\mathbf{r}}\psi_j(\mathbf{r},t),\\
\Pi_j(\mathbf{k},t) &= \frac{1}{\sqrt{\mathcal{V}}}\int d^dr\, e^{-i\mathbf{k}\cdot\mathbf{r}}\Pi_j(\mathbf{r},t),
\end{aligned} \tag{1.A.43}$$

where $\psi_j(-\mathbf{k},t) = \psi_j^*(\mathbf{k},t)$ and $\Pi_j(-\mathbf{k},t) = \Pi_j^*(\mathbf{k},t)$ for real fields $\psi_j(\mathbf{r},t)$ and $\Pi_j(\mathbf{r},t)$. For periodic boundary conditions, $k_x = \frac{2\pi}{L_x}n_x$ ($n_x \in \mathbb{Z}$) with L_x the system size in the x direction, and similar relations for k_y, etc. One then has[92]

$$\begin{aligned}
\Pi_j(\mathbf{k},t) &= \frac{1}{\sqrt{\mathcal{V}}}\int d^dr\, e^{-i\mathbf{k}\cdot\mathbf{r}}\frac{\delta L}{\delta\dot{\psi}_j(\mathbf{r},t)}\\
&= \frac{1}{\sqrt{\mathcal{V}}}\int d^dr\, e^{-i\mathbf{k}\cdot\mathbf{r}}\sum_{\mathbf{k}'}\frac{\partial L}{\partial\dot{\psi}_j(\mathbf{k}',t)}\frac{\delta\dot{\psi}_j(\mathbf{k}',t)}{\delta\dot{\psi}_j(\mathbf{r},t)}.
\end{aligned} \tag{1.A.44}$$

Since

$$\frac{\delta\dot{\psi}_j(\mathbf{k}',t)}{\delta\dot{\psi}_j(\mathbf{r},t)} = \frac{e^{-i\mathbf{k}'\cdot\mathbf{r}}}{\sqrt{\mathcal{V}}}, \tag{1.A.45}$$

[91] In the general case, the functions $\mathcal{F}_0$ and $\boldsymbol{\mathcal{F}}$ can depend explicitly on $\mathbf{r}$ and t.

[92] Note that as long as the system volume $\mathcal{V}$ is kept finite, $\mathbf{k}$ is not a continuous variable and $\partial L/\partial\dot{\psi}_j(\mathbf{k},t)$ is a usual derivative (not a functional derivative).

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we finally obtain

$$\begin{aligned}\Pi_j(\mathbf{k},t) &= \frac{1}{\mathcal{V}}\sum_{\mathbf{k}'}\int d^d r\, e^{-i(\mathbf{k}+\mathbf{k}')\cdot\mathbf{r}}\frac{\partial L}{\partial\dot{\psi}_j(\mathbf{k}',t)}\\ &= \sum_{\mathbf{k}'}\delta_{\mathbf{k}+\mathbf{k}',0}\frac{\partial L}{\partial\dot{\psi}_j(\mathbf{k}',t)}\\ &= \frac{\partial L}{\partial\dot{\psi}_j(-\mathbf{k},t)}.\end{aligned} \tag{1.A.46}$$

1.A.3.1 Symmetries and Noether's theorem

Suppose we describe the system with two fields, $\psi_j(x)$ and $\psi'_j(x)$, corresponding to Lagrangian densities $\mathcal{L}(\psi_j(x)),\partial_\mu\psi_j(x))$ and $\mathcal{L}'(\psi'_j(x),\partial_\mu\psi'_j(x))$, respectively.[93] Here $\mu = 0,x,\cdots$, $\partial_0=\partial_t$ and $\psi_j(x)=\psi_j(\mathbf{r},t)$. The transformation $\psi_j(x)\to\psi'_j(x)$ can arise from a genuine transformation of the field (such as a rotation for a N-component vector field $\boldsymbol{\psi}=(\psi_1,\cdots,\psi_N)$) or a mere transformation of the spacetime coordinates (such as a space translation $\psi_j(\mathbf{r},t)\to\psi'_j(\mathbf{r}',t)=\psi_j(\mathbf{r},t)$ with $\mathbf{r}'=\mathbf{r}+\mathbf{a}$). Since both fields refer to the same system, we must have $\mathcal{L}(\psi_j,\partial_\mu\psi_j)=\mathcal{L}'(\psi'_j,\partial_\mu\psi'_j)$ (up to a total time or space derivative), i.e. $S'[\psi']=S[\psi]$. If $\mathcal{L}'(\psi'_j,\partial_\mu\psi'_j)$ and $\mathcal{L}(\psi'_j,\partial_\mu\psi'_j)$ differ only by a total time or space derivative, i.e. $S'[\psi']=S[\psi']$, then the equation of motion of $\psi'_j(x)$ is identical to that of $\psi_j(x)$ and the system is invariant in the transformation $\psi_j(x)\to\psi'_j(x)$.

To derive Noether's theorem, we express the fact that the change

$$\delta S=\int dt\int d^d r\left\{\mathcal{L}\big(\psi'_j(x),\partial_\mu\psi'_j(x)\big)-\mathcal{L}\big(\psi_j(x),\partial_\mu\psi_j(x)\big)\right\}=0 \tag{1.A.47}$$

in the action $S[\psi]$ vanishes if the system is invariant in the transformation $\psi_j(x)\to\psi'_j(x)$ (this follows from the two conditions $S'[\psi']=S[\psi]$ and $S'[\psi']=S[\psi']$). Let us consider an infinitesimal (continuous) transformation,

$$\psi_j(x)\to\psi_j(x)+i\epsilon F_j(x)\qquad(\epsilon\to 0), \tag{1.A.48}$$

which satisfies (1.A.47). The Lagrangian density $\mathcal{L}$ is invariant up to a total derivative

$$\mathcal{L}(\psi_j,\partial_\mu\psi_j)\to\mathcal{L}(\psi_j,\partial_\mu\psi_j)+\epsilon\sum_\mu\partial_\mu\mathcal{F}_\mu(\psi_j), \tag{1.A.49}$$

where $\mu=0,x...$ Equations (1.A.48) and (1.A.49) imply

$$\sum_j\left[\frac{\partial\mathcal{L}}{\partial\psi_j(x)}iF_j(x)+\sum_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_j(x))}i\partial_\mu F_j(x)\right]=\sum_\mu\partial_\mu\mathcal{F}_\mu(x) \tag{1.A.50}$$

($\mathcal{F}_\mu(x)\equiv\mathcal{F}_\mu(\psi_j(x))$), i.e.

$$\begin{aligned}\sum_j\Bigg[&\frac{\partial\mathcal{L}}{\partial\psi_j(x)}iF_j(x)+\sum_\mu\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_j(x))}iF_j(x)\right)\\ &-\sum_\mu\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_j(x))}\right)iF_j(x)\Bigg]=\sum_\mu\partial_\mu\mathcal{F}_\mu(x).\end{aligned} \tag{1.A.51}$$

[93] See footnote 83 on page 123.

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If we assume that the field $\psi_j(x)$ satisfies the classical equations of motion (1.A.39), Eq. (1.A.51) becomes

$$\sum_{j,\mu} \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_j(x))} i F_j(x) \right) = \sum_\mu \partial_\mu \mathcal{F}_\mu(x). \tag{1.A.52}$$

We deduce the existence of a conserved (divergenceless) current $(j_0(x), \mathbf{j}(x))$,

$$\begin{aligned} \sum_\mu \partial_\mu j_\mu(x) &= 0, \\ j_\mu(x) &= -i \sum_j \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_j(x))} F_j(x) + \mathcal{F}_\mu(x). \end{aligned} \tag{1.A.53}$$

The continuity equation implies the existence of a conserved (time-independent) charge,

$$Q = \int d^d r \, j_0(x), \qquad \frac{dQ}{dt} = 0. \tag{1.A.54}$$

For a Lagrangian density $\mathcal{L}(\psi_j, \boldsymbol{\nabla}\psi_j, \dot{\psi}_j)$ which does not depend explicitly on time, an infinitesimal time translation $\psi_j(x) \to \psi_j(x) - \epsilon \dot{\psi}_j(x)$ gives $F_j(x) = i\dot{\psi}_j(x)$ and $\mathcal{F}_\mu = -\delta_{\mu,0}\mathcal{L}$. The density

$$j_0(x) = \sum_j \Pi_j(x) \dot{\psi}_j(x) - \mathcal{L} \equiv \mathcal{H} \tag{1.A.55}$$

is the Hamiltonian density and the conserved charge

$$\int d^d r \, j_0(x) = H \tag{1.A.56}$$

is the Hamiltonian. Similarly, for a space translation $\psi_j(x) \to \psi_j(x) - \epsilon \partial_\mu \psi_j(x)$ along the μ axis, $F_j(x) = i\partial_\mu \psi_j(x)$ and $\mathcal{F}_\nu = -\delta_{\nu,\mu}\mathcal{L}$. This gives

$$\mathbf{j}_0(x) = \sum_j \Pi_j(x) \boldsymbol{\nabla} \psi_j(x) \tag{1.A.57}$$

(we use the fact that the axis μ is arbitrary) and the conserved charge

$$\mathbf{P} = \int d^d r \, \mathbf{j}_0(x) = \int d^d r \sum_j \Pi_j(x) \boldsymbol{\nabla} \psi_j(x) \tag{1.A.58}$$

is naturally interpreted as the momentum carried by the field.[94] Finally, for a space rotation about an arbitrary axis $\mathbf{n}$, $\psi_j(x) \to \psi_j(x) - \epsilon(\mathbf{n} \times \mathbf{r}) \cdot \boldsymbol{\nabla} \psi_j(x)$ (we consider a three-dimensional system), we obtain

$$j_0(x) = \sum_j \Pi_j(x) (\mathbf{n} \times \mathbf{r}) \cdot \boldsymbol{\nabla} \psi_j(x) = \sum_j \Pi_j(x) \mathbf{n} \cdot (\mathbf{r} \times \boldsymbol{\nabla}) \psi_j(x), \tag{1.A.59}$$

which leads to the conservation of the angular momentum

$$\mathbf{J} = \int d^3 r \sum_j \Pi_j(x) (\mathbf{r} \times \boldsymbol{\nabla}) \psi_j(x). \tag{1.A.60}$$

[94] Equations (1.A.58) and (1.A.60) actually correspond to the opposite of the usual definitions of momentum and angular momentum. In the relativistic formalism, one finds that the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$ reintroduces the usual sign in the definition of $\mathbf{P}$ and $\mathbf{J}$. For instance, for a spacetime translation $x^\mu \to x^\mu + \epsilon^\mu$, Noether's theorem gives the conservation law $\sum_\mu \partial_\mu T^\mu{}_\nu = 0$ where $T^\mu{}_\nu = \frac{\partial}{\partial(\partial_\mu \psi)} \partial_\nu \psi - \delta^\mu{}_\nu \mathcal{L}$ is the stress-energy tensor, also called the energy-momentum tensor. The conserved densities are $T^{00} = T^0{}_0 = \mathcal{H}$ and $T^{0i} = -T^0{}_i = -\Pi(x)\partial_i \psi(x)$ $(i = x, y, z)$.

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The infinitesimal continuous transformation (1.A.48) can often be expressed with the generator G of the symmetry,

$$\psi'_j(x) = \psi_j(x) - i\epsilon G\psi(x). \tag{1.A.61}$$

For the examples discussed above, one has

$$\begin{aligned} &G \equiv P_\mu = -i\partial_\mu, \quad \text{(space translation in the } \mu \text{ direction)}, \\ &G \equiv L_{\mu\nu} = -i(x_\mu\partial_\nu - x_\nu\partial_\mu) \quad \text{(rotation in the plane } (\mu,\nu)). \end{aligned} \tag{1.A.62}$$

These examples refer to a mere transformation of the spacetime variable $x = (\mathbf{r}, t)$. But the field can also be transformed. For instance, in a space rotation (considering for simplicity a three-dimensional system), for a three-component field $\boldsymbol{\varphi}(x) = (\varphi_1(x), \varphi_2(x), \varphi_3(x))$ transforming as a vector, the total generator is

$$G = L_{\mu\nu} + S_{\mu\nu}, \tag{1.A.63}$$

where $S_{\mu\nu}$ is a skew-symmetric matrix of the Lie algebra $\mathfrak{so}(3)$. More generally, depending on the nature of the field (scalar, vector, tensor, etc.), the matrix $S_{\mu\nu}$ belongs to some representation of the group SO(3).

1.A.3.2 Classical electromagnetism

Classical electromagnetism in three-dimensional space follows from the Lagrangian

$$\begin{aligned} L = \frac{m}{2}\sum_i \dot{\mathbf{r}}_i^2 + \frac{\epsilon_0}{2}\int d^3r \left\{ \left[\boldsymbol{\nabla}\phi(\mathbf{r},t) + \partial_t \mathbf{A}(\mathbf{r},t)\right]^2 - c_l^2 \left[\boldsymbol{\nabla}\times\mathbf{A}(\mathbf{r},t)\right]^2 \right\} \\ - \int d^3r \left[\phi(\mathbf{r},t)\rho(\mathbf{r},t) - \mathbf{J}(\mathbf{r},t)\cdot\mathbf{A}(\mathbf{r},t)\right] \end{aligned} \tag{1.A.64}$$

(c_l denotes the velocity of light). L is a function of the positions $\mathbf{r}_i$ and velocities $\dot{\mathbf{r}}_i$ of the particles and a functional of the electromagnetic potentials $\phi(\mathbf{r},t)$ and $\mathbf{A}(\mathbf{r},t)$. The electric and magnetic fields can be expressed as $\mathbf{E} = -\boldsymbol{\nabla}\phi - \partial_t\mathbf{A}$ and $\mathbf{B} = \boldsymbol{\nabla}\times\mathbf{A}$. The matter-field coupling involves the charge and current densities, $\rho(\mathbf{r},t) = e\sum_i \delta(\mathbf{r}-\mathbf{r}_i)$ and $\mathbf{J}(\mathbf{r},t) = e\sum_i \dot{\mathbf{r}}_i\delta(\mathbf{r}-\mathbf{r}_i)$, where e is the charge of the particles. Using[95]

$$\begin{aligned} \frac{\partial L}{\partial \dot{\mathbf{r}}_i} &= m\dot{\mathbf{r}}_i + e\mathbf{A}(\mathbf{r}_i,t), \\ \frac{\partial L}{\partial \mathbf{r}_i} &= -e\boldsymbol{\nabla}\phi(\mathbf{r}_i,t) + e(\dot{\mathbf{r}}_i\cdot\boldsymbol{\nabla})\mathbf{A}(\mathbf{r}_i,t) + e\dot{\mathbf{r}}_i\times\big(\boldsymbol{\nabla}\times\mathbf{A}(\mathbf{r}_i,t)\big), \end{aligned} \tag{1.A.65}$$

we find that the Euler-Lagrange equation

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\mathbf{r}}_i} = \frac{\partial L}{\partial \mathbf{r}_i} \tag{1.A.66}$$

reproduces the Newton-Lorentz equation

$$m\ddot{\mathbf{r}}_i = e\mathbf{E}(\mathbf{r}_i,t) + e\dot{\mathbf{r}}_i\times\mathbf{B}(\mathbf{r}_i,t). \tag{1.A.67}$$

[95] We have used the vector identity $\boldsymbol{\nabla}(\mathbf{A}\cdot\mathbf{B}) = (\mathbf{A}\cdot\boldsymbol{\nabla})\mathbf{B} + (\mathbf{B}\cdot\boldsymbol{\nabla})\mathbf{A} + \mathbf{A}\times(\boldsymbol{\nabla}\times\mathbf{B}) + \mathbf{B}\times(\boldsymbol{\nabla}\times\mathbf{A})$ to obtain the second of Eqs. (1.A.65).

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Similarly, from

$$\begin{aligned} \frac{\delta L}{\delta \dot{\phi}(\mathbf{r},t)} &= 0, \\ \frac{\delta L}{\delta \phi(\mathbf{r},t)} &= -\epsilon_0[\boldsymbol{\nabla}^2\phi(\mathbf{r},t) + \boldsymbol{\nabla}\cdot\dot{\mathbf{A}}(\mathbf{r},t)] - \rho(\mathbf{r},t), \end{aligned} \tag{1.A.68}$$

and

$$\begin{aligned} \frac{\delta L}{\delta \dot{\mathbf{A}}(\mathbf{r},t)} &= \epsilon_0[\boldsymbol{\nabla}\phi(\mathbf{r},t) + \dot{\mathbf{A}}(\mathbf{r},t)], \\ \frac{\delta L}{\delta \mathbf{A}(\mathbf{r},t)} &= -\epsilon_0 c_l^2 \boldsymbol{\nabla}\times\big(\boldsymbol{\nabla}\times\mathbf{A}(\mathbf{r},t)\big) + \mathbf{J}(\mathbf{r},t), \end{aligned} \tag{1.A.69}$$

as well as the Euler-Lagrange equations

$$\begin{aligned} \frac{d}{dt}\frac{\delta L}{\delta \dot{\phi}(\mathbf{r},t)} &= \frac{\delta L}{\delta \phi(\mathbf{r},t)}, \\ \frac{d}{dt}\frac{\delta L}{\delta \dot{\mathbf{A}}(\mathbf{r},t)} &= \frac{\delta L}{\delta \mathbf{A}(\mathbf{r},t)}, \end{aligned} \tag{1.A.70}$$

we deduce the Maxwell equations

$$\begin{aligned} \boldsymbol{\nabla}\cdot\mathbf{E}(\mathbf{r},t) &= \frac{\rho(\mathbf{r},t)}{\epsilon_0}, \\ \boldsymbol{\nabla}\times\mathbf{B}(\mathbf{r},t) &= \mu_0\mathbf{J}(\mathbf{r},t) + \frac{1}{c_l^2}\frac{\partial\mathbf{E}(\mathbf{r},t)}{\partial t}. \end{aligned} \tag{1.A.71}$$

Appendix 1.B Review of quantum mechanics[96]

In this section, we review elementary aspects of quantum mechanics and introduce the concept of quantum fields which is discussed at length in Sec. 1.2.

1.B.1 The postulates of quantum mechanics

1.B.1.1 State space

To any physical system, we associate a Hilbert space $\mathcal{H}$ called the state space. A Hilbert space is a vector space with an inner product (or scalar product).[97] In Dirac's notation a vector Ψ is represented by a ket $|\Psi\rangle$. Physical states are represented by rays in the Hilbert space. A ray is a set of normalized vectors which differ only by a phase factor. Thus $|\Psi\rangle$ and $|\Psi'\rangle$ belong to the same ray if $|\Psi'\rangle = e^{i\alpha}|\Psi\rangle$ (α real). To each ket $|\Psi\rangle$, one can associate a bra $\langle\Psi|$ which is an element of the dual space $\mathcal{H}^*$ and such that the action of $\langle\Psi|$ on $|\Phi\rangle$ yields the scalar product $\langle\Psi|\Phi\rangle$ between $|\Psi\rangle$ and $|\Phi\rangle$.[98]

[96] See, for instance, Refs. [24–27].

[97] For a vector space to be a Hilbert space, it must also be complete: If a sequence of vectors is Cauchy, then it converges to some limit in the space.

[98] The dual space $\mathcal{H}^*$ is defined as the set of all linear functionals on $\mathcal{H}$.

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We can define coordinates in the state space by choosing a complete orthonormal basis, i.e. a set $\{|n\rangle\}$ of vectors satisfying

$$\langle n|m\rangle = \delta_{n,m} \tag{1.B.1}$$

and the closure relation

$$\sum_n |n\rangle\langle n| = 1, \tag{1.B.2}$$

where the symbol "1" in the rhs denotes the identity operator acting on $\mathcal{H}$.

1.B.1.2 Observables

With any measurable quantity A we associate an Hermitian operator $\hat{A}$ whose normalized eigenvectors $|n\rangle$ form an orthonormal basis of the Hilbert space $\mathcal{H}$.[99] Such an operator is called an observable. In a given basis $\{|n\rangle\}$, $\hat{A}$ is defined by its matrix elements $\langle n|\hat{A}|m\rangle$.

The only possible result of the measurement of the physical quantity A is one of the eigenvalues a_n of the operator $\hat{A}$ with probability $\mathcal{P}(a_n) = \langle\Psi|\hat{P}_{a_n}|\Psi\rangle$ where $\hat{P}_a$ is the projector onto the subspace associated with the eigenvalue a (we assume the state vector to be normalized, $\langle\Psi|\Psi\rangle = 1$).[100] The expectation value of the observable $\hat{A}$ is therefore

$$\langle\hat{A}\rangle = \sum_n \mathcal{P}(a_n)a_n = \sum_n \langle\Psi|\hat{P}_{a_n}|\Psi\rangle a_n = \langle\Psi|\hat{A}|\Psi\rangle \tag{1.B.3}$$

and its statistical fluctuations (of quantum origin) are given by the variance

$$\Delta A^2 = \langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2 = \langle\Psi|\big(\hat{A} - \langle\hat{A}\rangle\big)^2|\Psi\rangle. \tag{1.B.4}$$

If the eigenvalue a_n is measured then the state of the system after the measurement is given by the projection of the state vector onto the subspace associated with a_n: $|\Psi'\rangle \propto \hat{P}_{a_n}|\Psi\rangle$ (collapse of the state vector).

1.B.1.3 Coordinate and momentum representations

For a spinless particle moving in a d-dimensional space, a convenient basis is given by the eigenstates $|\mathbf{r}\rangle$ of the position operator $\hat{\mathbf{r}}$: $\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle$. These states are not normalizable and therefore not physical states. They satisfy

$$\langle\mathbf{r}|\mathbf{r}'\rangle = \delta(\mathbf{r}-\mathbf{r}') \quad \text{and} \quad \int d^d r\, |\mathbf{r}\rangle\langle\mathbf{r}| = 1. \tag{1.B.5}$$

Any state can be expanded in this basis as

$$|\Psi\rangle = \int d^d r\, \Psi(\mathbf{r})|\mathbf{r}\rangle, \tag{1.B.6}$$

where $\Psi(\mathbf{r}) = \langle\mathbf{r}|\Psi\rangle$ is the wavefunction in the coordinate representation. If the state is normalized,

$$\langle\Psi|\Psi\rangle = \int d^d r\, |\Psi(\mathbf{r})|^2 = 1, \tag{1.B.7}$$

[99] An operator $\hat{A}$ is Hermitian if it is equal to its adjoint $\hat{A}^\dagger$, where $\hat{A}^\dagger$ is defined by $\langle\Psi|\hat{A}^\dagger|\Phi\rangle = \langle\Phi|\hat{A}|\Psi\rangle^*$. To the ket $\hat{A}|\Psi\rangle$ corresponds the bra $\langle\Psi|\hat{A}^\dagger$.

[100] $\hat{P}_{a_n} = |n\rangle\langle n|$ when the eigenvalue a_n is not degenerate ($\hat{A}|n\rangle = a_n|n\rangle$).

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$|\Psi(\mathbf{r})|^2$ gives the probability density for the particle to be at point $\mathbf{r}$.

Another useful basis is given by the eigenstates $|\mathbf{p}\rangle$ of the momentum operator $\hat{\mathbf{p}}$: $\hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle$. They satisfy

$$|\mathbf{p}\rangle = \int d^d r \, \langle \mathbf{r}|\mathbf{p}\rangle |\mathbf{r}\rangle = \frac{1}{\sqrt{\mathcal{V}}} \int d^d r \, e^{i\mathbf{p}\cdot\mathbf{r}} |\mathbf{r}\rangle, \tag{1.B.8}$$

or the inverse relation

$$|\mathbf{r}\rangle = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{r}} |\mathbf{p}\rangle, \tag{1.B.9}$$

as well as the closure relation

$$\sum_{\mathbf{p}} |\mathbf{p}\rangle\langle\mathbf{p}| = 1. \tag{1.B.10}$$

The wavefunction in the $\mathbf{p}$ representation is the Fourier transform of $\Psi(\mathbf{r})$,

$$\Psi(\mathbf{p}) = \langle \mathbf{p}|\Psi\rangle = \frac{1}{\sqrt{\mathcal{V}}} \int d^d r \, e^{-i\mathbf{p}\cdot\mathbf{r}} \Psi(\mathbf{r}). \tag{1.B.11}$$

For periodic boundary conditions, $\Psi(\mathbf{r} + L\hat{\mathbf{x}}) = \Psi(\mathbf{r} + L\hat{\mathbf{y}}) = \cdots = \Psi(\mathbf{r})$,[101] $\mathbf{p}$ is a discrete variable which takes quantized values

$$p_\mu = \frac{2\pi}{L} n_\mu, \quad n_\mu \in \mathbb{Z} \quad (\mu = x, y, \cdots). \tag{1.B.12}$$

When the volume of the system tends to infinity, $\mathbf{p}$ becomes a continuous variable and

$$\lim_{\mathcal{V}\to\infty} \frac{1}{\mathcal{V}} \sum_{\mathbf{p}} = \int \frac{d^d p}{(2\pi)^d} \equiv \int_{\mathbf{p}}. \tag{1.B.13}$$

1.B.1.4 Canonical quantization

In the Hamiltonian formalism of classical mechanics, the system is described by N coordinates $q_i(t)$ and N conjugate momenta $p_i(t)$. In the quantum formalism, these variables become operators $\hat{q}_i(t), \hat{p}_i(t)$ acting on the state space $\mathcal{H}$. For any physical quantity $A(p_i, q_i, t)$ of the classical theory one then obtains an observable $\hat{A}(t) \equiv A(\hat{p}_i(t), \hat{q}_i(t), t)$ (Heisenberg's correspondence principle). When this correspondence rule yields products of the non-commuting operators $\hat{p}_i$ and $\hat{q}_i$, the order of the operators must be determined by additional conditions such as the Hermiticity of the observable $\hat{A}(t)$.

Moreover the Poisson bracket $[A, B]$ is replaced by the commutator of the two corresponding operators,

$$[A, B] \equiv \sum_i \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right) \to \frac{1}{i\hbar} (\hat{A}\hat{B} - \hat{B}\hat{A}) \equiv \frac{1}{i\hbar} [\hat{A}, \hat{B}]. \tag{1.B.14}$$

The factor i ensures that the rhs in (1.B.14) is Hermitian if both $\hat{A}$ and $\hat{B}$ are, and the factor $\hbar$ (now written explicitly) preserves the units of the Poisson braket. Equations (1.A.26) yield the equal-time commutation relations

$$[\hat{q}_i(t), \hat{q}_j(t)] = [\hat{p}_i(t), \hat{p}_j(t)] = 0, \qquad [\hat{q}_i(t), \hat{p}_j(t)] = i\hbar\delta_{i,j}. \tag{1.B.15}$$

[101] $\hat{\mathbf{x}}$ denotes the unit vector in the x direction, etc.

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The time evolution of these operators is deduced from (1.A.23),

$$\begin{aligned}\frac{d}{dt}\hat{q}_i(t) &= \frac{1}{i\hbar}[\hat{q}_i(t), \hat{H}(t)],\\ \frac{d}{dt}\hat{p}_i(t) &= \frac{1}{i\hbar}[\hat{p}_i(t), \hat{H}(t)],\end{aligned} \tag{1.B.16}$$

where $\hat{H}(t) \equiv H(\hat{p}_i(t), \hat{q}_i(t), t)$ is the quantum Hamiltonian. For an observable $\hat{A}(t) \equiv A(\hat{p}_i(t), \hat{q}_i(t), t)$, Eq. (1.A.24) yields

$$\frac{d\hat{A}(t)}{dt} = \frac{\partial \hat{A}(t)}{\partial t} + \frac{1}{i\hbar}[\hat{A}(t), \hat{H}(t)], \tag{1.B.17}$$

where $\partial\hat{A}/\partial t$ comes from the explicit time dependence of the operator $\hat{A}(t)$. From (1.B.17), we deduce that the time evolution of the expectation value $\langle\hat{A}(t)\rangle = \langle\Psi|\hat{A}(t)|\Psi\rangle$ of an observable is given by

$$\frac{d\langle\hat{A}(t)\rangle}{dt} = \left\langle\frac{\partial\hat{A}(t)}{\partial t}\right\rangle + \frac{1}{i\hbar}\langle[\hat{A}(t), \hat{H}(t)]\rangle \tag{1.B.18}$$

(Ehrenfest's equation). An observable which commutes with the Hamiltonian $\hat{H}$ and does not depend on time explicitly is a constant of motion.

So far we have discussed the Heisenberg picture of quantum mechanics where the state vector $|\Psi_H\rangle$ is time independent while the dynamics of the system is given by the time evolution of the observables $\hat{A}_H(t)$ [Eq. (1.B.17)]. Alternatively one can choose the Schrödinger picture where the state vector $|\Psi_S(t)\rangle$ is time dependent while the operators $\hat{A}_S(t)$ are not (except for a possible explicit time dependence). Both descriptions are related by a unitary transformation,

$$\begin{aligned}\hat{A}_H(t) &= \hat{U}^\dagger(t, t_0)\hat{A}_S(t)\hat{U}(t, t_0),\\ |\Psi_H\rangle &= \hat{U}^\dagger(t, t_0)|\Psi_S(t)\rangle\end{aligned} \tag{1.B.19}$$

($\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0) = \hat{U}(t, t_0)\hat{U}^\dagger(t, t_0) = 1$), which preserves the value of any matrix element

$$\langle\Psi_H|\hat{A}_H(t)|\Phi_H\rangle = \langle\Psi_H|\hat{U}^\dagger(t, t_0)\hat{A}_S(t)\hat{U}(t, t_0)|\Phi_H\rangle = \langle\Psi_S(t)|\hat{A}_S(t)|\Phi_S(t)\rangle. \tag{1.B.20}$$

t_0 is some arbitrary fixed time. If the two descriptions coincide at t_0 then $\hat{U}(t_0, t_0) = 1$ and the equation

$$|\Psi_S(t)\rangle = \hat{U}(t, t_0)|\Psi_H\rangle = \hat{U}(t, t_0)|\Psi_S(t_0)\rangle \tag{1.B.21}$$

shows that $\hat{U}(t, t_0)$ is the time-evolution operator in the Schrödinger picture. The unitary operator $\hat{U}(t, t_0)$ is further defined by its time evolution equation

$$i\hbar\frac{d}{dt}\hat{U}(t, t_0) = \hat{H}_S(t)\hat{U}(t, t_0), \tag{1.B.22}$$

where $\hat{H}_S(t)$ is the Hamiltonian in the Schrödinger picture. Equation (1.B.22) implies that the state vector $|\Psi_S(t)\rangle$ obeys the Schrödinger equation

$$i\hbar\frac{d}{dt}|\Psi_S(t)\rangle = \hat{H}_S(t)|\Psi_S(t)\rangle. \tag{1.B.23}$$

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From (1.B.19) and (1.B.22), one deduces

$$\frac{d}{dt}\hat{q}_{iH}(t) = \frac{1}{i\hbar}[\hat{q}_{iH}(t), \hat{H}_H(t)] + \hat{U}^\dagger(t,t_0)\frac{d\hat{q}_{iS}(t)}{dt}\hat{U}(t,t_0) \tag{1.B.24}$$

and a similar equation for $\hat{p}_{iH}(t)$. Comparing with (1.B.16), we conclude that the operators $\hat{q}_{iS}$ and $\hat{p}_{iS}$ are time independent in the Schrödinger picture. It follows that the time dependence of an observable $\hat{A}_S(t) \equiv A(\hat{p}_{iS}, \hat{q}_{iS}, t)$ in the Schrödinger picture can only be explicit. For an arbitrary observable, Eqs. (1.B.19) and (1.B.22) imply

$$\begin{aligned}\frac{d}{dt}\hat{A}_H(t) &= \frac{1}{i\hbar}[\hat{A}_H(t), \hat{H}_H(t)] + \hat{U}^\dagger(t,t_0)\frac{\partial\hat{A}_S(t)}{\partial t}\hat{U}(t,t_0)\\ &= \frac{1}{i\hbar}[\hat{A}_H(t), \hat{H}_H(t)] + \frac{\partial\hat{A}_H(t)}{\partial t}\end{aligned} \tag{1.B.25}$$

in agreement with the Heisenberg equation of motion (1.B.17). When the Hamiltonian $\hat{H}_S(t) \equiv \hat{H}$ in the Schrödinger picture is time independent, the time-evolution operator takes the form

$$\hat{U}(t,t_0) = \hat{U}(t-t_0) = e^{-\frac{i}{\hbar}\hat{H}(t-t_0)} \tag{1.B.26}$$

and the Hamiltonian $\hat{H}_H(t) = \hat{H}$ in the Heisenberg picture is also time independent.

1.B.2 Quantum fields

To quantize a classical field theory defined by its Hamiltonian density $\mathcal{H}(\Pi_j, \psi_j, \boldsymbol{\nabla}\psi_j)$ (Sec. 1.A.2), we promote the field $\psi_j(\mathbf{r},t)$ and its conjugate variable $\Pi_j(\mathbf{r},t)$ to operators acting on the state space $\mathcal{H}$ and satisfying the equal-time commutation relations

$$\begin{aligned}&[\hat{\psi}_j(\mathbf{r},t), \hat{\psi}_{j'}(\mathbf{r}',t)] = [\hat{\Pi}_j(\mathbf{r},t), \hat{\Pi}_{j'}(\mathbf{r}',t)] = 0,\\ &[\hat{\psi}_j(\mathbf{r},t), \hat{\Pi}_{j'}(\mathbf{r}',t)] = i\hbar\delta_{j,j'}\delta(\mathbf{r}-\mathbf{r}').\end{aligned} \tag{1.B.27}$$

In the Heisenberg picture, the quantum fields $\hat{\psi}_j(\mathbf{r},t)$ and $\hat{\Pi}_j(\mathbf{r},t)$ are time dependent while the state of the system is described by a time independent ket $|\Psi\rangle$. Alternatively one can use the Schrödinger picture as explained in the preceding section.

In Sec. 1.9 we use this procedure to quantize the classical electromagnetic field.[102] A similar approach can be used to quantize a system of particles.[103] Let us first consider a single particle. Its wavefunction satisfies the Schrödinger equation (1.B.23),

$$i\hbar\frac{\partial\Psi(\mathbf{r},t)}{\partial t} = \hat{H}\Psi(\mathbf{r},t). \tag{1.B.28}$$

We can see (1.B.28) as a wave equation for the matter, analogous to Maxwell's equations for electromagnetic waves, where $\Psi(\mathbf{r},t)$ is considered as a classical field (the Schrödinger field) rather than the wavefunction of a single particle. Equation (1.B.28) derives from a Lagrangian. The canonical quantization procedure leads to Hermitian conjugate quantum fields $\hat{\psi}(\mathbf{r},t)$ and $\hat{\psi}^\dagger(\mathbf{r},t)$ whose quantized excitations correspond to the particles of the system. While the standard commutation relations

$$\begin{aligned}&[\hat{\psi}(\mathbf{r},t), \hat{\psi}(\mathbf{r}',t)] = [\hat{\psi}^\dagger(\mathbf{r},t), \hat{\psi}^\dagger(\mathbf{r}',t)] = 0,\\ &[\hat{\psi}(\mathbf{r},t), \hat{\psi}^\dagger(\mathbf{r}',t)] = \delta(\mathbf{r}-\mathbf{r}')\end{aligned} \tag{1.B.29}$$

[102] An elementary example, the quantization of the classical harmonic string, is also discussed in Sec. 1.1.2.
[103] For a detailed discussion, see Refs. [24] (Chap. XIII) and [18] (Chap. II and exercise D_{II}.8).

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can be used for bosons, they would violate the Pauli principle for fermions. In this case, it is necessary to use anticommutators instead of commutators in (1.B.29). This quantization procedure is often (improperly) called "second quantization" as it seems to be based on a quantization of the Schrödinger equation, which is itself a quantum equation of motion. In fact, $\psi(\mathbf{r},t)$ should be seen as a classical field (not the wavefunction of a quantum particle) which is quantized only once.

While the classical field is the inescapable starting point when quantizing relativistic particles [7], a different quantization procedure, which does not rely on a classical field, can be used for non-relativistic particles. In this approach, one constructs the Fock space of the many-particle system directly from the one-particle Hilbert space used for the quantum mechanics of a single particle, and then directly defines the quantum fields $\hat{\psi}(\mathbf{r},t)$ and $\hat{\psi}^\dagger(\mathbf{r},t)$ by their action on the many-body states. This is the route followed in Sec. 1.2.

Appendix 1.C Review of statistical physics[104]

In this section, we review the basics of quantum statistical physics.

1.C.1 Macrostates

In quantum mechanics, the system is described by a normalized vector $|\Psi\rangle$ of the Hilbert space $\mathcal{H}$. Such a state, where the system is completely known or ideally prepared (*via* the control of a complete set of commuting observables), is called a microstate or a pure state. In statistical physics, one is interested in macroscopic systems with a very large number of degrees of freedom. These systems are never ideally prepared or completely known and are usually described by the specification of a small number of macroscopic variables. On the microscopic scale, a partially known or incompletely prepared system is described by a macrostate (or statistical mixture), i.e. by the set of possible microstates $|\Psi_\lambda\rangle$ and the corresponding probabilities q_λ. In this statistical description, a "system" denotes an element of a statistical ensemble of systems, all prepared under the same conditions. The central assumption of statistical physics is that the time average (over a characteristic time larger than microscopic time scales) of an observable for a given system coincides with the statistical average obtained from the statistical ensemble.

1.C.1.1 Density operator

All the information about the macrostate is contained in the density operator (or density matrix)

$$\hat{\rho} = \sum_\lambda |\Psi_\lambda\rangle q_\lambda \langle\Psi_\lambda|, \tag{1.C.1}$$

where the probabilities q_λ are positive and normalized: $\sum_\lambda q_\lambda = 1$. For instance, the expectation value of an observable $\hat{A}$ is given by

$$\langle\hat{A}\rangle = \sum_\lambda q_\lambda \langle\Psi_\lambda|\hat{A}|\Psi_\lambda\rangle = \mathrm{Tr}(\hat{\rho}\hat{A}). \tag{1.C.2}$$

The random character of the measurement has two origins, the probabilities q_λ on the one hand and the quantum nature of the microstates $|\Psi_\lambda\rangle$ on the other hand.

[104] This section closely follows Chaps. 2-5 of Ref. [28]. We write explicitly $\hbar$ and k_B.

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Two different statistical mixtures, $\{|\Psi_\lambda\rangle, q_\lambda\}$ and $\{|\Psi'_\mu\rangle, q'_\mu\}$, can lead to the same density operator and therefore the same physical reality. Thus the only meaningful representation of macrostates is given by density operators. A microstate $|\Psi\rangle$ corresponds to the density operator $\hat{\rho} = |\Psi\rangle\langle\Psi|$. Note that all state vectors $|\Psi\rangle$ belonging to the same ray (i.e. differing only by a phase factor) yield the same density operator.

The definition (1.C.1) gives the following properties:

1. $\hat{\rho}$ is Hermitian: $\hat{\rho}^\dagger = \hat{\rho}$.
2. $\hat{\rho}$ has unit trace: $\mathrm{Tr}(\hat{\rho}) = 1$.
3. $\hat{\rho}$ is non-negative: $\langle\Psi|\hat{\rho}|\Psi\rangle \geq 0, \quad \forall\, |\Psi\rangle \in \mathcal{H}$.

Conversely, any operator $\hat{\rho}$ satisfying these properties can be written in the basis of its eigenstates,

$$\hat{\rho} = \sum_m |m\rangle p_m \langle m| \qquad (p_m \in \mathbb{R}). \tag{1.C.3}$$

It follows from the properties (2) and (3) that $p_m \geq 0$ and $\sum_m p_m = 1$, which allows us to regard p_m as the probability of the microstate $|m\rangle$; $\hat{\rho}$ is a density operator. A necessary and sufficient condition to have a pure state is that $\hat{\rho}^2 = \hat{\rho}$.

1.C.1.2 Time evolution of the density operator

In the Schrödinger picture, the time evolution of the density operator is directly obtained from the Schrödinger equation satisfied by the states $|m\rangle$ [Eq. (1.C.3)],

$$i\hbar \frac{d\hat{\rho}}{dt} = [\hat{H}, \hat{\rho}] \tag{1.C.4}$$

(Liouville-von Neumann equation). The evolution of the expectation value of an observable follows from (1.C.4),

$$\frac{d}{dt}\mathrm{Tr}(\hat{\rho}\hat{A}) = \frac{1}{i\hbar}\mathrm{Tr}([\hat{H}, \hat{\rho}]\hat{A}) + \mathrm{Tr}\left(\hat{\rho}\frac{\partial \hat{A}}{\partial t}\right) = \frac{1}{i\hbar}\langle[\hat{A}, \hat{H}]\rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle, \tag{1.C.5}$$

in agreement with Ehrenfest's equation (1.B.18). In the Heisenberg picture, the states $|m\rangle$ are time independent and $d\hat{\rho}/dt = 0$.

> In statistical physics, one is often interested in a subsystem of the whole system. Knowing the density operator $\hat{\rho}$ of the system, it is possible to define a reduced density operator for the subsystem. Let us assume for instance that the system contains two subsystems a and b. The Hilbert space $\mathcal{H} = \mathcal{H}_a \otimes \mathcal{H}_b$ is then the tensor product of the Hilbert spaces of the subsystems. Let $\{|k_a, l_b\rangle\} = \{|k_a\rangle \otimes |l_b\rangle\}$ be a basis of $\mathcal{H}$. For an observable $\hat{A}_a \otimes 1_b$ acting only on $\mathcal{H}_a$, one has
>
> $$\begin{aligned}
> \langle \hat{A}_a \rangle &= \mathrm{Tr}[\hat{\rho}(\hat{A}_a \otimes 1_b)] \\
> &= \sum_{k_a, l_b, k'_a, l'_b} \langle k_a, l_b|\hat{\rho}|k'_a, l'_b\rangle\langle k'_a, l'_b|\hat{A}_a \otimes 1_b|k_a, l_b\rangle \\
> &= \sum_{k_a, l_b, k'_a} \langle k_a, l_b|\hat{\rho}|k'_a, l_b\rangle\langle k'_a|\hat{A}_a|k_a\rangle \\
> &= \mathrm{Tr}(\hat{\rho}_a \hat{A}_a),
> \end{aligned} \tag{1.C.6}$$

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where

$$\hat{\rho}_a = \mathrm{Tr}_b(\hat{\rho}) \tag{1.C.7}$$

is the density operator in the subsystem a, obtained by tracing out the degrees of freedom in the subsystem b. The definition of a density operator for a subsystem is particularly natural in the functional integral formalism where it relies on a partial functional integration of the degrees of freedom.

1.C.2 Statistical entropy – Boltzmann-Gibbs distribution

The knowledge of the system based on the density operator is incomplete. The missing information can be estimated by the statistical entropy (or von Neumann entropy)

$$S = -k_B \sum_{m=1}^{M} p_m \ln p_m = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho}), \tag{1.C.8}$$

where the p_m's are the probabilities of the states $|m\rangle$ that diagonalize the density operator. The factor k_B is *a priori* arbitrary. By choosing the Boltzmann constant, we ensure that at equilibrium the statistical entropy coincides with the thermodynamic entropy (see below). From the definition (1.C.8), we deduce the following properties:

1. S is a symmetric function of $p_1, \cdots, p_M$.
2. S is non-negative and vanishes when the system is in a pure state (one probability equals one and all others vanish): $S(1, 0, \cdots, 0) = 0$.
3. For M fixed, S is maximum when all probabilities are equal to $1/M$:

$$S\left(\frac{1}{M}, \cdots, \frac{1}{M}\right) = k_B \ln M. \tag{1.C.9}$$

4. If we divide the microstates $\{|\Psi_m\rangle\}$ into two groups of states, $A = \{|\Psi_1\rangle| \cdots |\Psi_N\rangle\}$ and $B = \{|\Psi_{N+1}\rangle| \cdots |\Psi_M\rangle\}$, then the entropy satisfies the following additivity property,

$$S = S(q_A, q_B) + q_A S\left(\frac{p_1}{q_A}, \cdots, \frac{p_N}{q_A}\right) + q_B S\left(\frac{p_{N+1}}{q_B}, \cdots, \frac{p_M}{q_B}\right), \tag{1.C.10}$$

where

$$q_A = \sum_{m=1}^{N} p_m \quad \text{and} \quad q_B = \sum_{m=N+1}^{M} p_m. \tag{1.C.11}$$

The first term in (1.C.10) corresponds to the missing information due to the choice between the sets of states A and B. The second term comes from the missing information once we know that the system is in a microstate of the set A (hence the factor q_A). p_m/q_A is the conditional probability giving the probability to find the system in the microstate $|\Psi_m\rangle$ knowing that $|\Psi_m\rangle$ belongs to the set A. The last term of (1.C.10) has a similar interpretation with the role of the sets A and B interchanged.

While these properties are a consequence of the definition (1.C.8), one can proceed the other way round and show that the expression (1.C.8) of the statistical entropy follows from the above properties, the most important one being the additivity postulate (1.C.10).

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1.C.2.1 The maximum statistical entropy principle

When some information about the system is known exactly (e.g. the energy and the volume of the system or the number of particles in a fluid), it is taken into account through the definition of the Hilbert space. If no other information is available, it is natural to use a statistical ensemble where the probabilities of the W possible microstates are equal. In macroscopic equilibrium, the macrostate is then defined by the density operator

$$\hat{\rho} = \sum_{m=1}^{W} |m\rangle \frac{1}{W} \langle m| \tag{1.C.12}$$

and the entropy is given by

$$S = k_B \ln W. \tag{1.C.13}$$

In many cases, however, the macrostate is characterized by the knowledge of the expectation values $\langle \hat{A}_i \rangle$ of some observables $\hat{A}_i$ such as the Hamiltonian $\hat{H}$ or the total number $\hat{N}$ of particles (while other properties such as the volume may be known exactly). This happens for instance when the system is put into contact with a large system with which it can exchange energy or particles. This large system plays the role of a reservoir which fixes the mean value of the energy $\langle \hat{H} \rangle$ or the mean total number of particles $\langle \hat{N} \rangle$ in the system we are interested in. The density operator must then satisfy the constraints

$$\langle \hat{A}_i \rangle = \mathrm{Tr}(\hat{\rho} \hat{A}_i). \tag{1.C.14}$$

The maximum statistical entropy principle states that the density operator $\hat{\rho}$ describing the system in macroscopic equilibrium is the one which maximizes the entropy while being compatible with the known information about the system. Any other choice would mean that we assume possessing information which is actually not available and consequently would lead to biased predictions.

1.C.2.2 Boltzmann-Gibbs distribution

We are thus led to maximize the entropy $S = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho})$ with the constraints $\mathrm{Tr}(\hat{\rho}) = 1$ and (1.C.14), and the fact that $\hat{\rho}$ must be Hermitian and non-negative. This can be done by introducing Lagrange multipliers λ_0, λ_i and requiring

$$-\mathrm{Tr}(\hat{\rho} \ln \hat{\rho}) - \sum_i \lambda_i \mathrm{Tr}(\hat{\rho} \hat{A}_i) - \lambda_0 \mathrm{Tr}(\hat{\rho}) \tag{1.C.15}$$

to be stationary. When $\hat{\rho}$ changes by $\delta\hat{\rho}$, (1.C.15) changes by

$$-\mathrm{Tr}\Big[\delta\hat{\rho}\Big(\ln \hat{\rho} + 1 + \sum_i \lambda_i \hat{A}_i + \lambda_0\Big)\Big], \tag{1.C.16}$$

so that the stationary condition gives

$$\hat{\rho} = e^{-1-\lambda_0 - \sum_i \lambda_i \hat{A}_i}. \tag{1.C.17}$$

The density operator $\hat{\rho}$ is Hermitian, since the observables $\hat{A}_i$ are, and non-negative. Since $\mathrm{Tr}(\hat{\rho}) = 1$, we can eliminate the Lagrange multiplier λ_0. Defining the partition function

$$Z = e^{1+\lambda_0} = \mathrm{Tr}\, e^{-\sum_i \lambda_i \hat{A}_i}, \tag{1.C.18}$$

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we can rewrite the density operator as

$$\hat{\rho} = \frac{1}{Z} e^{-\sum_i \lambda_i \hat{A}_i}. \tag{1.C.19}$$

This expression, which corresponds to the Boltzmann-Gibbs distribution, gives the general form of the density operator in thermodynamic equilibrium when the expectation values $\langle \hat{A}_i \rangle$ are given.

To complete the derivation, we must show that Eq. (1.C.19) corresponds to a maximum of the statistical entropy (and not to a minimum). We write the entropy as

$$S(\hat{\rho}) = -k_B \langle \ln \hat{\rho} \rangle = k_B \ln Z + k_B \sum_i \lambda_i \langle \hat{A}_i \rangle \tag{1.C.20}$$

and use the fact that two non-negative operators $\hat{X}$ and $\hat{Y}$ satisfy the inequality

$$\mathrm{Tr}(\hat{X} \ln \hat{Y}) - \mathrm{Tr}(\hat{X} \ln \hat{X}) \leq \mathrm{Tr}(\hat{Y}) - \mathrm{Tr}(\hat{X}). \tag{1.C.21}$$

For $\hat{X} = \hat{\rho}'$ an arbitrary density operator and $\hat{Y} = \hat{\rho}$, this gives

$$\mathrm{Tr}(\hat{\rho}' \ln \hat{\rho}) - \mathrm{Tr}(\hat{\rho}' \ln \hat{\rho}') \leq 0, \tag{1.C.22}$$

and therefore

$$\begin{aligned} S(\hat{\rho}') &\leq -k_B \mathrm{Tr}(\hat{\rho}' \ln \hat{\rho}) \\ &\leq k_B \ln Z + k_B \sum_i \lambda_i \mathrm{Tr}(\hat{\rho}' \hat{A}_i) \\ &\leq S(\hat{\rho}), \end{aligned} \tag{1.C.23}$$

where we have used $\mathrm{Tr}(\hat{\rho}' \hat{A}_i) = \mathrm{Tr}(\hat{\rho} \hat{A}_i) = \langle \hat{A}_i \rangle$ to obtain the last inequality. Thus the Boltzmann-Gibbs distribution provides a statistical entropy which is larger than that of any density operator satisfying the same constraints.

To determine the Lagrange multipliers λ_i as a function of the expectation values $\langle \hat{A}_i \rangle$, we use

$$\langle \hat{A}_i \rangle = \frac{1}{Z} \mathrm{Tr}\big(e^{-\sum_j \lambda_j \hat{A}_j} \hat{A}_i\big) = -\frac{\partial \ln Z}{\partial \lambda_i}. \tag{1.C.24}$$

In practice, we take the Lagrange multipliers λ_i as the parameters characterizing the equilibrium state rather than the expectation values $\langle \hat{A}_i \rangle$. (We discuss below their physical meaning when $\hat{A}_i$ corresponds to the Hamiltonian or the total number of particles.) The correlations and the statistical fluctuations of the constants of motion $\hat{A}_i$ can also be obtained from the derivatives of the partition function $Z(\lambda_i)$,

$$\langle \hat{A}_i \hat{A}_j \rangle - \langle \hat{A}_i \rangle \langle \hat{A}_j \rangle = \frac{\partial^2 \ln Z}{\partial \lambda_i \partial \lambda_j}. \tag{1.C.25}$$

Using (1.C.20) and (1.C.24), we can write the entropy as

$$S = k_B \ln Z - k_B \sum_i \lambda_i \frac{\partial \ln Z}{\partial \lambda_i}. \tag{1.C.26}$$

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If the parameters λ_i characterizing the equilibrium change by an infinitesimal amount $d\lambda_i$,

$$d\ln Z = -\sum_i \langle \hat{A}_i \rangle d\lambda_i \tag{1.C.27}$$

and, using (1.C.26),

$$dS = k_B d\ln Z - k_B \sum_i \left(d\lambda_i \frac{\partial \ln Z}{\partial \lambda_i} - \lambda_i d\langle \hat{A}_i \rangle \right) = k_B \sum_i \lambda_i d\langle \hat{A}_i \rangle. \tag{1.C.28}$$

While Z is a function of λ_i, the natural variables for the entropy are the expectation values $\langle \hat{A}_i \rangle$, and

$$\frac{\partial S}{\partial \langle \hat{A}_i \rangle} = k_B \lambda_i. \tag{1.C.29}$$

Equations (1.C.24), (1.C.26) and (1.C.29) show that $k_B \ln Z(\lambda_i)$ and $S(\langle \hat{A}_i \rangle)$ are related by a Legendre transform.

1.C.3 Statistical ensembles

In this section, we discuss the three most common statistical ensembles for a quantum fluid. The main results are summarized in Table 1.3.

1.C.3.1 Microcanonical ensemble

In the microcanonical ensemble, the number of particles N and the volume $\mathcal{V}$ of the system are known exactly, while the energy is assumed to be between E and $E+\Delta E$. ΔE is a small energy interval as compared to the accuracy to which E is known, but large with respect to the typical energy level spacing. The density operator and the entropy are then given by (1.C.12) and (1.C.13) where $W = \mathcal{D}(E)\Delta E$ is the number of possible microstates, with

$$\mathcal{D}(E) = \mathrm{Tr}\, \delta(E - \hat{H}) \tag{1.C.30}$$

the density of states of the system. Note that the dependence of $\hat{\rho}$ and S on $E, \mathcal{V}, N$ is not explicit but comes from that of W. The variables conjugate to E, N and $\mathcal{V}$

$$\left.\frac{\partial S}{\partial E}\right|_{N,\mathcal{V}} = \frac{1}{T}, \qquad \left.\frac{\partial S}{\partial N}\right|_{E,\mathcal{V}} = -\frac{\mu}{T}, \qquad \left.\frac{\partial S}{\partial \mathcal{V}}\right|_{E,N} = \frac{P}{T}, \tag{1.C.31}$$

are related to the temperature T, the chemical potential μ and the pressure P. These relations yield the usual thermodynamic definition of μ and P,

$$\mu = \left.\frac{\partial E}{\partial N}\right|_{S,\mathcal{V}} = -\frac{\left.\frac{\partial S}{\partial N}\right|_{E,\mathcal{V}}}{\left.\frac{\partial S}{\partial E}\right|_{N,\mathcal{V}}}, \qquad P = -\left.\frac{\partial E}{\partial \mathcal{V}}\right|_{S,N} = \frac{\left.\frac{\partial S}{\partial \mathcal{V}}\right|_{E,N}}{\left.\frac{\partial S}{\partial E}\right|_{N,\mathcal{V}}}. \tag{1.C.32}$$

1.C.3.2 Canonical ensemble

In the canonical ensemble, the number of particles and the volume are known exactly but the energy $E = \langle \hat{H} \rangle$ is known only on average. The corresponding Lagrange multiplier is denoted by β, and is related to the thermodynamic absolute temperature T by $\beta = 1/k_B T$

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(see the discussion on statistical and thermodynamic entropies on page 148). The density operator is given by

$$\hat{\rho} = \frac{1}{Z_C} e^{-\beta \hat{H}}, \tag{1.C.33}$$

where

$$Z_C(\beta, N, \mathcal{V}) = \operatorname{Tr} e^{-\beta \hat{H}} \tag{1.C.34}$$

is the canonical partition function. The entropy reads

$$S(E, N, \mathcal{V}) = k_B \ln Z_C + k_B \beta E \tag{1.C.35}$$

and satisfies

$$\left.\frac{\partial S}{\partial E}\right|_{N,\mathcal{V}} = k_B \beta \tag{1.C.36}$$

(see Eqs. (1.C.20) and (1.C.29)). In the canonical ensemble, it is traditional to use as thermodynamic potential the Helmholtz free energy

$$F(T, N, \mathcal{V}) = E - TS = -\frac{1}{\beta} \ln Z_C(\beta, N, \mathcal{V}) \tag{1.C.37}$$

defined as the Legendre transform of the energy. Its differential reads

$$dF = -S dT + \mu dN - P d\mathcal{V}, \tag{1.C.38}$$

where

$$S = -\left.\frac{\partial F}{\partial T}\right|_{N,\mathcal{V}}, \quad \mu = \left.\frac{\partial E}{\partial N}\right|_{S,\mathcal{V}} = \left.\frac{\partial F}{\partial N}\right|_{T,\mathcal{V}}, \quad P = -\left.\frac{\partial E}{\partial \mathcal{V}}\right|_{S,N} = -\left.\frac{\partial F}{\partial \mathcal{V}}\right|_{T,N}. \tag{1.C.39}$$

1.C.3.3 Grand canonical ensemble

In the grand canonical ensemble, both the energy $E = \langle \hat{H} \rangle$ and the number of particles $N = \langle \hat{N} \rangle$ are known on average. When the number of particles is not fixed, one must use a Hilbert space (the Fock space) which is constructed from the direct sum of the N-particle spaces (Sec. 1.2.1). The density operator

$$\hat{\rho} = \frac{1}{Z_G} e^{-\beta \hat{H} + \alpha \hat{N}} \tag{1.C.40}$$

is expressed in terms of the two Lagrange multipliers β and $-\alpha$, the grand canonical partition function being defined by

$$Z_G(\beta, \alpha, \mathcal{V}) = \operatorname{Tr} e^{-\beta \hat{H} + \alpha \hat{N}}. \tag{1.C.41}$$

The equilibrium entropy is given by

$$S(E, N, \mathcal{V}) = k_B \ln Z_G + k_B \beta E - k_B \alpha N \tag{1.C.42}$$

and satisfies

$$\left.\frac{\partial S}{\partial E}\right|_{N,\mathcal{V}} = k_B \beta, \qquad \left.\frac{\partial S}{\partial N}\right|_{E,\mathcal{V}} = -k_B \alpha. \tag{1.C.43}$$

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From these equations, we can relate the chemical potential

$$\mu = \left.\frac{\partial E}{\partial N}\right|_{S,\mathcal{V}} = -\frac{\left.\frac{\partial S}{\partial N}\right|_{E,\mathcal{V}}}{\left.\frac{\partial S}{\partial E}\right|_{N,\mathcal{V}}} = \frac{\alpha}{\beta} \tag{1.C.44}$$

to the Lagrange multipliers, so that $\alpha = \beta\mu = \mu/k_BT$. In the grand canonical ensemble it is traditional to use as thermodynamic potential the grand potential

$$\Omega(T,\mu,\mathcal{V}) = E - TS - \mu N = -\frac{1}{\beta}\ln Z_G(\beta,\alpha,\mathcal{V}) \tag{1.C.45}$$

defined as a double Legendre transform of the energy. Its differential reads

$$d\Omega = -SdT - Nd\mu - Pd\mathcal{V}, \tag{1.C.46}$$

where

$$S = -\left.\frac{\partial\Omega}{\partial T}\right|_{\mu,\mathcal{V}}, \qquad N = -\left.\frac{\partial\Omega}{\partial\mu}\right|_{T,\mathcal{V}}, \qquad P = -\left.\frac{\partial\Omega}{\partial\mathcal{V}}\right|_{T,\mu}. \tag{1.C.47}$$

The extensivity of $\ln Z_G$ and Ω then implies

$$\Omega(T,\mu,\mathcal{V}) = -P\mathcal{V}. \tag{1.C.48}$$

Apart from the sign, the grand potential per unit volume is equal to the pressure.

1.C.3.4 Other examples of ensembles

One can easily extend the preceding discussion to other physical observables. Let us for example consider a system of N interacting particles that can freely move in space; the expectation value of its total momentum $\hat{\mathbf{P}} = \sum_i \hat{\mathbf{p}}_i$ can then be nonzero. Denoting by $-\beta\mathbf{v}$ the Lagrange multiplier associated with $\hat{\mathbf{P}}$ (we shall see that $\mathbf{v}$ is the velocity of the system), the partition function reads

$$Z(T,N,\mathcal{V},\mathbf{v}) = \mathrm{Tr}\, e^{-\beta(\hat{H}-\hat{\mathbf{P}}\cdot\mathbf{v})}, \qquad \hat{H} = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m} + \hat{V}, \tag{1.C.49}$$

where we assume that the interacting part of the Hamiltonian, $\hat{V}$, depends only on the particles' positions. Equation (1.C.49) shows that the Hamiltonian in the rest frame of the fluid (i.e. the moving frame) is $\hat{H} - \hat{\mathbf{P}}\cdot\mathbf{v}$ with $\hat{H}$ and $\hat{\mathbf{P}}$ the Hamiltonian and total momentum, respectively, in the laboratory frame.

Using

$$\hat{H} - \hat{\mathbf{P}}\cdot\mathbf{v} = \sum_i \frac{(\hat{\mathbf{p}}_i - m\mathbf{v})^2}{2m} - \frac{1}{2}Nm\mathbf{v}^2 + \hat{V}, \tag{1.C.50}$$

one finds

$$Z(T,N,\mathcal{V},\mathbf{v}) = Z_0(T,N,\mathcal{V})e^{\beta\frac{N}{2}m\mathbf{v}^2}, \tag{1.C.51}$$

since we can transform $\hat{\mathbf{p}}_i - m\mathbf{v}$ back to $\hat{\mathbf{p}}_i$ by a unitary transformation which does not change the trace. Hence

$$F(T,N,\mathcal{V},\mathbf{v}) = -\frac{1}{\beta}\ln Z(T,N,\mathcal{V},\mathbf{v}) = F_0(T,N,\mathcal{V}) - \frac{N}{2}m\mathbf{v}^2, \tag{1.C.52}$$

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Table 1.3: Table of thermodynamic potentials.

Ensemble	Microcanonical	Canonical	Grand canonical
Variables	$E, N, \mathcal{V}$	$T, N, \mathcal{V}$	$T, \mu, \mathcal{V}$
Partition function	W	$Z_C = \mathrm{Tr}\, e^{-\beta \hat{H}}$	$Z_G = \mathrm{Tr}\, e^{-\beta(\hat{H}-\mu\hat{N})}$
Density operator	$\hat{\rho} = \sum_m \lvert m\rangle \frac{1}{W} \langle m\rvert$	$\hat{\rho} = \frac{1}{Z_C} e^{-\beta \hat{H}}$	$\hat{\rho} = \frac{1}{Z_G} e^{-\beta(\hat{H}-\mu\hat{N})}$
Thermodynamic potential	Entropy	Helmholtz free energy	Grand potential
	$S(E, N, \mathcal{V}) = k_B \ln W$	$F(T, N, \mathcal{V}) = -\frac{1}{\beta} \ln Z_C = E - TS$	$\Omega(T, \mu, \mathcal{V}) = -\frac{1}{\beta} \ln Z_G = E - TS - \mu N$
	$dS = \frac{1}{T}(dE - \mu dN + P d\mathcal{V})$	$dF = -SdT + \mu dN - P d\mathcal{V}$	$d\Omega = -SdT - N d\mu - P d\mathcal{V}$

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where $F_0(T,N,\mathcal{V}) = -\frac{1}{\beta}\ln Z_0(T,N,\mathcal{V})$ is the free energy for $\mathbf{v}=0$. Equation (1.C.24) gives

$$\langle\hat{\mathbf{P}}\rangle = -\frac{\partial}{\partial\mathbf{v}}F(T,N,\mathcal{V},\mathbf{v})\bigg|_{T,N,\mathcal{V}} = Nm\mathbf{v}, \tag{1.C.53}$$

which shows that $\mathbf{v}$ is the velocity of the system. From (1.C.52) we deduce the chemical potential

$$\mu = \frac{\partial F}{\partial N}\bigg|_{T,\mathcal{V},\mathbf{v}} = \frac{\partial F_0}{\partial N}\bigg|_{T,\mathcal{V}} - \frac{1}{2}m\mathbf{v}^2 \equiv \mu_0 - \frac{1}{2}m\mathbf{v}^2 \tag{1.C.54}$$

and the grand potential $\Omega = F - \mu N$ in the grand canonical ensemble,

$$\begin{aligned}\Omega(T,\mu,\mathcal{V},\mathbf{v}) &= F_0(T,N,\mathcal{V}) - \frac{N}{2}m\mathbf{v}^2 - \left(\mu_0 - \frac{1}{2}m\mathbf{v}^2\right)N \\ &= \Omega_0(T,\mu_0,\mathcal{V}) \\ &= \Omega_0\left(T,\mu+\frac{1}{2}m\mathbf{v}^2,\mathcal{V}\right).\end{aligned} \tag{1.C.55}$$

Equation (1.C.53) gives

$$\langle\hat{\mathbf{P}}\rangle = -\frac{\partial}{\partial\mathbf{v}}\Omega(T,\mu,\mathcal{V},\mathbf{v})\bigg|_{T,\mu,\mathcal{V}} = Nm\mathbf{v}. \tag{1.C.56}$$

Another example is provided by a system which can freely rotate and whose expectation value of its angular momentum $\hat{\mathbf{J}}$ can therefore be nonzero. In that case the Lagrange multiplier associated with $\hat{\mathbf{J}}$ is $-\beta\boldsymbol{\omega}$, where $\boldsymbol{\omega}$ is the angular velocity of the system, and the equilibrium density operator in the rest frame of the fluid can be written in the form

$$\frac{1}{Z}e^{-\beta(\hat{H}-\hat{\mathbf{J}}\cdot\boldsymbol{\omega})}, \tag{1.C.57}$$

where $\hat{H}$ and $\hat{\mathbf{J}}$ are the Hamiltonian and angular momentum, respectively, in the laboratory frame.

1.C.3.5 Thermodynamic limit – Equivalence between ensembles

In the thermodynamic limit, the size of the system becomes very large with respect to the typical microscopic scales and can be considered to be infinite. Extensive variables are proportional to the volume. For instance, the entropy reads

$$S(E,N,\mathcal{V}) = \mathcal{V}s\left(\frac{E}{\mathcal{V}},\frac{N}{\mathcal{V}}\right), \tag{1.C.58}$$

where s reaches a finite limit for $\mathcal{V}\to\infty$ with $E/\mathcal{V}$,$N/\mathcal{V}$ or α,β constant.

Let us consider a system in equilibrium in the canonical ensemble. The density operator $\hat{\rho}$ can be written in terms of the eigenstates $\{|m\rangle,\epsilon_m\}$ of the Hamiltonian,

$$\hat{\rho} = \sum_m |m\rangle p_m\langle m|, \tag{1.C.59}$$

where

$$p_m = \frac{e^{-\beta\epsilon_m}}{Z_C} \tag{1.C.60}$$

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is the probability for the system to be in the microstate $|m\rangle$. Thus the system has a probability $p(E)dE$ to have an energy in the interval $[E, E+dE]$,

$$p(E) = \frac{e^{-\beta E}}{Z_C}\mathcal{D}(E), \tag{1.C.61}$$

where $\mathcal{D}(E)$ is the density of states [Eq. (1.C.30)]. $\mathcal{D}(E)$ is related to the microcanonical entropy $S(E)$ *via* $S(E) = k_B \ln[\mathcal{D}(E)\Delta E]$, where ΔE is the energy uncertainty in the microcanonical ensemble. This enables us to rewrite $p(E)$ as

$$p(E) = \frac{e^{-\beta E + k_B^{-1} S(E)}}{Z_C \Delta E} \propto e^{-\beta E + k_B^{-1} S(E)}. \tag{1.C.62}$$

In the thermodynamic limit, $\mathcal{V} \to \infty$, we thus obtain

$$p(E) \propto \exp\left\{-\mathcal{V}\left[\beta\frac{E}{\mathcal{V}} - \frac{S(E)}{k_B \mathcal{V}}\right]\right\}, \tag{1.C.63}$$

with both $S/\mathcal{V}$ and $E/\mathcal{V}$ taking a finite limit (independent of $\mathcal{V}$) when $\mathcal{V} \to \infty$. $p(E)$ is then strongly peaked about the most probable value $\bar{E}$ and can be approximated by a Gaussian probability distribution[105]

$$p(E) \propto \exp\left\{\frac{1}{2k_B}\frac{\partial^2 S}{\partial E^2}\bigg|_{E=\bar{E}}(E-\bar{E})^2\right\}. \tag{1.C.64}$$

$\bar{E}$ is defined by

$$\frac{1}{k_B}\frac{\partial S}{\partial E}\bigg|_{E=\bar{E}} - \beta = 0 \tag{1.C.65}$$

and therefore corresponds to the energy of the system in the microcanonical ensemble (recall that $S(E)$ is the entropy in the microcanonical ensemble and note that $\partial^2 S/\partial E^2|_{E=\bar{E}} < 0$) [Eq. (1.C.31)]. The statistical fluctuations of the energy in the canonical ensemble are characterized by the variance

$$\Delta E^2 = \langle E^2\rangle - \bar{E}^2 \sim \left(\left|\frac{\partial^2 S}{\partial E^2}\right|_{E=\bar{E}}\right)^{-1}. \tag{1.C.66}$$

Since $S \propto \mathcal{V}$ and $E \propto \mathcal{V}$, we deduce $\partial^2 S/\partial E^2 \propto 1/\mathcal{V}$ and $\Delta E \propto \sqrt{\mathcal{V}}$. In the thermodynamic limit, the relative fluctuations

$$\frac{\Delta E}{\bar{E}} \propto \frac{1}{\sqrt{\mathcal{V}}} \tag{1.C.67}$$

of the energy become negligible and the microcanonical and canonical descriptions are equivalent.

Let us now consider a fluid in the grand canonical ensemble,

$$\hat{\rho} = \frac{1}{Z_G} e^{-\beta(\hat{H}-\mu\hat{N})} = \frac{1}{Z_G}\sum_m |m\rangle e^{-\beta(\epsilon_m - \mu N_m)}\langle m|, \tag{1.C.68}$$

[105] See Sec. 1.7.1 for a mathematical justification of this approximation.

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where we have used the basis $\{|m\rangle\}$ which diagonalizes the Hamiltonian $\hat{H}$ and the particle number operator $\hat{N}$. The probability to have N particles in the system is

$$p(N) = \frac{1}{Z_G} \sum_{\substack{m \\ (N_m = N)}} e^{-\beta(\epsilon_m - \mu N_m)} = e^{\beta\mu N} \frac{Z_C}{Z_G} = \frac{1}{Z_G} e^{-\beta(F-\mu N)}, \tag{1.C.69}$$

where $Z_C(\beta, N, \mathcal{V})$ is the canonical partition function and $F(T, N, \mathcal{V}) = -\frac{1}{\beta} \ln Z_C$ the Helmholtz free energy. In the thermodynamic limit, N is large and can be considered as a continuous variable. Moreover, since F/N takes a finite limit when $N \to \infty$, we can factorize out a factor N in the argument of the exponential in (1.C.69), and approximate $p(N)$ by the Gaussian distribution function[105]

$$p(N) \propto \exp\left\{ -\frac{\beta}{2} \frac{\partial^2 F}{\partial N^2}\bigg|_{N=\bar{N}} (N - \bar{N})^2 \right\}, \tag{1.C.70}$$

where the most probable value $\bar{N}$ is defined by

$$\mu - \frac{\partial F}{\partial N}\bigg|_{N=\bar{N}} = 0. \tag{1.C.71}$$

$\bar{N}$ is therefore such that the canonical chemical potential $\partial F/\partial N$ [Eq. (1.C.39)] coincides with the grand canonical potential μ. The statistical fluctuations of N are given by

$$\Delta N^2 \sim \left(\beta \frac{\partial^2 F}{\partial N^2}\bigg|_{N=\bar{N}} \right)^{-1} \propto \bar{N}, \tag{1.C.72}$$

so that the relative fluctuations

$$\frac{\Delta N}{\bar{N}} \propto \frac{1}{\sqrt{\bar{N}}} \tag{1.C.73}$$

become negligible when $\bar{N} \to \infty$. The canonical and grand canonical descriptions are equivalent in the thermodynamic limit.

Statistical entropy *vs* **thermodynamic entropy.** For an isolated system, the Liouville-von Neumann equation (1.C.4) ensures that the energy is conserved,

$$\frac{dE}{dt} = \frac{1}{i\hbar} \langle [\hat{H}, \hat{H}] \rangle = 0. \tag{1.C.74}$$

If the system is not isolated, its energy varies by an amount

$$dE = \mathrm{Tr}(d\hat{\rho}\hat{H}) + \mathrm{Tr}(\hat{\rho} d\hat{H}) \tag{1.C.75}$$

in an infinitesimal transformation where the Hamiltonian and the density operator vary by $d\hat{H}$ and $d\hat{\rho}$, respectively. When $\hat{\rho}$ commutes with the Hamiltonian, it can be written as

$$\hat{\rho} = \sum_m |m\rangle p_m \langle m| \tag{1.C.76}$$

in the basis $\{|m\rangle, \epsilon_m\}$ of the eigenstates of $\hat{H}$, where p_m is the probability of the microstate $|m\rangle$. The energy of the equilibrium state is then $E = \sum_m p_m \epsilon_m$ and its variation (1.C.75) reads

$$dE = \sum_m (dp_m \epsilon_m + p_m d\epsilon_m). \tag{1.C.77}$$

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The quantity

$$\delta W = \sum_m p_m d\epsilon_m = \mathrm{Tr}(\hat{\rho} d\hat{H}) \tag{1.C.78}$$

corresponds to the work received by the system in the infinitesimal transformation. On the other hand, the energy variation

$$\delta Q = \sum_m dp_m \epsilon_m = \mathrm{Tr}(d\hat{\rho}\hat{H}) \tag{1.C.79}$$

due to the change in the probabilities p_m is the heat transferred to the system.

Let us consider a system in equilibrium in the canonical ensemble. In a quasi-static transformation, the system remains in equilibrium and its statistical entropy changes by[106]

$$\begin{aligned} dS &= -k_B \mathrm{Tr}(d\hat{\rho} \ln \hat{\rho}) \\ &= k_B \mathrm{Tr}[d\hat{\rho}(\ln Z + \beta \hat{H})] \\ &= k_B \beta \mathrm{Tr}(d\hat{\rho}\hat{H}) \\ &= k_B \beta \delta Q. \end{aligned} \tag{1.C.80}$$

This result must be compared with Clausius' equation $\delta Q = T dS_{\mathrm{th}}$ where S_{th} is the thermodynamic entropy at equilibrium. If we choose $k_B\beta = 1/T$ (with T the thermodynamic absolute temperature), then $dS = dS_{\mathrm{th}}$, which implies that the statistical entropy S coincides with the thermodynamic entropy up to an additive constant [28].

1.C.4 The classical limit

In the Hamiltonian formalism, a classical microstate is represented by a point $\{q_i, p_i\}$ in phase space. A classical macrostate is characterized by the probability law for the classical microstates,

$$\rho_N(q_1, p_1, \cdots, q_N, p_N) d\tau_N, \tag{1.C.81}$$

where ρ_N is the phase density (or density in phase) and $d\tau_N$ the volume element in phase space. ρ_N is real, positive and normalized. A classical observable A then becomes a random variable and its mean value is given by

$$\langle A \rangle = \int d\tau_N \, \rho_N(q_1, p_1, \cdots, q_N, p_N) A(q_1, p_1, \cdots, q_N, p_N). \tag{1.C.82}$$

The normalization of $d\tau_N$ does not matter, since ρ_N is normalized. However, when one derives (1.C.82) as a limiting case of quantum statistical physics, one finds

$$d\tau_N = \frac{1}{N!} \prod_{i=1}^{N} \frac{d^3 q_i d^3 p_i}{h^3} \tag{1.C.83}$$

for a three-dimensional fluid of N particles. The factor h^{-3N} makes $d\tau_N$ dimensionless, while the factor $1/N!$ is associated with the indistinguishability of the particles. This suggests that even in classical mechanics, two points in phase space which differ from one another by a permutation of the indices of the particles represent the same physical microstate.[107]

[106] Since the density operator satisfies $\mathrm{Tr}(\hat{\rho}) = 1$, $\mathrm{Tr}(d\hat{\rho}) = 0$.

[107] For arbitrary conjugate variables (q_i, p_i), $N!$ is replaced by a factor $\mathcal{S}$ equal to the number of configurations in phase space which describe the same situation if we take into account the indistinguishability of the particles.

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The time evolution of the quantum density operator was derived from the Schrödinger equation and the conservation of the probability q_λ of the microstate $|\psi_\lambda\rangle$. Similarly, the probability $\rho_N d\tau_N$ is conserved along the classical trajectory defined by Hamilton's equations of motion (1.A.21). Moreover, the volume of a region in phase space remains constant when one follows the region during the temporal evolution (Liouville's theorem).[108] It follows that both $d\tau_N$ and ρ_N are conserved along the classical trajectories. From

$$\rho_N(q_i + \dot{q}_i dt, p_i + \dot{p}_i dt, t + dt) = \rho_N(q_i, p_i, t) \qquad (dt \to 0), \tag{1.C.84}$$

we then deduce

$$\frac{\partial \rho_N}{\partial t} + \sum_i \left(\frac{\partial \rho_N}{\partial q_i}\dot{q}_i + \frac{\partial \rho_N}{\partial p_i}\dot{p}_i \right) = 0, \tag{1.C.85}$$

i.e.

$$\frac{\partial \rho_N}{\partial t} = [H, \rho_N] \tag{1.C.86}$$

(Liouville's equation), where [,] denotes the Poisson bracket.

The classical statistical entropy is defined as

$$S = -k_B \sum_N \int d\tau_N \, \rho_N \ln \rho_N, \tag{1.C.87}$$

which can be shown to be the classical limit of (1.C.8). If the number of particles in the system is well defined, the sum over N in (1.C.87) is absent. The phase density of the system in equilibrium is obtained from the maximum entropy principle. This leads to the classical Boltzmann-Gibbs distribution

$$\rho_N = \frac{1}{Z} e^{-\sum_i \lambda_i A_i}, \tag{1.C.88}$$

where the partition function is defined by

$$Z = \sum_N \int d\tau_N \, e^{-\sum_i \lambda_i A_i}. \tag{1.C.89}$$

The observables A_i in (1.C.88) and (1.C.89) correspond to the (known) constants of motion and the λ_i's are Lagrange multipliers introduced to enforce the constraints $\langle A_i \rangle = \text{const}$. For example, the classical canonical ensemble has a phase density

$$\rho_N = \frac{1}{Z_C} e^{-\beta H_N} \tag{1.C.90}$$

with a partition function

$$Z_C(\beta, N, \mathcal{V}) = \int d\tau_N \, e^{-\beta H_N}, \tag{1.C.91}$$

while the classical grand canonical ensemble is defined by

$$\begin{aligned} \rho_N &= \frac{1}{Z_G} e^{-\beta H_N + \alpha N}, \\ Z_G(\beta, \alpha, \mathcal{V}) &= \sum_N \int d\tau_N \, e^{-\beta H_N + \alpha N} = \sum_N e^{\alpha N} Z_C(\beta, N, \mathcal{V}), \end{aligned} \tag{1.C.92}$$

where $\alpha = \beta\mu = \mu/k_B T$.

[108] Liouville's theorem follows from the fact that the infinitesimal transformation $q_i \to q_i + \dot{q}_i dt$, $p_i \to p_i + \dot{p}_i dt$ has a unit Jacobian, which is a direct consequence of Hamilton's equations of motion (1.A.21).

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Classical ideal gas. For N non-interacting classical particles in three dimensions with Hamiltonian $H = \sum_{i=1}^N \mathbf{p}_i^2/2m$, the canonical partition function reads

$$Z_C(\beta, N, \mathcal{V}) = \frac{1}{N!} \int \prod_{i=1}^N \frac{d^3q_i d^3p_i}{h^3} \exp\left\{-\beta \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m}\right\} = \frac{z^N}{N!}, \tag{1.C.93}$$

where

$$z = \int \frac{d^3q d^3p}{h^3} e^{-\beta \mathbf{p}^2/2m} = \frac{\mathcal{V}}{\lambda_T^3} \tag{1.C.94}$$

is the partition function of a single particle and $\lambda_T = (2\pi\hbar^2\beta/m)^{1/2}$ the thermal de Broglie wavelength. We then deduce the free energy $F(T, N, \mathcal{V}) = -\frac{1}{\beta} \ln Z_C(\beta, N, \mathcal{V})$ and the pressure

$$P = -\left.\frac{\partial F}{\partial \mathcal{V}}\right|_{T,N} = \frac{N}{\beta \mathcal{V}} \tag{1.C.95}$$

and recover the equation of state $P\mathcal{V} = Nk_BT$ of the classical ideal gas.

The grand canonical partition function is simply

$$Z_G(\beta, \alpha, \mathcal{V}) = \sum_{N=0}^\infty \frac{z^N}{N!} e^{\alpha N} = \exp(ze^\alpha). \tag{1.C.96}$$

This gives the grand potential

$$\Omega(T, \mu, \mathcal{V}) = -\frac{\mathcal{V}}{\beta \lambda_T^3} e^{\beta\mu} \tag{1.C.97}$$

and the mean number of particles

$$N = -\left.\frac{\partial \Omega}{\partial \mu}\right|_{T,\mathcal{V}} = -\beta\Omega, \tag{1.C.98}$$

and in turn the equation of state $P\mathcal{V} = -\Omega = Nk_BT$.

1.C.5 Non-interacting quantum gases

In this section, we consider a system of non-interacting quantum particles (bosons or fermions) in equilibrium in the grand canonical ensemble. We denote by $\{|\alpha\rangle, \epsilon_\alpha\}$ an orthonormal basis of one-body states which diagonalizes the one-particle Hamiltonian. The many-particle Hilbert space $\mathcal{H}$ (Fock space) is obtained from the direct sum $\oplus_{N=0}^\infty \mathcal{H}_N$ of the N-particle Hilbert spaces $\mathcal{H}_N$. An orthonormal basis in the Fock space is formed by the states[109]

$$|n_{\alpha_1} \cdots n_{\alpha_i} \cdots\rangle \tag{1.C.99}$$

obtained by putting n_{α_i} particles in the one-body state $|\alpha_i\rangle$ and symmetrizing or antisymmetrizing the many-body states according to the quantum statistics of the particles. These states are eigenstates of the Hamiltonian and the total number of particles operator,

$$\begin{aligned} \hat{H}|n_{\alpha_1} \cdots n_{\alpha_i} \cdots\rangle &= \left(\sum_j n_{\alpha_j} \epsilon_{\alpha_j}\right) |n_{\alpha_1} \cdots n_{\alpha_i} \cdots\rangle, \\ \hat{N}|n_{\alpha_1} \cdots n_{\alpha_i} \cdots\rangle &= \left(\sum_j n_{\alpha_j}\right) |n_{\alpha_1} \cdots n_{\alpha_i} \cdots\rangle. \end{aligned} \tag{1.C.100}$$

[109] See Sec. 1.2.1 for more details.

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While for bosons there is no restriction on the occupation numbers n_α, for fermions they are constrained by the Pauli principle and can take only the value 0 or 1. Using the basis $\{|n_{\alpha_1}\cdots n_{\alpha_i}\cdots\rangle\}$, we easily calculate the grand canonical partition function,

$$Z = \mathrm{Tr}\, e^{-\beta(\hat{N}-\mu\hat{N})} = \sum_{\{n_\alpha\}} e^{-\beta\sum_\alpha n_\alpha\xi_\alpha}, \tag{1.C.101}$$

where $\xi_\alpha = \epsilon_\alpha - \mu$. From (1.C.101), we obtain

$$Z = \sum_{n_{\alpha_1},n_{\alpha_2},\cdots} e^{-\beta n_{\alpha_1}\xi_{\alpha_1}} e^{-\beta n_{\alpha_2}\xi_{\alpha_2}}\cdots = \prod_\alpha \sum_{n_\alpha} e^{-\beta n_\alpha\xi_\alpha} = \prod_\alpha \left(1-\zeta e^{-\beta\xi_\alpha}\right)^{-\zeta}, \tag{1.C.102}$$

where $\zeta = 1$ for bosons and $\zeta = -1$ for fermions. We deduce that the grand potential reads

$$\Omega = \frac{\zeta}{\beta}\sum_\alpha \ln\left(1-\zeta e^{-\beta\xi_\alpha}\right). \tag{1.C.103}$$

The mean number of particles is given by

$$N = -\left.\frac{\partial\Omega}{\partial\mu}\right|_T = \sum_\alpha \frac{1}{e^{\beta\xi_\alpha}-\zeta} \equiv \sum_\alpha n_\zeta(\xi_\alpha), \tag{1.C.104}$$

where $n_+(\xi_\alpha) = n_B(\xi_\alpha)$ (Bose-Einstein distribution function) and $n_-(\xi_\alpha) = n_F(\xi_\alpha)$ (Fermi-Dirac distribution function). From the grand potential, one can also deduce the mean energy

$$E = \left.\frac{\partial\beta\Omega}{\partial\beta}\right|_\mu + \mu N = \sum_\alpha \epsilon_\alpha n_\zeta(\xi_\alpha) \tag{1.C.105}$$

and the entropy

$$\begin{aligned} S = -\left.\frac{\partial\Omega}{\partial T}\right|_\mu &= \frac{1}{T}\sum_\alpha\left[\xi_\alpha n_\zeta(\xi_\alpha) - \frac{\zeta}{\beta}\ln\left(1-\zeta e^{-\beta\xi_\alpha}\right)\right] \\ &= -k_B\sum_\alpha\left\{n_\zeta(\xi_\alpha)\ln n_\zeta(\xi_\alpha) - \zeta[1+\zeta n_\zeta(\xi_\alpha)]\ln[1+\zeta n_\zeta(\xi_\alpha)]\right\}. \end{aligned} \tag{1.C.106}$$

Quantum ideal gas. The grand potential of non-interacting spin-1/2 fermions read

$$\Omega = -\frac{2}{\beta}\sum_{\mathbf{k}}\ln\left(1+e^{-\beta(\epsilon_{\mathbf{k}}-\mu)}\right) = -2\frac{\mathcal{V}}{\beta}\int_0^\infty d\epsilon\,\rho(\epsilon)\ln\left(1+e^{-\beta(\epsilon-\mu)}\right) \tag{1.C.107}$$

(a factor 2 comes from the spin degeneracy), where $\epsilon_{\mathbf{k}} = \mathbf{k}^2/2m$ and

$$\rho(\epsilon) = \int_{\mathbf{k}}\delta(\epsilon-\epsilon_{\mathbf{k}}) = \frac{(2m)^{3/2}}{4\pi^2}\sqrt{\epsilon} \tag{1.C.108}$$

is the density of states per spin and unit volume (we assume a three-dimensional system). An elementary calculation gives the pressure $P = -\Omega/\mathcal{V}$ and the density $n = N/\mathcal{V} = \partial P/\partial\mu$,

$$P = \frac{2}{\beta\lambda_T^3}f_{5/2}(z), \qquad n = \frac{2}{\lambda_T^3}f_{3/2}(z), \tag{1.C.109}$$

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where $z = e^{\beta\mu}$, λ_T is the thermal de Broglie wavelength defined after (1.C.94) and

$$f_p(z) = \frac{1}{\Gamma(p)} \int_0^\infty dx \frac{x^{p-1}}{z^{-1}e^x + 1}. \tag{1.C.110}$$

In the low-temperature dilute limit, $\mu < 0$ and $T \ll |\mu|$ (or, equivalently, $n\lambda_T^3 \ll 1$), using $f_p(z) \to z$ for $z \to 0$ one recovers the equation of state $P = nk_BT$ of the classical ideal gas.

Similarly, for non-interacting spin-zero bosons (with chemical potential $\mu < 0$), one has

$$\Omega = \frac{\mathcal{V}}{\beta} \int_0^\infty d\epsilon\, \rho(\epsilon) \ln\bigl(1 - e^{-\beta(\epsilon-\mu)}\bigr) \tag{1.C.111}$$

and one finds

$$P = \frac{1}{\beta\lambda_T^3} g_{5/2}(z), \qquad n = \frac{1}{\lambda_T^3} g_{3/2}(z), \tag{1.C.112}$$

where

$$g_p(z) = \frac{1}{\Gamma(p)} \int_0^\infty dx \frac{x^{p-1}}{z^{-1}e^x - 1}. \tag{1.C.113}$$

Again, noting that $g_p(z) \to z$ for $z \to 0$, one recovers the ideal classical gas in the low-temperature dilute limit $\mu < 0$ and $T \ll |\mu|$. On the other hand, we see that the equation for the density cannot be satisfied at low temperatures when $g_{3/2}(z \to 1^-)/\lambda_T^3$ becomes smaller than n, i.e. below the temperature

$$T_c = \frac{2\pi\hbar^2}{mk_B} \left(\frac{n}{\zeta(3/2)}\right)^{2/3} \tag{1.C.114}$$

(we use $g_{3/2}(1) = \zeta(3/2)$). This signals Bose–Einstein condensation (BEC), that is the occupation of the one-body zero-momentum state by a macroscopic number $N_0 = n_0\mathcal{V}$ of particles. The equation for the density becomes (with $\mu \to 0^-$)

$$n = n_0 + \frac{1}{\mathcal{V}} \sum_{\mathbf{k}\neq 0} n_B(\epsilon_{\mathbf{k}}) = n_0 + \frac{\zeta(3/2)}{\lambda_T^3} = n_0 + n \left(\frac{T}{T_c}\right)^{3/2}, \tag{1.C.115}$$

which gives the condensate density

$$n_0 = n \left[1 - \left(\frac{T}{T_c}\right)^{3/2}\right]. \tag{1.C.116}$$

Whereas the non-interacting fermion gas provides us in some cases with a good starting point for understanding the effect of interactions (e.g. in the case of a Fermi liquid, see Chap. 4), the non-interacting boson system is somewhat pathological and perturbation theory about the non-interacting limit is ill behaved.[110]

Appendix 1.D Functionals and functional derivatives

A functional is a function that maps functions to numbers. Its argument is conventionally written in square brackets rather than parentheses. The derivative of a functional $F[\phi]$ is defined in such a way that if one discretizes[111] the function $\phi(\mathbf{x})$ ($\mathbf{x}$ denotes a d-dimensional

[110] For instance, the non-interacting Bose gas at zero temperature has an infinite compressibility, but any nonzero interactions make the compressibility finite (see Chap. 7).

[111] The continuous variable $\mathbf{x}$ then becomes a discrete variable defined on a d-dimensional lattice with lattice spacing a. The continuous formulation is recovered when a tends to zero. For example, for a one-dimensional variable, $x \to x_i = ia$ and $\phi(x) \to \phi(x_i) \equiv \phi_i$, with i integer.

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variable), one recovers the usual rules for the derivation of ordinary functions. We then impose

$$\frac{\delta\phi(\mathbf{x})}{\delta\phi(\mathbf{y})} = \delta(\mathbf{x}-\mathbf{y}), \tag{1.D.1}$$

which is the analog of $\partial\phi_i/\partial\phi_j = \delta_{i,j}$, and the standard rules

$$\begin{aligned}
&\frac{\delta}{\delta\phi(\mathbf{x})}\left(F_1[\phi]+F_2[\phi]\right) = \frac{\delta F_1[\phi]}{\delta\phi(\mathbf{x})} + \frac{\delta F_2[\phi]}{\delta\phi(\mathbf{x})},\\
&\frac{\delta}{\delta\phi(\mathbf{x})}F_1[\phi]F_2[\phi] = \frac{\delta F_1[\phi]}{\delta\phi(\mathbf{x})}F_2[\phi] + F_1[\phi]\frac{\delta F_2[\phi]}{\delta\phi(\mathbf{x})},\\
&\frac{\delta}{\delta\phi(\mathbf{x})}f\big(\phi(\mathbf{y})\big) = f'\big(\phi(\mathbf{y})\big)\frac{\delta\phi(\mathbf{y})}{\delta\phi(\mathbf{x})} = f'\big(\phi(\mathbf{y})\big)\delta(\mathbf{x}-\mathbf{y}).
\end{aligned} \tag{1.D.2}$$

Equation (1.D.1) also implies

$$\frac{\delta}{\delta\phi(\mathbf{y})}\boldsymbol{\nabla}\phi(\mathbf{x}) = \boldsymbol{\nabla}\frac{\delta\phi(\mathbf{x})}{\delta\phi(\mathbf{y})} = \boldsymbol{\nabla}\delta(\mathbf{x}-\mathbf{y}) \equiv \frac{\partial}{\partial\mathbf{x}}\delta(\mathbf{x}-\mathbf{y}). \tag{1.D.3}$$

For a functional $F[\phi]$ that can be expressed as an integral of a function of $\phi(\mathbf{x})$ and $\boldsymbol{\nabla}\phi(\mathbf{x})$,

$$F[\phi] = \int d^d x\, f\big(\phi(\mathbf{x}), \boldsymbol{\nabla}\phi(\mathbf{x})\big), \tag{1.D.4}$$

one then finds

$$\begin{aligned}
\frac{\delta F[\phi]}{\delta\phi(\mathbf{x})} &= \int d^d y \left(\frac{\partial f}{\partial\phi(\mathbf{y})}\frac{\delta\phi(\mathbf{y})}{\delta\phi(\mathbf{x})} + \frac{\partial f}{\partial(\boldsymbol{\nabla}\phi(\mathbf{y}))}\cdot\frac{\delta(\boldsymbol{\nabla}\phi(\mathbf{y}))}{\delta\phi(\mathbf{x})}\right)\\
&= \int d^d y \left(\frac{\partial f}{\partial\phi(\mathbf{y})}\delta(\mathbf{x}-\mathbf{y}) + \frac{\partial f}{\partial(\boldsymbol{\nabla}\phi(\mathbf{y}))}\cdot\frac{\partial}{\partial\mathbf{y}}\delta(\mathbf{x}-\mathbf{y})\right)\\
&= \frac{\partial f}{\partial\phi(\mathbf{x})} - \boldsymbol{\nabla}\cdot\frac{\partial f}{\partial(\boldsymbol{\nabla}\phi(\mathbf{x}))},
\end{aligned} \tag{1.D.5}$$

where the last result is obtained by integrating the second term by parts assuming that the field $\phi(\mathbf{x})$ vanishes at the boundaries of its domain of definition or satisfies periodic boundary conditions.

The functional $F[\phi]$ and its functional derivative can be expanded as

$$\begin{aligned}
F[\phi] &= F[0] + \sum_{n=1}^{\infty}\frac{1}{n!}\int d^d x_1\cdots d^d x_n\, F^{(n)}(\mathbf{x}_1,\cdots,\mathbf{x}_n)\phi(\mathbf{x}_1)\cdots\phi(\mathbf{x}_n),\\
\frac{\delta F[\phi]}{\delta\phi(\mathbf{x})} &= F^{(1)}(x) + \sum_{n=2}^{\infty}\frac{1}{(n-1)!}\int d^d x_1\cdots d^d x_{n-1}\, F^{(n)}(\mathbf{x}_1,\cdots,\mathbf{x}_{n-1},\mathbf{x})\\
&\qquad\qquad \times\phi(\mathbf{x}_1)\cdots\phi(\mathbf{x}_{n-1}),
\end{aligned} \tag{1.D.6}$$

where the function $F^{(n)}(\mathbf{x}_1,\cdots,\mathbf{x}_n)$ is symmetric under the exchange $\mathbf{x}_i \leftrightarrow \mathbf{x}_j$. We deduce

$$F[\phi+u] = F[0] + \sum_{n=1}^{\infty}\frac{1}{n!}\int d^d x_1\cdots d^d x_n\, F^{(n)}(\mathbf{x}_1,\cdots,\mathbf{x}_n)$$

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$$\begin{aligned}
&\times \big(\phi(\mathbf{x}_1)+u(\mathbf{x}_1)\big)\cdots\big(\phi(\mathbf{x}_n)+u(\mathbf{x}_n)\big)\\
&= F[\phi] + \int dx\, F^{(1)}(x)u(x) + \sum_{n=2}^{\infty}\frac{1}{(n-1)!}\int d^dx_1\cdots d^dx_{n-1}\\
&\qquad \times F^{(n)}(\mathbf{x}_1,\cdots,\mathbf{x}_{n-1},\mathbf{x})\phi(\mathbf{x}_1)\cdots\phi(\mathbf{x}_{n-1})u(\mathbf{x}) + \mathcal{O}(u^2)\\
&= F[\phi] + \int d^dx\,\frac{\delta F[\phi]}{\delta\phi(\mathbf{x})}u(\mathbf{x}) + \mathcal{O}(u^2).
\end{aligned} \tag{1.D.7}$$

More generally, one finds

$$F[\phi+u] = F[\phi] + \sum_{n=1}^{\infty}\frac{1}{n!}\int d^dx_1\cdots d^dx_n\,\frac{\delta^n F[\phi]}{\delta\phi(\mathbf{x}_1)\cdots\delta\phi(\mathbf{x}_n)}u(\mathbf{x}_1)\cdots u(\mathbf{x}_n). \tag{1.D.8}$$

1.D.1 Grassmann variables

Similar rules can be defined for a functional $F[\psi^*,\psi]$ of a Grassmannian function $\psi(\mathbf{x})$ and its complex conjugate $\psi^*(\mathbf{x})$, provided the anticommuting properties of $\psi(\mathbf{x})$ and $\psi^*(\mathbf{x})$ are properly taken into account. For instance,

$$\begin{aligned}
\frac{\delta}{\delta\psi(\mathbf{x})}\int d^dy\,\psi^*(\mathbf{y})\psi(\mathbf{y}) &= -\psi^*(\mathbf{x}),\\
\frac{\delta}{\delta\psi^*(\mathbf{x})}\int d^dy\,\psi^*(\mathbf{y})\psi(\mathbf{y}) &= \psi(\mathbf{x}).
\end{aligned} \tag{1.D.9}$$

In particular, one has

$$\begin{aligned}
F[\psi^*+u^*,\psi+u] &= F[\psi^*,\psi]\\
&\quad + \int d^dx\left(u^*(\mathbf{x})\frac{\delta F[\psi^*,\psi]}{\delta\psi^*(\mathbf{x})} + u(\mathbf{x})\frac{\delta F[\psi^*,\psi]}{\delta\psi(\mathbf{x})}\right) + \mathcal{O}(u^2),
\end{aligned} \tag{1.D.10}$$

where the Grassmann numbers $u^*(\mathbf{x})$ and $u(\mathbf{x})$ must appear on the left of the functional derivatives in agreement with the chain rule for derivation with anticommuting variables [Eq. (1.240)].

Appendix 1.E Gaussian integrals and Wick's theorem

1.E.1 Real variables

We consider the multi-dimensional integral

$$\begin{aligned}
Z(J) &= \int_{-\infty}^{\infty} dx_1\cdots dx_n\, e^{-\frac{1}{2}\sum_{i,j}x_iA_{ij}x_j+\sum_i J_ix_i}\\
&= \int_{-\infty}^{\infty} dx_1\cdots dx_n\, e^{-\frac{1}{2}x^TAx+J^Tx},
\end{aligned} \tag{1.E.1}$$

where x denotes the column vector $(x_1,\cdots,x_n)^T$. The n-dimensional real matrix A is assumed to be symmetric and positive definite. It can be diagonalized by an orthogonal transformation: $A = O^TDO$, with O an orthogonal matrix ($O^TO = OO^T = 1$) and D

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a diagonal matrix with non-negative elements ($D_{ii} > 0$). With the change of variables $y = O(x - A^{-1}J)$ (Jacobian $|\det 0| = 1$), we obtain

$$-\frac{1}{2}x^T Ax + J^T x = -\frac{1}{2}y^T Dy + \frac{1}{2}J^T A^{-1} J \tag{1.E.2}$$

and $Z(J)$ reduces to a product of Gaussian integrals,[112]

$$\begin{aligned} Z(J) &= e^{\frac{1}{2}J^T A^{-1}J} \prod_{i=1}^{n} \int_{-\infty}^{\infty} dy\, e^{-\frac{1}{2}y^2 D_{ii}} \\ &= e^{\frac{1}{2}J^T A^{-1}J} \prod_{i=1}^{n} \left(\frac{2\pi}{D_{ii}}\right)^{1/2} \\ &= e^{\frac{1}{2}J^T A^{-1}J} (2\pi)^{n/2} (\det A)^{-1/2}, \end{aligned} \tag{1.E.3}$$

using $\det D = \det A$.

1.E.1.1 Wick's theorem

The mean value

$$\langle x_{i_1} \cdots x_{i_k} \rangle = \frac{\int_{-\infty}^{\infty} dx_1 \cdots dx_n\, x_{i_1} \cdots x_{i_k}\, e^{-\frac{1}{2}x^T Ax}}{\int_{-\infty}^{\infty} dx_1 \cdots dx_n\, e^{-\frac{1}{2}x^T Ax}} \tag{1.E.4}$$

can be obtained from the generating function (1.E.1),

$$\begin{aligned} \langle x_{i_1} \cdots x_{i_k} \rangle &= \frac{1}{Z(0)} \frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_k}} Z(J)\Big|_{J=0} \\ &= \frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_k}} e^{\frac{1}{2}J^T A^{-1}J}\Big|_{J=0}. \end{aligned} \tag{1.E.5}$$

We deduce

$$\begin{aligned} \langle x_{i_1} x_{i_2} \rangle &= A^{-1}_{i_1 i_2}, \\ \langle x_{i_1} x_{i_2} x_{i_3} x_{i_4} \rangle &= A^{-1}_{i_1 i_2} A^{-1}_{i_3 i_4} + A^{-1}_{i_1 i_3} A^{-1}_{i_2 i_4} + A^{-1}_{i_1 i_4} A^{-1}_{i_2 i_3}, \end{aligned} \tag{1.E.6}$$

and more generally

$$\begin{aligned} \langle x_{i_1} \cdots x_{i_{2k}} \rangle &= \sum_{\substack{\text{all possible pairings} \\ P \text{ of } \{i_1, \cdots, i_{2k}\}}} \langle x_{i_{P(1)}} x_{i_{P(2)}} \rangle \cdots \langle x_{i_{P(2k-1)}} x_{i_{P(2k)}} \rangle \\ &= \sum_{\substack{\text{all possible pairings} \\ P \text{ of } \{i_1, \cdots, i_{2k}\}}} A^{-1}_{i_{P(1)} i_{P(2)}} \cdots A^{-1}_{i_{P(2k-1)} i_{P(2k)}}. \end{aligned} \tag{1.E.7}$$

This result holds for any Gaussian probability distribution $e^{-\frac{1}{2}x^T Ax}$ with zero mean $\langle x_i \rangle = 0$. In field theory, it is known as Wick's theorem and underlies perturbation theory (Sec. 1.5).

[112] Recall that $\int_{-\infty}^{\infty} dy\, e^{-\frac{a}{2}y^2} = \sqrt{2\pi/a}$ for $a > 0$.

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1.E.2 Complex variables

Now we consider the multidimensional integral

$$\begin{aligned} Z(J^*,J) &= \int_{-\infty}^{\infty} \prod_{i=1}^{n} \frac{dz_i^* dz_i}{2i\pi}\, e^{-\sum_{i,j} z_i^* A_{ij} z_j + \sum_i (J_i^* z_i + \text{c.c.})} \\ &= \int_{-\infty}^{\infty} \prod_{i=1}^{n} \frac{dz_i^* dz_i}{2i\pi}\, e^{-z^\dagger A z + (J^\dagger z + \text{c.c.})} \end{aligned} \tag{1.E.8}$$

over the complex variables z_i and z_i^* $(i = 1, \cdots, n)$, where

$$\frac{dz_i^* dz_i}{2i\pi} = \frac{1}{\pi} d\Re(z_i) d\Im(z_i) \tag{1.E.9}$$

and $z^\dagger = (z_1^*, \cdots, z_n^*)$. Let us first assume the complex matrix A to be positive definite Hermitian ($A^\dagger = A$). We can then proceed as in the real case. A can be diagonalized by a unitary transformation: $A = U^\dagger D U$ with U a unitary matrix ($UU^\dagger = U^\dagger U = 1$) and D a diagonal matrix with non-negative real elements ($D_{ii} > 0$). The change of variables

$$z' = U(z - A^{-1}J), \qquad z'^\dagger = (z^\dagger - J^\dagger A^{-1}) U^\dagger \tag{1.E.10}$$

has a Jacobian $|\det U|^2 = 1$, and leads to[113]

$$Z(J^*,J) = e^{J^\dagger A^{-1} J} \int_{-\infty}^{\infty} \prod_{i=1}^{n} \frac{dz_i'^* dz_i'}{2i\pi} e^{-z'^\dagger D z'} = e^{J^\dagger A^{-1} J} (\det A)^{-1} \tag{1.E.11}$$

(using $\det D = \det A$). It can be shown that (1.E.11) holds for any complex matrix with a positive definite Hermitian part $\frac{1}{2}(A + A^\dagger)$.[114]

1.E.2.1 Wick's theorem

The average value of a monomial can be obtained from the generating function $Z(J^*, J)$,

$$\begin{aligned} \langle z_{i_1} z_{i'_1}^* \cdots z_{i_k} z_{i'_k}^* \rangle &= \frac{\int_{-\infty}^{\infty} \prod_{i=1}^{n} \frac{dz_i^* dz_i}{2i\pi}\, z_{i_1} z_{i'_1}^* \cdots z_{i_k} z_{i'_k}^*\, e^{-z^\dagger A z}}{\int_{-\infty}^{\infty} \prod_{i=1}^{n} \frac{dz_i^* dz}{2i\pi} e^{-z^\dagger A z}} \\ &= \frac{1}{Z(0,0)} \frac{\partial}{\partial J_{i_1}^*} \frac{\partial}{\partial J_{i'_1}} \cdots \frac{\partial}{\partial J_{i_k}^*} \frac{\partial}{\partial J_{i'_k}} Z(J^*,J)\Big|_{J^*=J=0} \\ &= \frac{\partial}{\partial J_{i_1}^*} \frac{\partial}{\partial J_{i'_1}} \cdots \frac{\partial}{\partial J_{i_k}^*} \frac{\partial}{\partial J_{i'_k}} e^{J^\dagger A^{-1} J}\Big|_{J^*=J=0}. \end{aligned} \tag{1.E.12}$$

Note that only monomials with equal number of z_i and $z_{i'}^*$ factors have a non-vanishing average. From (1.E.12), we obtain

$$\langle z_{i_1} z_{i'_1}^* \cdots z_{i_k} z_{i'_k}^* \rangle = \prod_{j=1}^{k} \frac{\partial}{\partial J_{i_j}^*} \left(\sum_{l_1} J_{l_1}^* A_{l_1 i'_1}^{-1} \right) \cdots \left(\sum_{l_k} J_{l_k}^* A_{l_k i'_k}^{-1} \right) e^{J^\dagger A^{-1} J}\Big|_{J^*=J=0}$$

[113] We use $\int_{-\infty}^{\infty} \frac{dz^* dz}{2i\pi} e^{-a|z|^2} = a^{-1}$ for $a > 0$.

[114] This can be understood by noting that, contrary to the case of the real Gaussian integral, the result (1.E.11) is an algebraic function of the matrix elements A_{ij} and can therefore be analytically continued to any complex matrix with a positive definite Hermitian part [4].

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$$= \sum_{P\in S_k} A^{-1}_{i_{P(1)}i'_1} \cdots A^{-1}_{i_{P(k)}i'_k}, \tag{1.E.13}$$

where the sum is over all permutations P of $\{1, \cdots, k\}$.

1.E.3 Grassmannian integrals

In order to establish the analogous results for Grassmann variables, we first need to derive the law of change of variables in a Grassmannian integral.

1.E.3.1 Change of variables

Let us consider the integral

$$I = \int d\theta f(\theta) \tag{1.E.14}$$

with θ a Grassmann variable, and perform the change of variables $\theta = a\theta' + b$ with a a c-number and b a Grassmann number. Since integration and derivation are identical for Grassmann numbers (Sec. 1.3.2), we obtain

$$I = \frac{\partial}{\partial\theta} f(\theta) = \frac{\partial\theta'}{\partial\theta}\frac{\partial f}{\partial\theta'} = \frac{1}{a}\frac{\partial f}{\partial\theta'} = \frac{1}{a}\int d\theta' f(a\theta' + b) \tag{1.E.15}$$

using the chain rule for differentiation of Grassmann variables [Eq. (1.240)]. The Jacobian of the transformation is therefore a^{-1} instead of a for ordinary variables.

This result can be generalized to the multidimensional integral

$$\begin{aligned} I &= \int d\theta_1 \cdots d\theta_n \, f(\theta_1 \cdots \theta_n) \\ &= \prod_{i=1}^n \frac{\partial}{\partial\theta_i} f(\theta_1 \cdots \theta_n) \\ &= \prod_{i=1}^n \left(\sum_j \frac{\partial\theta'_j}{\partial\theta_i}\frac{\partial}{\partial\theta'_j} \right) f(\theta_1 \cdots \theta_n). \end{aligned} \tag{1.E.16}$$

For an affine transformation $\theta'_i = \sum_j M_{ij}\theta_j + B_i$, with M_{ij} a c-number and B_i a Grassmann number, we obtain

$$I = \prod_{i=1}^n \left(\sum_j M_{ji} \frac{\partial}{\partial\theta'_j} \right) f(\theta_1 \cdots \theta_n). \tag{1.E.17}$$

Since the differential operators $\partial/\partial\theta'_j$ anticommute, the differential operator acting on $f(\theta_1 \cdots \theta_n)$ is proportional to $\partial_{\theta'_1} \cdots \partial_{\theta'_n}$, and the prefactor is easily seen to be the determinant of the matrix M,

$$I = \det\left(\frac{\partial\theta'_i}{\partial\theta_j}\right) \prod_{i=1}^n \frac{\partial}{\partial\theta'_i} f(\theta_1 \cdots \theta_n) = \det\left(\frac{\partial\theta'_i}{\partial\theta_j}\right) \int \prod_{i=1}^n d\theta'_i \, f(\theta_1 \cdots \theta_n), \tag{1.E.18}$$

i.e.

$$\prod_{i=1}^n d\theta_i = \det\left(\frac{\partial\theta'_i}{\partial\theta_j}\right) \prod_{i=1}^n d\theta'_i. \tag{1.E.19}$$

The Jacobian is the inverse of the determinant $\det(\partial\theta_i/\partial\theta'_j)$.

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1.E.3.2 Gaussian integrals

The one-dimensional Gaussian integral over a pair of complex Grassmann variables reads

$$\int d\theta^* d\theta \, e^{-a\theta^*\theta} = \int d\theta^* d\theta (1 - a\theta^*\theta) = a \tag{1.E.20}$$

for any c-number a. This result is easily generalized to the multidimensional Gaussian integral

$$Z = \int \prod_{i=1}^{n} d\theta_i^* d\theta_i \, e^{-\sum_{i,j} \theta_i^* A_{ij} \theta_j}, \tag{1.E.21}$$

where A is an arbitrary complex matrix. With the linear transformation $\theta_i' = \sum_j A_{ij}\theta_j$, we obtain

$$Z = (\det A) \int \prod_{i=1}^{n} d\theta_i^* d\theta_i' \, e^{-\sum_i \theta_i^* \theta_i'} = (\det A) \prod_{i=1}^{n} \int d\theta_i^* d\theta_i' (1 - \theta_i^* \theta_i') = \det A. \tag{1.E.22}$$

Note that for ordinary complex variables, one would obtain $(\det A)^{-1}$.

The generalized Gaussian integral

$$Z(J^*, J) = \int \prod_{i=1}^{n} d\theta_i^* d\theta_i \, e^{-\theta^\dagger A \theta + (J^\dagger \theta + \text{c.c.})} = (\det A) \, e^{J^\dagger A^{-1} J} \tag{1.E.23}$$

(J_i^* and J_i are anticommuting numbers) is easily calculated using the affine change of variables $\theta = \theta' + A^{-1}J$ and $\theta^\dagger = \theta'^\dagger + J^\dagger A^{-1}$.

1.E.3.3 Wick's theorem

As with complex variables, the average value

$$\begin{aligned} \langle \theta_{i_1} \cdots \theta_{i_k} \theta_{i_k'}^* \cdots \theta_{i_1'}^* \rangle &= \frac{\int \Pi_{i=1}^n d\theta_i^* d\theta_i \, \theta_{i_1} \cdots \theta_{i_k} \theta_{i_k'}^* \cdots \theta_{i_1'}^* \, e^{-\theta^\dagger A \theta}}{\int \Pi_{i=1}^n d\theta_i^* d\theta_i \, e^{-\theta^\dagger A \theta}} \\ &= \frac{(-1)^k}{Z(0,0)} \frac{\partial}{\partial J_{i_1}^*} \cdots \frac{\partial}{\partial J_{i_k}^*} \frac{\partial}{\partial J_{i_k'}} \cdots \frac{\partial}{\partial J_{i_1'}} Z(J, J^*) \Big|_{J^* = J = 0} \end{aligned} \tag{1.E.24}$$

can be obtained from the generating function (1.E.23). This yields

$$\begin{aligned} \langle \theta_{i_1} \cdots \theta_{i_k} \theta_{i_k'}^* \cdots \theta_{i_1'}^* \rangle &= \frac{\partial}{\partial J_{i_1}^*} \cdots \frac{\partial}{\partial J_{i_k}^*} \left(\sum_{l_k} J_{l_k}^* A_{l_k i_k'}^{-1} \right) \cdots \left(\sum_{l_1} J_{l_1}^* A_{l_1 i_1'}^{-1} \right) e^{J^\dagger A^{-1} J} \Big|_{J^* = J = 0} \\ &= \sum_{P \in S_k} \epsilon_P A_{i_{P(k)} i_k'}^{-1} \cdots A_{i_{P(1)} i_1'}^{-1}, \end{aligned} \tag{1.E.25}$$

where ϵ_P denotes the signature of the permutation P.

1.E.3.4 "Real" Grassmannian Gaussian integrals

We can also consider integrals of the type

$$Z = \int d\theta_1 \cdots d\theta_{2n} \, e^{-\frac{1}{2}\theta^T A \theta}, \tag{1.E.26}$$

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where A is an antisymmetric matrix: $A_{ij} = -A_{ji}$.[115] Expanding the exponential in (1.E.26) and retaining the only non-vanishing terms, we obtain

$$\begin{aligned} Z &= \frac{(-1)^n}{2^n n!} \int d\theta_1 \cdots d\theta_{2n} \Big(\sum_{i,j=1}^{2n} \theta_i A_{ij} \theta_j \Big)^n \\ &= \frac{1}{2^n n!} \sum_{P \in S_{2n}} \epsilon_P A_{P(1),P(2)} \cdots A_{P(2n-1),P(2n)}. \end{aligned} \tag{1.E.27}$$

The quantity in the rhs is called the Pfaffian of the matrix A,

$$Z = \mathrm{Pf}(A). \tag{1.E.28}$$

The Pfaffian is related in a simple way to the determinant. To see this, we consider the square of Z,

$$Z^2 = \int d\theta_1 \cdots d\theta_{2n} d\theta'_1 \cdots d\theta'_{2n}\, e^{-\frac{1}{2}\theta^T A \theta - \frac{1}{2}\theta'^T A \theta'}. \tag{1.E.29}$$

The change of variables

$$\begin{aligned} \eta_k &= \frac{1}{\sqrt{2}}(\theta_k + i\theta'_k), \\ \eta^*_k &= \frac{1}{\sqrt{2}}(\theta_k - i\theta'_k) \end{aligned} \tag{1.E.30}$$

gives

$$\begin{aligned} d\theta_1 \cdots d\theta_{2n} d\theta'_1 \cdots d\theta'_{2n} &= (-1)^n d\theta_1 d\theta'_1 \cdots d\theta_{2n} d\theta'_{2n} \\ &= (-1)^n J d\eta^*_1 d\eta_1 \cdots d\eta^*_{2n} d\eta_{2n}, \end{aligned} \tag{1.E.31}$$

where J is the Jacobian of the transformation $\theta_k, \theta'_k \to \eta^*_k, \eta_k$. Since

$$\det \left(\frac{\partial(\eta^*_k, \eta_k)}{\partial(\theta_k, \theta'_k)} \right) = \det \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} = i, \tag{1.E.32}$$

we find $J = i^{2n} = (-1)^n$ and therefore

$$Z^2 = \int d\eta^*_1 d\eta_1 \cdots d\eta^*_{2n} d\eta_{2n}\, e^{-\eta^\dagger A \eta} = \det(A), \tag{1.E.33}$$

i.e.

$$\mathrm{Pf}(A)^2 = \det A. \tag{1.E.34}$$

In most cases of interest, we can use $Z = \pm(\det A)^{1/2}$, since the sign of Z either does not matter (e.g. when calculating average values $\langle \theta_{i_1} \cdots \theta_{i_k} \rangle$) or can be easily determined.

The result for the generalized Gaussian integral

$$Z(J) = \int d\theta_1 \cdots d\theta_{2n}\, e^{-\frac{1}{2}\theta^T A \theta + J^T \theta} = \pm \sqrt{\det A}\, e^{-\frac{1}{2} J^T A^{-1} J} \tag{1.E.35}$$

[115] If A is not antisymmetric, only its antisymmetric part contributes since the product $\theta_i \theta_j$ is antisymmetric.

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(note the minus sign in the exponential) is derived from the change of variables $\theta = \theta' - A^{-1}J$. Equation (1.E.35) implies that Wick's theorem for real Grassmann variables reads

$$\langle \theta_{i_1} \cdots \theta_{i_{2k}} \rangle = \sum_{\substack{\text{all possible pairings} \\ P \text{ of } \{i_1,\cdots,i_{2k}\}}} \epsilon_P A^{-1}_{i_{P(1)} i_{P(2)}} \cdots A^{-1}_{i_{P(2k-1)} i_{P(2k)}}. \tag{1.E.36}$$

Except for the signature ϵ_P of the permutation, this result is similar to the one obtained for real variables [Eq. (1.E.7)].

1.E.4 Gaussian functional integrals

Since functional integrals can be defined as the continuum limit of ordinary integrals (Secs. 1.1 and 1.4), the preceding results generalize straightforwardly to Gaussian functional integrals, i.e.

$$\begin{aligned} Z[J] &= \int \mathcal{D}[\phi]\, e^{-\frac{1}{2}\int d^dx d^dy\, \phi(\mathbf{x}) A(\mathbf{x},\mathbf{y}) \phi(\mathbf{y}) + \int d^dx\, J(\mathbf{x})\phi(\mathbf{x})} \\ &= (\det A)^{-1/2}\, e^{\frac{1}{2}\int d^dx d^dy\, J(\mathbf{x}) A^{-1}(\mathbf{x},\mathbf{y}) J(\mathbf{y})} \end{aligned} \tag{1.E.37}$$

for a real field $\phi(\mathbf{x})$ coupled to a real external "source" $J(\mathbf{x})$, and

$$\begin{aligned} Z[J^*, J] &= \int \mathcal{D}[\psi^*, \psi]\, e^{-\int d^dx d^dy\, \psi^*(\mathbf{x}) A(\mathbf{x},\mathbf{y}) \psi(\mathbf{y}) + \int d^dx\, [J^*(\mathbf{x})\psi(\mathbf{x}) + \text{c.c.}]} \\ &= (\det A)^{-\zeta}\, e^{\int d^dx d^dy\, J^*(\mathbf{x}) A^{-1}(\mathbf{x},\mathbf{y}) J(\mathbf{y})} \end{aligned} \tag{1.E.38}$$

for a complex ($\zeta = 1$) or Grassmann ($\zeta = -1$) field which couples to an external source of the same type. For a (real) Grassmannian field $\psi(\mathbf{x})$ coupled to a Grassmannian source $J(\mathbf{x})$, one finds

$$\begin{aligned} Z[J] &= \int \mathcal{D}[\psi]\, e^{-\frac{1}{2}\int d^dx d^dy\, \psi(\mathbf{x}) A(\mathbf{x},\mathbf{y}) \psi(\mathbf{y}) + \int d^dx\, J(\mathbf{x})\psi(\mathbf{x})} \\ &= \pm(\det A)^{1/2}\, e^{-\frac{1}{2}\int d^dx d^dy\, J(\mathbf{x}) A^{-1}(\mathbf{x},\mathbf{y}) J(\mathbf{y})} \end{aligned} \tag{1.E.39}$$

(note the minus sign in the exponential). Equations (1.E.37-1.E.39) hold up to a multiplicative constant which depends on the precise definition of the functional integral measure $\mathcal{D}[\cdot]$.

The inverse A^{-1} of the operator A is defined by

$$\int d^dz\, A(\mathbf{x},\mathbf{z}) A^{-1}(\mathbf{z},\mathbf{y}) = \int d^dz\, A^{-1}(\mathbf{x},\mathbf{z}) A(\mathbf{z},\mathbf{y}) = \delta(\mathbf{x}-\mathbf{y}). \tag{1.E.40}$$

$\det A$ denotes the determinant of the operator A. A useful identity is

$$\det A = \exp(\ln \det A) = \exp(\operatorname{Tr} \ln A), \tag{1.E.41}$$

where $\operatorname{Tr} A$ denotes the trace of the operator A. When A can be written as $A = 1 + K$,

$$\operatorname{Tr} \ln A = \operatorname{Tr} \ln(1+K) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \operatorname{Tr} K^n \tag{1.E.42}$$

(provided K^n exists for any n), where

$$\operatorname{Tr} K^n = \int d^dx_1 \cdots d^dx_n\, K(\mathbf{x}_1,\mathbf{x}_2) K(\mathbf{x}_2,\mathbf{x}_3) \cdots K(\mathbf{x}_n,\mathbf{x}_1). \tag{1.E.43}$$

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1.E.4.1 Wick's theorem

The various forms of Wick's theorem derived in the previous sections translate into

$$\begin{aligned}
\langle\phi(\mathbf{x}_1)\cdots\phi(\mathbf{x}_{2k})\rangle &= \sum_{\substack{\text{all possible pairings}\\ P \text{ of } \{\mathbf{x}_1,\cdots,\mathbf{x}_{2k}\}}} A^{-1}\big(\mathbf{x}_{P(1)},\mathbf{x}_{P(2)}\big)\cdots A^{-1}\big(\mathbf{x}_{P(2k-1)},\mathbf{x}_{P(2k)}\big),\\
\langle\psi(\mathbf{x}_1)\cdots\psi(\mathbf{x}_k)\psi^*(\mathbf{x}'_k)\cdots\psi^*(\mathbf{x}'_1)\rangle &= \sum_{P\in S_k} \zeta^P A^{-1}\big(\mathbf{x}_1,\mathbf{x}'_{P(1)}\big)\cdots A^{-1}\big(\mathbf{x}_k,\mathbf{x}'_{P(k)}\big),\\
\langle\psi(\mathbf{x}_1)\cdots\psi(\mathbf{x}_{2k})\rangle &= \sum_{\substack{\text{all possible pairings}\\ P \text{ of } \{\mathbf{x}_1,\cdots,\mathbf{x}_{2k}\}}} \epsilon_P A^{-1}\big(\mathbf{x}_{P(1)},\mathbf{x}_{P(2)}\big)\cdots A^{-1}\big(\mathbf{x}_{P(2k-1)},\mathbf{x}_{P(2k)}\big),
\end{aligned} \tag{1.E.44}$$

where the averages are obtained from the generating functionals (1.E.37), (1.E.38) and (1.E.39), respectively. ζ^P equals one for bosons and the signature of the permutation P for fermions. The last line in (1.E.44) holds for Grassmann variables.

Finally we note that Wick's theorem for complex variables can be generalized from the result

$$\begin{aligned}
Z[J^*,J] &= \int \mathcal{D}[\psi^*,\psi]\, e^{-\frac{1}{2}\int d^dx d^dy (\psi^*(\mathbf{x}),\psi(\mathbf{x}))A(\mathbf{x},\mathbf{y})\binom{\psi(\mathbf{y})}{\psi^*(\mathbf{y})}+\int d^dx (J^*(\mathbf{x}),J(\mathbf{x}))\binom{\psi(\mathbf{x})}{\psi^*(\mathbf{x})}}\\
&= \pm(\det A)^{-1/2} e^{\frac{1}{2}\int d^dx d^dy (J^*(\mathbf{x}),J(\mathbf{x}))A^{-1}(\mathbf{x},\mathbf{y})\binom{J(\mathbf{x})}{J^*(\mathbf{x})}}.
\end{aligned} \tag{1.E.45}$$

Contrary to the case previously discussed, the matrix $A(\mathbf{x},\mathbf{y})$ in (1.E.45) can connect $\psi(\mathbf{x})$ and $\psi(\mathbf{y})$, or $\psi^*(\mathbf{x})$ and $\psi^*(\mathbf{y})$. Thus the averages $\langle\psi(\mathbf{x})\psi(\mathbf{y})\rangle$ and $\langle\psi^*(\mathbf{x})\psi^*(\mathbf{y})\rangle$ can be nonzero. Wick's theorem is otherwise unchanged: The average of a product of ψ and ψ^* fields is given by the sum of all possible pairings.

Appendix 1.F Matsubara frequency sums

In this Appendix, we describe the standard methods to carry out summations over Matsubara frequencies ω_n and discuss explicitly a few examples.

General method

We consider the Matsubara sum

$$S = \frac{1}{\beta}\sum_{\omega_n} f(i\omega_n), \tag{1.F.1}$$

where f is some function. To compute S, we consider the integral

$$I = \int_{\mathcal{C}} \frac{dz}{2i\pi} f(z) n_\zeta(z) \tag{1.F.2}$$

along some path $\mathcal{C}$ in the complex plane, where

$$n_\zeta(z) = \frac{1}{e^{\beta z}-\zeta} = \begin{cases} n_B(z) & \text{(bosons)},\\ n_F(z) & \text{(fermions)}. \end{cases} \tag{1.F.3}$$

Since $n_\zeta(z)$ has simple poles at the imaginary frequencies $z = i\omega_n$ (with residue ζ/β), the integral I will be related to S if the integration path $\mathcal{C}$ is suitably chosen.

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Example 1

As a first example, we consider the sum

$$S = \frac{1}{\beta}\sum_{\omega_n} \frac{e^{i\omega_n \eta}}{i\omega_n - \xi} \qquad (\eta \to 0^+). \tag{1.F.4}$$

S can be calculated from the integral

$$I = \oint_{\mathcal{C}} \frac{dz}{2i\pi} \frac{e^{z\eta}}{z-\xi} n_\zeta(z) \tag{1.F.5}$$

along a circle $\mathcal{C}$ of radius $R \to \infty$ around the origin $z = 0$ in the complex plane. In the limit $|z| \to \infty$, the function $e^{z\eta} n_\zeta(z)$ is exponentially suppressed[116] so that the integral I vanishes for $R \to \infty$. Now we can also evaluate I by the residue theorem. Besides the poles of $n_\zeta(z)$ at the Matsubara frequencies ω_n, there is a pole at $z = \xi$, and

$$I = \sum_{\omega_n} \mathrm{Res}(i\omega_n) + \mathrm{Res}(\xi) = \zeta S + n_\zeta(\xi) = 0, \tag{1.F.6}$$

i.e.

$$\frac{1}{\beta}\sum_{\omega_n} \frac{e^{i\omega_n \eta}}{i\omega_n - \xi} = \begin{cases} -n_B(\xi) & \text{(bosons)}, \\ n_F(\xi) & \text{(fermions)}. \end{cases} \tag{1.F.7}$$

Example 2

As a second example, we consider the grand potential of non-interacting bosons or fermions (Sec. 1.4.2),

$$\Omega_0 = \frac{\zeta}{\beta}\sum_{\alpha,\omega_n} \ln[\beta(-i\omega_n + \xi_\alpha)] e^{i\omega_n \eta}, \tag{1.F.8}$$

which can be calculated from the integral

$$I_\alpha = \oint_{\mathcal{C}} \frac{dz}{2i\pi} n_\zeta(z) \ln[\beta(-z+\xi_\alpha)] e^{z\eta}. \tag{1.F.9}$$

The contour $\mathcal{C}$ is chosen such as to avoid the branch cut of the function $\ln[\beta(-z+\xi_\alpha)]$ (Fig. 1.17). For non-interacting bosons, the chemical potential is always smaller than $\min_\alpha(\epsilon_\alpha)$. When $\mu \to \min_\alpha(\epsilon_\alpha)$, there is BEC. The condition $\xi_\alpha = \epsilon_\alpha - \mu > 0$ implies that there is no overlap between the branch cut of $\ln[\beta(-z+\xi_\alpha)]$ and the Matsubara frequency $\omega_{n=0} = 0$. Again, the function $e^{z\eta} n_\zeta(z)$ ensures that the circular parts of $\mathcal{C}$ do not contribute to I_α in the limit $R \to \infty$. We therefore obtain

$$\begin{aligned} I_\alpha &= \int_{\xi_\alpha^-}^{\infty} \frac{d\epsilon}{2i\pi} n_\zeta(\epsilon) \ln[\beta(-\epsilon - i\eta + \xi_\alpha)] + \int_{\infty}^{\xi_\alpha^-} \frac{d\epsilon}{2i\pi} n_\zeta(\epsilon) \ln[\beta(-\epsilon + i\eta + \xi_\alpha)] \\ &= \int_{\xi_\alpha^-}^{\infty} \frac{d\epsilon}{2i\pi} n_\zeta(\epsilon) [\ln(-\epsilon - i\eta + \xi_\alpha) - \text{c.c.}], \end{aligned} \tag{1.F.10}$$

[116] Note that this property depends crucially on the factor $e^{z\eta}$ when $\Re(z) < 0$.

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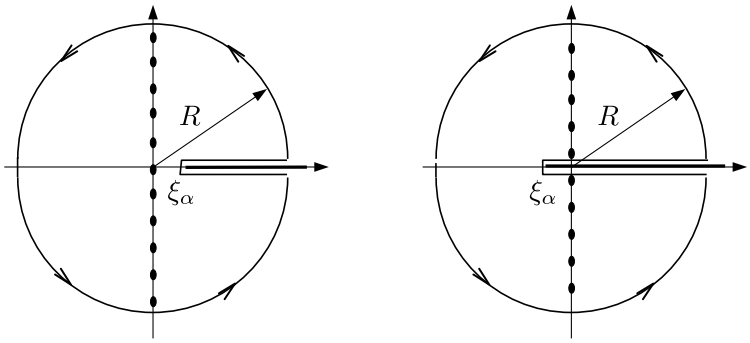


Figure 1.17: Integration contour $\mathcal{C}$ in Eq. (1.F.9) for bosons (left) and fermions (right). The thick line on the real axis shows the branch cut of the function $\ln(-z+\xi_\alpha)$ and the black dots the positions of the imaginary frequencies $i\omega_n$. In the bosonic case, the branch cut is always located on the positive part of the real axis.

where $\xi_\alpha^- = \xi_\alpha - 0^+$.[117] Using

$$n_\zeta(\epsilon) = \frac{\zeta}{\beta}\frac{d}{d\epsilon}\ln\left|1-\zeta e^{-\beta\epsilon}\right| \tag{1.F.11}$$

and integrating by parts,

$$\begin{aligned} I_\alpha &= -\frac{\zeta}{\beta}\int_{\xi_\alpha^-}^{\infty}\frac{d\epsilon}{2i\pi}\ln\left|1-\zeta e^{-\beta\epsilon}\right|\frac{d}{d\epsilon}[\ln(-\epsilon-i\eta+\xi_\alpha)-\text{c.c.}] \\ &= -\frac{\zeta}{\beta}\int_{\xi_\alpha^-}^{\infty}\frac{d\epsilon}{2i\pi}\ln\left|1-\zeta e^{-\beta\epsilon}\right|[-2i\pi\delta(\epsilon-\xi_\alpha)] \\ &= \frac{\zeta}{\beta}\ln\left(1-\zeta e^{-\beta\xi_\alpha}\right) \end{aligned} \tag{1.F.12}$$

since $\xi_\alpha > 0$ for bosons ($\zeta = 1$). On the other hand, the residue theorem gives

$$I_\alpha = \frac{\zeta}{\beta}\sum_{\omega_n}\ln(-i\omega_n+\xi_\alpha)e^{i\omega_n\eta}. \tag{1.F.13}$$

From (1.F.12) and (1.F.13), we finally deduce

$$\Omega_0 = \frac{\zeta}{\beta}\sum_\alpha \ln\left(1-\zeta e^{-\beta\xi_\alpha}\right), \tag{1.F.14}$$

which is the familiar result for non-interacting particles (Sec. 1.C.5).

Example 3

In some cases, it is possible to compute the Matsubara sum more directly. For instance, the particle-hole susceptibility (Lindhard function) encountered in Chaps. 4 and 5 reads

$$\chi^0_{nn}(q) = -\frac{2}{\beta\mathcal{V}}\sum_k G_0(k)G_0(k+q), \tag{1.F.15}$$

[117] It should be understood that the limit $\eta \to 0^+$ is taken at fixed $\xi_\alpha - \xi_\alpha^- > 0$. This ensures that the small vertical part of the integration contour does not contribute and the boundary term in (1.F.12) vanishes.

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where $G_0(k) = (i\omega_n - \xi_{\mathbf{k}})^{-1}$ is the non-interacting Green function and ω_n a fermionic Matsubara frequency. $q = (\mathbf{q}, i\omega_\nu)$ and ω_ν is a bosonic Matsubara frequency. Adding the factor $e^{i\omega_n \eta}$, which obviously does not change the result, we find

$$\begin{aligned}\chi^0_{nn}(q) &= -\frac{2}{\beta\mathcal{V}} \sum_k \frac{e^{i\omega_n\eta}}{i\omega_\nu + \xi_{\mathbf{k}} - \xi_{\mathbf{k}+\mathbf{q}}} \left(\frac{1}{i\omega_n - \xi_{\mathbf{k}}} - \frac{1}{i\omega_n + i\omega_\nu - \xi_{\mathbf{k}+\mathbf{q}}} \right) \\ &= -2 \int_{\mathbf{k}} \frac{n_F(\xi_{\mathbf{k}}) - n_F(\xi_{\mathbf{k}+\mathbf{q}})}{i\omega_\nu + \xi_{\mathbf{k}} - \xi_{\mathbf{k}+\mathbf{q}}},\end{aligned} \tag{1.F.16}$$

using (1.F.7).

Similarly the pairing susceptibility

$$\chi_0(q) = \frac{1}{\beta\mathcal{V}} \sum_k G_0(k) G_0(q-k) \tag{1.F.17}$$

which arises in the theory of superconductivity [Eq. (7.230)] is given by

$$\begin{aligned}\chi_0(q) &= \frac{1}{\beta\mathcal{V}} \sum_k \frac{e^{i\omega_n\eta}}{i\omega_\nu - \xi_{\mathbf{k}} - \xi_{\mathbf{q}-\mathbf{k}}} \left(\frac{1}{i\omega_n - \xi_{\mathbf{k}}} + \frac{1}{i\omega_\nu - i\omega_n - \xi_{\mathbf{q}-\mathbf{k}}} \right) \\ &= \int_{\mathbf{k}} \frac{n_F(\xi_{\mathbf{k}}) - n_F(-\xi_{\mathbf{q}-\mathbf{k}})}{i\omega_\nu - \xi_{\mathbf{k}} - \xi_{\mathbf{q}-\mathbf{k}}} \\ &= \int_{\mathbf{k}} \frac{n_F(\xi_{\mathbf{k}}) + n_F(\xi_{\mathbf{q}-\mathbf{k}}) - 1}{i\omega_\nu - \xi_{\mathbf{k}} - \xi_{\mathbf{q}-\mathbf{k}}}.\end{aligned} \tag{1.F.18}$$

Suppose that instead of $\chi^0_{nn}(q)$ [Eq. (1.F.15)], we wish to calculate

$$\tilde{\chi}^0_{nn}(q) = -\frac{2}{\beta\mathcal{V}} \sum_k G(k) G(k+q) \tag{1.F.19}$$

for a fermion system, where $G(k) = [i\omega_n - \xi_{\mathbf{k}} - \Sigma(k)]^{-1}$. The standard method starts from the integral

$$I = \oint_{\mathcal{C}} \frac{dz}{2i\pi} n_F(z) G(\mathbf{k}, z) G(\mathbf{k}+\mathbf{q}, z + i\omega_\nu). \tag{1.F.20}$$

Since $G(\mathbf{k}, z)$ can have branch cuts on the real axis $\Im(z) = 0$ (Sec. 3.5), the contour $\mathcal{C}$ is chosen as shown in Fig. 1.18. The circular parts of the contour do not contribute when the radius R of the circle is sent to infinity, and we find

$$\begin{aligned}I = &\int_{-\infty}^{\infty} \frac{d\epsilon}{2i\pi} n_F(\epsilon) \left[G^R(\mathbf{k}, \epsilon) - G^A(\mathbf{k}, \epsilon) \right] G(\mathbf{k}+\mathbf{q}, \epsilon + i\omega_\nu) \\ &+ \int_{-\infty}^{\infty} \frac{d\epsilon}{2i\pi} n_F(\epsilon) \left[G^R(\mathbf{k}+\mathbf{q}, \epsilon) - G^A(\mathbf{k}+\mathbf{q}, \epsilon) \right] G(\mathbf{k}, \epsilon - i\omega_\nu),\end{aligned} \tag{1.F.21}$$

where $G^R(\mathbf{k}, \epsilon) = G^A(\mathbf{k}, \epsilon)^*$ denotes the retarded Green function $G(\mathbf{k}, i\omega_n \to \epsilon + i\eta)$. $G(\mathbf{k}, z)$ being analytic for $\Im(z) \neq 0$ (Sec. 3.5), the residue theorem gives

$$I = -\frac{1}{\beta} \sum_{\omega_n} G(k) G(k+q), \tag{1.F.22}$$

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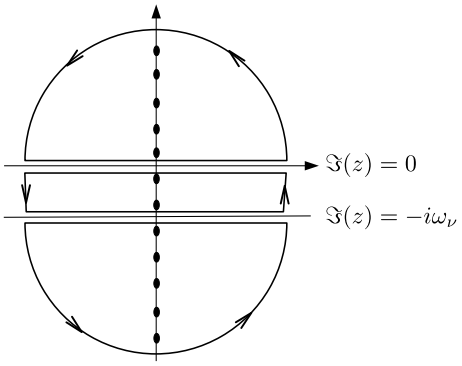


Figure 1.18: Integration contour $\mathcal{C}$ in Eq. (1.F.20).

which leads to

$$\tilde{\chi}^0_{nn}(q) = -2\int_{\mathbf{k}}\int_{-\infty}^{\infty} d\epsilon\, n_F(\epsilon)\left[A(\mathbf{k},\epsilon)G(\mathbf{k}+\mathbf{q},\epsilon+i\omega_\nu) + A(\mathbf{k}+\mathbf{q},\epsilon)G(\mathbf{k},\epsilon-i\omega_\nu)\right], \tag{1.F.23}$$

where $A(\mathbf{k},\epsilon) = -\frac{1}{\pi}\Im[G^R(\mathbf{k},\epsilon)]$ is the spectral function of the one-particle Green function (Sec. 3.5).

Alternatively, one can use the spectral representation (3.239) to obtain

$$\begin{aligned}\frac{1}{\beta}\sum_{\omega_n} G(k)G(k+q) &= \frac{1}{\beta}\sum_{\omega_n}\int_{-\infty}^{\infty} d\epsilon d\epsilon' \frac{A(\mathbf{k},\epsilon)A(\mathbf{k}+\mathbf{q},\epsilon')}{(i\omega_n-\epsilon)(i\omega_n+i\omega_\nu-\epsilon')} \\ &= \int_{-\infty}^{\infty} d\epsilon d\epsilon' A(\mathbf{k},\epsilon)A(\mathbf{k}+\mathbf{q},\epsilon')\frac{n_F(\epsilon)-n_F(\epsilon')}{i\omega_\nu+\epsilon-\epsilon'},\end{aligned} \tag{1.F.24}$$

where we have used (1.F.7). Using again the spectral representation, Eq. (1.F.24) can be rewritten as

$$\frac{1}{\beta}\sum_{\omega_n} G(k)G(k+q) = \int_{-\infty}^{\infty} d\epsilon\, n_F(\epsilon)[A(\mathbf{k},\epsilon)G(\mathbf{k}+\mathbf{q},\epsilon+i\omega_\nu) + A(\mathbf{k}+\mathbf{q},\epsilon)G(\mathbf{k},\epsilon-i\omega_\nu)], \tag{1.F.25}$$

which leads to (1.F.23).

Example 4

Finally, we note that it is sometimes possible to express Matsubara sums in terms of special functions. For example, the coefficient of the quartic term in the Ginzburg-Landau expansion of the free energy of a superconductor [Eq. (7.284)] depends on the Matsubara sum

$$S = \frac{1}{\beta}\sum_{\omega_n}\frac{1}{|\omega_n|^3} = \frac{1}{4\pi^3T^2}\sum_{n=0}^{\infty}\frac{1}{\left(n+\frac{1}{2}\right)^3} = \frac{1}{4\pi^3T^2}\zeta\left(3,\frac{1}{2}\right), \tag{1.F.26}$$

where

$$\zeta(z,q) = \sum_{n=0}^{\infty}\frac{1}{(n+q)^z} \tag{1.F.27}$$

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is the generalized Riemann Zeta function. Using $\zeta(z,1/2)=(2^z-1)\zeta(z)$ where

$$\zeta(z)=\sum_{n=1}^{\infty}\frac{1}{n^z} \tag{1.F.28}$$

is the Riemann Zeta function, we finally obtain

$$S=\frac{7\zeta(3)}{4\pi^3T^2}, \tag{1.F.29}$$

with $\zeta(3)\simeq 1.2021$.

Appendix 1.G Perturbation theory in the operator formalism

In this section we briefly discuss perturbation theory in the operator formalism.[118] We consider a quantum system with a time-independent (grand canonical, i.e. including the term $-\mu\hat{N}$) Hamiltonian $\hat{H}=\hat{H}_0+\hat{V}$ where $\hat{H}_0$ is the non-interacting part while $\hat{V}$ is due to the interactions between particles.

1.G.1 Zero-temperature formalism

1.G.1.1 Interaction representation

In the Schrödinger representation the operators $\hat{O}_S$ are time independent (in the absence of time-dependent external fields) and the state vectors evolve in time according to $|\Psi_S(t)\rangle=\hat{U}_S(t,t')|\Psi_S(t')\rangle$ where $\hat{U}_S(t,t')=e^{-i\hat{H}(t-t')}$ is the unitary time-evolution operator and $\hat{H}$ the (time-independent) Hamiltonian. Alternatively, one can use the Heisenberg picture where the state vectors $|\Psi_H\rangle\equiv|\Psi(0)\rangle$ are time-independent while the operators, defined by $\hat{O}_H(t)=e^{i\hat{H}t}\hat{O}_Se^{-i\hat{H}t}$, become time dependent (Appendix 1.B). In the interaction representation both state vectors and operators are time dependent,

$$\begin{aligned}|\Psi_I(t)\rangle&=e^{i\hat{H}_0t}e^{-i\hat{H}t}|\Psi_S(0)\rangle,\\ \hat{O}_I(t)&=e^{i\hat{H}_0t}\hat{O}_Se^{-i\hat{H}_0t}.\end{aligned} \tag{1.G.1}$$

These three representations coincide at time $t=0$ and are equivalent since

$$\langle\hat{O}\rangle=\langle\Psi_S(t)|\hat{O}_S|\Psi_S(t)\rangle=\langle\Psi_H|\hat{O}_H(t)|\Psi_H\rangle=\langle\Psi_I(t)|\hat{O}_I(t)|\Psi_I(t)\rangle. \tag{1.G.2}$$

We may now define a time-evolution propagator

$$\hat{U}_I(t,t')=e^{i\hat{H}_0t}e^{-i\hat{H}(t-t')}e^{-i\hat{H}_0t'} \tag{1.G.3}$$

in the interaction representation such that $|\Psi_I(t)\rangle=\hat{U}_I(t,t')|\Psi_I(t')\rangle$. This propagator satisfies the equation

$$i\partial_t\hat{U}_I(t,t')=\hat{V}_I(t)\hat{U}_I(t,t')\quad\text{with}\quad\hat{U}_I(t,t)=1. \tag{1.G.4}$$

This equation can be solved by iteration,

$$\hat{U}_I(t,t')=\hat{U}_I(t',t')-i\int_{t'}^{t}dt_1\hat{V}_I(t_1)\hat{U}_I(t_1,t')$$

[118] For more details see Refs. [19–21].

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$$= 1 - i\int_{t'}^{t} dt_1 \hat{V}_I(t_1) + (-i)^2 \int_{t'}^{t} dt_1 \hat{V}_I(t_1) \int_{t'}^{t_1} dt_2 \hat{V}_I(t_2) + \cdots \tag{1.G.5}$$

Each term in this expansion can be rewritten in a simpler form using the time-ordering operator T, introduced in Sec. 1.2.3, which orders the operators from right to left in ascending time order and adds a factor $\zeta = \pm$ for each permutation of operators.[119] For instance, for the second-order term, one has

$$\begin{aligned}
&\int_{t'}^{t} dt_1 \hat{V}_I(t_1) \int_{t'}^{t_1} dt_2 \hat{V}_I(t_2) \\
&\quad = \frac{1}{2}\int_{t'}^{t} dt_1 \hat{V}_I(t_1) \int_{t'}^{t_1} dt_2 \hat{V}_I(t_2) + \frac{1}{2}\int_{t'}^{t_1} dt_2 \hat{V}_I(t_2) \int_{t'}^{t_2} dt_1 \hat{V}_I(t_1) \\
&\quad = \frac{1}{2}\int_{t'}^{t} dt_1 \int_{t'}^{t} dt_2 [\Theta(t_1 - t_2)\hat{V}_I(t_1)\hat{V}_I(t_2) + \Theta(t_2 - t_1)\hat{V}_I(t_2)\hat{V}_I(t_1)] \\
&\quad \equiv \frac{1}{2}\int_{t'}^{t} dt_1 \int_{t'}^{t} dt_2\, T[\hat{V}_I(t_1)\hat{V}_I(t_2)].
\end{aligned} \tag{1.G.6}$$

This yields a compact expression of the propagator,

$$\hat{U}_I(t,t') = T \exp\left\{-i\int_{t'}^{t} dt''\, \hat{V}_I(t'')\right\}, \tag{1.G.7}$$

which satisfies the following properties

$$\hat{U}_I^\dagger(t,t') = \hat{U}_I(t',t), \qquad \hat{U}_I(t,t'')\hat{U}_I(t'',t') = \hat{U}_I(t,t'). \tag{1.G.8}$$

1.G.1.2 Gell-Mann and Low theorem

We now assume the interaction to be switched on adiabatically, i.e.

$$\hat{H}(t) = \hat{H}_0 + e^{-\epsilon|t|}\hat{V}, \tag{1.G.9}$$

where $\epsilon \to 0^+$. It is readily verified that Eq. (1.G.7) is still satisfied provided that we replace $\hat{V}_I(t)$ by $e^{-\epsilon|t|}\hat{V}_I(t)$. We shall denote the propagator by $\hat{U}_{I,\epsilon}(t,t')$ to emphasize that it depends on ϵ. At very large times, both in the past and in the future, the Hamiltonian reduces to $\hat{H}_0$, which we know how to solve. Suppose that in the distant past ($t \to -\infty$) the system is in the ground state $|\Phi_0\rangle$ of $\hat{H}_0$. Since the interaction is switched on adiabatically, one can expect the system at time t to be an eigenstate $|\Psi_0\rangle$ of the "instantaneous" Hamiltonian $\hat{H}(t)$.

The Gell-Mann and Low theorem [30] states that if the quantity

$$\lim_{\epsilon\to 0^+} \frac{\hat{U}_{I,\epsilon}(0,-\infty)|\Phi_0\rangle}{\langle\Phi_0|\hat{U}_{I,\epsilon}(0,-\infty)|\Phi_0\rangle} = \lim_{\epsilon\to 0^+} \frac{|\Psi_{0,\epsilon}\rangle}{\langle\Phi_0|\Psi_{0,\epsilon}\rangle} \tag{1.G.10}$$

exists to all orders in perturbation theory, then it is an eigenstate of $\hat{H}(0) \equiv \hat{H} = \hat{H}_0 + \hat{V}$,

$$\lim_{\epsilon\to 0^+} \hat{H}\frac{|\Psi_{0,\epsilon}\rangle}{\langle\Phi_0|\Psi_{0,\epsilon}\rangle} = \lim_{\epsilon\to 0^+} E\frac{|\Psi_{0,\epsilon}\rangle}{\langle\Phi_0|\Psi_{0,\epsilon}\rangle}. \tag{1.G.11}$$

[119] The interaction Hamiltonian $\hat{V}$ is necessarily a "bosonic" operator; for fermionic particles it consists of a product of an even number of creation/annihilation operators.

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An essential point of the theorem is that the limit $\epsilon \to 0^+$ does not separately exist for the numerator and the denominator in (1.G.10) and we cannot directly identify $\lim_{\epsilon\to 0^+} |\Psi_{0,\epsilon}\rangle$ with an eigenstate $|\Psi_0\rangle$ of $\hat{H}$ [20]. The denominator in (1.G.11) serves to cancel an infinite phase $\propto 1/\epsilon$ which appears in $|\Psi_{0,\epsilon}\rangle$. If $|\Phi_0\rangle$ is the ground state of $\hat{H}_0$, the corresponding eigenstate of $\hat{H}$ is usually the ground state but this is by no means necessary; in the following we assume that the state that develops out from $|\Phi_0\rangle$ is non-degenerate and the ground state of $\hat{H}$.

The theorem equally applies to

$$\lim_{\epsilon\to 0^+} \frac{\hat{U}_{I,\epsilon}(0,\infty)|\Phi_0\rangle}{\langle\Phi_0|\hat{U}_{I,\epsilon}(0,\infty)|\Phi_0\rangle} = \lim_{\epsilon\to 0^+} \frac{|\tilde{\Psi}_{0,\epsilon}\rangle}{\langle\Phi_0|\tilde{\Psi}_{0,\epsilon}\rangle}. \tag{1.G.12}$$

Here the system is in the state $|\Phi_0\rangle$ for $t \to \infty$ and evolves (backwards in time) to an eigenstate of $\hat{H}$ when $t = 0$. If the ground state of $\hat{H}$ is non-degenerate the two states, obtained from $|\Psi(t=-\infty)\rangle = |\Phi_0\rangle$ and $|\Psi(t=\infty)\rangle = |\Phi_0\rangle$, must coincide at $t = 0$ up to a phase factor. But the equality

$$\langle\Phi_0|\frac{|\Psi_{0,\epsilon}\rangle}{\langle\Phi_0|\Psi_{0,\epsilon}\rangle} = \langle\Phi_0|\frac{|\tilde{\Psi}_{0,\epsilon}\rangle}{\langle\Phi_0|\tilde{\Psi}_{0,\epsilon}\rangle} = 1 \tag{1.G.13}$$

implies that the phase factor necessarily vanishes. Thus

$$\lim_{\epsilon\to 0^+} \frac{|\Psi_{0,\epsilon}\rangle}{\langle\Phi_0|\Psi_{0,\epsilon}\rangle} = \lim_{\epsilon\to 0^+} \frac{|\tilde{\Psi}_{0,\epsilon}\rangle}{\langle\Phi_0|\tilde{\Psi}_{0,\epsilon}\rangle}. \tag{1.G.14}$$

The unitary operator $\hat{U}_{I,\epsilon}$ preserves the norm of the states. Thus, if $|\Phi_0\rangle$ is normalized, so are $|\Psi_{0,\epsilon}\rangle$ and $|\tilde{\Psi}_{0,\epsilon}\rangle$. Since these two state vectors become colinear with the interacting ground state $|\Psi_0\rangle$ in the limit $\epsilon \to 0^+$, they defer only by a (actually diverging) phase. This implies that the expectation value of an operator $\hat{O}$ in the interacting ground state can be written as

$$\begin{aligned}\lim_{\epsilon\to 0^+} \frac{\langle\tilde{\Psi}_{0,\epsilon}|\hat{O}|\Psi_{0,\epsilon}\rangle}{\langle\tilde{\Psi}_{0,\epsilon}|\Psi_{0,\epsilon}\rangle} &= \lim_{\epsilon\to 0^+} \frac{\langle\Phi_0|\hat{U}_{I,\epsilon}(\infty,0)\hat{O}_I(0)\hat{U}_{I,\epsilon}(0,-\infty)|\Phi_0\rangle}{\langle\Phi_0|\hat{U}_{I,\epsilon}(\infty,-\infty)|\Phi_0\rangle} \\ &= \lim_{\epsilon\to 0^+} \frac{\langle\Phi_0|T[\hat{U}_{I,\epsilon}(\infty,-\infty)\hat{O}_I(0)]|\Phi_0\rangle}{\langle\Phi_0|\hat{U}_{I,\epsilon}(\infty,-\infty)|\Phi_0\rangle},\end{aligned} \tag{1.G.15}$$

where we have used $\hat{O} = \hat{O}_I(0)$ and (1.G.8) as well as the definition of the time-ordering operator T. Since the expressions in (1.G.15) consist of a ratio, the divergent phase factors cancel and it is permissible to take the limit $\epsilon \to 0^+$.

These results allow us to write the energy of the interacting system as

$$\begin{aligned}E &= E_0 + \lim_{\epsilon\to 0^+} \frac{\langle\tilde{\Psi}_{0,\epsilon}|\hat{V}|\Psi_{0,\epsilon}\rangle}{\langle\tilde{\Psi}_{0,\epsilon}|\Psi_{0,\epsilon}\rangle} \\ &= E_0 + \lim_{\epsilon\to 0^+} \frac{\langle\Phi_0|T[\hat{U}_{I,\epsilon}(\infty,-\infty)\hat{V}_I(0)]|\Phi_0\rangle}{\langle\Phi_0|\hat{U}_{I,\epsilon}(\infty,-\infty)|\Phi_0\rangle}.\end{aligned} \tag{1.G.16}$$

Similarly we can write the real-time one-particle propagator defined in Sec. 1.2.3 as

$$iG(\alpha t, \alpha' t') = \langle\Psi_0|T[\hat{\psi}_{\alpha,H}(t)\hat{\psi}^\dagger_{\alpha',H}(t')]|\Psi_0\rangle$$

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$$= \lim_{\epsilon\to 0^+} \frac{\langle\Phi_0|\hat{U}_{I,\epsilon}(\infty,0)T[\hat{\psi}_{\alpha,H}(t)\hat{\psi}^\dagger_{\alpha',H}(t')]\hat{U}_{I,\epsilon}(0,-\infty)|\Phi_0\rangle}{\langle\Phi_0|\hat{U}_{I,\epsilon}(\infty,-\infty)|\Phi_0\rangle}, \tag{1.G.17}$$

where $\hat{\psi}^{(\dagger)}_{\alpha,H}(t) = e^{i\hat{H}t}\hat{\psi}^{(\dagger)}_\alpha e^{-i\hat{H}t}$ is defined in the Heisenberg representation. Using

$$\hat{\psi}_{\alpha,H}(t)\hat{\psi}^\dagger_{\alpha',H}(t') = \hat{U}_{I,\epsilon}(0,t)\hat{\psi}_{\alpha,I}(t)\hat{U}_{I,\epsilon}(t,t')\hat{\psi}^\dagger_{\alpha',I}(t')\hat{U}_{I,\epsilon}(t',0), \tag{1.G.18}$$

one finds

$$iG(\alpha t,\alpha' t') = \lim_{\epsilon\to 0^+} \frac{\langle\Phi_0|T[\hat{\psi}_{\alpha,I}(t)\hat{\psi}^\dagger_{\alpha',I}(t')\hat{U}_{I,\epsilon}(\infty,-\infty)]|\Phi_0\rangle}{\langle\Phi_0|\hat{U}_{I,\epsilon}(\infty,-\infty)|\Phi_0\rangle}. \tag{1.G.19}$$

1.G.1.3 Wick's theorem

This theorem states that the expectation value of a time-ordered product of operators in the non-interacting ground state is given by the sum of all complete contractions, i.e.

$$\begin{aligned}\langle\Phi_0|T[\hat{\psi}_{\alpha_1,I}(t_1)\cdots\hat{\psi}_{\alpha_n,I}(t_n)\hat{\psi}^\dagger_{\alpha'_n,I}(t'_n)\cdots\hat{\psi}^\dagger_{\alpha'_1,I}(t'_1)]|\Phi_0\rangle \\ = \sum_{P\in S_n}\zeta^P\langle\Phi_0|T[\hat{\psi}_{\alpha_1,I}(t_1)\hat{\psi}^\dagger_{\alpha'_{P(1)},I}(t'_{P(1)})]|\Phi_0\rangle\cdots \\ \cdots\langle\Phi_0|T[\hat{\psi}_{\alpha_n,I}(t_n)\hat{\psi}^\dagger_{\alpha'_{P(n)},I}(t'_{P(n)})]|\Phi_0\rangle,\end{aligned} \tag{1.G.20}$$

where the sum is over all permutations P of $\{1,\cdots,n\}$ and ζ^P is equal to one for bosons ($\zeta=1$) and to the signature of P for fermions ($\zeta=-1$), and

$$\langle\Phi_0|T[\hat{\psi}_{\alpha,I}(t)\hat{\psi}^\dagger_{\alpha',I}(t')]|\Phi_0\rangle = iG_0(\alpha t,\alpha' t') \tag{1.G.21}$$

is the non-interacting one-particle propagator.

In the basis where the Hamiltonian $\hat{H}_0 = \sum_\alpha \xi_\alpha\hat{\psi}^\dagger_\alpha\hat{\psi}_\alpha$ is diagonal, one easily obtains $\hat{\psi}_{\alpha,I}(t) = e^{-i\xi_\alpha t}\hat{\psi}_\alpha$ and $\hat{\psi}^\dagger_{\alpha,I}(t) = e^{i\xi_\alpha t}\hat{\psi}^\dagger_\alpha$, so that

$$iG_0(\alpha t,\alpha' t') = \delta_{\alpha,\alpha'}e^{-i\xi_\alpha(t-t')}[\Theta(t-t')(1+\zeta n_\alpha)+\Theta(t'-t)\zeta n_\alpha], \tag{1.G.22}$$

where $n_\alpha = \langle\hat{\psi}^\dagger_\alpha\hat{\psi}_\alpha\rangle$ is given by the Fermi-Dirac function $n_F(\xi_F, T=0)$ for fermions and the Bose-Einstein function $n_B(\xi_\alpha, T=0)$ for bosons. In the latter case $n_B(\xi_\alpha,T=0) = \lim_{\beta\to\infty}(e^{\beta\xi_\alpha}-1)^{-1}$ is defined only for a negative chemical potential μ (assuming $\min_\alpha\epsilon_\alpha = 0$ and that the ϵ_α's form a continuum), which corresponds to the vacuum state (no particles in the ground state). As discussed in Sec. 1.8.2 perturbation theory for finite-density boson systems requires to take into account the BEC of the particles in the minimum-energy one-particle state from the outset.

1.G.1.4 Perturbation theory and Feynman diagrams

Perturbation theory is obtained by expanding

$$\hat{U}_{I,\epsilon}(\infty,-\infty) = \sum_{n=0}^\infty \frac{(-i)^n}{n!}\int_{-\infty}^\infty dt_1\cdots dt_n\, e^{-\epsilon(|t_1|+\cdots+|t_n|)}T[\hat{V}_I(t_1)\cdots\hat{V}_I(t_n)] \tag{1.G.23}$$

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in series. From now on the discussion is similar to that of Sec. 1.5. When computing a physical quantity, e.g. the ground state energy E or the one-particle propagator $G(\alpha t, \alpha' t')$, one uses Wick's theorem to represent each term in the expansion by Feynman diagrams where the non-interacting propagator is presented by a directed line and the interaction by a vertex with two incoming and two outgoing lines. Moreover the sole role of $\langle\Phi_0|\hat{U}_I(\infty,-\infty)|\Phi_0\rangle$ in the denominators of Eqs. (1.G.16) and (1.G.19) is to remove the non-connected diagrams. Thus both E and $G(\alpha t, \alpha' t')$ (as well as higher-order Green functions) are given by connected diagrams (the connected diagrams contributing to E are the so-called vacuum fluctuation diagrams). Once the precise diagrammatic rules are established (see Sec. 1.5), whether they were derived from the operator formalism or the functional integral one does not matter.

1.G.2 Finite-temperature formalism

In the finite-temperature formalism, the interaction picture is defined by

$$\begin{aligned} \hat{O}_I(\tau) &= e^{\tau\hat{H}_0}\hat{O}_S e^{-\tau\hat{H}_0}, \\ \hat{U}_I(\tau,\tau') &= e^{\tau\hat{H}_0}e^{-(\tau-\tau')\hat{H}}e^{-\tau'\hat{H}_0}, \end{aligned} \tag{1.G.24}$$

where $\tau, \tau' \in [0,\beta]$. $\hat{U}_I$ satisfies the equation

$$\partial_\tau \hat{U}_I(\tau,\tau') = -\hat{V}_I(\tau)\hat{U}_I(\tau,\tau') \quad \text{with} \quad \hat{U}_I(\tau,\tau) = 1 \tag{1.G.25}$$

and can therefore be expressed as

$$\hat{U}_I(\tau,\tau') = T_\tau \exp\left\{-\int_{\tau'}^{\tau} d\tau''\, \hat{V}_I(\tau'')\right\}, \tag{1.G.26}$$

where T_τ is the imaginary-time-ordering operator [Eq. (1.194)]. One therefore has

$$\begin{aligned} \hat{U}_I(\tau,\tau') &= \hat{U}_I(\tau-\tau'), \\ \hat{U}_I(\tau,\tau'')\hat{U}_I(\tau'',\tau') &= \hat{U}_I(\tau,\tau'), \\ \hat{U}_I(\tau,\tau') &= \hat{U}_I^{-1}(\tau',\tau). \end{aligned} \tag{1.G.27}$$

All physical quantities can be expressed as a function of $\hat{U}_I$. The partition function Z is defined by

$$\frac{Z}{Z_0} = \frac{\mathrm{Tr}\, e^{-\beta\hat{H}}}{Z_0} = \frac{1}{Z_0}\mathrm{Tr}[e^{-\beta\hat{H}_0}\hat{U}_I(\beta,0)] \equiv \langle\hat{U}_I(\beta,0)\rangle_0, \tag{1.G.28}$$

where $Z_0 = \mathrm{Tr}\, e^{-\beta\hat{H}_0}$ and $\langle\cdots\rangle_0 = Z_0^{-1}\mathrm{Tr}(e^{-\beta\hat{H}_0}\cdots)$. The one-particle Green function reads

$$\begin{aligned} G(\alpha\tau,\alpha'\tau') &= -\langle T_\tau[\hat{\psi}_{\alpha,H}(\tau)\hat{\psi}^\dagger_{\alpha',H}(\tau')]\rangle \\ &= -\frac{1}{Z}\mathrm{Tr}\{e^{-\beta\hat{H}}T_\tau[e^{\tau\hat{H}}\hat{\psi}_\alpha e^{-(\tau-\tau')\hat{H}}\hat{\psi}^\dagger_{\alpha'}e^{-\tau'\hat{H}}]\} \\ &= -\frac{\langle T_\tau[\hat{U}_I(\beta,0)\hat{\psi}_{\alpha,I}(\tau)\hat{\psi}^\dagger_{\alpha',I}(\tau')]\rangle_0}{\langle\hat{U}_I(\beta,0)\rangle_0}, \end{aligned} \tag{1.G.29}$$

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where $\hat{\psi}^{(\dagger)}_{\alpha,H}(\tau) = e^{\tau\hat{H}}\hat{\psi}^{(\dagger)}_{\alpha}e^{-\tau\hat{H}}$ are creation and annihilation operators defined in the Heisenberg picture.

Perturbation theory is obtained by expanding

$$\hat{U}_I(\beta,0) = \sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\int_0^\beta d\tau_1\cdots\int_0^\beta d\tau_n\, T_\tau[\hat{V}_I(\tau_1)\cdots\hat{V}_I(\tau_n)] \tag{1.G.30}$$

and using Wick's theorem

$$\begin{aligned}&\langle T_\tau[\hat{\psi}_{\alpha_1}(\tau_1)\cdots\hat{\psi}_{\alpha_n}(\tau_n)\hat{\psi}^\dagger_{\alpha'_n}(\tau'_n)\cdots\hat{\psi}^\dagger_{\alpha'_1}(\tau'_1)]\rangle_0\\ &\quad=\sum_{P\in S_n}\zeta^P\langle T_\tau[\hat{\psi}_{\alpha_1}(\tau_1)\hat{\psi}^\dagger_{\alpha'_{P(1)}}(\tau'_{P(1)})]\rangle_0\cdots\langle T_\tau[\hat{\psi}_{\alpha_n}(\tau_n)\hat{\psi}^\dagger_{\alpha'_{P(n)}}(\tau'_{P(n)})]\rangle_0,\end{aligned} \tag{1.G.31}$$

where

$$-\langle T_\tau[\hat{\psi}_\alpha(\tau)\hat{\psi}^\dagger_{\alpha'}(\tau')]\rangle_0 = G_0(\alpha\tau,\alpha'\tau') \tag{1.G.32}$$

is the non-interacting Green function in imaginary time. In the diagonal basis where $\hat{H}_0 = \sum_\alpha \xi_\alpha\hat{\psi}^\dagger_\alpha\hat{\psi}_\alpha$,

$$G_0(\alpha\tau,\alpha'\tau') = -\delta_{\alpha,\alpha'}e^{-\xi_\alpha(\tau-\tau')}[\Theta(\tau-\tau')(1+\zeta n_\alpha)+\Theta(\tau'-\tau)\zeta n_\alpha]. \tag{1.G.33}$$

From now on the discussion is similar to that of Secs. 1.5 and 1.G.1. Both $\ln(Z/Z_0)$ (linked cluster theorem, Sec. 1.5.2) and $G(\alpha\tau,\alpha'\tau')$, as well as higher-order Green functions, are given by the sum of connected diagrams. We refer to Sec. 1.5 for a detailed discussion of the Feynman diagram rules.

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Guide to the bibliography

- Many aspects of Feynman's path integral [1, 2] are reviewed in Schulman's book [3] (see also Refs. [4, 5]).
- Most textbooks on quantum field theory (e.g. Refs. [5–8]) discuss the functional integral formalism, perturbation theory, as well as the effective action (the generating functional of 1PI vertices).
- Functional integrals in non-relativistic systems and their applications to condensed-matter physics are discussed in [9, 10].
- The Luttinger-Ward functional (the generating functional of 2PI vertices) was introduced in Refs. [11–15] in condensed-matter physics and generalized to relativistic field theories in [16]. For a general introduction see Ref. [17].
- The quantization of the electromagnetic field is presented in detail in [18].
- The "traditional" approach (based on the operator formalism) to non-relativistic quantum field theory is described in Refs. [19–21].

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Bibliography

[1] R. P. Feynman, *Space-time approach to non-relativistic quantum mechanics*, Rev. Mod. Phys. **20**, 367 (1948).

[2] R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals* (McGrawHill, New York, 1965).

[3] L. S. Schulman, *Techniques and Applications of Path Integration* (Dover, 2005).

[4] J. Zinn-Justin, *Intégrale de Chemin en Mécanique Quantique: Introduction (EDP Sciences, 2003).*

[5] H. Kleinert, Path Integrals in Quantum Mechanics, Statistics, Polymer Physics, and Financial Markets (World Scientific, 2005).

[6] J. Zinn-Justin, Quantum Field Theory and Critical Phenomena, Third Edition (Clarendon Press, Oxford, 1996).

[7] M. E. Peskin and D. V. Schroeder, An Introduction to Quantum Field Theory (Westview Press, 1995).

[8] S. Weinberg, The Theory of Quantum Fields, Vols. I and II (Cambridge University Press, 1995).

[9] J. W. Negele and H. Orland, Quantum Many-Particle Systems (Addison-Wesley, 1988).

[10] A. Altland and B. Simons, Condensed-Matter Field Theory (Cambridge University Press, 2006).

[11] J. M. Luttinger, Ground-state energy of a many-fermion system. II, Phys. Rev. **118***, 1417 (1960).*

[12] G. Baym and L. P. Kadanoff, Conservation laws and correlation functions, Phys. Rev. **124***, 287 (1961).*

[13] G. Baym, Self-consistent approximations in many-body systems, Phys. Rev. **127***, 1391 (1962).*

[14] C. De Dominicis and P. C. Martin, Stationary entropy principle and renormalization in normal and superfluid systems. I. Algebraic formulation, J. Math. Phys. **5***, 14 (1964).*

[15] C. De Dominicis and P. C. Martin, Stationary entropy principle and renormalization in normal and superfluid systems. II. Diagrammatic formulation, J. Math. Phys. **5***, 31 (1964).*

[16] J. M. Cornwall, R. Jackiw and E. Tomboulis, Effective action for composite operators, Phys. Rev. D **10***, 2428 (1974).*

[17] G. Stefanucci and R. van Leeuwen, Nonequilibrium Many-Body Theory of Quantum Systems: A Modern Introduction (Cambridge University Press, 2013).

[18] C. Cohen-Tannoudji, J. Dupont-Roc, and G. Grynberg, Photons and Atoms (Wiley, 1989).

[19] A. A. Abrikosov, L. P. Gor'kov, and I. E. Dzyaloshinski, Methods of Quantum Field Theory in Statistical Physics (Dover, 1975).

[20] A. L. Fetter and J. D. Walecka, Quantum Theory of Many-Particle Systems (Dover, 2003).

[21] G. D. Mahan, Many-Particle Physics, 2nd edition (Plenum press, New York, 1990).

[22] H. Goldstein, Classical Mechanics (Addison-Wesley, Reading, Massachussets, 1950).

[23] T. Kibble, Classical Mechanics (Imperial College Press, London, 2004).

[24] L. I. Schiff, Quantum Mechanics (McGraw-Hill, 1955).

[25] A. Messiah, Quantum Mechanics, Vols. I and II (North-Holland, Amsterdam, 1962).

[26] C. Cohen-Tannoudji, B. Diu, and F. Laloë, Quantum Mechanics, Vols. I and II (Wiley and Herman, Paris, 1977).

[27] R. Shankar, Principles of Quantum Mechanics, second edition (Plenum Press, New York, 1994).

[28] R. Balian, From Microphysics to Macrophysics, Vols. I and II (Springer-Verlag, 1991).

[29] R. J. Glauber, Coherent and incoherent states of the radiation field, Phys. Rev. **131***, 2766 (1963).*

[30] M. Gell-Mann and F. Low, Bound states in quantum field theory, Phys. Rev. **84***, 350 (1951).*

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